TS5000 Robot Controller

TP5000 Operation Manual

(Compatible Controllers: TS5000 Series)

Ver. 10

March 4, 2024

IMPORTANT

- Ensure that this manual is provided to the actual final user of the industrial robot.
- Be sure to read this manual before operating the industrial robot.
- After reading through this manual, keep it nearby for future reference.

Shibaura Machine

The controller supported by this instruction manual is the TS5000.

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Introduction

Thank you for purchasing this SHIBAURA MACHINE robot system.
This manual describes the operating method for teaching and operating the robot controller TS5000 using the teach pendant TP5000 and the details of the TP5000 screens.
For the operating method during maintenance, refer to the "Robot Manual" and "Controller Manual" according to the robot and controller to be used.

This manual consists of the following parts.

Part 1: Basic Operation
This describes the TP5000 specifications and basic operating methods.

Part 2: Operating Procedures
This describes the TP5000 operating methods focusing on the operating procedures.

Part 3: Screen Details
This describes the details of the TP5000 screens.

Manufacturer

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Please contact the client at purchase time if the customer has purchased this controller outside Japan.

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Usage Scope

This system was designed for supporting automation of a wide range of operations, including transportation operations such as workpiece movement and alignment and assembly operations such as part insertion and screw tightening.
We ask that customers use this system for the purpose of supporting automation of these operations and incorporating into automation equipment.

This system was not designed for use in operations where contact is made with people (operations in medical treatment, nursing care, etc.), for incorporating into equipment used for sustaining human life (life support devices, infant incubators, etc.), for incorporating into equipment used for transporting people (automobile and other transportation equipment, airplanes and other aerospace equipment, roller coasters and other amusement park equipment), for equipment used for protecting human life or the human body (safety devices, etc.), or similar equipment.
This system was also not designed for incorporating into equipment for the purpose of controlling high amounts of heat (nuclear control equipment, combustion furnace control equipment, etc.).
In cases where a fault or malfunction of this product could directly threaten human life or cause an injury, its use in operations or incorporating into equipment is prohibited. If this product is used, the customer assumes all responsibility for ensuring safe operation.

Operating Procedures

This describes the operating conditions for TS5000. Usage under different conditions could expose the user to unexpected risks. Be sure to follow the required conditions.

1. Main Specifications of Product Refer to the robot manual for your robot.
2. Usage Restrictions Tables 1 to 6 each show the "Restricted Specifications Sheet," in which the usage restrictions are described.

Table 1. Restricted Specifications Sheet

1. Purpose and application for using product (usage restrictions)	
1) Intended use	To make the robot perform operations for specific purposes (item transport, assembly, inspection, machining, etc.) (1) Transportation [1] Obtain a transportation route to the unpacking site. [2] Use a forklift or other equipment to lift the packing box. [3] Move the packing box to the unpacking site. (2) Unpacking [1] Obtain the space required for unpacking. [2] Cut the packing straps and unpack. [3] Cut the packing straps securing the robot. [4] Peel off the tape from the controller packing box, and unpack. [5] Check that the robot and controller are not damaged. [6] Check the accessories and their quantities. - When transferring the robot to the installation site [7] Dispose of, or store and manage the packing materials. - When storing the robot [7] Secure in place in the same way as the state at delivery, and repack. [8] Move the packing box to the storage site. (3) Transfer - Robot [1] Check the orientation of the robot during transfer. [2] Check that the robot fixing jigs are mounted. [3] If using a crane, mount the supplied eyebolts. [4] Check the tightness of the robot fixing jig fixing bolts. [5] Use a crane or cart to transfer to the installation site. - Controller [1] Use a cart to transfer to the installation site.

Table 1. Restricted Specifications Sheet (continued)

1. Purpose and application for using product (usage restrictions)	
1) Intended use (continued)	(4) Installation [1] Use bolts to secure the robot to the frame and other parts. [2] Fabricate any needed cables. [3] Remove the robot fixing jigs and eyebolts. [4] Install the controller and connect it to the robot. [5] Press the robot by hand to move to a location where it is easy to perform work. [6] Mount the end effector. [7] Connect the required external devices. (5) Teaching [1] Move the robot to the desired position or orientation by guiding it with a teach pendant or pushing it by hand. [2] Teach the position where the robot will operate. [3] Create the operation sequence (program, etc.). [4] Make sure that the signals and data are transferred to and from the external equipment. [5] Check operation at low speed (250 mm/sec or less). [6] While gradually increasing the operating speed, make sure that the operation matches the specific purpose. (6) Operation [1] Start robot operation. [2] Make sure that operation is being performed to match the specific purpose at the specified speed. [3] Stop robot operation. (7) Repacking [1] Remove the end effector. [2] Move the robot to the position and orientation where the fixing jig can be mounted by guiding it with a teach pendant or pushing it by hand. [3] Mount the fixing jig. [4] Turn off the power, and remove the wiring. [5] Obtain the space needed for packing. [6] Use a crane to transfer the robot, and mount the supplied eyebolts. [7] Transfer the robot and controller to the packing site. [8] Pack the robot and controller so that they are in the original delivery state. [9] Obtain a transportation route to the transportation equipment (truck, etc.). [10] Use a forklift to lift the packing box. [11] Load the packing box onto the transportation equipment.

Table 1. Restricted Specifications Sheet (continued)

1. Purpose and application for using product (usage restrictions)	
1) Intended use (continued)	(8) Maintenance Work performed when the primary power is off: - Robot [1] Check the bolts for any looseness and backlash and re-tighten them. [2] Make sure that the cable clamp is not loose. [3] Check for any damage. [4] Clean off any dust. [5] Inspect and apply corrosion inhibitor. [6] Check for any wear on the cable and air tube. [7] Check the operation of each joint and check origin position for misalignment. - Controller [1] Replace the memory backup batteries in the controller. Work performed when the primary power is on: [1] Check the grease state and apply grease (when the servo power is off). [2] Check for any abnormal vibrations and sounds during operation (when the servo power is on). [3] Check the teaching point for any misalignment (when the servo power is on). [4] Replace the backup batteries for the robot position detector (when the servo power is off). (9) Disposal [1] Remove the end effector. [2] Move the robot to the position and orientation where the fixing jig can be mounted by guiding it with a teach pendant or pushing it by hand. [3] Mount the fixing jig. [4] Turn off the power, and remove the wiring. [5] Use a crane to transfer the robot, and mount the supplied eyebolts. [6] Transfer the robot and controller to the disposal site. [7] Dispose of using a method compliant with your country’s laws and regulations.

Table 1. Restricted Specifications Sheet (continued)

Purpose and application for using product (usage restrictions)	
2) Foreseeable misuse	- Use for work which applies excessive impact to the robot or to which excessive impact is applied by the robot - Modification of robot, controller, and optional parts and energization with these covers removed - Operation in other than the specified installation state (floor, suspended, and wall-mounted) - Operation on a frame or other structure that does not satisfy the specified strength - Operation under conditions not satisfying the specified installation requirements - Entry into the hazard zone of the robot by mistakenly assuming that operations would have stopped although the stop is just for waiting for an operation signal during operation - Usage under conditions where the specification values are exceeded or usage under improper settings - Stop of the robot caused by power off during operation or when the servo power is on - Unintended operation caused by inadvertently pressing of a button - Operating without checking operation in MANUAL mode - Operating without checking the tightness of the robot and hand set bolts - Unintended operation due to improper wiring - Unintended operation due to incorrect program selection - Riding on or leaning against the robot or hanging on the arm
3) Unexpected start	- Malfunction due to excessive noise - Robot start due to start signal unexpectedly sent from peripheral equipment to the robot - Unintended start due to abnormal communication data - Malfunction due to voltage fluctuation - Power on due to powering after power outage - Malfunction due to failure or other cause

Table 2. Restricted Specifications Sheet

2. Replacement or other modification of program components (time restrictions)	
1) Mechanical restrictions	- Before operating the robot, perform operation at a low speed (20% or less of maximum speed) for five (5) minutes
2) Electrical restrictions	- Replace consumable parts as required or periodically by conducting daily inspection, regular inspection (every 3 and 6 months and every year) and overhauls (every 5 years).

Table 3. Restricted Specifications Sheet

3 Movable range of product (restriction on space)		
1) Work envelope	1st axis 2nd axis 3rd axis 4th axis Additional axis - -	Compliant with 1.4 Specifications Table Compliant with 1.4 Specifications Table Compliant with 1.4 Specifications Table Compliant with 1.4 Specifications Table Compliant with individual specifications. - -
2) Interface	Transportation: The packing box shall be able to be transferred by a transportation worker. Unpacking: The packing box shall be able to be carried in. Space of at least 2 m^3 shall be secured. Transfer: The robot shall be able to be hoisted by a crane. The robot shall be able to be lifted by two or more people and loaded onto a cart or similar device. There shall be no obstacles on the route from the unpacking site to the installation site. Installation: The product shall be installed at a location suitable for the outline and work envelope of the robot. The product shall be installed at a location suitable for the outline and that provides adequate ventilation of the controller. The cable shall be installed at a location where it will not be stepped on by operators. Teaching: Teaching work shall be able to be performed from outside the hazard zone of the robot. Space shall be secured to prevent a worker from being caught in the robot and peripheral equipment if an entry is made into the dangerous range. Operation: A fence or enclosure shall be installed to prevent any operator from approaching the hazard zone of the robot. The robot shall be able to be operated from outside the hazard zone of the robot. An operator in the fence or enclosure shall be visible from the operation location. Repacking: Work shall be able to be performed to remove the hand, mount the fixing jig, separate the robot and controller, and remove the robot and controller from the frame. The robot shall be able to be moved from the installation site to the packing site. The repacking work shall be able to be performed. A forklift or other equipment shall be able to be used to lift the packing box. Maintenance: Space required for maintenance shall be obtained. After maintenance, operation shall be able to be checked from outside the hazard zone of the robot. In the dangerous range, space shall be secured to prevent a worker from being caught in the robot and peripheral equipment. Disposal: Work shall be able to be performed to remove the hand, mount the fixing jig, separate the robot and controller, and remove the robot and controller from the frame. The robot shall be able to be moved from the installation site to the disposal location.	

Table 3. Restricted Specifications Sheet (continued)

3. Movable range of product (restriction on space)	
3) Work environment	- A fence, enclosure, or other protection shall be provided that surrounds the hazard zone of the robot to prevent people from entering the zone during operation. - A display indicating the robot operation status shall be provided to indicate whether the robot is operating or stopped. - Lighting shall be provided for enabling visual checks by operators. - To prevent operators from slipping, the floor shall not be covered with oil or similar substances. - No noises that impair attention shall be emitted.

Table 4. Restricted Specifications Sheet

4. Life cycle of product
In the product life cycle of assembly, shipment testing, transportation, unpacking, transfer, installation, teaching, operation, repacking, maintenance, and disposal, risk assessment is carried out for the life cycle of “transfer and installation” (transportation, unpacking, transfer, installation, and repacking), “use” (teaching, operation, and maintenance), and “disposal.”

Table 5. Restricted Specifications Sheet

5. Person who approaches the machine		
Applicable person	Knowledge and requirements	
1) Transportation worker	Knowledge	Contents of the instruction manual
	Requirement	Person who has received training for forklifts and other transfer equipment work as defined by the respective country's laws and regulations and who has been permitted to perform the work by the employer
2) Unpacking worker	Knowledge	Contents of the instruction manual
	Requirement	—
3) Transfer worker	Knowledge	Contents of the instruction manual
	Requirement	Person who has received training for crane work as defined by the respective country's laws and regulations and who has been permitted to perform the work by the employer
4) Installation worker	Knowledge	Contents of the instruction manual, operating procedures for target robot
	Requirement	When guiding a robot, a person who has received training for robot teaching work as defined by the respective country's laws and regulations and who has been permitted to perform the work by the employer
5) Teaching worker Work supervisor	Knowledge	Contents of the instruction manual, operation procedures for target robot, work regulations, and robot operations
	Requirement	Person who has received training for robot teaching work as defined by the respective country's laws and regulations and who has been permitted to perform the work by the employer
6) Operator	Knowledge	Contents of the instruction manual, operating procedures for target robot, work regulations, and robot operations
	Requirement	—
7) Repacking worker	Knowledge	Contents of the instruction manual, operating procedures for target robot
	Requirement	Person who has received training for crane work as defined by the respective country's laws and regulations and who has been permitted to perform the work by the employer When guiding a robot, a person who has received training for robot teaching work as defined by the respective country's laws and regulations and who has been permitted to perform the work by the employer

Table 5. Restricted Specifications Sheet (continued)

5. Person who approaches the machine		
Applicable person	Knowledge and requirements	
8) Maintenance worker Work supervisor	Knowledge	Contents of the instruction manual, operating procedures for target robot, work regulations, and robot operations Storage procedures for target robot
	Requirement	Person who has received training for robot testing work as defined by the respective country's laws and regulations and who has been permitted to perform the work by the employer
9) Disposal worker	Knowledge	Contents of instruction manual, operating procedures for target robot, and waste disposal methods
	Requirement	Person who has received training for crane work as defined by the respective country's laws and regulations and who has been permitted to perform the work by the employer When guiding a robot, a person who has received training for robot teaching work as defined by the respective country's laws and regulations and who has been permitted to perform the work by the employer
10) People passing nearby Other workers	- Measures shall be taken to prevent entry into the fence, enclosure, or other protection surrounding the hazard zone of the robot. - Measures shall be taken to make the robot stop operation when someone approaches the hazard zone. - If another worker enters the hazard zone of the robot to perform work, robot operation shall be stopped, the drive power turned off, and a teaching worker or maintenance worker shall supervise the situation to ensure safety.	

Table 6. Restricted Specifications Sheet

6. Other

Warranty

This product is provided by SHIBAURA MACHINE after strict testing and inspection and verification that it satisfies our performance standards.
SHIBAURA MACHINE provides the following warranty in the event that a fault occurs.

1. Warranty Period

The warranty period of this product is one of the following periods, whichever is shortest.

(1) Period of 18 months from SHIBAURA MACHINE factory shipping
(2) Period of 12 months from customer installation

2. Warranty Terms and Conditions

(1) The warranty applies to this product. The warranty scope is the specifications and functions described in the product specifications, catalog, instruction manual, and other documents. SHIBAURA MACHINE will not be liable for any secondary or incidental damages arising as a result of faults in this product.
(2) Within the warranty period of this product, SHIBAURA MACHINE will repair free of charge only those faults that arise under operation and use in accordance with the instruction manual provided with this product.
A fee will be charged for all repairs of faults that occur after the warranty period has expired.

3. Disclaimers

The following cases are outside the scope of the warranty.

(1) Faults and damage due to usage that did not follow the instruction manual or due to negligent usage.
(2) Problems due to deterioration over time or usage wear (natural fading of the paint coating, deterioration of consumable parts[1])
(3) Problems due to sensory experiences (such as occurrence of noise that does not affect functionality)
(4) When the product was modified or disassembled without the approval of SHIBAURA MACHINE
(5) Faults and damage due to inadequate inspection and maintenance or improper repairs
(6) Faults and damage due to natural disasters, fires, and other external factors
(7) Internal data such as programs and points created or modified by the customer
(8) When this product was purchased in Japan and taken outside the country

1) Consumable parts refer to the maintenance replacement parts described in the target controller manual.

4. Notes

(1) SHIBAURA MACHINE cannot guarantee the basic performance of this product when it is used in an environment outside the specifications.
(2) SHIBAURA MACHINE will not be liable for any injuries, damage, faults, or other problems that arise by failing to follow the danger, warning, caution and notice information provided in this instruction manual.
(3) Be aware that the danger, warning, caution, notice, and other safety information in this

instruction manual apply to the safety risks foreseen by SHIBAURA MACHINE, and they do not cover every possible scenario.

Main standards, laws, and regulations pertaining to the safety of industrial robots

The international industrial standard ISO/DIS 12100 "Safety of machinery" describes the risk reduction measures for machines and equipment.
Step 1) Inherently safe design measures: Limitations on force, speed, energy, etc.
Step 2) Safeguarding and/or complementary protective measures: Installation of safety fence, etc.
Added protective measures: Installation of emergency stop devices, etc.
Step 3) Information for use: Warning signage, alarms, instruction manuals, etc.

Based on this, the guidelines known as "Guide 51" were defined by classifying the various standards for the ISO/IEC international standards in a hierarchical structure.
The safety standards for industrial robots are standardized as type C in the individual safety standards for machinery.

- ISO 10218 and ISO 10218-1

Be sure to follow the standards, laws, regulations established in the country where the industrial robot is used.

Cautions on Safety

The instruction manual for the controller contains important safety information for ensuring the safe and correct operation of this device and for preventing hazards to the operators and others and property damage.

Make sure that you well understand the following details (indications and symbols) before reading this manual. Always observe the information that is noted.

[Explanation of indications]

Indication	Meaning of indication
⚠DANGER	Indicates an imminently hazardous situation which, if not avoided, will result in death or serious injury.
⚠WARNING	Indicates a potentially hazardous situation which, if not avoided, could result in death or serious injury.
⚠CAUTION	Indicates a hazardous situation which, if not avoided, could result in minor or moderate injury[1].
NOTICE	Indicates a situation which could result in physical damage[2] only. Indicates advice for proceeding work.

1) Minor or moderate Injuries refer to injuries, burns and electric shocks, etc., which do not require hospitalization or long-term medical treatment.
2) Physical damage refers to damages due to destruction of assets or resources.

⚠DANGER
● To ensure that the procedures from transportation to disposal of the controller are performed safely, be sure to carefully read the safety manual described in the controller manual. Performing the operation that differs from the contents described in this manual could harm persons.

Service Network

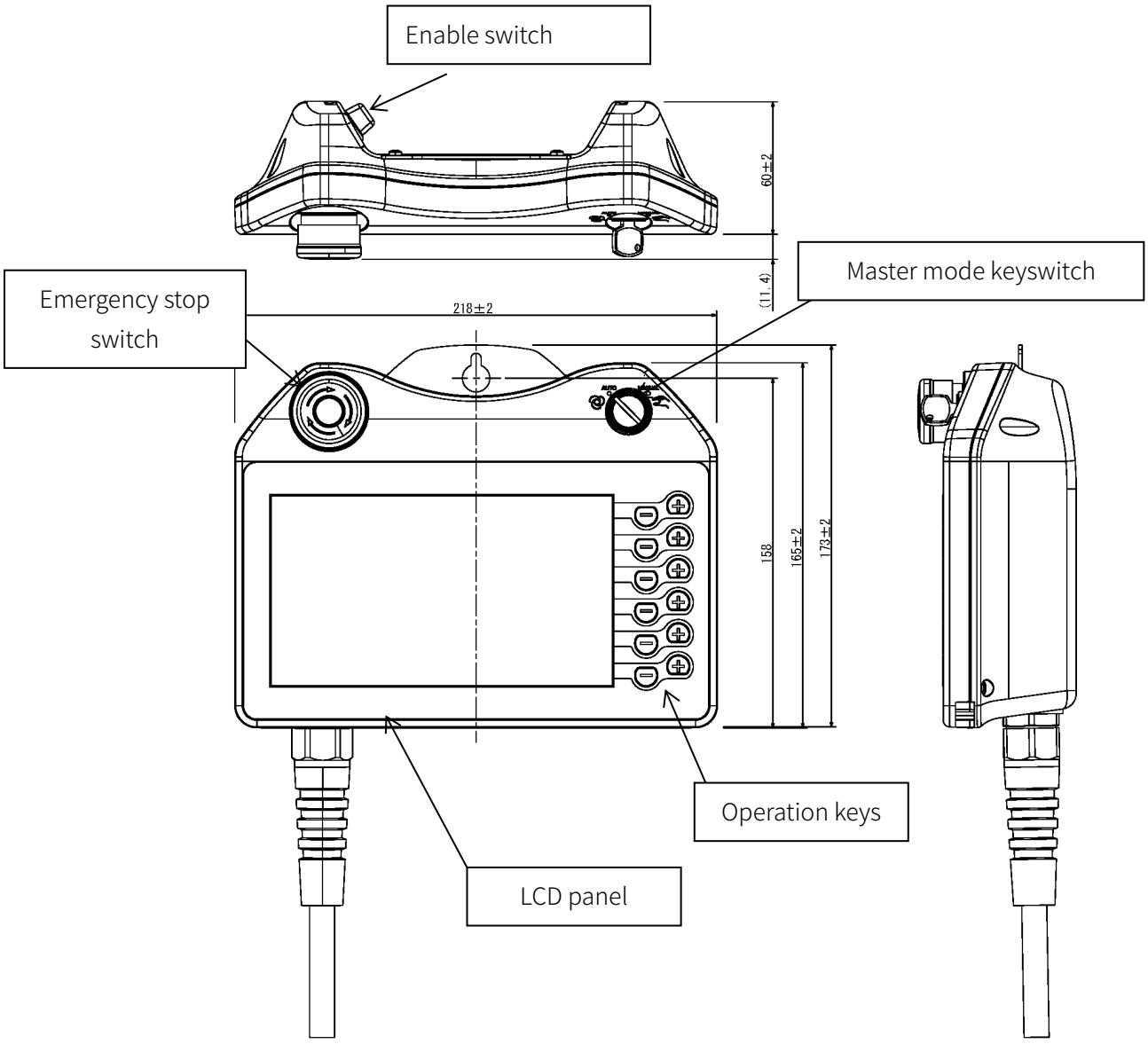
Please refer to our distributor's handouts

Basic Operation

This describes the basic matters for operating the TP5000.

1. TP5000

1.1 TP5000 External View



1.2 TP5000 Specifications

No.	Item	Specifications	Notes
1	Ambient operating temperature	0°C to 40°C	
2	Ambient storage temperature	-20°C to 60°C	
3	Ambient operating humidity	10% to 90% RH (no condensation, wet-bulb temperature of 39°C or less)	
4	Ambient storage humidity	10% to 90% RH (no condensation, wet-bulb temperature of 39°C or less)	
5	Atmospheric pressure resistance (operating altitude)	800 to 1114 hPa (2000 m max.)	
6	Vibration resistance	Compliant with JIS B 3502 and IEC61131-2	
7	Protection rating	IP65	
8	Weight	800 g max. (excluding device connection cables)	
9	Dimensions	218(W)×173(H)×60(D) mm	Not including cables, and protrusions such as switches,
10	Cable length	5 m (standard), 10m, 15m (options)	
11	Live cable connection and disconnection	Allowed	
12	Longevity	TFT backlight: 50,000 hours Touch panel: Touch input 10 million times	

MEMO

Pulling the TP5000 out from the TP connector of the TS5000 will set the TP5000 to the same state as when the emergency stop switch is pressed.

1.3 Basic Operation of TP5000

This section describes the operations performed on the TP5000.

1.3.1 Tap

This refers to the action of touching and removing your finger from the LCD panel.
Operations where "tapping" is mentioned in this manual are activated by touching and removing your finger from the specified location on the screen.

> *MEMO*
>
> *The screen changes while the button on the screen is touched.*
>
> *If the tap operation is not working for you, check that the displayed button has changed before removing your finger from the button.*

1.3.2 Press

This refers to the action of pressing an operation key.
Operations where "pressing" is mentioned in this manual are activated when the specified operation key is pressed.

1.4 TP5000 Screen Layout

The TP5000 screen layout is shown below.



For the description of the number element in the figure, see the following table.

No. in Figure	Name	Description
1	Status bar	This shows the robot statuses and current settings. Tapping a button opens a sub-window under the status bar. While the sub-window is shown, tapping a location other than the sub-window hides the sub-window.
2	Main window	This provides display and setting of various information about the robot. The displayed screen varies depending on the operation.
3	Keyboard	Tapping the keys enters characters into the text box being edited. Also, the special keys perform operation specific to the screen.
4	Operation key indicators	They show the operations when operation keys are pressed.

1.4.1 Status Bar

The status bar shows the various statuses (master mode, alarm state, servo-on state, etc.) and current settings for the robot.
Some displayed information varies depending on the movement mode.
Status bars can be one of three types: the "button" type which is used to change the settings or open the setting screen when tapped, the "panel" type which shows the statuses and settings, and the "panel button" type which has the features of both the "button" and "panel" types.
For details on the menus, see the following table.

Status Bar (Operation Mode: Running Mode)



Status Bar (Operation Mode: Manual Guidance Mode)



No. in Figure	Name	Type	Description	
1	Master Mode Panel	Panel	Displayed content	Shows the master mode of the controller. MANUAL : MANUAL mode AUTO : AUTO mode
2	Operation Status Panel	Panel	Displayed content	This shows the robot's operation status. RUN : Executing program FEED HOLD : Pausing STOP RESET : Stopping initialization of program execution STOP RETRY : Aborting program execution STOP CONT : Stopping program execution

No. in Figure	Name	Type	Description	
3	Alarm button	Panel button	Displayed content	This indicates whether an alarm is generated.  Lit: Alarm is generated  Not lit: Alarm is not generated.
			When tapped	Opens the Alarm window.
4	Select Execution Program button	Panel button	Displayed content	Registered to execution program. Shows the file name and current execution line.
			When tapped	Opens the Select Execution Program screen. (Top screen only)
5	Reset Operation button	Button	When tapped	Opens the Reset Operation window.
6	Execution Mode button	Panel button	Displayed content	Shows the execution mode.
			When tapped	Shows the Execution Mode Setting window.
7	Override button	Panel button	Displayed content	Shows the override setting value.
			When tapped	Opens the Override Setting window.
8	Servo ON button	Panel button	Displayed content	Shows the Servo ON state. Lit: Servo ON state. Not lit: Servo OFF state.
			When tapped	In Servo ON state: Changes to Servo OFF state In Servo OFF state: Changes to Servo ON state
9	Operation Mode button	Panel button	Displayed content	Shows the TP5000 operation mode.  : Running mode  : Manual guidance mode
			When tapped	Switches the operation mode.

<table>
<tr><th>No. in Figure</th><th>Name</th><th>Type</th><th colspan="2">Description</th></tr>
<tr><td rowspan="2">10</td><td rowspan="2">Jog Setting button</td><td rowspan="2">Panel button</td><td>Displayed content</td><td>Opens the jog settings.

Guidance type
 

Guidance coordinates

Guidance speed
(When the guidance type is inching, this shows the setting for the guidance amount.)</td></tr>
<tr><td>When tapped</td><td>Opens the Jog Setting window.</td></tr>
<tr><td rowspan="2">11</td><td rowspan="2">Tool Setting button</td><td rowspan="2">Panel button</td><td>Displayed content</td><td>Shows the tool coordinates name that is currently set.</td></tr>
<tr><td>When tapped</td><td>Opens the Tool Setting window.</td></tr>
<tr><td>12</td><td>Message Panel</td><td>Panel</td><td>Displayed content</td><td>This shows character strings printed from the TS5000 to the TP5000 or messages about TP5000 operations.</td></tr>
</table>

1.4.2 Main Window

The main window provides display and setting of various information about the robot.
The displayed screen varies depending on the operation.
For details on the screens shown in the main window, see 11 Utility Window Details.



1.4.3 Keyboard

This section describes the keyboard displayed on the lower part of the screen and the basic text editing method.

The keyboard is always displayed on the lower part of the screen.
The keyboard can be used to enter characters or numbers into the text box being edited.

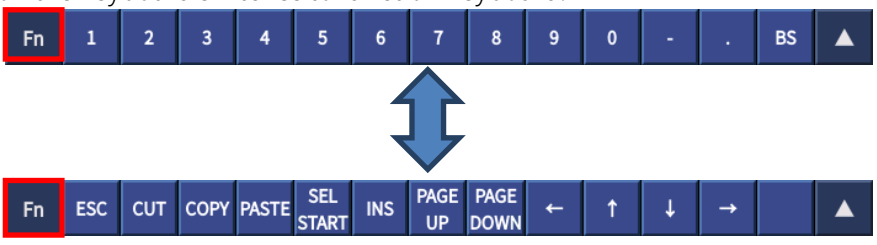
Two types of keyboards can be displayed: “single-line keyboard” where only one line appears or “full keyboard” where the entire keyboard appears.

Single-line keyboard



Tapping ▲ on the keyboard switches to full keyboard.

Tapping Fn on the keyboard switches to function keyboard.

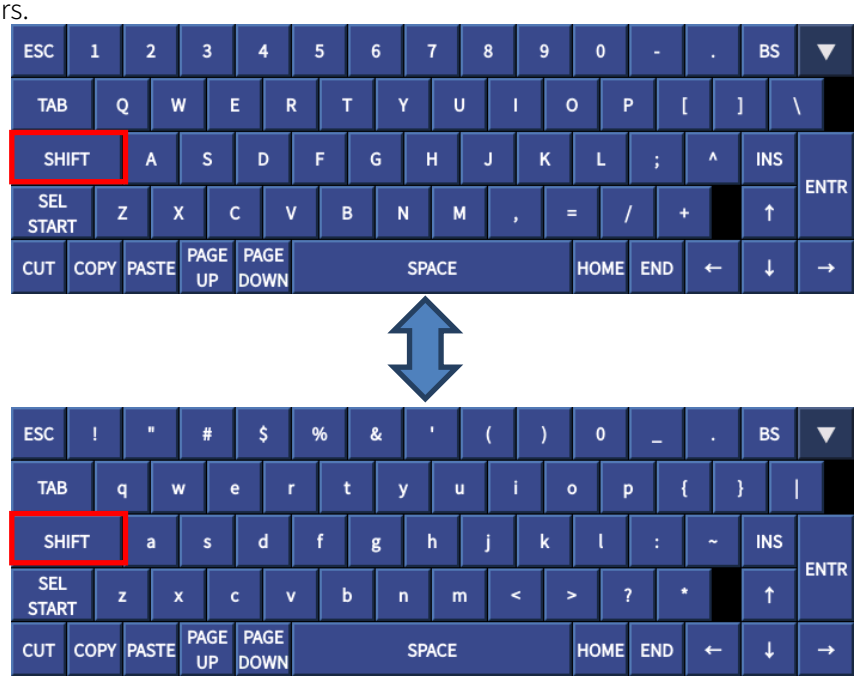


Full keyboard

Keyboard



Tapping the SHIFT key switches between numbers and symbols and between uppercase and lowercase letters.



Tapping  switches to single-line keyboard.

Text editing

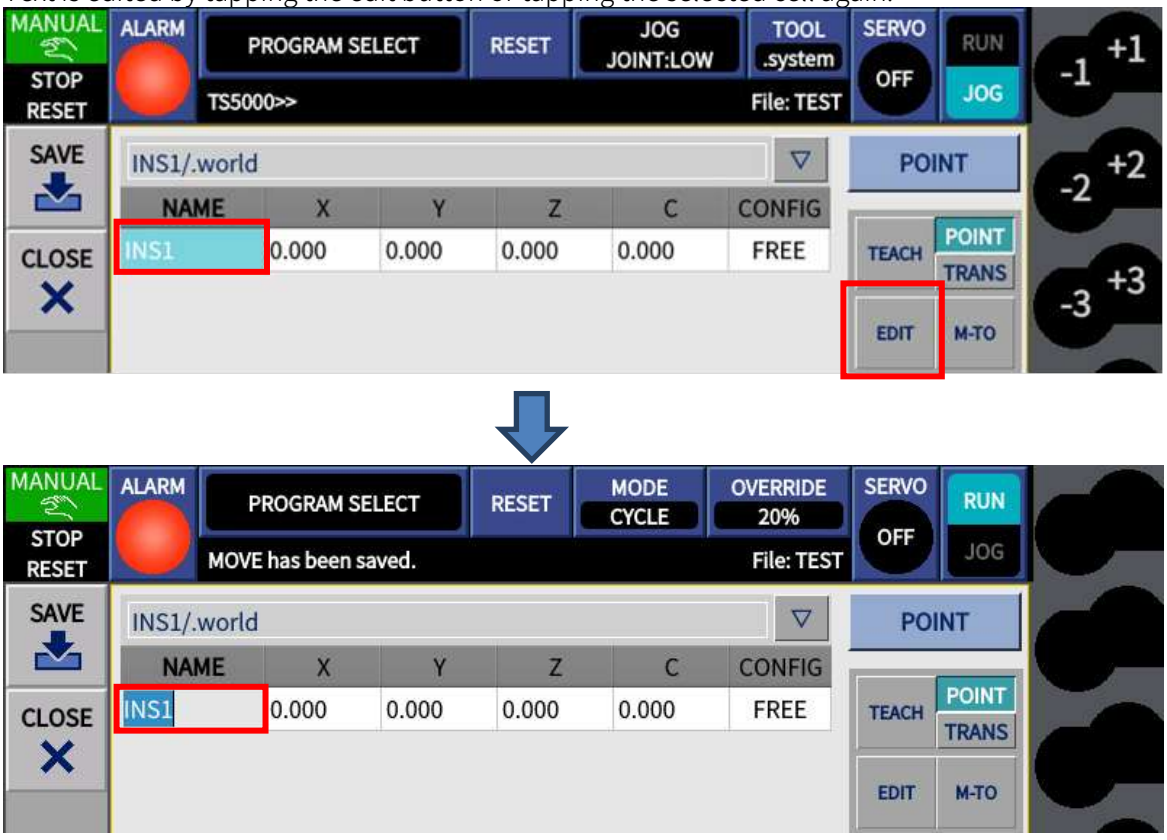
Text editing uses the keyboard displayed on the lower part of the screen.

For description of keyboard operation, see "1.4.3 Keyboard."

Also, the special keys perform operation specific to the screen. For details, see the descriptions of the respective screens.

1 Placing in editing state

Text is edited by tapping the edit button or tapping the selected cell again.



MEMO

On the screen without the edit button, tap the selected cell again.

2 Entering text

In editing state, the keyboard is operated to enter text.

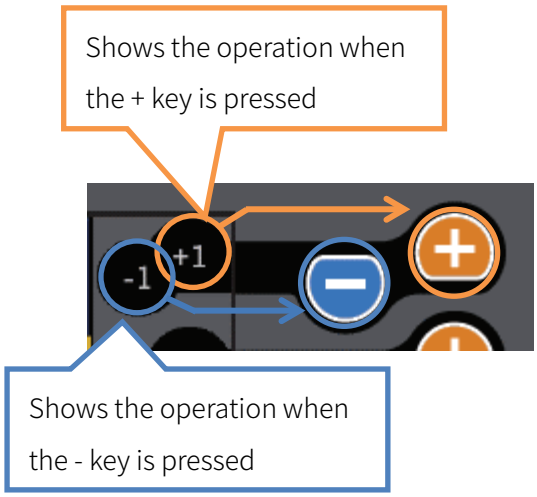
3 Saving text

Text is saved by tapping the position other than the selected cell in the screen.

> *MEMO*
>
> *If an invalid value is entered to save text, text appears red.*
>
> *Re-enter a valid value.*

1.4.4 Operation key indicators

They show the operations that are executed when the operation keys are pressed.
The correspondence between the operation keys and display section is shown below.



The correspondence between the key indicators and the operations when the operation keys are pressed is shown in the table below.

Indicator	Operation when operation key is pressed
	While the key is held down, the joint axis corresponding to the key moves.
	While the key is held down, the robot moves in the axial direction corresponding to the key for the guidance coordinates (world coordinates, tool coordinates, work coordinates).
	This executes the program. * The + key alone is enabled.
	This stops execution of the program. * Both the + and - keys are enabled.
	This aborts execution of a program. * Both the + and - keys are enabled.
	This pauses movement of the robot and execution of the program. When operated in a paused state, this deactivates it. * Both the + and - keys are enabled.
	While the key is held down, the robot performs PTP movement toward the set target position.

Indicator	Operation when operation key is pressed
	While the key is held down, the robot performs linear interpolation movement toward the set target position.
	While the key is held down, the robot performs bypass movement using PTP movement toward the set target position.
	While the key is held down, the robot performs bypass movement using linear interpolation movement toward the set target position.
	The DO1 to DO4 statements that was set on the DO screen are executed. * The + key alone is enabled. For details on the DO screen, see “11.10 DO Screen.”

MEMO

The robot will not move even if a displayed operation key is tapped.

Operating Procedures

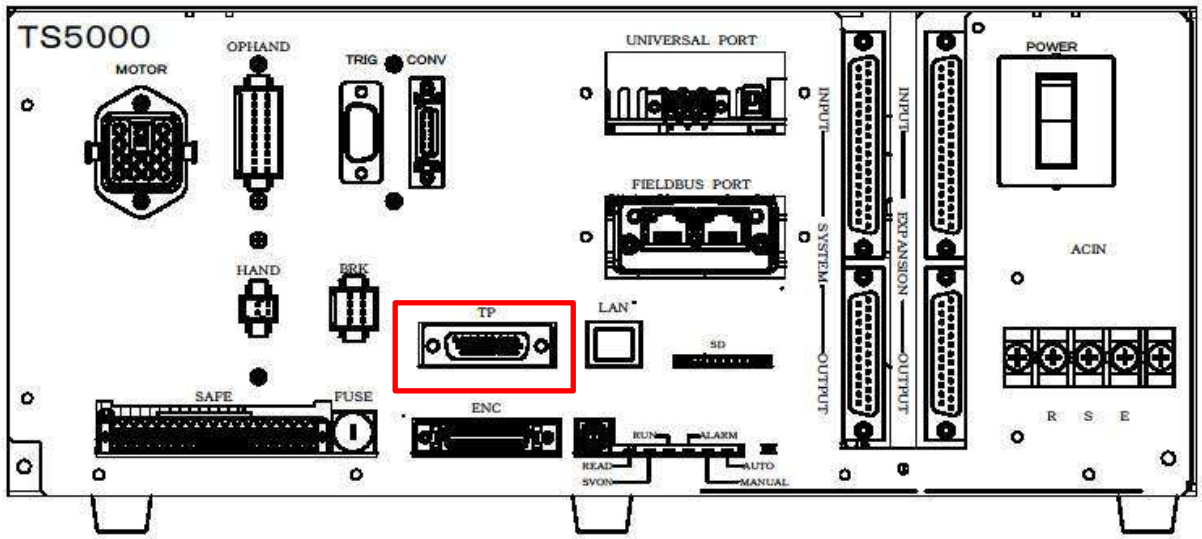
This describes the operations that can be performed using the TP5000, focusing on the operating procedures.

2. TP5000 Startup

This chapter describes how to connect the TP5000 and TS5000 and the operating procedures until the TP5000 is powered on.

2.1 Connecting with TS5000

Connect the connector of teach pendant TP5000 to the TP connector of the TS5000 (see the red frame in the following figure).



2.2 Starting TP5000

With the TS5000 connected, powering the TS5000 on automatically starts up the TP5000.

After starting the TP5000, the message "Now booting" appears during the startup process, and then the top screen appears.
The displayed status bar varies depending on the master mode (MANUAL or AUTO) at startup.
The screens at startup are as shown below.

> *MEMO*
> *While the TS5000 controller firmware is being updated, the TP5000 does not start up. If the TP5000 is connected during startup, the message "Communication Error" appears.*
> *After the TS5000 controller has been completely updated, remove the SD card from the controller and restart the controller.*

Screen Displayed at Startup (MANUAL Mode)



Screen Displayed at Startup (AUTO Mode)



MEMO

If the status of the TP5000 master mode key and the controller master mode are different when the controller is powered on, the alarm on "8-283 Master mode key operation error" is generated.

Use the master mode key to switch the master mode.

3. Operations in MANUAL Mode

This chapter describes the operations performed in MANUAL mode.
MANUAL mode is used in teaching the robot and confirming operations.
The speed of the robot tip in MANUAL mode is limited to a reduced speed (250 m/sec or less).

Use the following procedure to switch from AUTO mode to MANUAL mode.

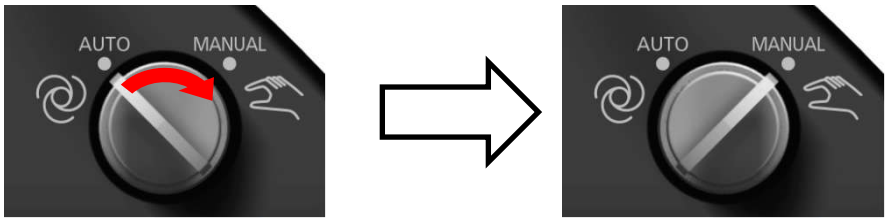
Switching Procedure to MANUAL Mode

1 Insert the key provided with the teach pendant into the master mode key switch.



2 Set the master mode keyswitch to the MANUAL position.

Master mode is switched to MANUAL.



MEMO

- *If the master mode is switched during Servo ON, the status changes to Servo OFF.*
- *If the robot is operating when master mode is switched, the robot performs a deceleration stop, and the status changes to Servo OFF.*
- *If the master mode keyswitch is switched on other screens than the top screen the top screen, an alarm is generated.*

3 Remove the key if you want to prevent other workers from switching the master mode.



3.1 Manual Guidance

This chapter describes manual guidance of the robot where the TP5000 is used.

3.1.1 Overview

In this manual, "manual guidance" refers to the movement of the robot in the axial direction of a coordinate system by using the teach pendant. (This does not include test operation, direct execution, or teaching point movement.)
Manual guidance can be performed only when the operation mode is manual guidance mode.

Manual guidance requires Setting Guidance Type, Setting Guidance Coordinates, and Setting Guidance Speed (or Setting Guidance Amount). These are set in the manual guidance setting window. These are set in the manual guidance setting window.

If a tool is mounted on the tip of the robot, Setting Manual Guidance Load is required.

3.1.2 Switching to Manual Guidance Mode

This describes how to switch the operation mode to manual guidance mode.
The current operation mode setting is displayed in the status bar. (See the figure below.)



Manual Guidance Mode



Operation Mode



Switch Procedure to Manual Guidance Mode

1 Tap the movement mode selector button in the status bar.

This switches to manual guidance mode when the movement mode is operation mode.

• Screen for manual guidance mode



• Screen for operation mode



MEMO

The displayed operation keys change to those corresponding to manual guidance mode so that the execution mode setting button and override setting button in the status bar change to the manual guidance setting button and tool setting button.

3.1.3 Setting Guidance Type

This describes how to set the guidance type.
The guidance types include jog and inching.

• Jog
The robot moves while the operation key is held down.
The robot is stopped when the operation key is released.
Pressing the + operation key moves the robot in the + direction of the guidance coordinates, and pressing the - operation key moves the robot in the - direction.
The robot moves at the speed that is set for the guidance speed.
In particular, setting the guidance type to Jog and performing manual guidance is called "jog operation".

• Inching
When the operation key is pressed, the robot moves only the movement amount that is set for the guidance amount and then comes to a stop.
Pressing the + operation key moves the robot in the + direction of the guidance coordinates, and pressing the - operation key moves the robot in the - direction.

* Even if the operation key is held down, the robot moves by the amount for single pressing of the key.
Even if the key is pressed during movement of the robot, the robot does not move in series.

In particular, setting the guidance type to Inching and performing manual guidance is called "inching operation".

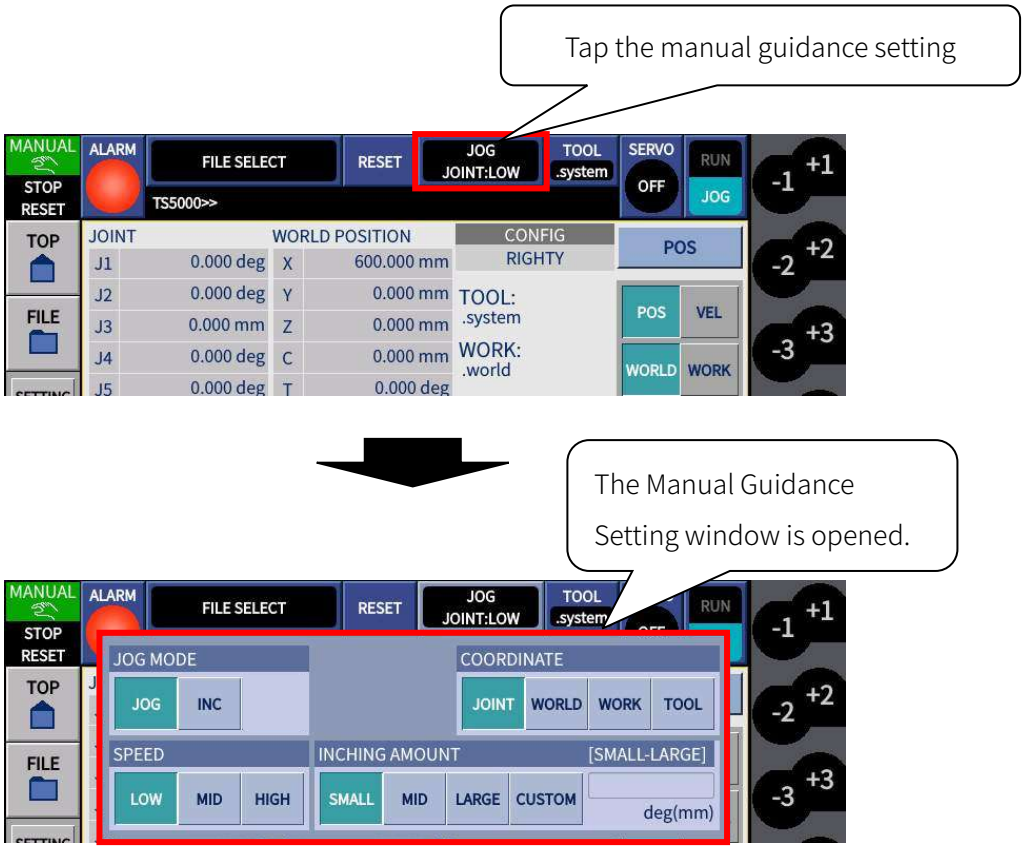
MEMO

Simultaneous pressing of the operation keys is not supported.

How to Set Guidance Type

1 Tap the manual guidance setting button (indicated by red box) in the status bar.

This opens the Manual Guidance Setting window.

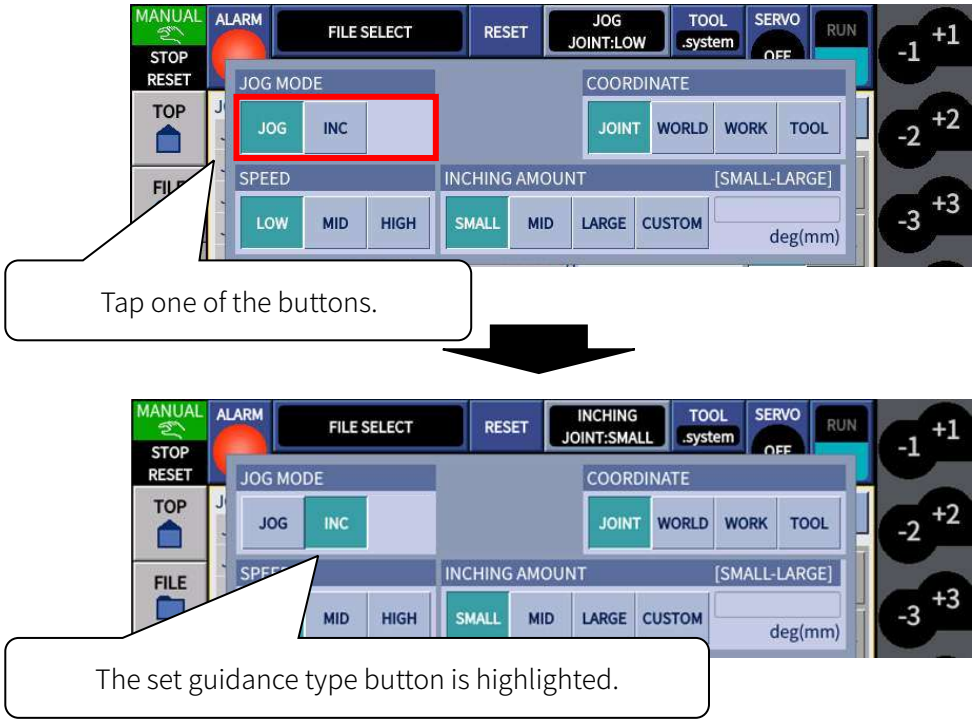


2 Tap the guidance type button (indicated by red box) to be set for the guidance type.

The guidance type is set corresponding to the tapped button, and the displayed button changes.

In this case, the set guidance type is displayed for the manual guidance setting button, and the displayed operation keys change to those for the guidance type.
(The displayed operation keys for jog and inching are identical.)

(In the example in the figure below, the guidance type is set to inching.)



3.1.4 Setting Guidance Coordinates

This describes how to set the guidance coordinates.
The guidance coordinates include joint coordinates, world coordinates, work coordinates, and tool coordinates.

• Joint coordinates
The robot moves for each joint axis.

• World coordinates
The robot moves on the Cartesian coordinate system.

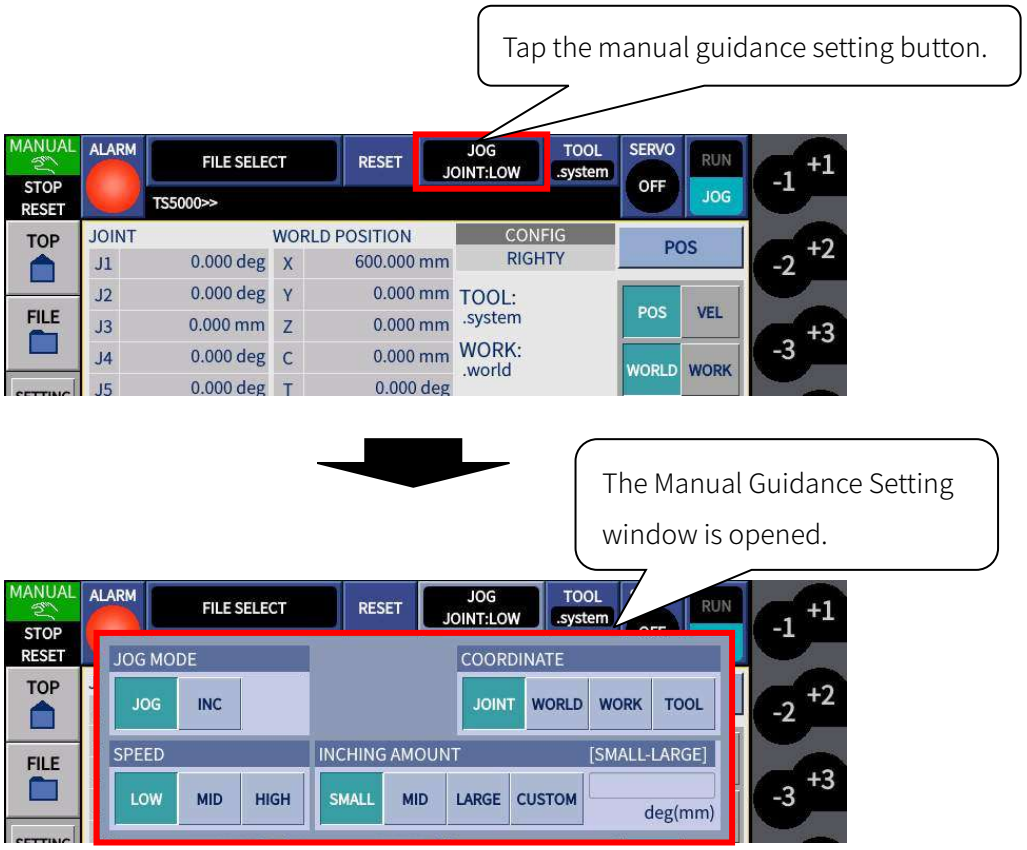
• Work coordinates
The robot moves on the work coordinate system that is currently set.

• Tool coordinates
The robot moves on the coordinate system with the tool tip as the origin.

How to Set Guidance Coordinates

1 Tap the manual guidance setting button (indicated by red box) in the status bar.

This opens the Manual Guidance Setting window.

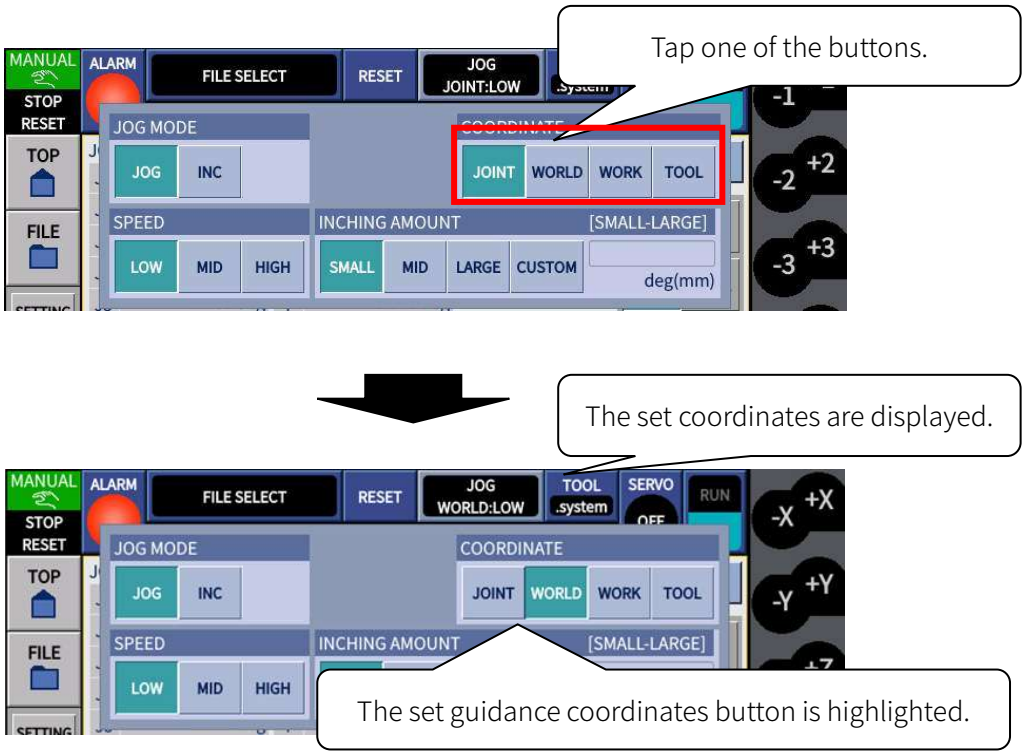


2 Tap the guidance coordinates button (indicated by red box) to be set for the guidance coordinates.

The guidance coordinates are set corresponding to the tapped button, and the displayed button changes.

In this case, the guidance coordinates set are displayed for the manual guidance setting button, and the displayed operation keys change to the ones for the guidance coordinates.

(In the example in the figure below, the guidance coordinates are set to world coordinates.)



3.1.5 Setting Guidance Speed

This describes how to set the guidance speed.

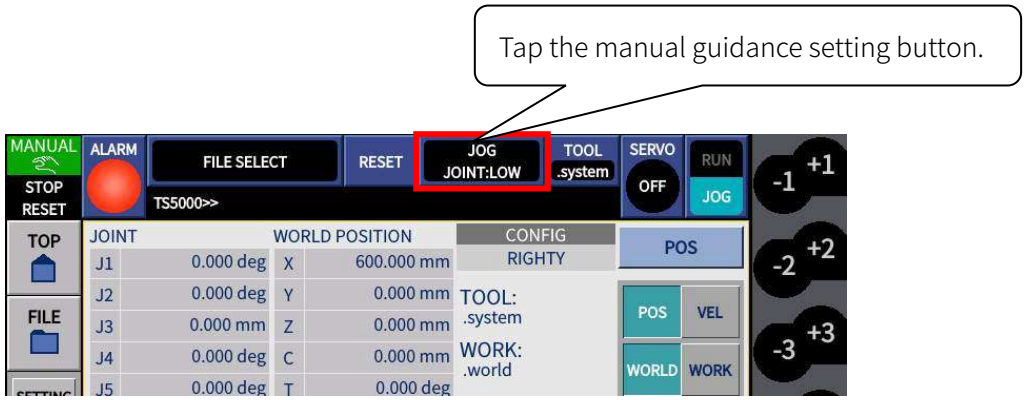
The guidance speed is the movement speed of the robot when the robot makes movement by jog operation.

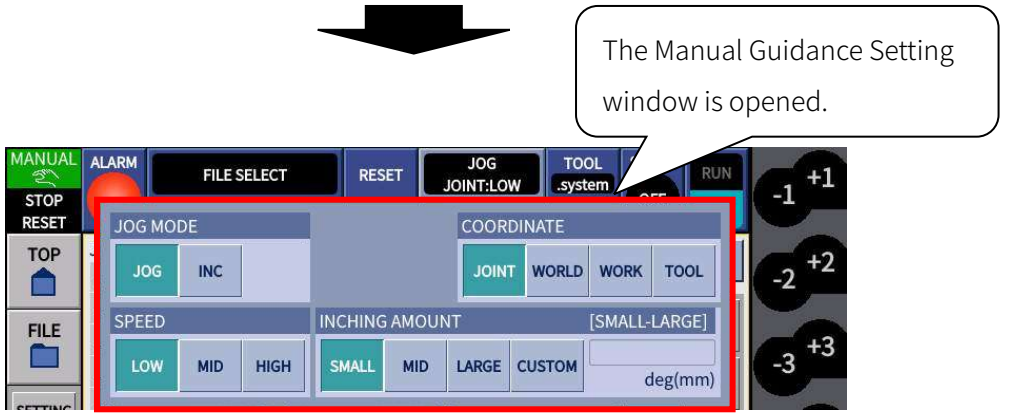
The available guidance speeds are LOW (low speed), MID (medium speed), and HIGH (high speed).

How to Set Guidance Speed

1 Tap the manual guidance setting button (indicated by red box) in the status bar.

This opens the Manual Guidance Setting window.

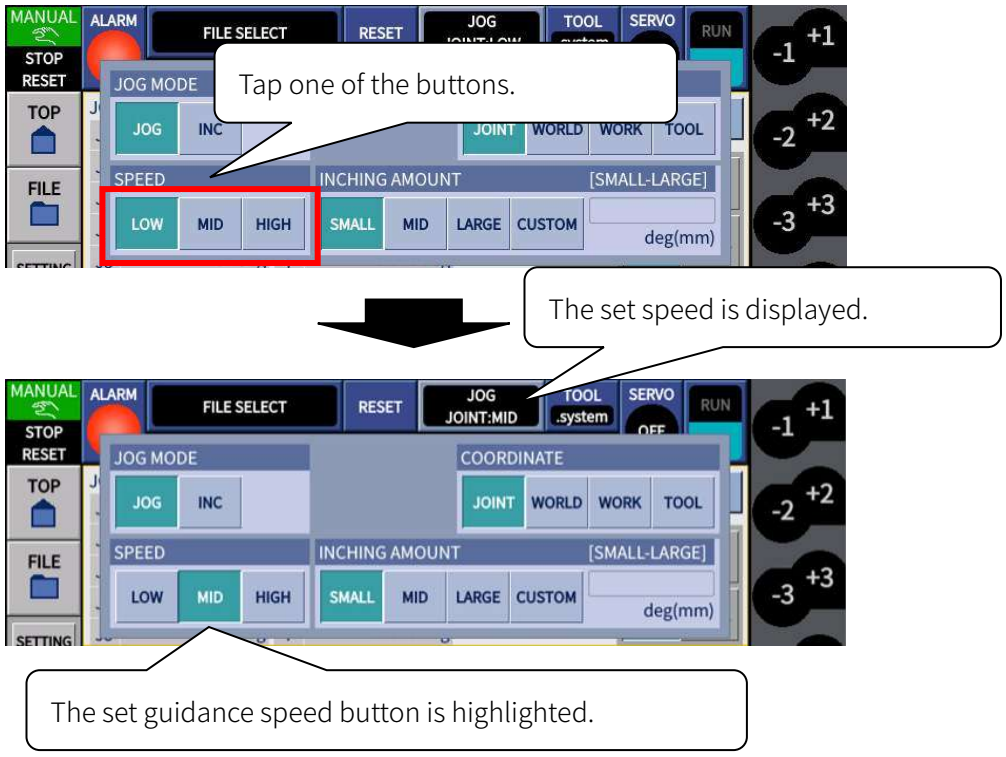




2 Tap the guidance speed button (indicated by red box) to be set for the guidance speed.

The guidance speed is set corresponding to the tapped button, and the displayed button changes.

Also, the guidance speed (LOW, MID, HIGH) is indicated in the manual guidance setting buttons.
(In the example in the figure below, the guidance speed is set to MID.)



MEMO

The guidance speed is predetermined by the manufacturer, but it can also be set to the guidance amount desired by the customer.

For details, see the "U005" section in the STEA1166 Parameter Manual.

3.1.6 Setting Guidance Amount

This describes how to set the guidance amount.

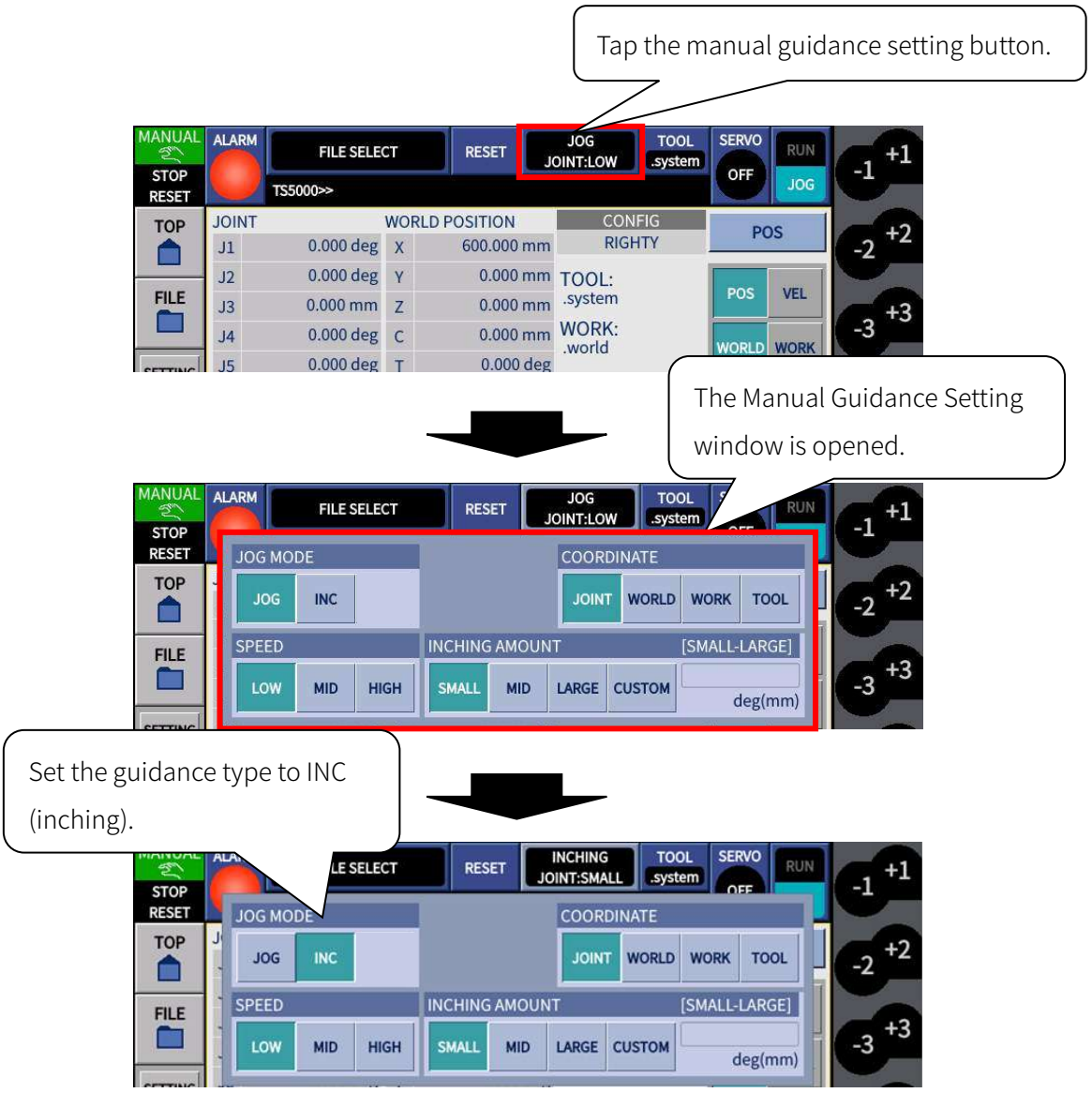
The guidance amount is the amount of movement of the robot when the robot moves by inching operation.

The available guidance amounts are SMALL, MID (medium), LARGE, and CUSTOM.

How to Set Guidance Amount

1 Tap the manual guidance setting button (indicated by red box) in the status bar.

This opens the Manual Guidance Setting window.



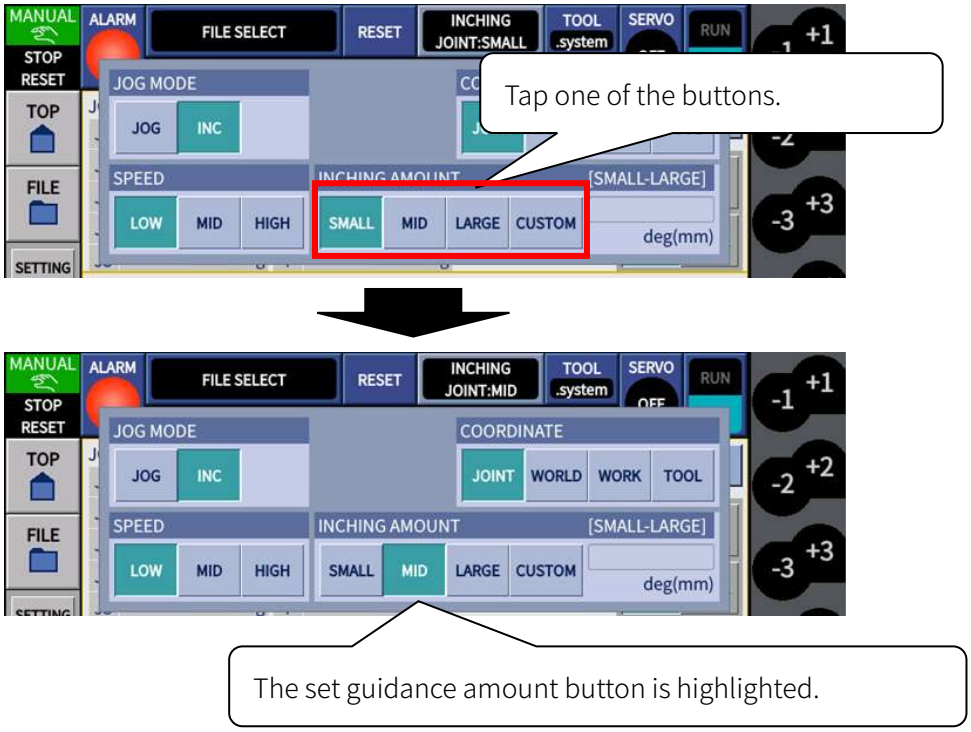
1) Setting the guidance amount to SMALL, MID, or LARGE

1. Tap the guidance amount button (indicated by red box) to be set for the guidance amount.

The guidance amount is set corresponding to the tapped button, and the displayed button changes.

Also, the set guidance amount (SMALL, MID, LARGE) is indicated in the manual guidance setting buttons.

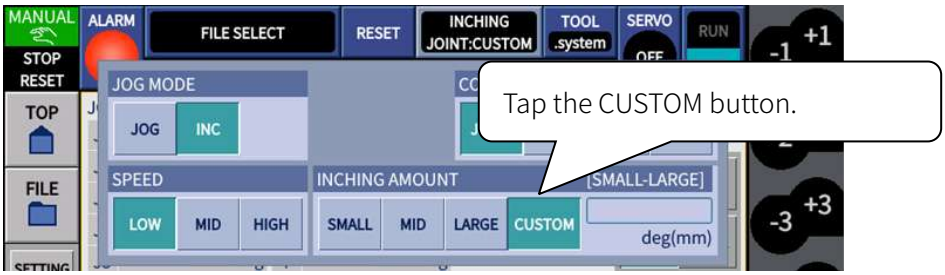
(In the example in the figure below, the guidance amount is set to MID.)



2) Setting the guidance amount to CUSTOM

1. Tap the CUSTOM button in the Manual Guidance Setting window.

The cursor appears in the text box to indicate that the content can be edited.



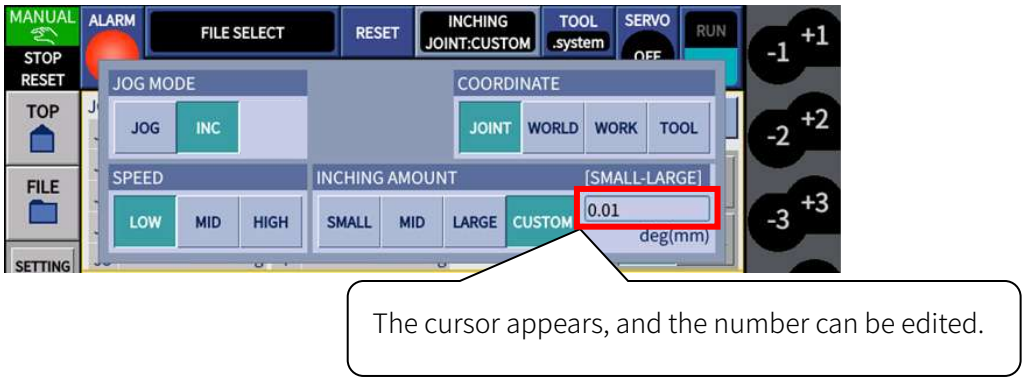
MEMO

In this state, the setting for the guidance amount is reset.

Inching cannot be performed until one of the guidance amount buttons is pressed again.

2. Use the keyboard to enter a custom number, and then press the Enter key or tap any location outside the text box in the Guidance window.

This saves the edited changes, and the cursor disappears.



MEMO

- *If a non-numerical entry is made into the text box, the characters in the text box appear red.*
  When numeric values outside the range are set, inching operation cannot be performed.
- *The guidance speed is predetermined by the manufacturer, but it can also be set to the guidance amount desired by the customer.*
  For details, see the "U005" section in the STEA1166 Parameter Manual.

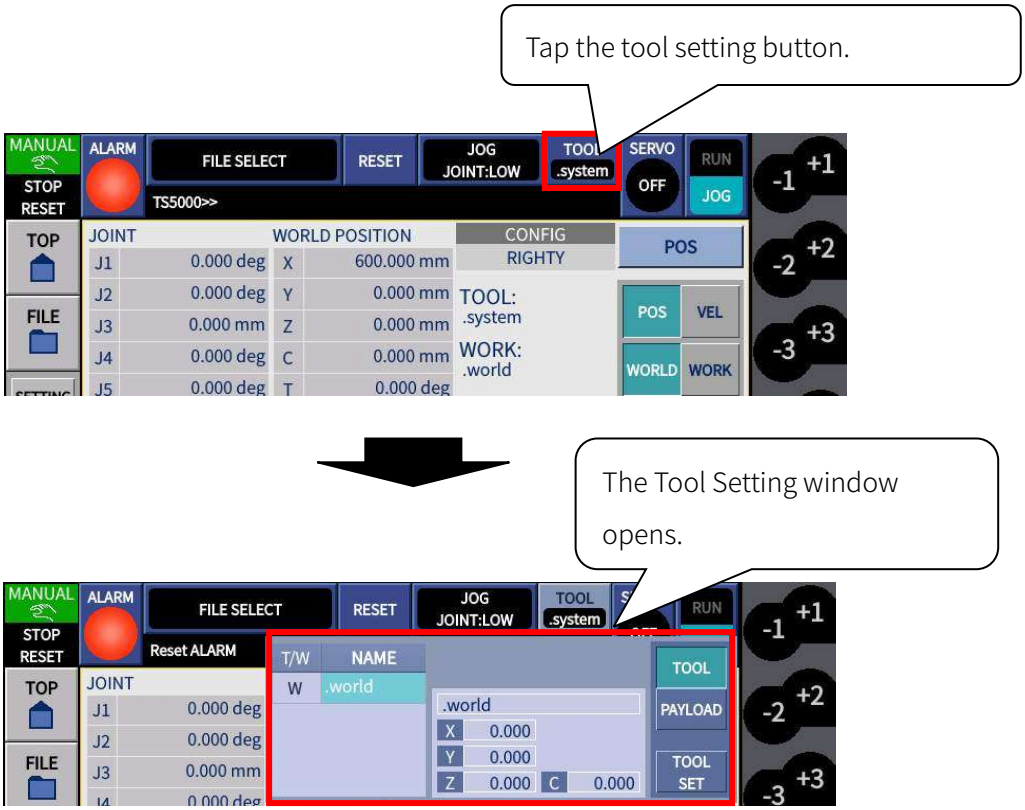
3.1.7 Setting Manual Guidance Load

This describes how to set the load for manual guidance.

How to Set Manual Guidance Load

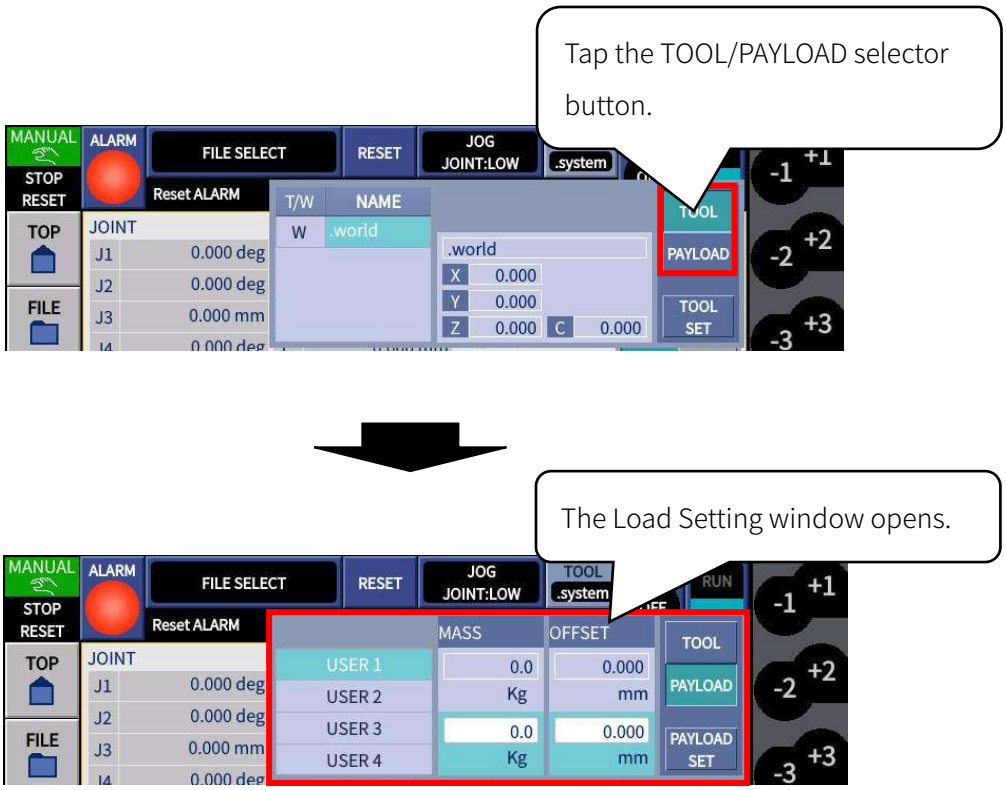
1 Tap the tool setting button in the status bar.

The Tool Setting window opens.



2 Tap the TOOL/PAYLOAD selector button.

This opens the Load Setting window.



1) To use USER load:

1 Tap one of USER1 to USER4.

The setting values that are registered to the tapped button (USER1 to USER4) are displayed.

2 Tap the MASS or OFFSET text box.

The cursor appears enable edition.



3 Enter the load information to be set into the text box, and then press the Enter key or tap any location outside the text box in the Guidance window.

This saves the edited changes, and the cursor disappears.

MEMO

The registered values of the selected USER1 to USER4 are updated.

4 Tap the SET button.

The value in the text box is set for the manual load.



MEMO

The values set for the loads for USER1 to USER4 can be saved.

The values set for USER1 to USER4 are retained even when the controller is turned off.

2) To use the load that is registered to the program:

1. Tap the USER/PROG selector button.

The setting load that is registered to the selected program is displayed.

The setting load that is registered to the tapped load is displayed.



MEMO

For details on how to register the load, see "4.3.13 Editing PAYLOAD-type Data."

The setting value of the load that is registered to the program cannot be changed.

2. Tap the SET button.

The value in the text box is set for the manual load.



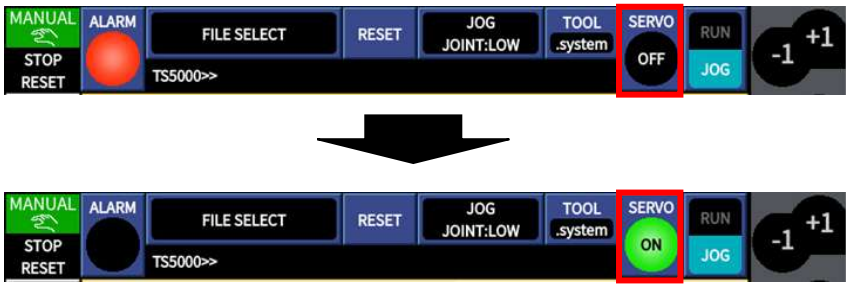
3.1.8 Manual Guidance

This describes how to perform manual guidance.

How to Perform Manual Guidance

1 Tap the SERVO ON button while pressing down the enable switch by one level.

This turns on the servo, and the SERVO ON button indicator is changed.



2 Press the operation key.

The robot moves under the conditions based on the guidance settings (guidance type, coordinates, speed or amount).

MEMO

- *The servo cannot be set ON if an alarm at level 8 or higher has been generated.*
- *If the SVOFF signals are OFF, the servo cannot be set ON.*
- *The robot does not move if two or more keys are pressed simultaneously.*
- *If the STOP and BREAK signals are OFF, the servo can be set ON, but the robot will not move even if the Guide key is pressed.*

⚠DANGER

● Be sure to always check that the tool coordinates are set correctly.
Not doing so may result in an accidental collision with a robot in unexpected motion.

● Before setting ON the servo, be sure to exit the maximum robot space. Failure to do so may result in an accidental collision with a robot in unexpected motion.

⚠ WARNING

● Before setting ON the robot servo, be sure to check that no person exists within the movable range of the robot. Failure to do so may cause the robot in motion to collide with the person existing within the movable range.

- When entering the movable range, leave the safety door open, carry the teach pendant placed in the MANUAL mode, and indicate that work is underway.
- The LED lamp on the robot can be used to check the servo status.
- Perform manual guidance in a state where the enable switch or emergency stop switch can always be accessed.

NOTICE

● Before starting manual guidance, be sure to always check the selected guidance mode, guidance coordinates, guidance speed, and guidance amount.

Failure to do so may cause the robot in unexpected motion to collide with the peripheral device.

3.2 Test Operation

This section describes how to perform test operation using the TP5000.

3.2.1 Overview

Test operation refers to executing programs in MANUAL mode.
The files used for executing programs are called execution programs.

Test operation requires Selecting Execution Program, Setting Execution Mode, and Setting Override.
The setting procedures are described in detail in the sections "3.2.3. Selecting Execution Program", "3.2.4. Setting Execution Mode", and "3.2.5. Setting Override".

MEMO

- *Test operation can be performed only when the operation mode is running mode.*

3.2.2 Switching to Operation Mode

This describes how to switch the operation mode to running mode.

The current movement mode setting is displayed in the status bar. (See the figure below.)



Manual Guidance Mode



Running Mode



How to Switch to Running Mode

1 Tap the operation mode selector button in the status bar.

This switches to manual guidance mode when the operation mode is running mode, and switches to running mode when the operation mode is manual guidance mode.

> *MEMO*
>
> *The displayed operation keys change to those corresponding to running mode so that the manual guidance setting button and tool setting button in the status bar change to the execution mode setting button and override setting button.*

• Screen for running mode



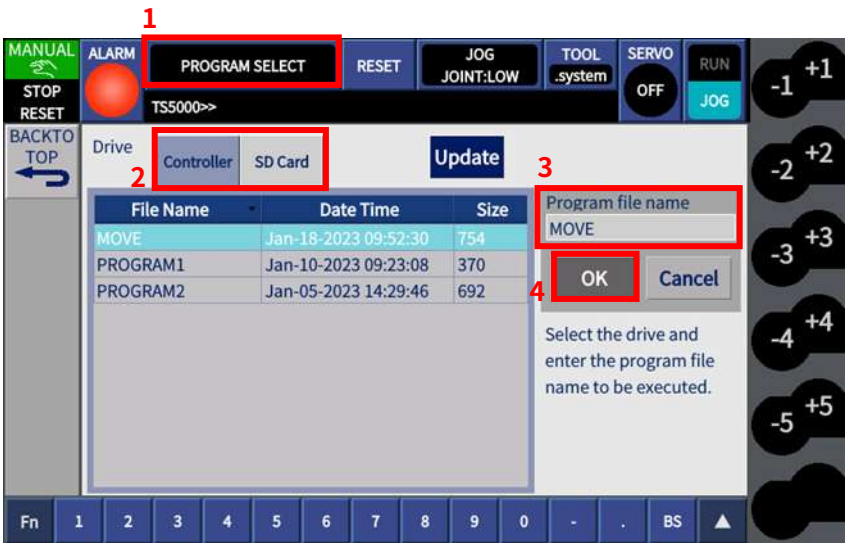
• Screen for manual guidance mode



3.2.3 Selecting Execution Program

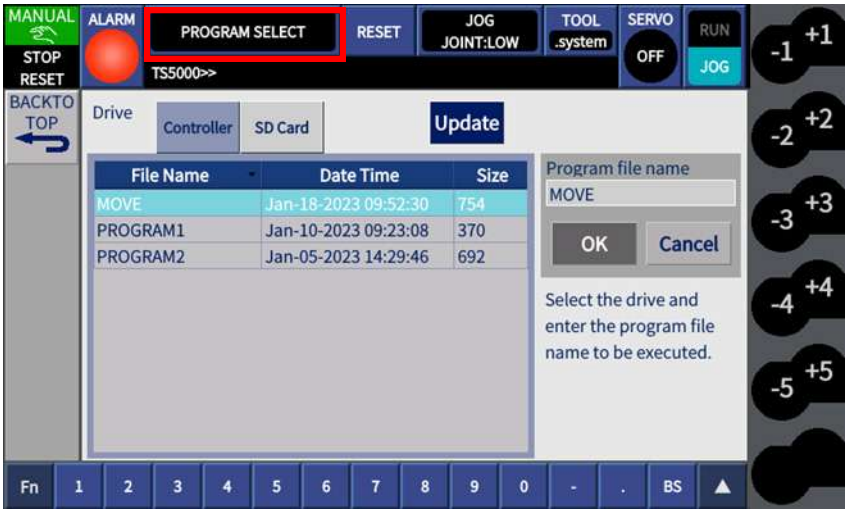
Test operation requires selection of a single program file as an execution program.
This describes how to select the execution program.

How to Select Execution Program



1 Tap the FILE SELECT button in the status bar.

This opens the File Select screen.



2 Tap the medium (controller, SD card) for selection where the program file to be executed is stored.

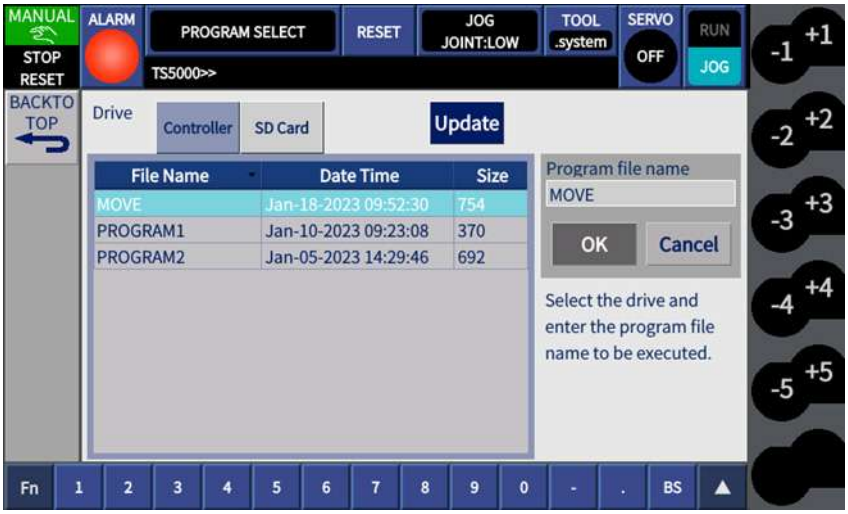
This displays a list of the files stored on the selected medium.

MEMO

When the File Select screen is opened, "Controller" is selected.

3 Select and tap the file name to be executed from the file list.

The tapped file name appears.



MEMO

• When there are a large number of stored files, the files are displayed over multiple pages.

To navigate through the pages, tap the "↑" and "↓" keys or "PageUp" and "PageDown" on the keyboard.

• Tapping File Name at the top of the file list will sort by file name.

4 Tap the OK button.

The file having the displayed name is analyzed, and a syntax check (compile) is performed.

After the compiling is completed, the selected file is stored as an execution program.



MEMO

• If a syntax error is found, the Alarm "1-160 Compile error" is generated.

• When the execution program is selected, the program is reset. If the same program is re-selected, the program is also reset.

3.2.4 Setting Execution Mode

This describes how to set the execution mode.
The execution modes include continuous operation mode, cycle operation mode, segment operation mode, and step operation mode.

• Continuous operation mode
This repeatedly executes the execution program.

• Cycle operation mode
This executes one cycle of the execution program.

• Segment operation mode
This executes the execution program until the execution of the next running command is completed.

• Step operation mode
This executes the execution program step by step.

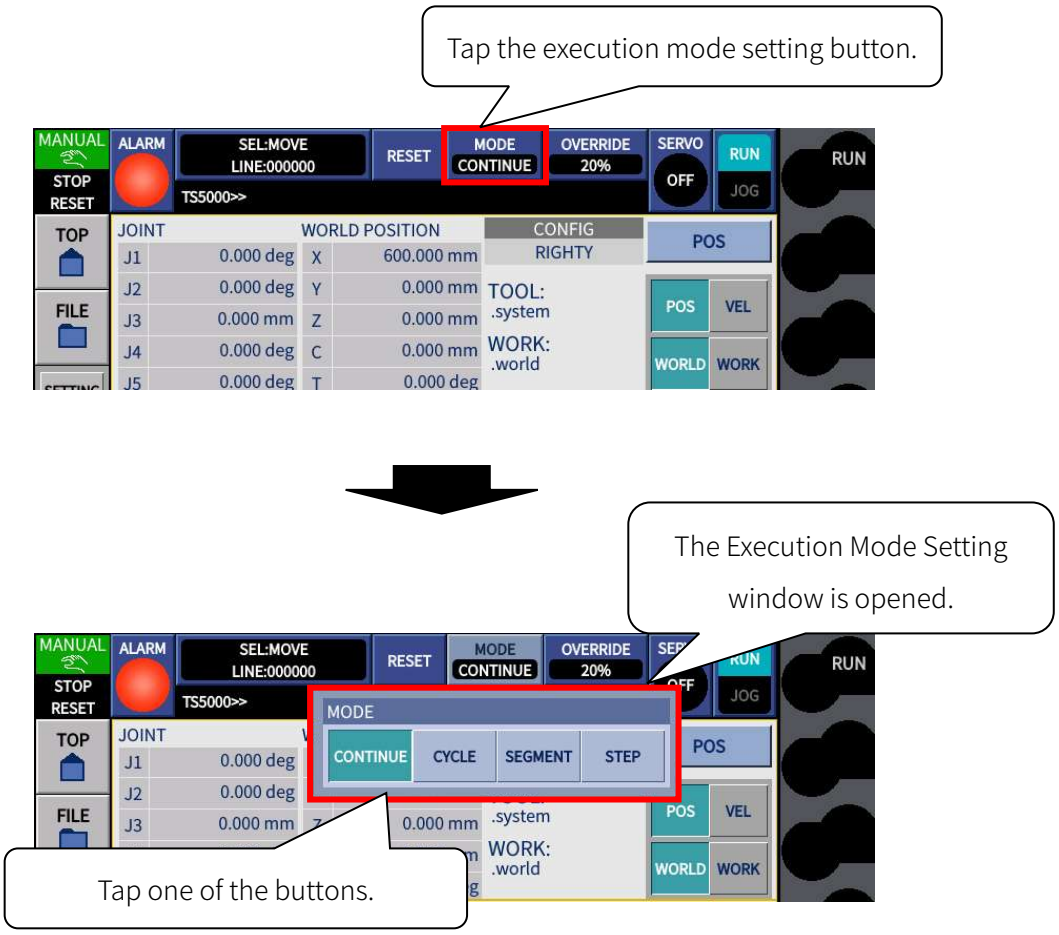
MEMO

- *During program execution in continuous operation, cycle operation, or segment operation mode, the execution mode can be switched to continuous operation, cycle operation, or segment operation mode.*
- *During program execution in step operation mode, the execution mode cannot be switched to another execution mode.*

Execution Mode Setting Procedure

1 Tap the execution mode setting button (indicated by red box) in the status bar.

The Execution Mode Setting window is opened.

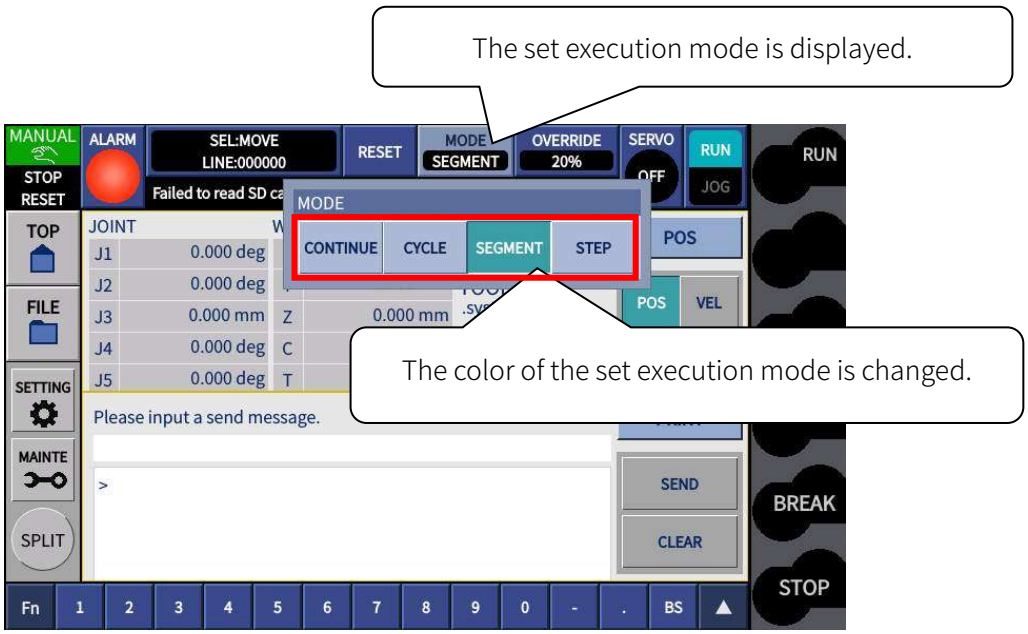


2 Tap the button for the mode to be set for the execution mode (indicated by red box).

The execution mode is set to the mode corresponding to the tapped button, and the displayed button changes.

MEMO

The set execution mode is displayed for the execution mode setting button.



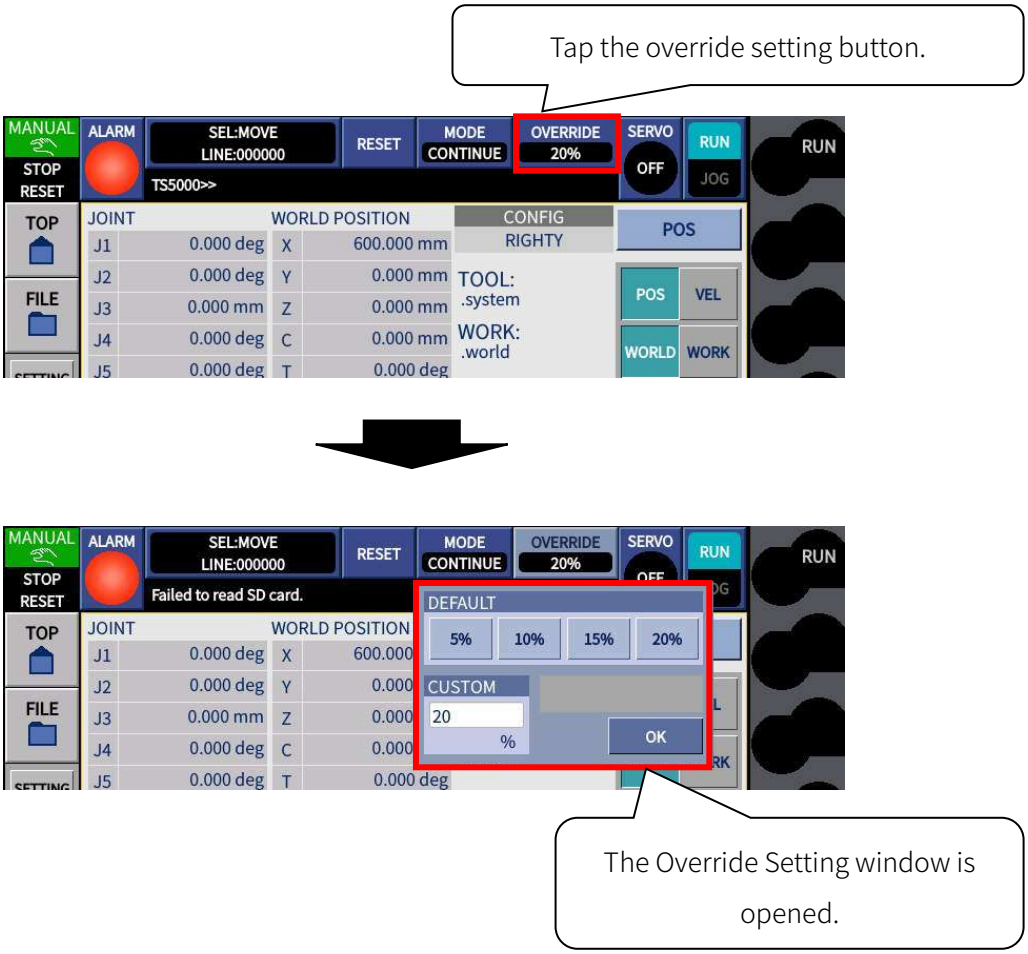
3.2.5 Setting Override

This describes how to set the override.

How to Set Override

1 Tap the override setting button (indicated by red box) in the status bar.

The Override Setting window is opened.



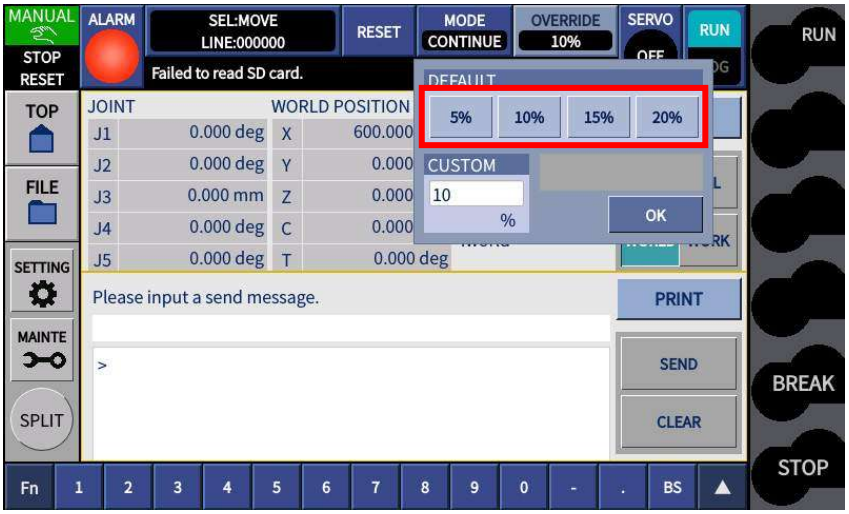
2 Enter the override value to be set into the text box in the Override Setting window.

Data can be entered into text boxes using one of the two methods below.

Entering Default Override Value

1. Tap one of the override input buttons (indicated by red box).

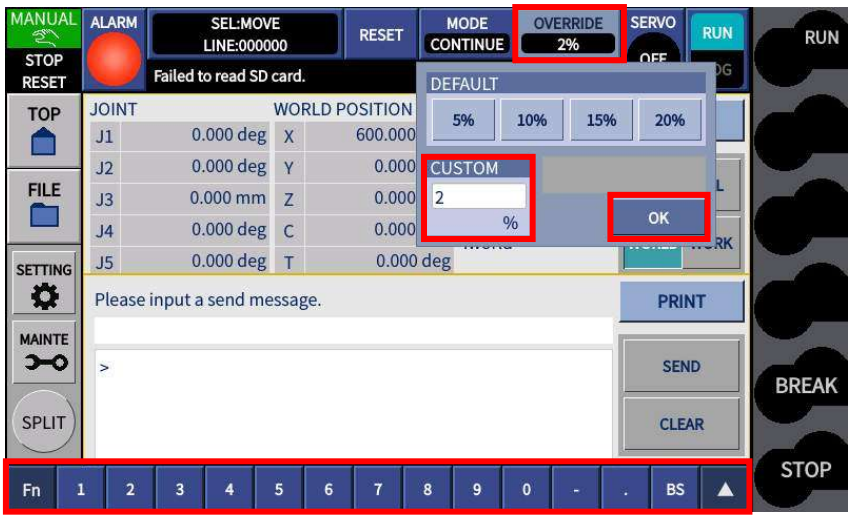
The value for the tapped button is input to the text box.



MEMO

The value that was already input to the text box is overwritten.

Entering Custom Value



1. Tap the text box in the override setting window.

The cursor appears in the text box to indicate that the content can be edited.

2. **Use the keyboard to enter a custom value.**

3. **After editing is completed, press the Enter key or tap any location outside the text box in the Override Setting window.**

   This saves the edited changes, and the cursor disappears.

MEMO

If a non-numerical entry is made into the text box, the characters in the text box appear in red.

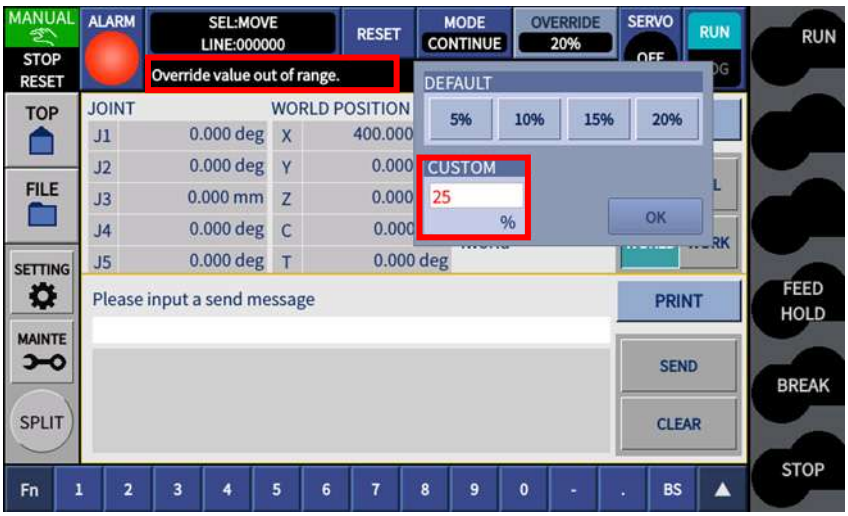
4. **Tap the SET button.**

   The value in the text box is set for the override.

   Also, the override setting value is displayed on the override setting button.

MEMO

- *In MANUAL mode, the override value can be set from 1 to 20.*
  When a value outside this range is entered into the text box, the override does not change even if the SET button is tapped.
- *The override can be set even during program execution.*



3.2.6 Executing Program

This describes how to execute the execution program.

MEMO

Before program execution, check that the execution program has been selected and that the execution mode and override have been set correctly.

How to Execute File

1 Tap the SERVO ON button while pressing down the enable switch by one level.

This turns on the servo, and the SERVO ON button indicator is changed.

MEMO

• The servo cannot be set ON if an alarm at level 8 or higher has been generated.

• If the SVOFF signals are OFF, the servo cannot be set ON.

2 Press the displayed operation key that is assigned to "RUN".

This starts program execution from the current line.

⚠ DANGER

●Be sure to switch the master key mode to MANUAL, and carry the teach pendant. Failure to do so may result in an accidental collision with the robot activated by the third person.

⚠ WARNING

●Do not enter the hazard zone during test operation. Doing so may result in an accidental collision with a robot.

●Do not leave the equipment that may hinder work in the work space. Doing so may cause the worker to trip over it and fall. Concentrating on teaching may divert attention to the surrounding and result in an accidental collision with the peripheral device.

NOTICE

- Do not place an obstacle within the operating range of the robot. Doing so may result in an accidental collision with the robot.
- Be sure to check the program execution number before test operation. Failure to check it may interfere with the peripheral device, resulting in device breakdown.
- Be sure to turn off the main power after turning off the servo power during robot stop. Turning off the main power during robot motion may cause the arm to idle and collide with the peripheral device.
- Perform program execution in a state where the enable switch or emergency stop switch can always be accessed. Not doing so may result in an accidental collision with a robot in unexpected motion.
- Before starting manual guidance, be sure to check the selected guidance mode, guidance coordinates, guidance speed, and guidance amount. Failure to check them may interfere with the peripheral device, resulting in device breakdown.

MEMO

During program execution, the operation status panel in the status bar changes as shown below

Screen when program execution is stopped



Screen during program execution



MEMO

The program is not executed even if the above operation is performed in the following cases.

- *When an alarm at level 2 or higher has been generated*
- *When the STOP signal is OFF*
- *When the BREAK signal is OFF*
- *When the signal FEEDHOLD function is enabled and 0 is written to the register G12C*
- *When Servo ON is not set*
- *When an execution program has not been selected*

3.2.7 Stopping and Aborting Program Execution

This describes how to stop program execution.

Program execution can be stopped by a stop or abort operation.

Each procedure is described in detail below.

Stop: Stops program execution once execution of the current command is completed.

Abort: Stops program execution immediately. In this state, the robot decelerates to a stop.

3.2.7.1 Stopping Program Execution

This describes how to stop program execution.

How to Stop Program Execution

1 Press the displayed operation key that is assigned to "STOP".

The program is stopped once execution of the current command is completed.

MEMO

While a program is stopped, the status indicator changes as shown below.

Screen during program execution



Screen when program execution is stopped



MEMO

In addition to the above operation, the following operations can be used to stop program execution.

• Setting the STOP signal to OFF during program execution

(For details, refer to the STA1167 Interface Manual.)

• Sending the communication command "SOP1" to the HOST port of the controller

(For details, refer to the STA1164 Communication Manual.)

• Executing the STOP command in SCOL2 language

(For details, refer to STA1163 STA1163-0 Robot Language Manual (SCOL2).)

3.2.7.2 Aborting (BREAK) Program Execution

This describes how to abort program execution.

How to Abort Program Execution

1 Press the displayed operation key that is assigned to "BREAK".

This aborts execution of a program and immediately decelerates the robot in motion to a stop.

MEMO

After aborting, when program execution is performed, program execution is performed again from the execution line where execution was aborted.

MEMO

In addition to the above operation, the following operations can be used to abort program execution.

• Setting the BREAK signal to OFF during program execution

(For details, refer to the STA1167 Interface Manual.)

• Sending the communication command "BRK1" to the HOST port of the controller

(For details, refer to the STA1164 Communication Manual.)

3.2.8 Pausing (FEEDHOLD) Program Execution

This describes how to pause/resume program execution.

How to Pause Program Execution

1 Press the displayed operation key that is assigned to "FEEDHOLD".

This pauses the program execution.

MEMO

When program execution is paused, the operation status panel in the status bar changes as shown below.

Screen during program execution



Screen while program execution is paused



MEMO

Even while program execution is paused, you can perform the stopping/aborting operation.

If such operation is performed, the program execution stops immediately.

MEMO

In addition to the above operation, the following operation can be used to pause program execution.

• Enabling [P006] and set the FEEDHOLD (register G12C) to OFF during program execution

(For details, refer to STA1166 Parameter Manual and STA1168 TCMINI (Internal PLC) Manual.)

How to Resume Program Execution

1 Press the displayed operation key that is assigned to "FEEDHOLD".

This resumes the program execution.

MEMO

In addition to the above operation, the following operation can be used to pause program execution.

• Enabling [P006] and set the FEEDHOLD (register G12C) to OFF during program execution

(For details, refer to STA1166 Parameter Manual and STA1168 TCMINI (Internal PLC) Manual.)

NOTICE

- Resume program execution in a state where the emergency stop switch can always be accessed.
  Failure to do so may cause the robot in unexpected motion to collide with the peripheral device.

3.2.9 Reset

This describes the reset operations.
The following four reset operations are available.

• Program reset (PROG)
This resets the execution line and variables of the file being executed.

• Cycle reset (CYCLE)
This resets the variables except for the global variables of the file being executed and moves the current execution line to RCYCLE.

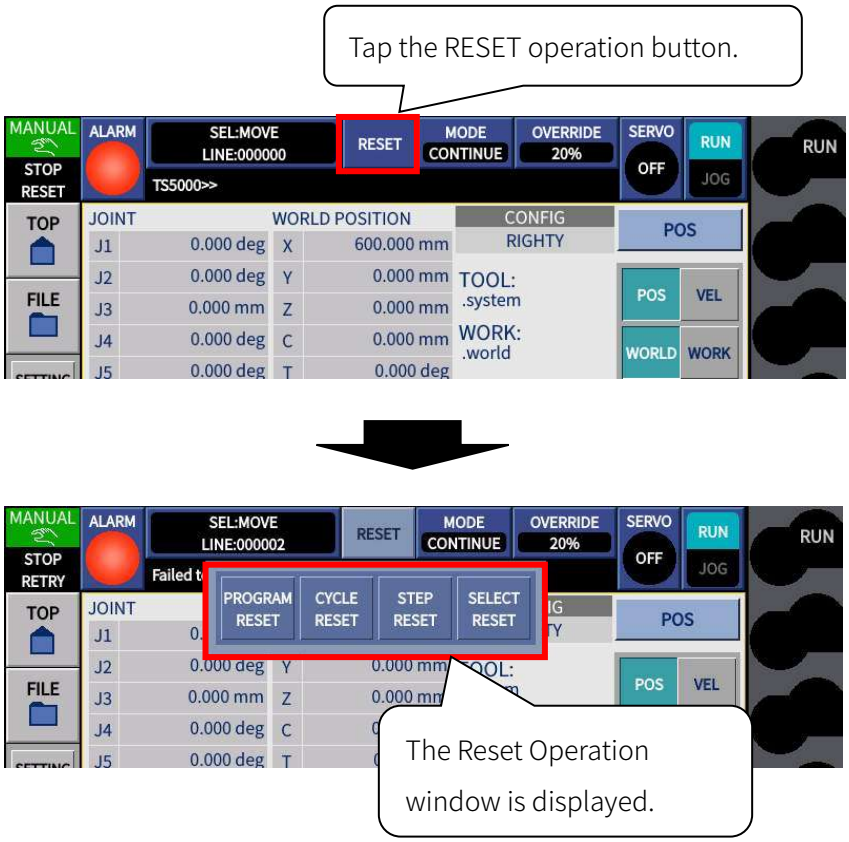
• Step (STEP)
This resets the variables except for the global variables of the file being executed and resets the execution line.

• Select reset (SEL)
This resets the execution programs that are currently registered so that no execution programs are registered.

Reset Procedure

1 Tap the RESET operation button (indicated by red box) in the status bar.

This opens the Reset Operation window.



2 Tap the RESET operation button (indicated by red box) for the reset operation to be performed.

The reset operation for the tapped button is performed.



3.2.10 Monitoring of Execution Programs

This describes how to monitor the content of the execution program.

The content of the execution program is displayed on the LDISP screen.

How to Open LDISP Screen

1 Tap the Utility menu of the main window.

The Utility Menu Switching window is displayed.



2 Tap the LDISP menu button.

The LDISP screen is opened in the main window.



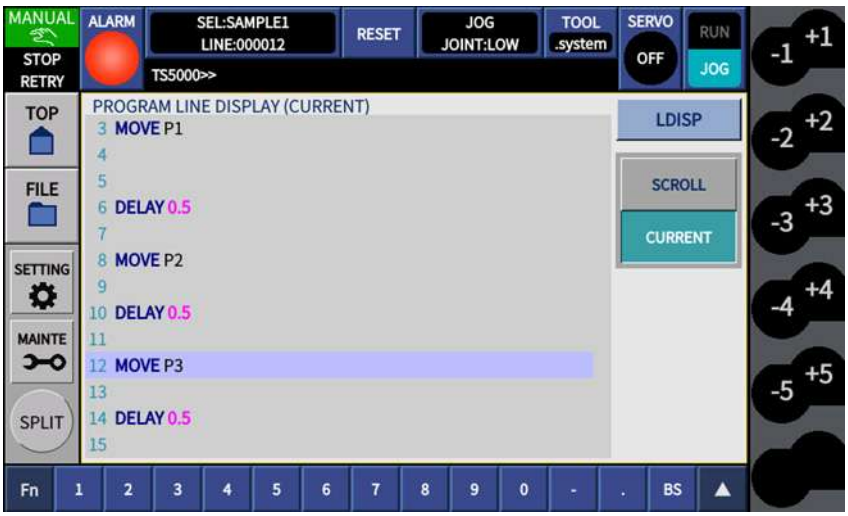
The content of the execution program can be displayed on the LDISP screen using one of the following two methods.

- Display the content in synchronism with the line currently being executed (synchronous display)
- Display any selected line (asynchronous display)

> *MEMO*
>
> *Tapping the SCROLL/CURRENT button can switch between synchronous display and asynchronous display.*

Synchronous Display of Execution Program

In synchronous display, the screen is scrolled based on the line currently being executed. The line currently being executed is highlighted in blue.



Asynchronous Display of Execution Program

In asynchronous display, the cursor can be moved to scroll the screen regardless of the execution line.

The line where the cursor is located changes is highlighted in yellow.

The line currently being executed is highlighted in blue.



3.2.11 Display and Overwriting of Global Variable

This describes how to display and overwrite the global variables in the execution program.

Global variables refer to variables defined in the global section (area enclosed by GLOBAL and END) or the data section (area enclosed by DATA and END) in the program file.
Global variables can be displayed and overwritten from the WATCH screen.
For details on the WATCH screen, see "11.8. WATCH Screens" in this manual.

How to Open WATCH Screen

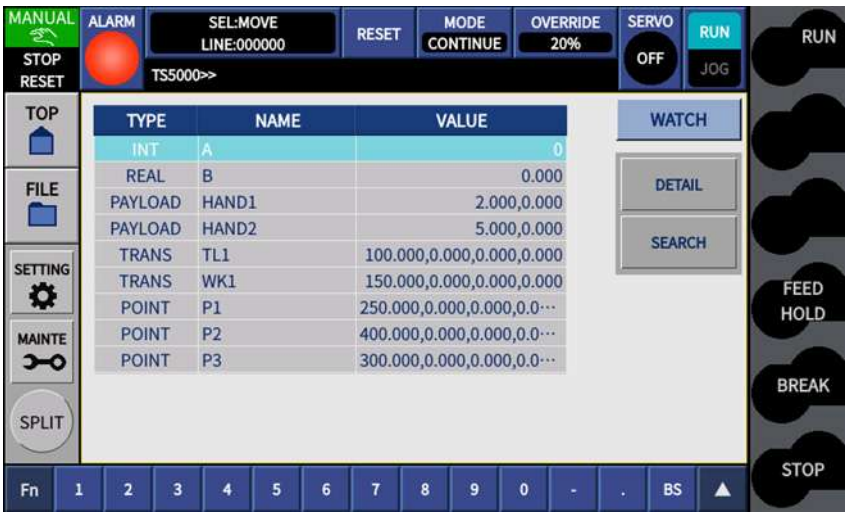
1 Tap the Utility menu of the main window.

The Utility Menu Switching window is displayed.



2 Tap the WATCH menu button.

The WATCH screen is opened in the main window.



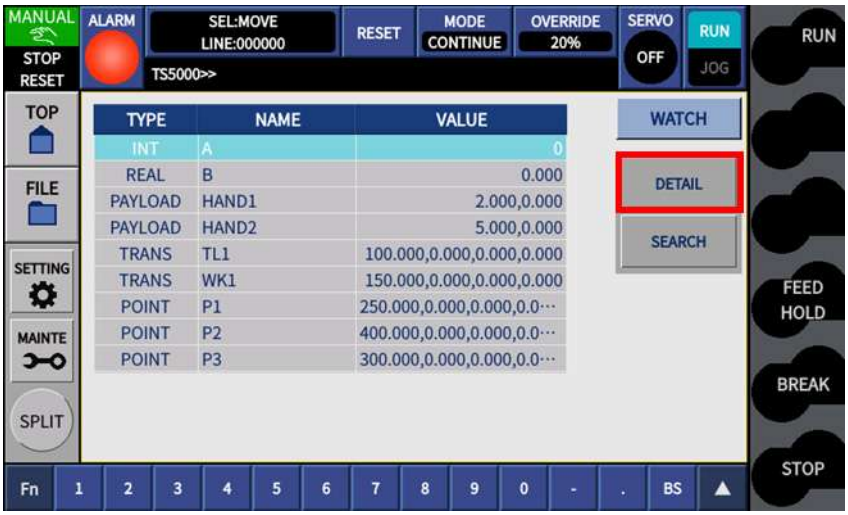
Global Variable Details Display Procedure

1 From the list, select the variable whose details you want to view or whose value you want to overwrite.

The background color and text color change for the selected line.

2 Tap the DETAIL button.

Details for the selected global variable are displayed.



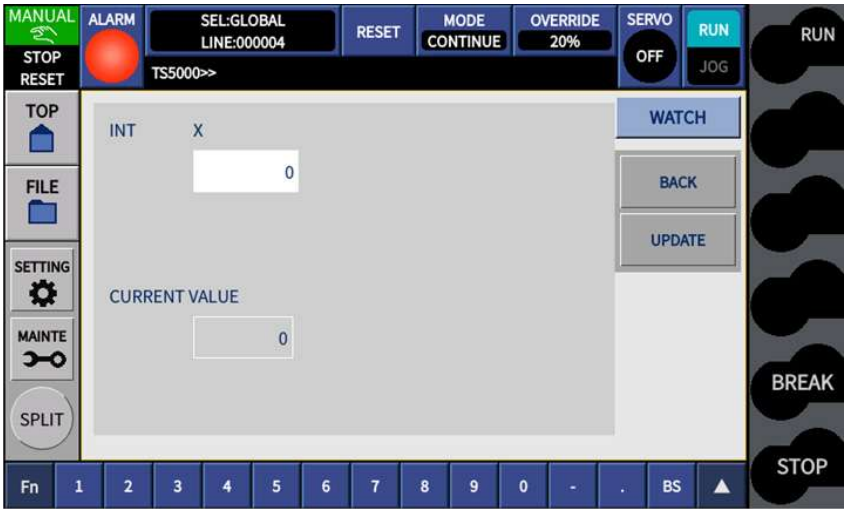
MEMO

For array-type variables, after editing the text box showing the element number, tap the SHOW button to display the desired variable.

Global Variable Overwrite Procedure

1 Tap the text box.

The cursor appears so that the number can be edited.



2 Use the keyboard to enter the overwriting value.

3 After editing, press the Enter key or tap a location on the screen outside the text box.

This saves the edited changes.

* If the value in the text box is outside the range of -9999.999 to 9999.999, the entered value is not set, and the characters in the text box appear in red.

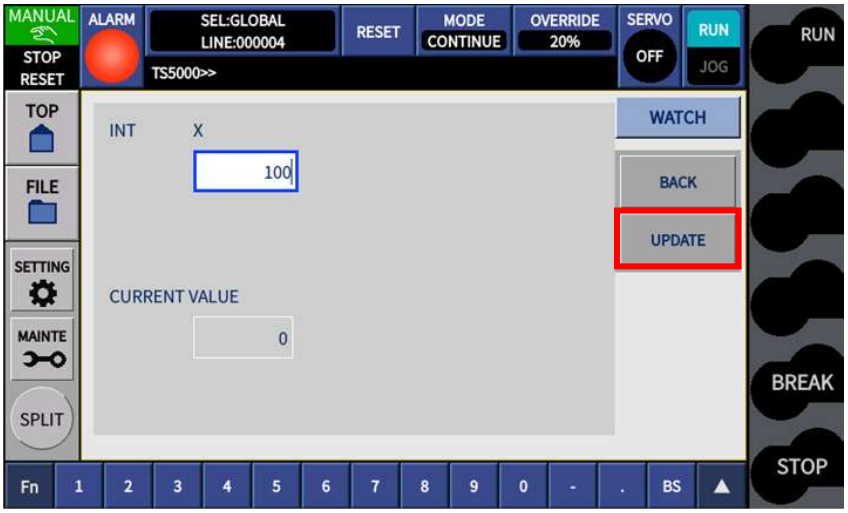
At this stage, the value for the global variable being used in the program is not overwritten yet.

To overwrite the value, perform procedure 5.

4 Edit all the elements for the value to be overwritten by performing procedures 1 to 3.

5 Tap the UPDATE button.

The value of the currently-selected variable is overwritten by the value in the text box.



MEMO

- *To prevent the robot from running out of control due to erroneous input, for position-type and coordinate-type variables, overwriting to element values that exceed ±3 mm (or deg) from their current value is not allowed.*
- *The overwritten value is a temporary change, and the value is reset by performing a program reset. To permanently apply the overwritten value, consider either editing the file or using a memory variable. For details on memory variables, refer to the section "MEMORY" in "STA1163 Robot Language Manual (SCOL2)."*

NOTICE

●Before overwriting a target global variable, confirm the effect that the variable will have on the program. Failure to check it may cause the robot to collide with the peripheral device.

3.3 Direct Execution

This describes how to perform direct execution.

Direct execution is a function that enables execution of one line of commands in SCOL2 language without using test operation.

Direct execution is performed from the DO screen.

How to Open DO Screen

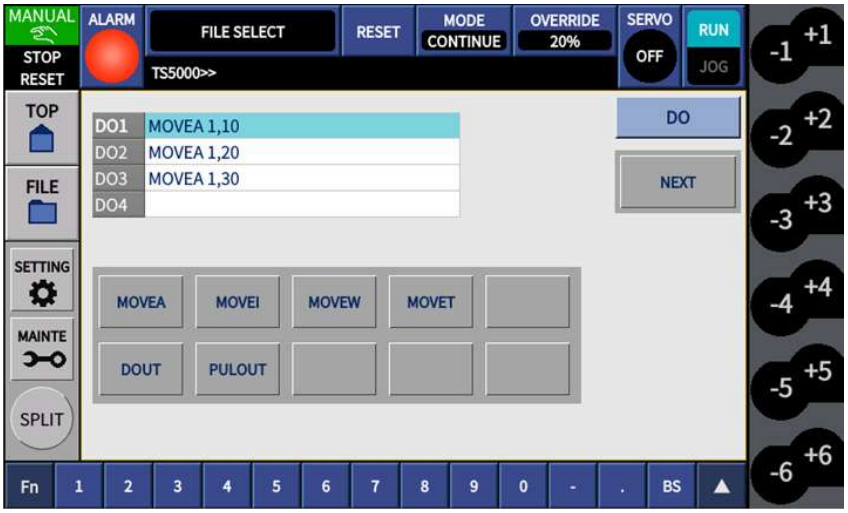
1 Tap the Utility menu of the main window.

The Utility Menu Switching window is displayed.



2 Tap the DO menu button.

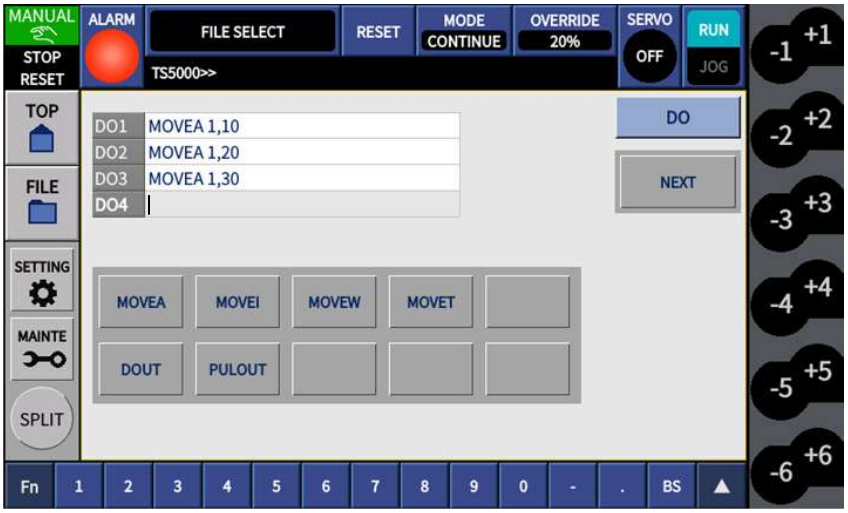
The DO (setting) screen is opened in the main window.



Direct Execution Procedure

1 Tap the text box.

The cursor appears so that the number can be edited.



2 Use the keyboard to enter the character string for direct execution, and either press the Enter key or tap a location on the screen outside the text box and buttons.

This saves the edited changes.

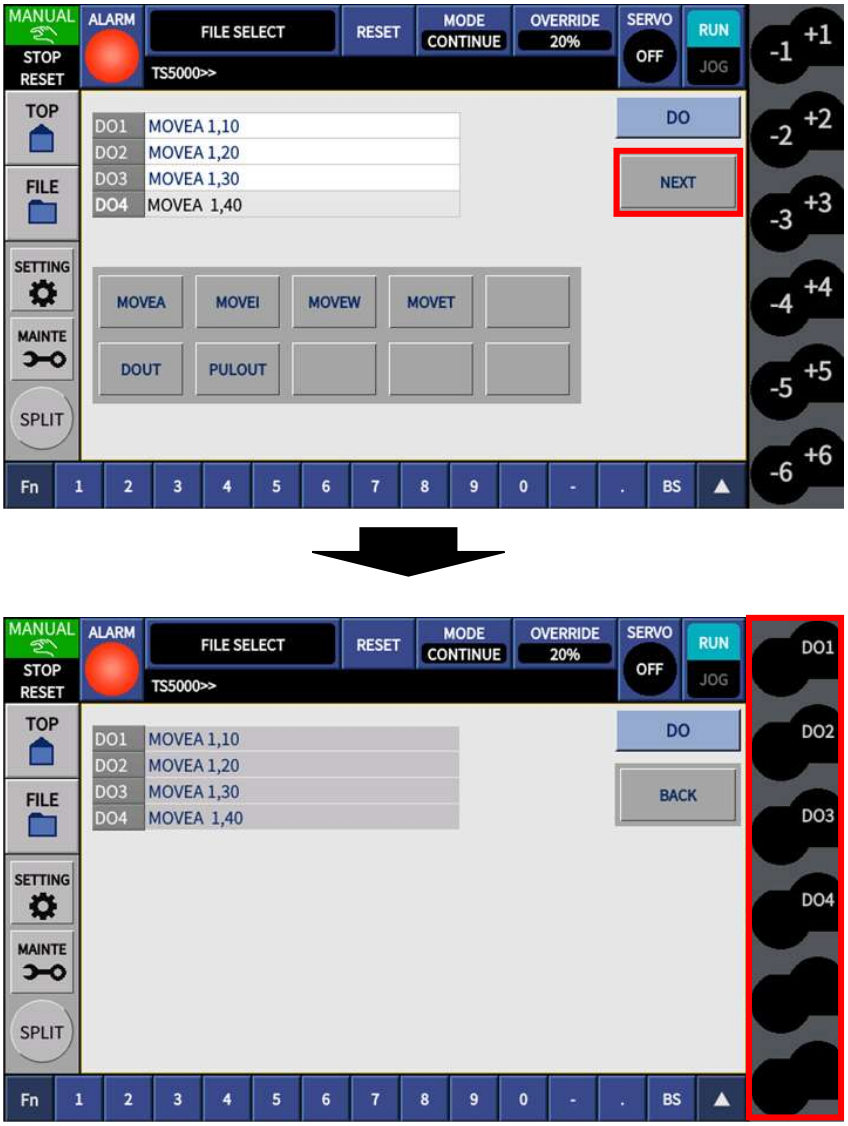
MEMO

The character strings in the DO1-DO4 text boxes are retained even when the controller is turned off.

3 Tap the NEXT button.

The DO (execution) screen is opened.

The displayed operation keys change as shown below.



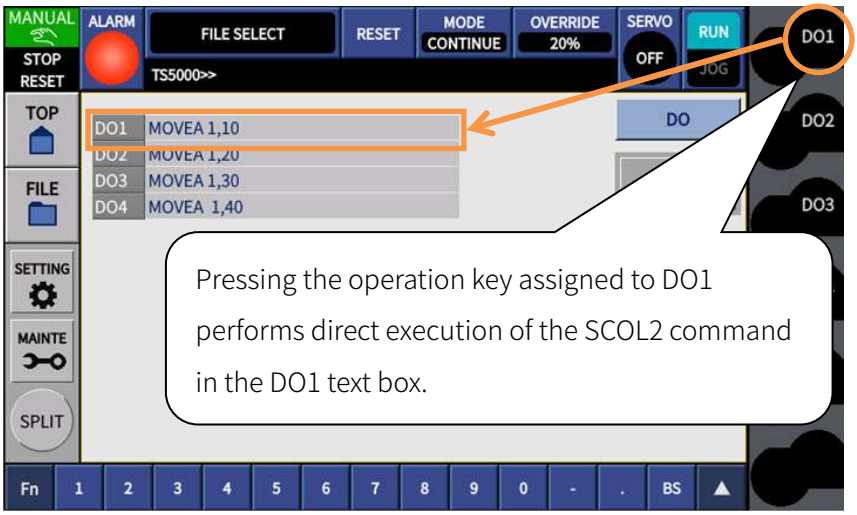
MEMO

The displayed operation keys do not change while the DO (execution) screen is open.

To change the displayed operation keys, either tap the BACK button or switch the displayed screen of the Utility Window from the DO screen.

4 Set Servo ON, and press the operation key where the character string for direct execution is assigned.

Perform direct execution of the character string in the text box.



MEMO

Direct execution is not performed even if the above operation is performed in the following cases.

- *When an alarm at level 2 or higher has been generated*
- *When the STOP signal is OFF*
- *When the BREAK signal is OFF*
- *When the signal FEEDHOLD function is enabled and 0 is written to the register G12C*
- *When Servo ON is not set*
- *When the SCOL2 command in the text box has a syntax error*

⚠ WARNING

●Before setting ON the robot servo, be sure to check that no person exists within the movable range of the robot. Failure to do so may cause the robot in motion to collide with the person existing within the movable range.

• When entering the movable range, leave the safety door open, carry the teach pendant placed in the MANUAL mode, and indicate that work is underway.

• The LED lamp of the robot can be used to check the servo status.

NOTICE

- Before performing direct execution, check that the character string to be executed has been entered correctly. Failure to check it may cause robot to make an unexpected motion, resulting in a collision with the robot and peripheral device.

4. File Editing and Operations

This chapter describes the file editing and operating procedures.

The execution programs used by the TS5000 have the following structure.

Execution program
- Global section
- Program section
- Data section

Global variables are defined in the global section, operation commands for the robot are contained in the program section, and POINT-type or TRANS-type data is described in the data section.

4.1 Opening File Edit Screen

This describes how to open the file edit screen.

How to Open File Edit Screen

1 Tap the FILE button.

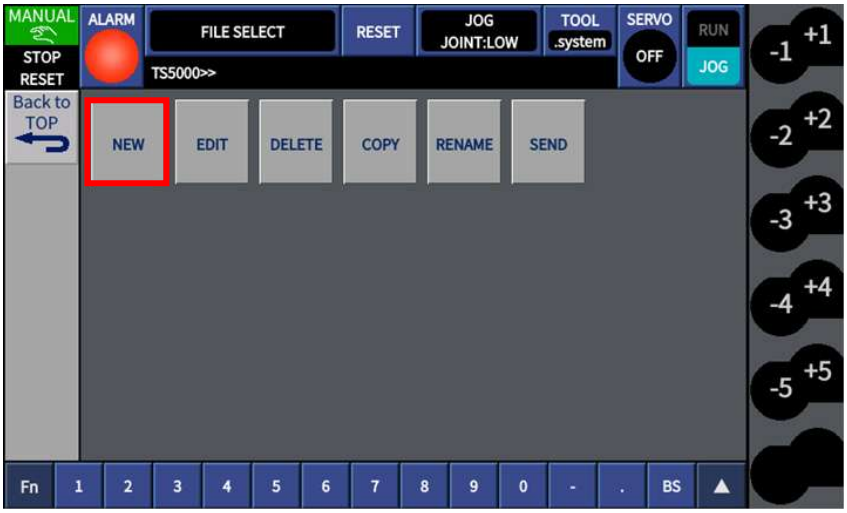
The file operation menu screen is opened.



2 To create a new file, tap the NEW button.

To edit an existing file, tap the EDIT button.

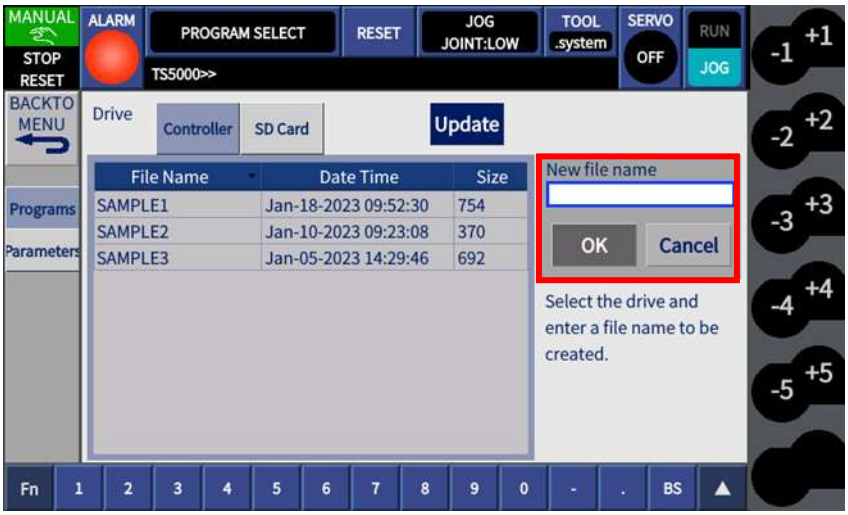
A file list is displayed.



1) To create a new file

1. Tap the text box.

The cursor appears so that the number can be edited.



2. Enter the name of the new file.

MEMO

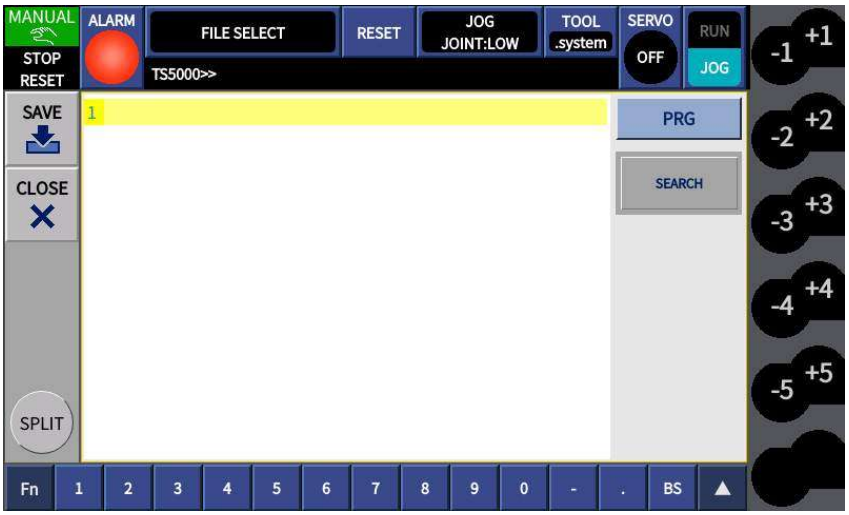
- *The file naming rules are as follows.*

  File name consists of a maximum of 8 alphanumeric characters.

  The extension can use a maximum of 3 characters.
- *Tapping a file displayed in the file list enters the file name into the text box.*

3. After entry of the file name is complete, tap the OK button.

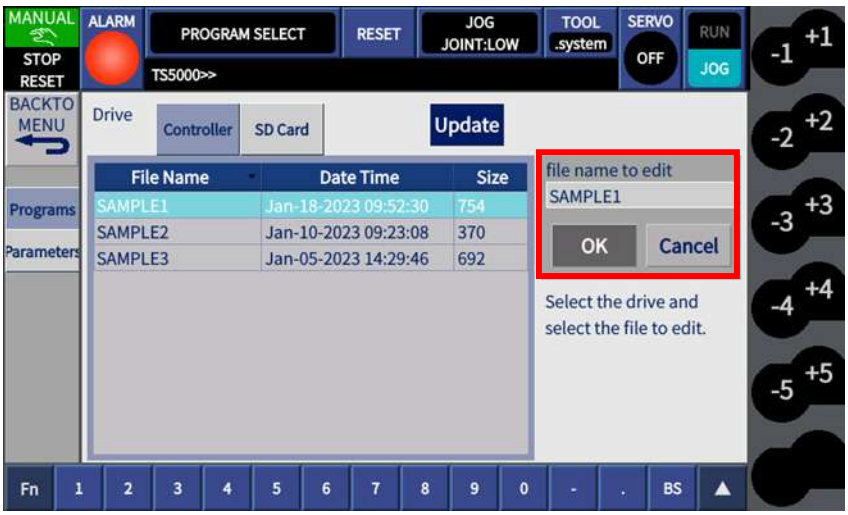
The new file is created, and the file edit screen is opened.



2) To edit an existing file

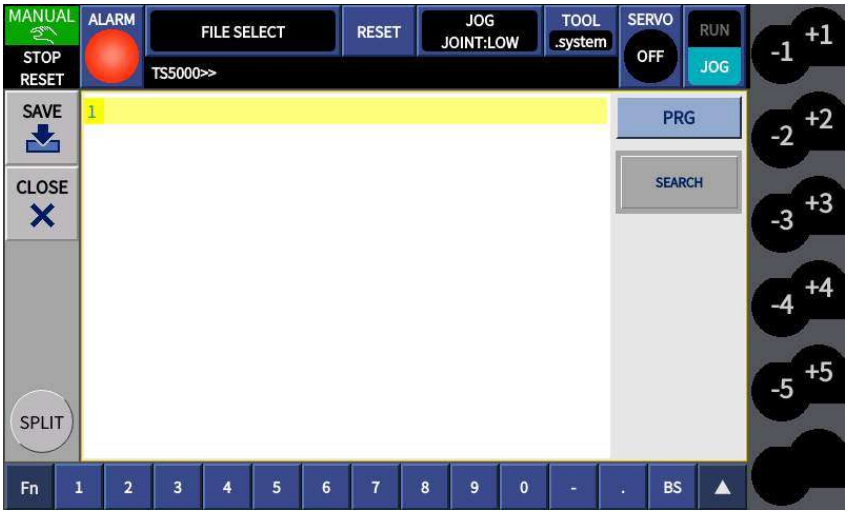
1. Select and tap the file to be edited from the file list.

The name of the file selected is displayed in the box on the right side.



2. Confirm the name of the selected file, and tap the OK button.

The selected file is opened, and the file edit screen is opened.



4.2 Editing Global Section and Program Section

This describes the editing operations for the global section and program section.

4.2.1 Opening PRG Screen

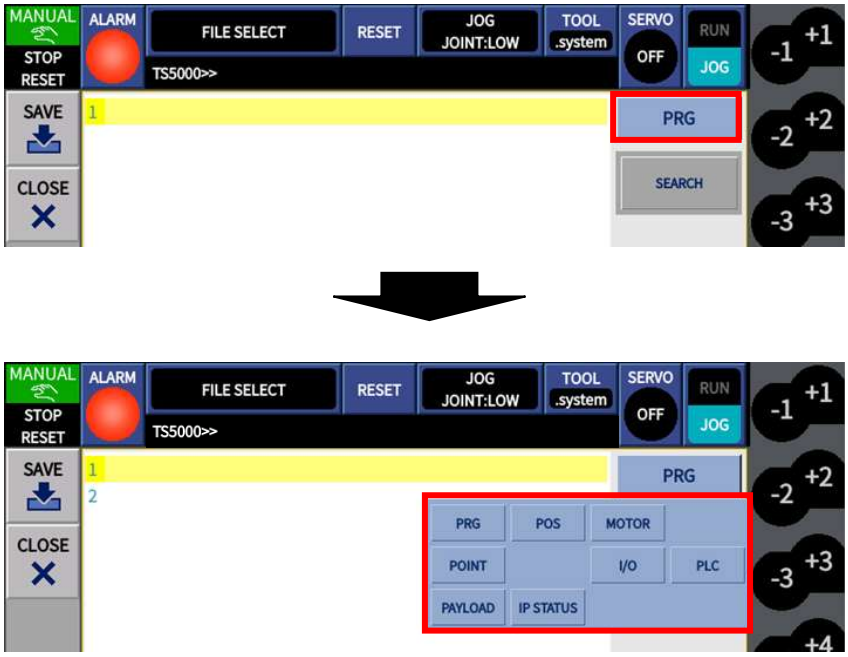
This describes how to open the PRG screen.

The global section and program section can be edited in the PRG screen.

How to Open PRG Screen

1 Tap the Utility menu button in the main window.

The Utility Menu Switching window is displayed.



2 Tap the PRG button in the menu.

The PRG screen is opened.

4.2.2 Editing Global Section and Program Section

Tap the letters, numbers, or symbols keys on the keyboard to enter characters at the cursor position.

Create a program by following the syntax rules for SCOL2 language.

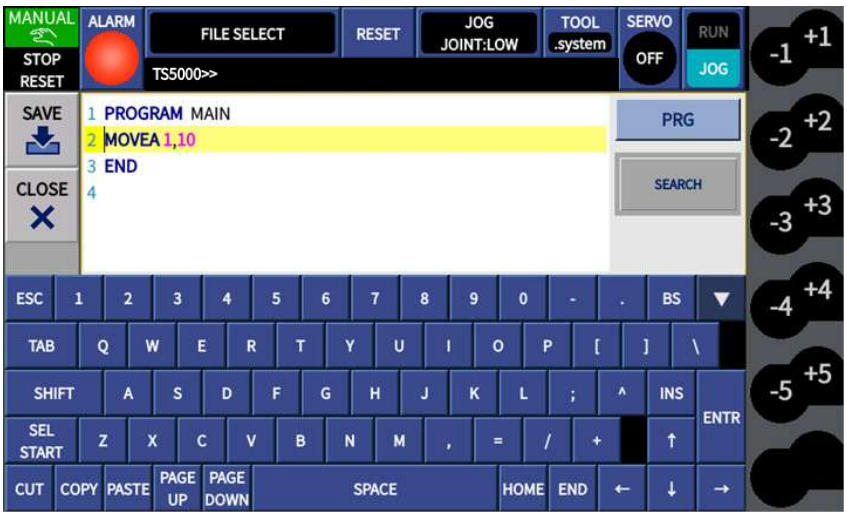
For details on the syntax rules for SCOL2 language, see the separate "STA1163 Robot Language (SCOL2) Manual".

4.2.3 Cutting Character String

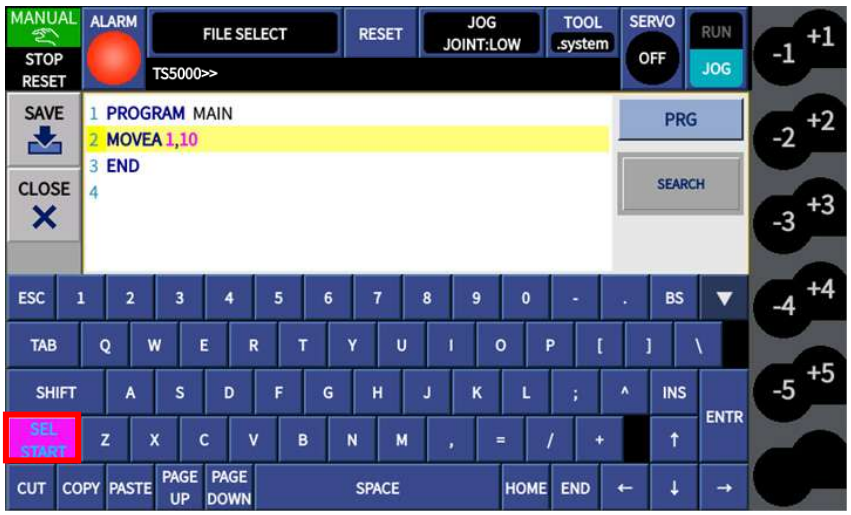
This describes how to cut a character string when editing the program section.

How to Cut Character String

1 Move the cursor to the top of the character string to be cut in the program section edit screen.

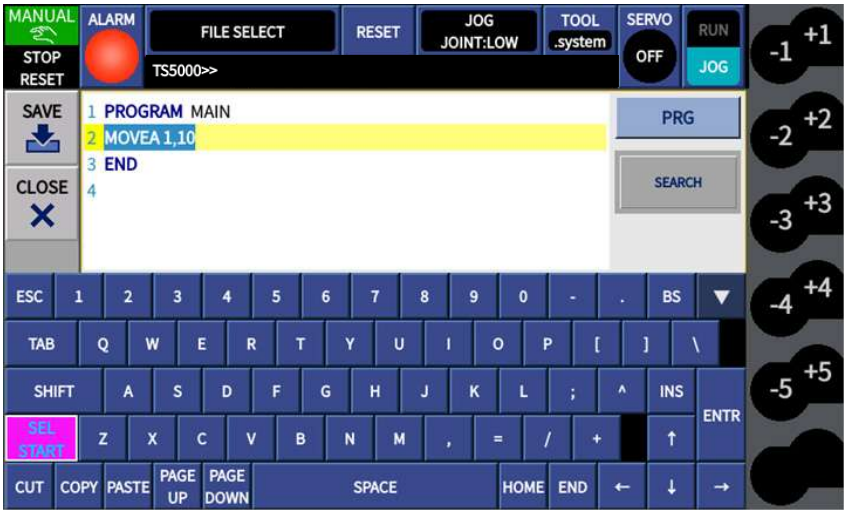


2 Tap the SEL START key on the keyboard.



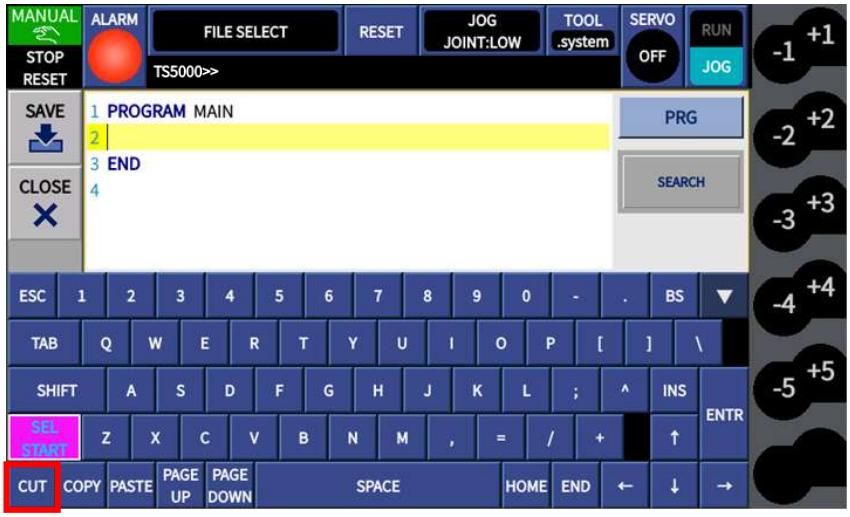
3 Move the cursor to the end of the character string to be cut.

The background color of the selected area changes.



4 Tap the CUT key on the keyboard.

This cuts the selected area.

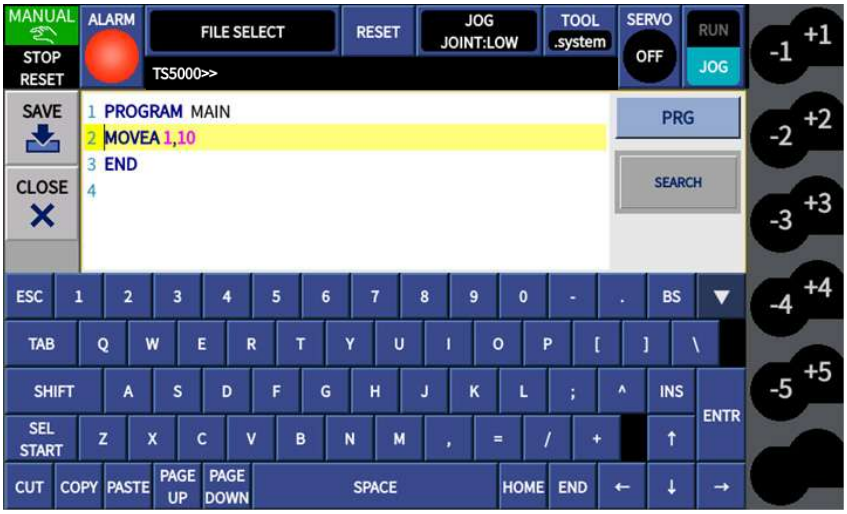


4.2.4 Copying Character String

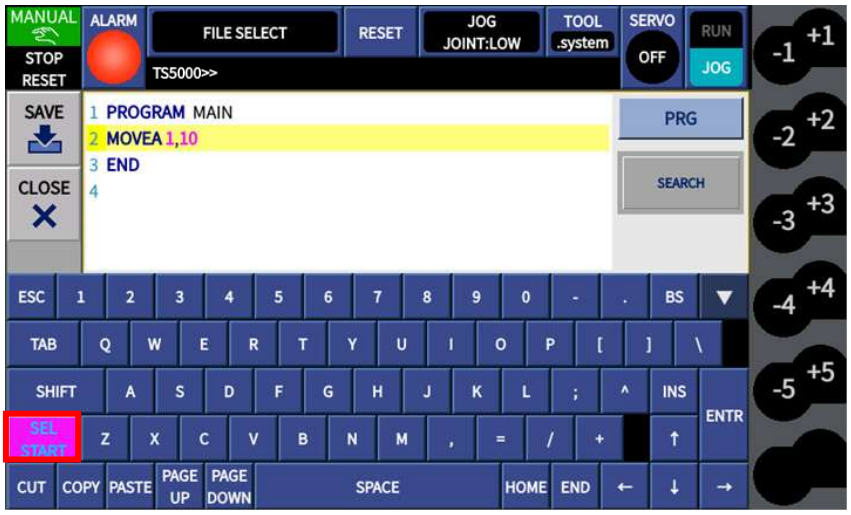
This describes how to copy a character string during edition of the program section.

How to Copy Character String

1 Move the cursor to the top of the character string to be copied in the program section edit screen.

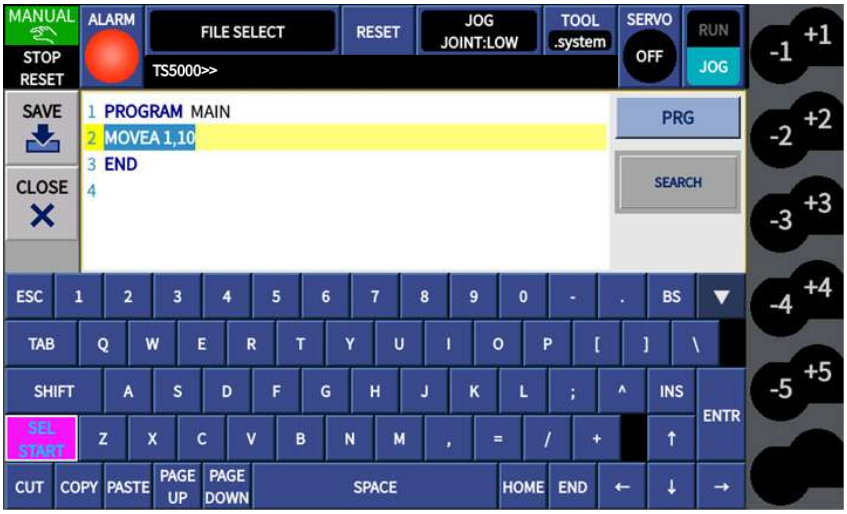


2 Tap the SEL START key on the keyboard.



3 Move the cursor to the end of the character string to be copied.

The background color of the selected area changes.



4 Tap the COPY key on the keyboard.

This copies the selected area.



4.2.5 Pasting Character String

This describes how to paste a character string during edition of the program section.

How to Paste Character String

1 Move the cursor to the position where the character string will be pasted in the program section edit screen.



2 Tap the PASTE key on the keyboard.

The cut or copied character string is inserted at the cursor position.



4.3 Editing Data Section

This section describes how to edit the data section.

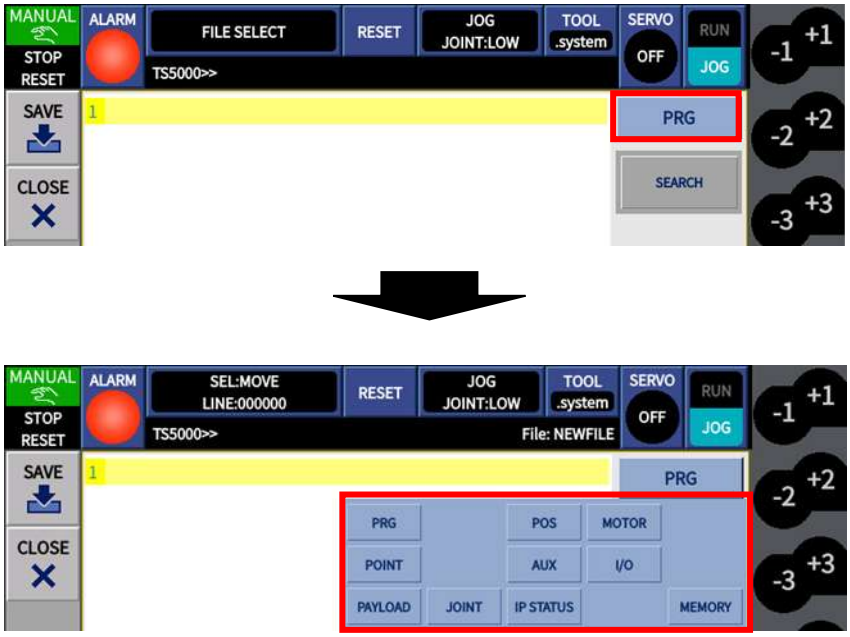
The data section is roughly divided into three data types: POINT-type data, coordinate-type data, and load-type data.

4.3.1 Opening POINT (Edit) Screen

How to Open POINT (Edit) Screen

1 Tap the Utility menu button in the main window of the file edit screen.

The Utility menu button is located at the position in the red frame in the following figure.



Tapping the button opens the Utility Menu Switching windows.

2 Tap the POINT button in the menu.

The POINT (Edit) screen is opened.



How to Open POINT (Edit) Screen When TRANS (Edit) Screen Is Open

1 Tap the POINT-TRANS selector button.

The TRANS (Edit) screen switches to the POINT (Edit) screen.

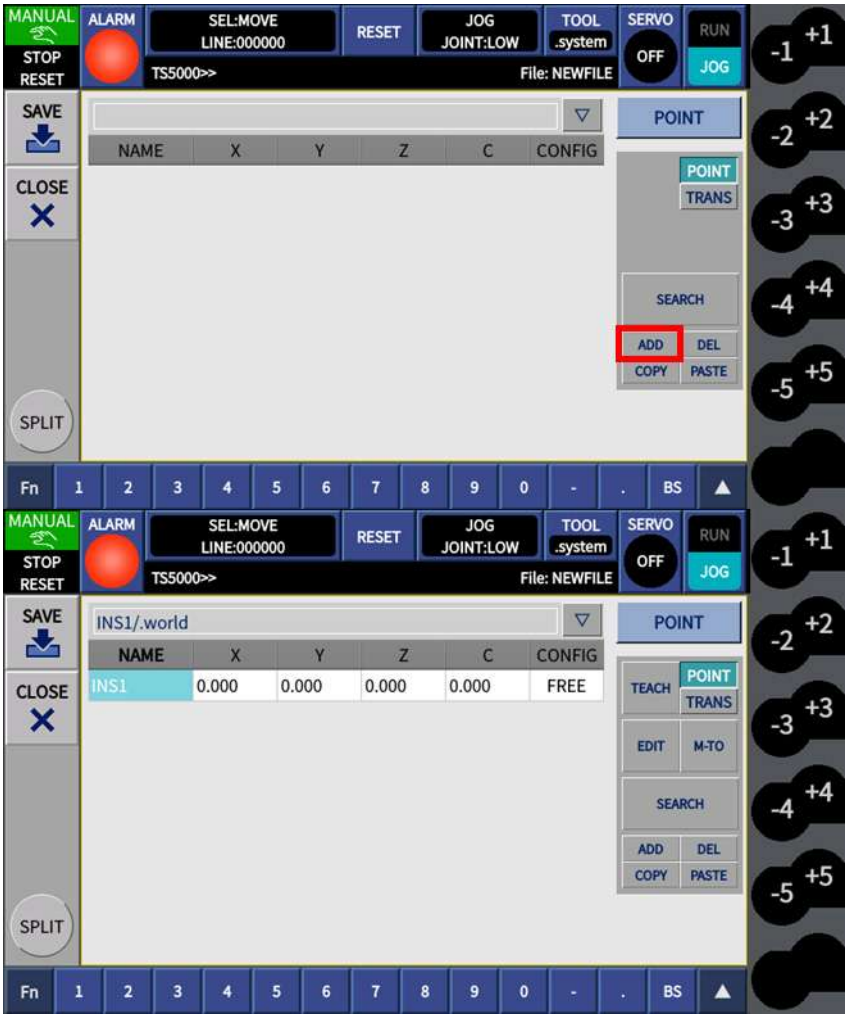


4.3.2 Editing POINT-type Data

How to Add POINT-type Data

1 Tap the ADD button.

A line is added.



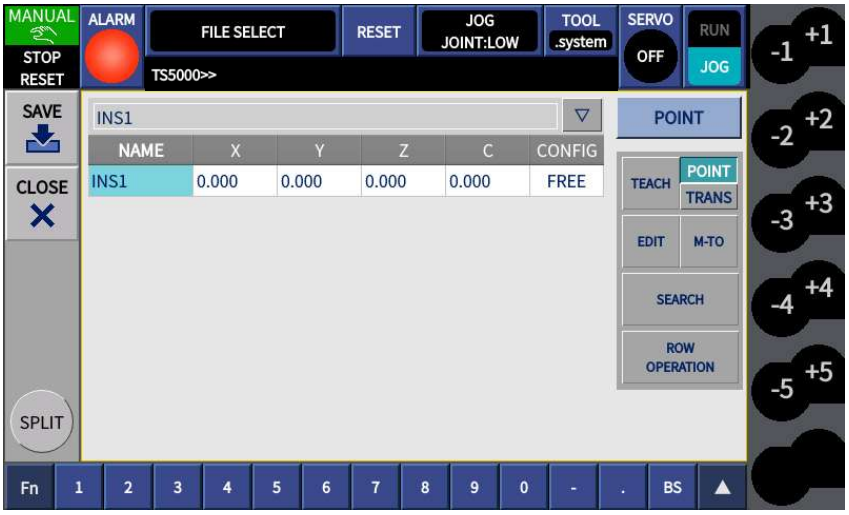
MEMO

Line Operation can be used to add, delete, copy, or paste lines.

How to Edit POINT-type Data Name

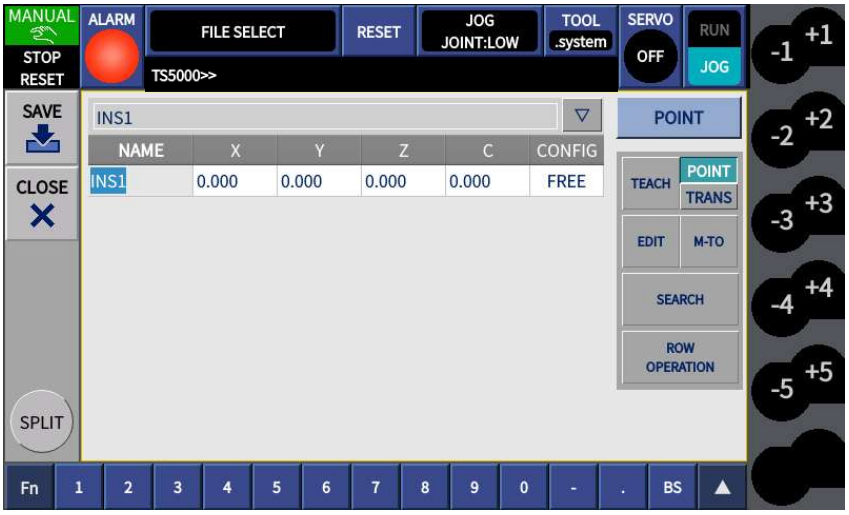
1 Tap the cell of the item to be edited.

The cell is selected and highlighted.



2 Tap the selected cell.

The cursor appears at the end of the cell so that the content can be edited.



3 Tap the character and number keys on the keyboard to enter the desired content.

MEMO

Tapping the ESC key erases all of the characters in the cell. If no characters are entered in the cell, editing is canceled, and the cell is selected.

4 After editing is completed, press the Enter key or tap a location on the screen outside the cell being edited.

This saves the edited changes, and the cell is selected.

MEMO

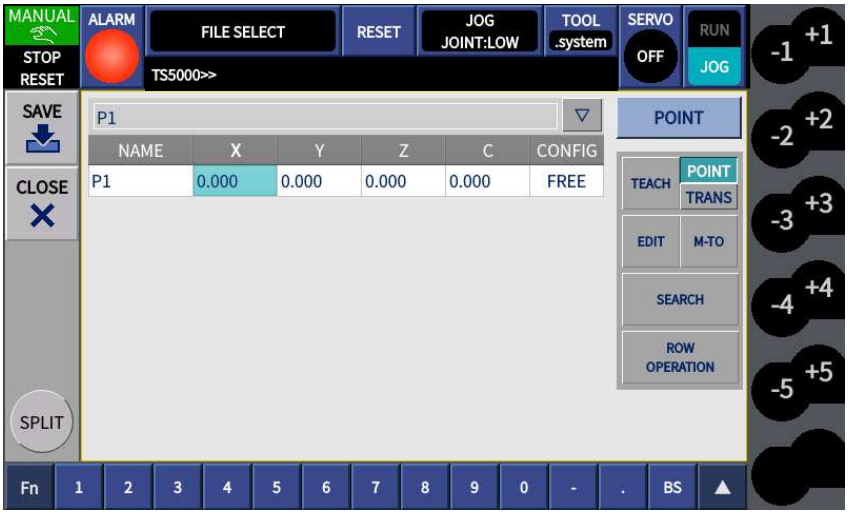
The character string that can be set for the POINT-type data name consists of a maximum of 20 alphanumeric characters starting with a letter.

If the character string exceeds 20 characters or includes symbols, the entered character string is not saved, and the content in the cell appears in red text.

How to Edit Coordinate Value of POINT-type Data

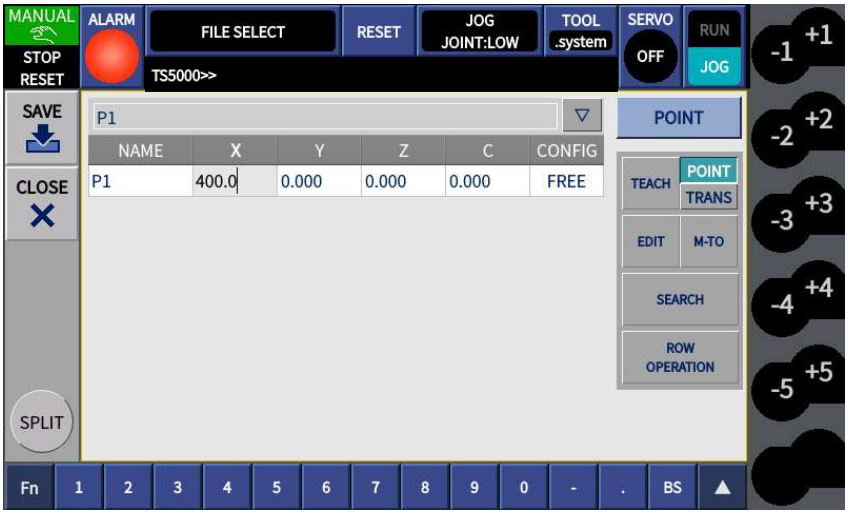
1 Tap the cell of the item to be edited.

The cell is selected and highlighted.



2 Tap the selected cell.

The cursor appears at the end of the cell so that the content can be edited.



3 Tap the number and symbol ("-" (minus) and "." (period)) keys on the keyboard to enter the number.

MEMO

Tapping the ESC key erases all of the characters in the cell. If no characters are entered in the cell, editing is canceled, and the cell is selected.

4 After editing is completed, press the Enter key or tap a location on the screen outside the cell being edited.

This saves the edited changes, and the cell is selected.

MEMO

The value input in the cell must be in the range from -9999.999 to 9999.999. If a value outside this range or a non-numerical character is entered, the entry is not set, and the content in the cell appears in red text.

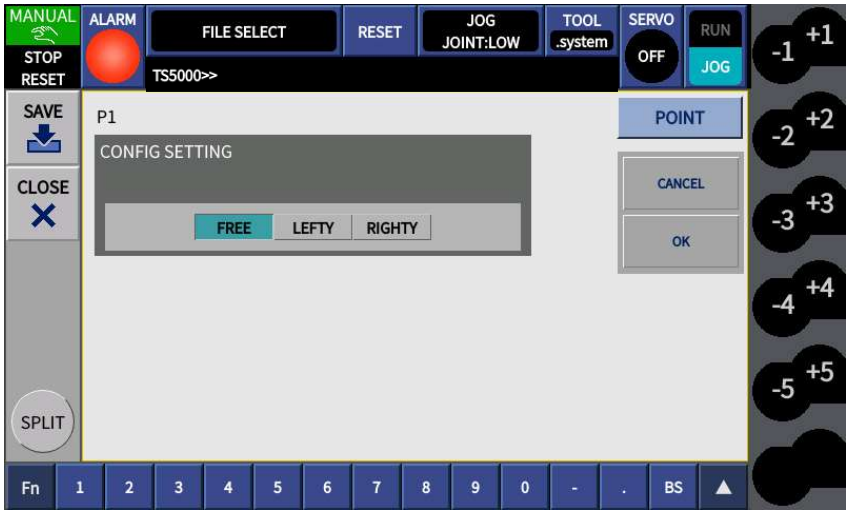
To Edit POINT-type Data Configuration

1 Tap the CONFIG cell of the POINT-type data to be edited.



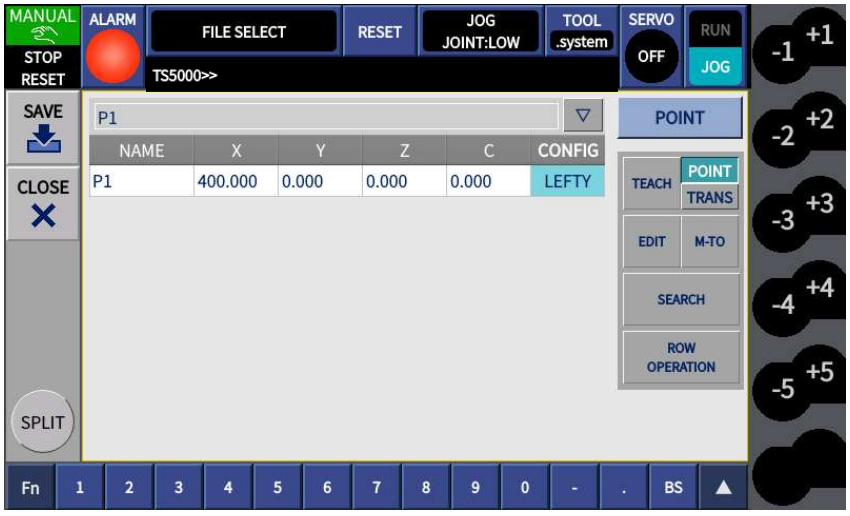
2 Tap the EDIT button.

The CONFIG setting screen is opened.



3 Tap the CONFIG button to be set, and tap the OK button.

The CONFIG of the POINT-type data for the selected line is overwritten.



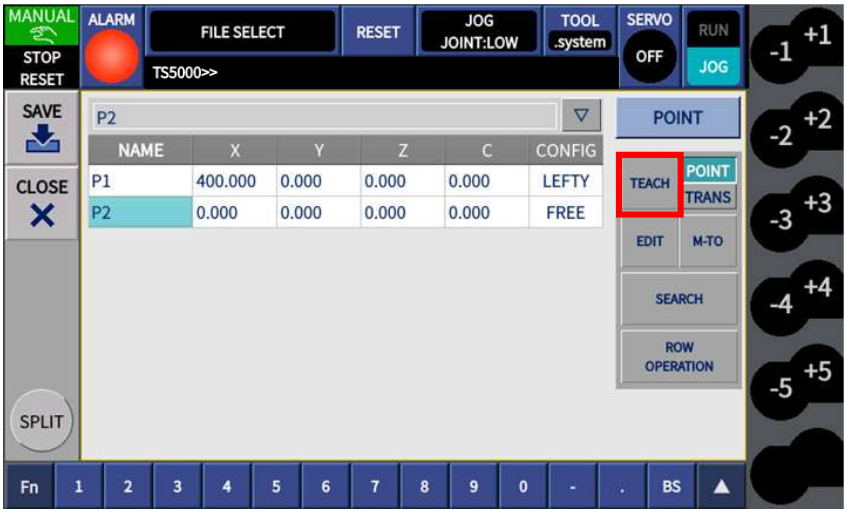
4.3.3 Teaching Current Position

This describes how to teach the current position for the robot TCP (tool center position) to the POINT-type data.

How to Teach Current Position

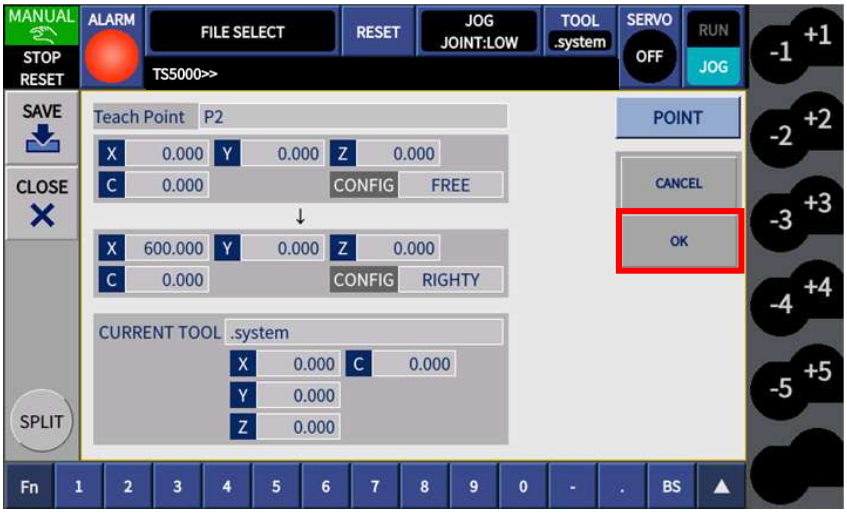
1 Tap the TEACH button on the POINT (Edit) screen.

The POINT (Teach) screen is opened.



2 Tap the OK button.

The selected POINT-type data value is overwritten by the current position of the TCP.



4.3.4 Moving Teaching Point

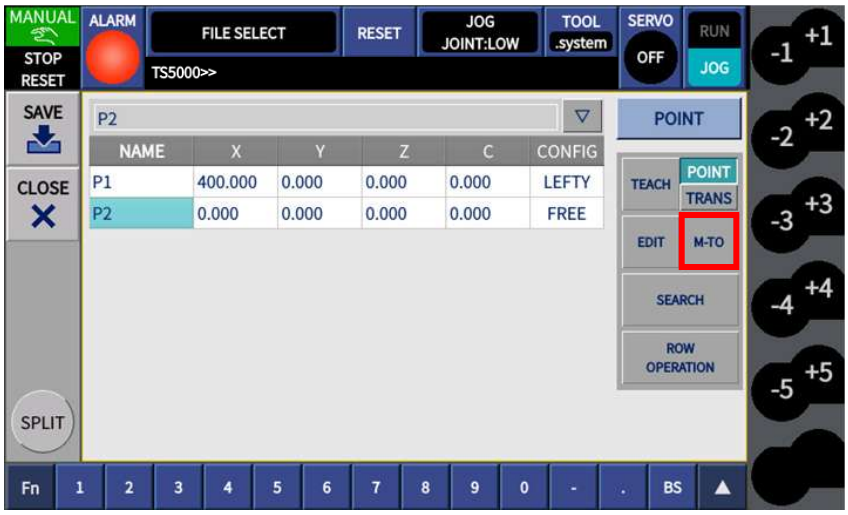
This describes how to move the robot TCP (tool center position) to the selected POINT-type data position.

The teaching point can be moved using PTP movement and linear interpolation movement.

How to Move Teaching Point

1 Tap the M-TO button on the POINT (Edit) screen.

The POINT (Teaching Point Movement Setting) screen is opened.

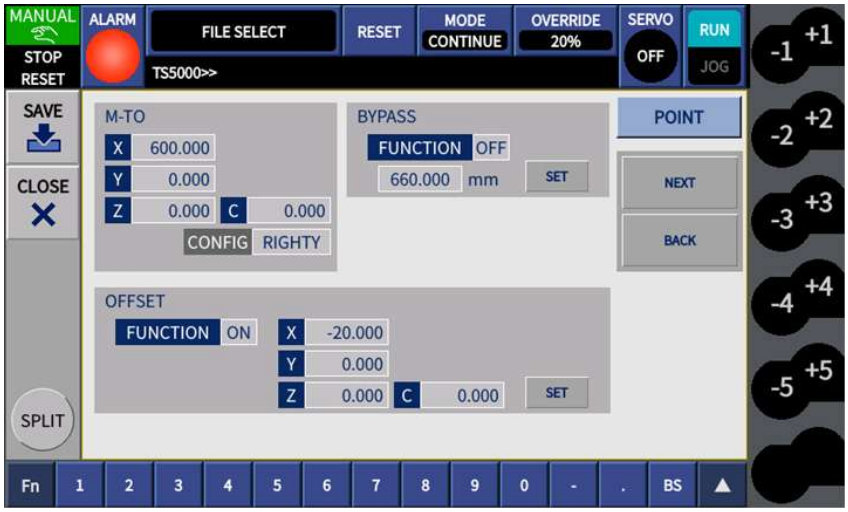


2 Confirm the coordinates of the teaching point, and tap the NEXT button.

The POINT (Teaching Point Movement Execution) screen is opened, and the displayed operation keys are changed.

The robot target position incorporating the offset and bypass settings is displayed on the POINT (Teaching Point Movement Execution) screen.

Check that the displayed target position is correct.



MEMO

For details on the offset and bypass settings, see "4.3.5. Moving Teaching Point to Offset Position" and "4.3.6. Moving Teaching Point by Bypass Movement"

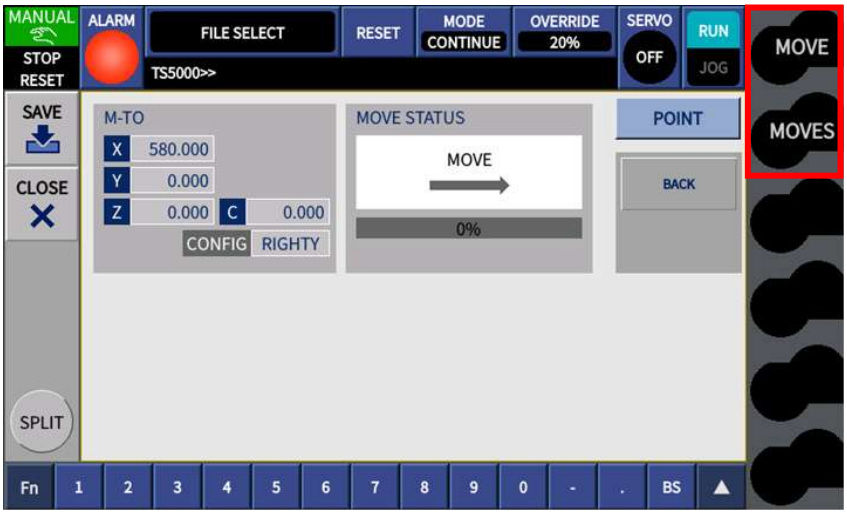
3 Tap the SERVO ON button while pressing down the enable switch by one level.

This turns on the servo, and the SERVO ON button indicator is changed.

4 Press the MOVE TO key or MOVES TO operation key.

The robot moves toward the target position while the MOVE TO or MOVES TO key is held down.

(The robot performs PTP movement when the MOVE TO key is pressed, and performs linear interpolation movement when the MOVES TO key is pressed.)



⚠ WARNING

- Do not enter the hazard zone during test operation. Doing so may result in an accidental collision with a robot.
- Do not allow a person other than an operator to approach the device. Doing so may result in an accidental collision with a robot.
- Perform teaching point movement in a state where the enable switch or emergency stop switch can always be accessed. Failure to do so may cause the robot to collide with the peripheral device.

NOTICE

- Check that the target position is set correctly. Failure to check it may cause the robot in unexpected motion to collide with the peripheral device.
- Be sure to check that the tool coordinates are set correctly. Failure to check it may cause the robot in unexpected motion to collide with the peripheral device.

4.3.5 Moving Teaching Point to Offset Position

This describes how to move the teaching point to the position that was offset from the POINT-type data.

How to Make Teaching Point Movement to Offset Position from POINT-type Data

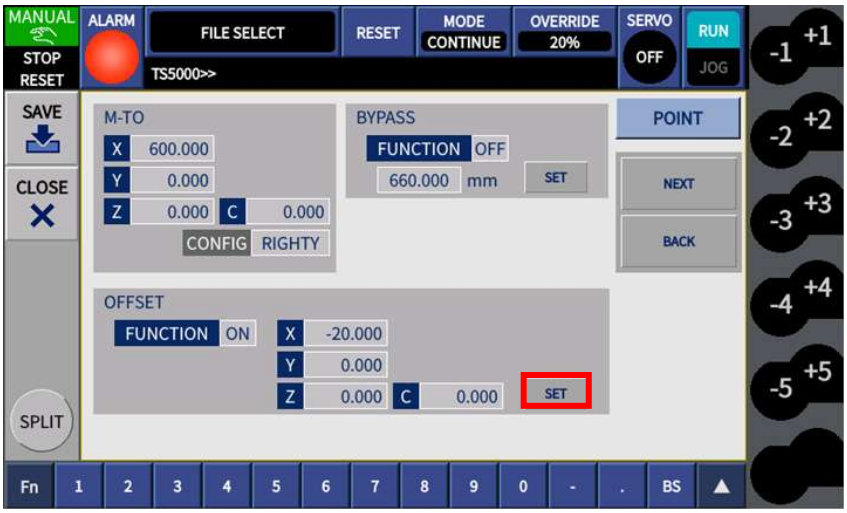
1 Tap the M-TO button on the POINT (Edit) screen.

The POINT (Teaching Point Movement Setting) screen is opened.

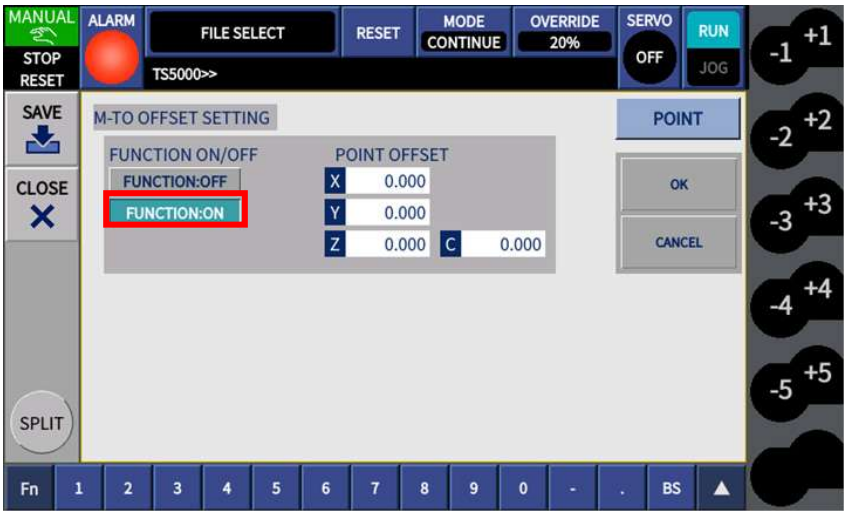


2 Tap the SET button in the OFFSET setting box.

The POINT (Offset Setting) screen is opened.

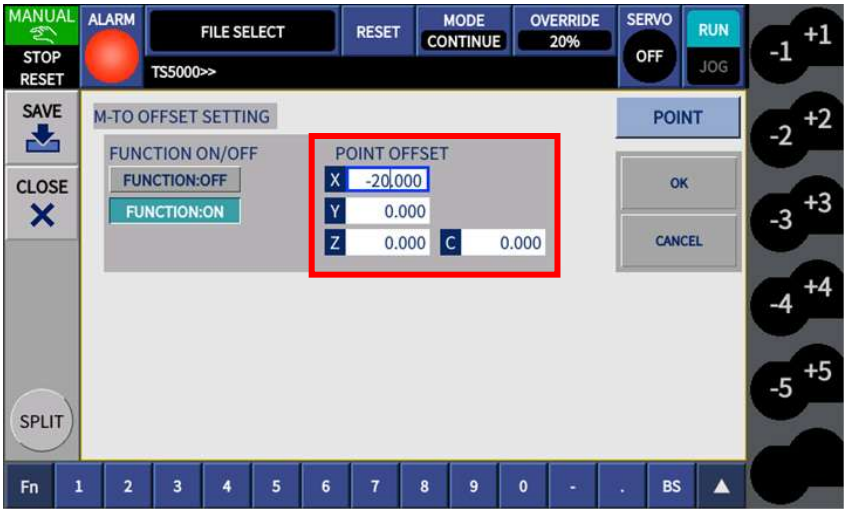


3 Tap "FUNCTION: ON".



4 Tap the offset input text box.

The cursor appears to allow the number to be edited.



5 Use the keyboard to enter a custom offset amount, and then press the Enter key or tap any location outside the text box.

This saves the edited changes, and the cursor disappears.

> *MEMO*
>
> *If a non-numerical entry is made into the text box, the characters in the text box appear in red.*

6 When the coordinate settings are completed, tap the OK button.

7 Confirm the coordinates of the teaching point, and tap the NEXT button.

The POINT (Teaching Point Movement Execution) screen is opened, and the displayed operation keys are changed.

The robot target position incorporating the offset and bypass settings is displayed on the POINT (Teaching Point Movement Execution) screen.

(The 1st via point position is displayed when the bypass setting is enabled.)

Check that the displayed target position is correct.

MEMO

For details on the bypass settings, see "4.3.6. Moving Teaching Point by Bypass Movement".

8 Tap the SERVO ON button while pressing down the enable switch by one level.

This turns on the servo, and the SERVO ON button indicator is changed.

9 Press the MOVE TO key or MOVES TO key.

The robot moves toward the target position while the MOVE TO or MOVES TO operation key is held down.

(The robot performs PTP movement when the MOVE TO key is pressed, and makes linear interpolation movement when the MOVES TO key is pressed.)

⚠ WARNING

- Do not enter the hazard zone during test operation. Doing so may result in an accidental collision with a robot.
- Do not allow a person other than an operator to approach the device. Doing so may result in an accidental collision with a robot.
- Perform teaching point movement in a state where the enable switch or emergency stop switch can always be accessed. Failure to do so may cause the robot to collide with the peripheral device.

NOTICE

- Check that the target position is set correctly. Failure to check it may cause the robot in unexpected motion to collide with the peripheral device.
- Be sure to check that the tool coordinates are set correctly. Failure to check it may cause the robot in unexpected motion to collide with the peripheral device.

4.3.6 Moving Teaching Point by Bypass Movement

This describes how to move the teaching point to the POINT-type data position by bypass movement. The bypass movement refers to passing through the offset position (via position) in the Z-axis direction to the target position when moving to the target position.
If an obstacle is located between the current position and the target position, the bypass movement can be used to avoid the obstacle and perform teaching point movement.

The teaching point can be moved by bypass movement using PTP movement and linear interpolation movement.

How to Move Teaching Point by Bypass Movement

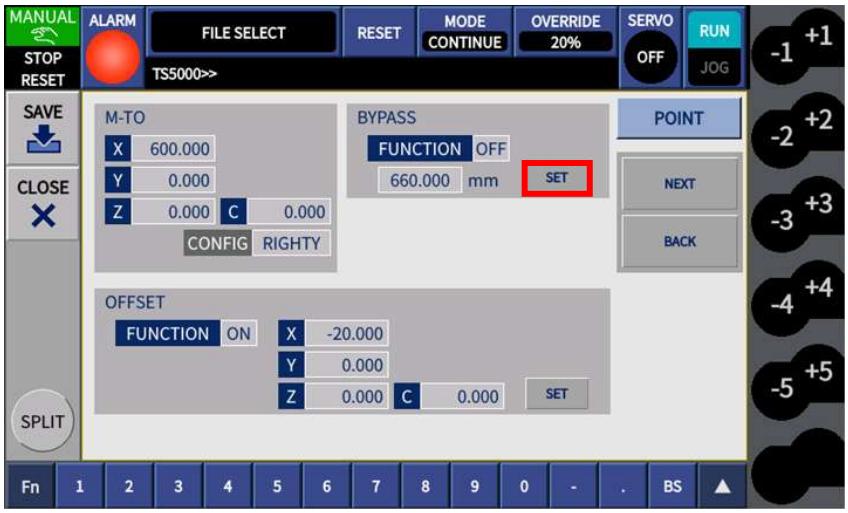
1 Tap the M-TO button on the POINT (Edit) screen.

The POINT (Teaching Point Movement Setting) screen is opened.

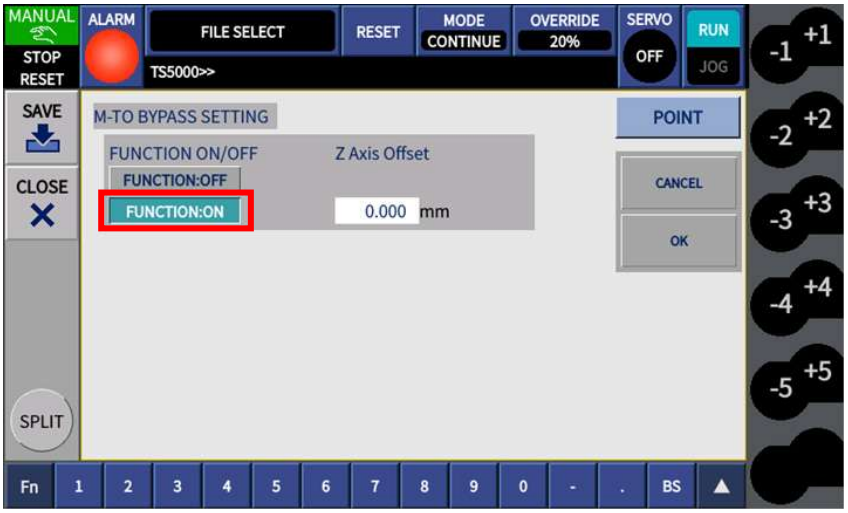


2 Tap the SET button in the BYPASS setting box.

The POINT (Bypass Setting) screen is opened.

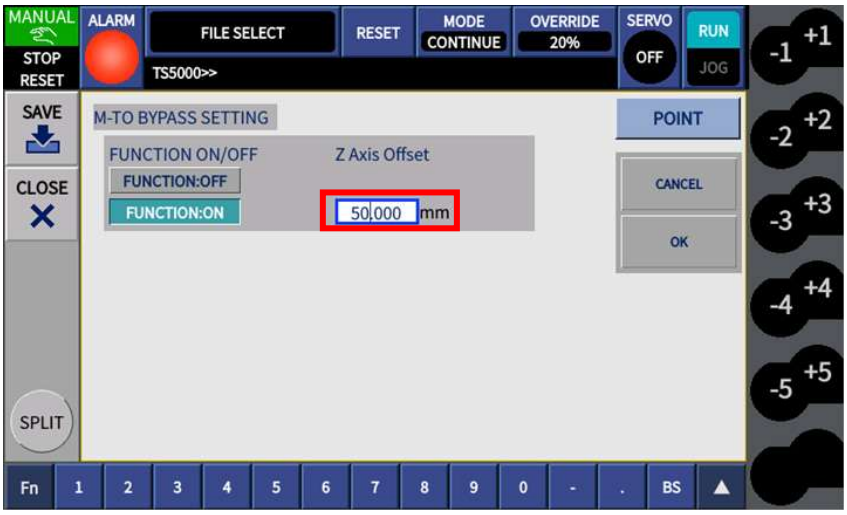


3 Tap “FUNCTION: ON”.



4 Tap the Z-axis offset input text box.

The cursor appears to allow the number to be edited.



5 Use the keyboard to enter the Z-axis offset amount, and then press the Enter key or tap any location outside the text box.

This saves the edited changes, and the cursor disappears.

MEMO

- *Specify the Z-axis offset as an absolute position.*
- *If a non-numerical entry is made into the text box, the characters in the text box appear in red.*

6 When the offset amount setting is completed, tap the OK button.

7 Confirm the coordinates of the teaching point, and tap the NEXT button.

The POINT (Bypass Execution) screen is opened, and the displayed operation keys are changed.

MEMO

For details on the offset settings, see "4.3.5. Moving Teaching Point to Offset Position".

The 1st via point position for the bypass movement incorporating the offset and bypass settings is displayed on the POINT (Bypass Execution) screen.

Check that the displayed target position is correct.

8 Tap the SERVO ON button while pressing down the enable switch by one level.

This turns on the servo, and the SERVO ON button indicator is changed.

9 Press the BYPASS MOVE TO key or BYPASS MOVES TO key.

The robot moves toward the target position while the BYPASS MOVE TO or BYPASS MOVES TO operation key is held down.



MEMO

(The robot performs PTP movement when the BYPASS MOVE TO key is pressed, and performs linear interpolation movement when the BYPASS MOVES TO key is pressed.)

⚠ WARNING

- Do not enter the hazard zone during test operation. Doing so may result in an accidental collision with a robot.
- Do not allow a person other than an operator to approach the device. Doing so may result in an accidental collision with a robot.
- Perform teaching point movement in a state where the enable switch or emergency stop switch can always be accessed. Failure to do so may cause the robot to collide with the peripheral device.

NOTICE

- Check that the target position is set correctly. Failure to check it may cause the robot in unexpected motion to collide with the peripheral device.
- Be sure to check that the tool coordinates are set correctly. Failure to check it may cause the robot in unexpected motion to collide with the peripheral device.

4.3.7 Opening TRANS (Edit) Screen

How to Open TRANS (Edit) Screen

1 Tap the POINT/TRANS selector button on the POINT (Edit) screen.

The POINT (Edit) screen switches to the TRANS (Edit) screen.

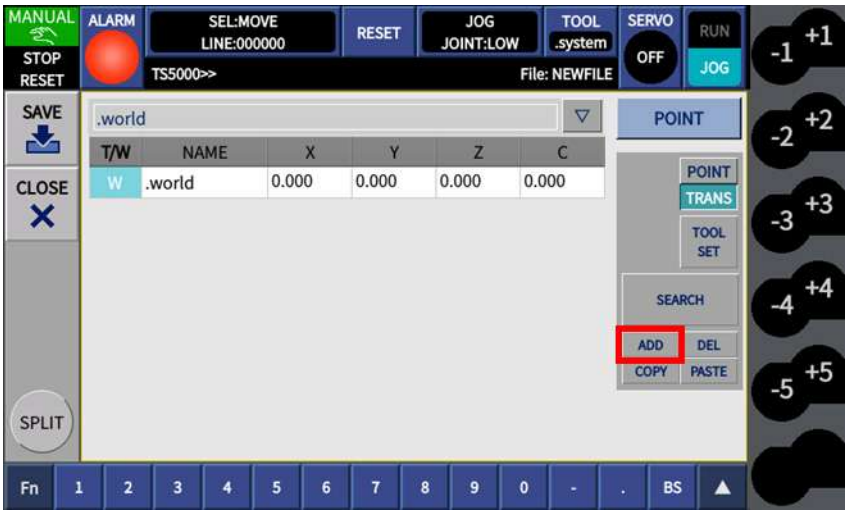
> *MEMO*
>
> *For details on how to open the POINT (Edit) screen, see "4.3.1. Opening POINT (Edit) Screen".*

4.3.8 Editing TRANS-type Data

How to Add TRANS-type Data

1 Tap the LINE OPERATION button.

An operation list is displayed.



2 Tap the ADD button.

A line is added.



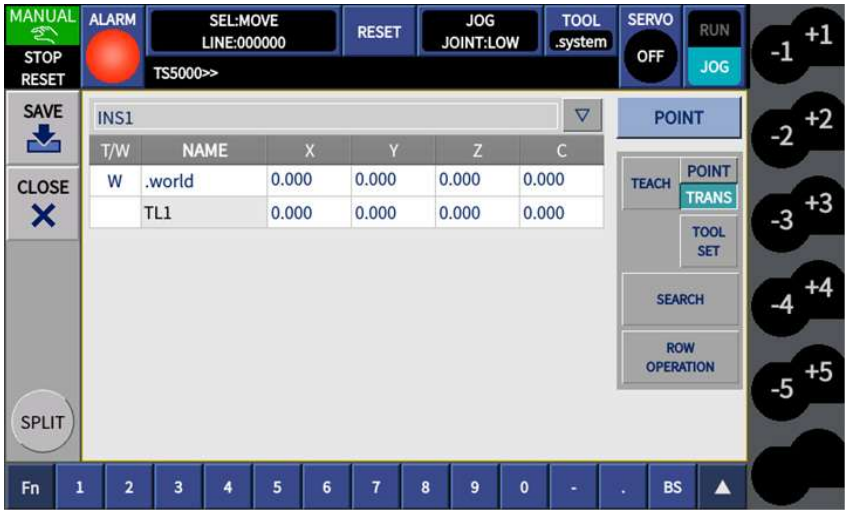
How to Edit TRANS-type Data Name

1 Tap the cell of the item to be edited on the TRANS (Edit) screen.

The cell is selected and highlighted.

2 Tap the selected cell.

The cursor appears at the end of the cell so that the content can be edited.



3 Tap the character and number keys on the keyboard to enter the desired content.

MEMO

Tapping the ESC key erases all of the characters in the cell. If no characters are entered in the cell, editing is canceled, and the cell is selected.

4 After editing is completed, press the Enter key or tap a location on the screen outside the cell being edited.

This saves the edited changes, and the cell is selected.

MEMO

The character string that can be set for the POINT-type data name consists of a maximum of 20 alphanumeric characters starting with a letter. If the character string exceeds 20 characters or includes symbols, the entered character string is not saved, and the content in the cell appears in red text.

How to Edit Coordinate Value of TRANS-type Data

1 Tap the cell of the item to be edited on the TRANS (Edit) screen.

The cell is selected and highlighted.

2 Tap the selected cell.

The cursor appears at the end of the cell so that the content can be edited.

3 Tap the number and symbol ("-" (minus) and "." (period)) keys on the keyboard to enter the number.

MEMO

Tapping the ESC key erases all of the characters in the cell. If no characters are entered in the cell, editing is canceled, and the cell is selected.

4 After editing is completed, press the Enter key or tap a location on the screen outside the cell being edited.

This saves the edited changes, and the cell is selected.

MEMO

The value input in the cell must be in the range from -9999.999 to 9999.999. If a value outside this range or a non-numerical character is entered, the entry is not saved, and the content in the cell appears in red text.

4.3.9 Teaching TRANS-type Data

This describes how to teach TRANS-type data.

TRANS-type data can be taught using one of the two methods below.

- 1-point teaching
- 2-point teaching

> *MEMO*
>
> *Please refer to the attached "STA1163 LANGUAGE MANUAL" for more information on TRANS types.*

4.3.9.1 1-point Teaching

This generates the TRANS-type data using the point that was taught as the origin.

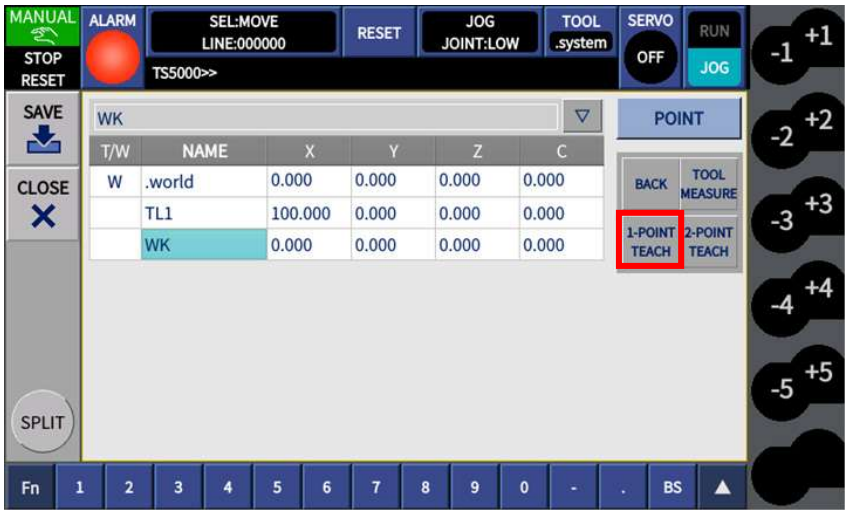
1-point Teaching Procedure

1 Tap the TEACH button on the TRANS (Edit) screen.



2 Tap the 1-POINT TEACH button.

The TRANS (1-POINT TEACH) screen is opened.



3 Move the robot TCP to the origin of the TRANS-type data to be generated, and tap the OK button.

The displayed TRANS-type data value is overwritten by the TRANS-type data value where the robot current position is set as the origin.



4.3.9.2 2-point Teaching

Two points (origin and point on coordinate axis) are taught to generate the TRANS-type data.

How to Perform 2-point Teaching

1 Tap the TEACH button on the TRANS (Edit) screen.



2 Tap the 2-POINT TEACH button.

The TRANS (2-POINT TEACH) screen is opened.



3 Enter the origin of the TRANS-type data to be generated and coordinate values of a point on the coordinate system to the lines for ORIGIN and DIRECTION.

The coordinate values can be entered by teaching the current position of the robot TCP or by using existing POINT-type data.



How to Teach Current Position of Robot TCP

1. **Tap a cell of the line where the coordinate value will be changed.**

2. **Move the robot TCP to the position of the point to be taught, and tap the SET button.**

   The coordinate value of the line is overwritten by the current position of the robot TCP.

How to Input Existing POINT-type Data

1. **Tap a cell of the line where the coordinate value will be changed.**

2. **Tap the READ button.**

   A list of the POINT-type data in the .world coordinate system is displayed.

3. **Tap the line of the POINT-type data to be used as the teaching point.**

   The tapped line is selected.

4. **Tap the SET button.**

   The coordinate value in the line is overwritten by the coordinate value of the selected POINT-type data.

4 Tap the CREATE button.

This returns to the TRANS (Edit) screen, and the line where the cursor is located is overwritten by the values of the TRANS-type data generated from the two teaching points.



4.3.10 Setting Tool Coordinate

This describes how to set the TRANS-type data to the tool coordinates.

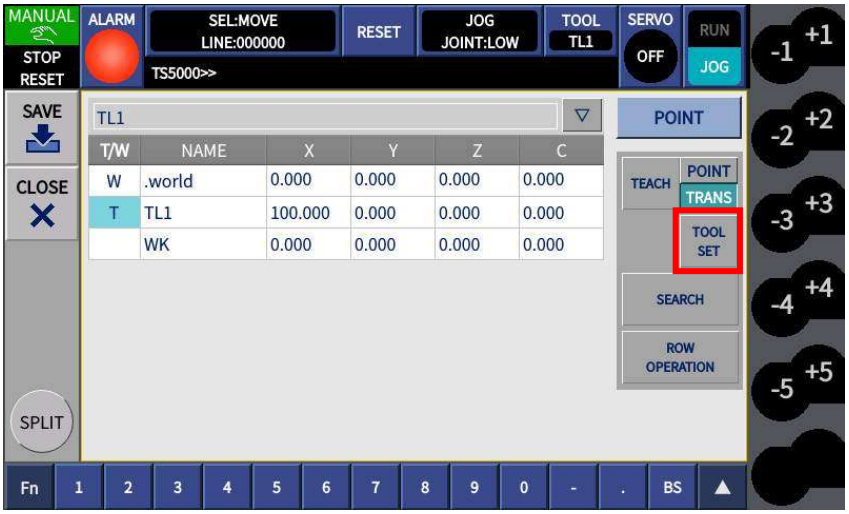
How to Set to Tool Coordinate

1 Move the cursor to the line in the TRANS-type data to be set to the tool coordinates on the TRANS (Edit) screen.



2 Tap the tool coordinate setting button.

The TRANS-type data in the line where the cursor is located is set to the tool coordinates.



MEMO

- *TRANS-type data that is set to the tool coordinates is indicated by a T in the column at the left end of the line.*
- *".system" cannot be registered to the tool coordinates.*

4.3.11 Setting Work Coordinate

This describes how to set the TRANS-type data to the work coordinates.

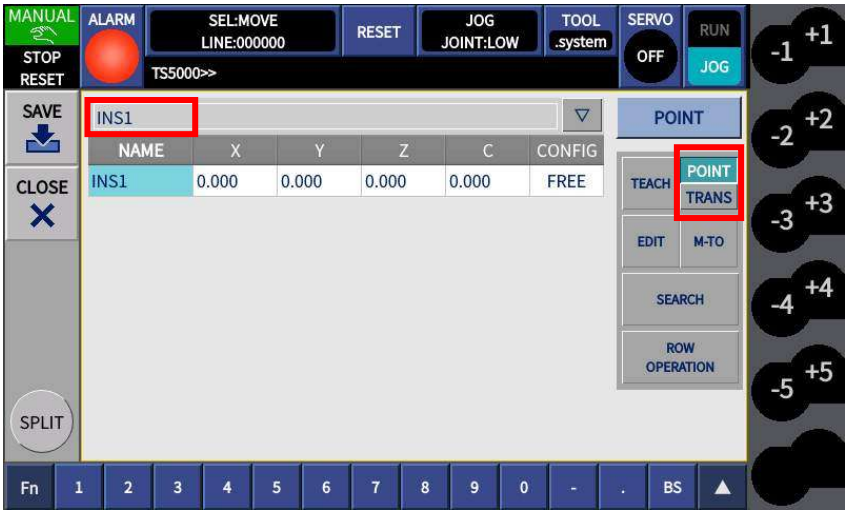
How to Set to Work Coordinate

1 Move the cursor to the line in the TRANS-type data to be set to the work coordinates on the TRANS (Edit) screen.



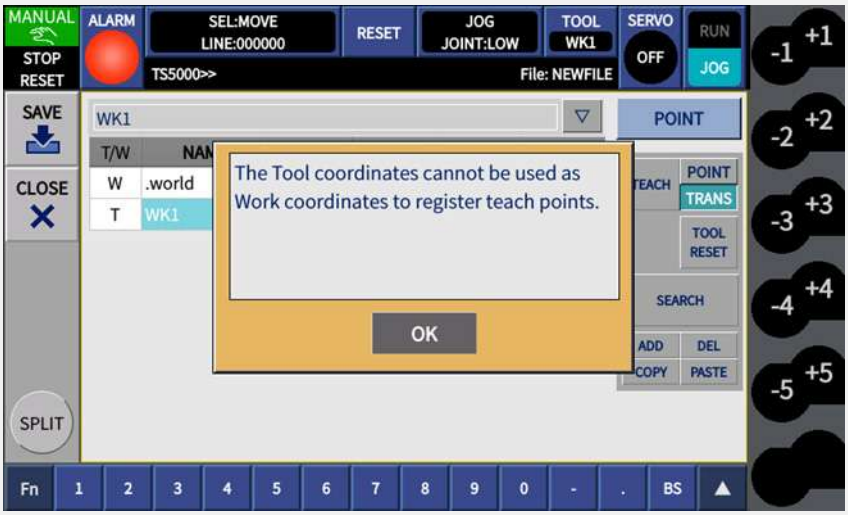
2 Tap the POINT/TRANS selector button.

The TRANS-type data in the line where the cursor is located is set to the work coordinates, and the POINT (Edit) screen is opened.



MEMO

• If the selected TRANS-type data is set as tool, the screen cannot be switched to the POINT screen.



4.3.12 Opening PAYLOAD (Edit) Screen

How to Open PAYLOAD (Edit) Screen

1 Tap the Utility menu in the main window of the file edit screen.

The Utility Menu Switching window is displayed.

2 Tap the PAYLOAD button in the menu.

The PAYLOAD (Edit) screen is opened.



4.3.13 Editing PAYLOAD-type Data

How to Add TRANS-type Data

1 Tap the LINE OPERATION button.

An operation list is displayed.



2 Tap the ADD button.

A line is added.



How to Edit PAYLOAD-type Data Name

1 Tap the cell of the item to be edited on the PAYLOAD (Edit) screen.

The cell is selected and highlighted.



2 Tap the selected cell.

The cursor appears at the end of the cell so that the content can be edited.



3 Tap the character and number keys on the keyboard to enter the desired content.

MEMO

Tapping the ESC key erases all of the characters in the cell. If no characters are entered in the cell, editing is canceled, and the cell is selected.

4 After editing is completed, press the Enter key or tap a location on the screen outside the cell being edited.

This saves the edited changes, and the cell is selected.

MEMO

The character string that can be set for the PAYLOAD-type data name consists of a maximum of 20 alphanumeric characters starting with a letter. If the character string exceeds 20 characters or includes symbols, the entered character string is not saved, and the content in the cell appears in red text.

How to Edit PAYLOAD-type Data Value

1 Tap the cell of the item to be edited on the PAYLOAD (Edit) screen.

The cell is selected and highlighted.



2 Tap the selected cell.

The cursor appears at the end of the cell so that the content can be edited.



3 Tap the number and symbol ("-" (minus) and "." (period)) keys on the keyboard to enter the number.

MEMO

Tapping the ESC key erases all of the characters in the cell. If no characters are entered in the cell, editing is canceled, and the cell is selected.

4 After editing is completed, press the Enter key or tap a location on the screen outside the cell being edited.

This saves the edited changes, and the cell is selected.

MEMO

The value input in the cell must be in the range from -9999.999 to 9999.999. If a value outside this range or a non-numerical character is entered, the entry is not saved, and the content in the cell appears in red text.

4.4 File Saving and Edition Termination

This describes how to save the edited file and terminate the editing process.

4.4.1 Saving Files and Terminating Edition

1 Tap the SAVE button on the file edit screen.

This opens the SAVE menu.



2 Tap the OK button.

The file is saved.



3 Tap the CLOSE button.

This terminates the file editing process.

After terminating the file editing process, the file operation menu screen is opened.



4.4.2 Discarding Changes to File

1 Tap the CLOSE button on the file edit screen.

The NOSAVE menu is displayed.

2 Tap the OK button.

The file changes are discarded, and the file editing process is terminated. The file content is restored to the state before editing was performed.

After terminating the file editing process, the file operation menu screen is opened.



4.4.3 Saving Files

1 Tap the SAVE button on the file edit screen.

This opens the SAVE menu.



2 Tap the OK button.

The file is saved.



4.5 File Operations

File operations can be performed in MANUAL mode only.

4.5.1 Copying Files

This section describes how to copy files saved in the controller and how to copy files in the SD card inserted into the controller.

For details on how to copy files between the controller and SD card, see "4.5.4 Transferring Files."

Copying to Same Drive

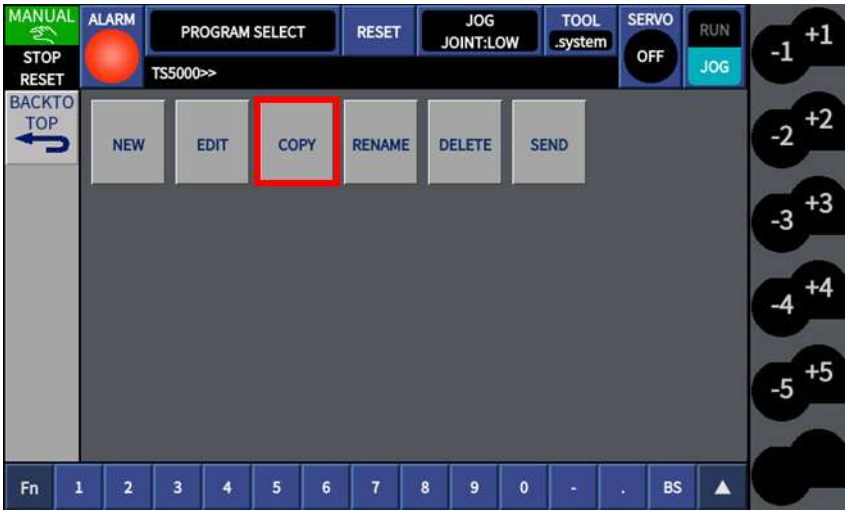
1 Tap the FILE button on the top screen.

The file operation menu screen is opened.



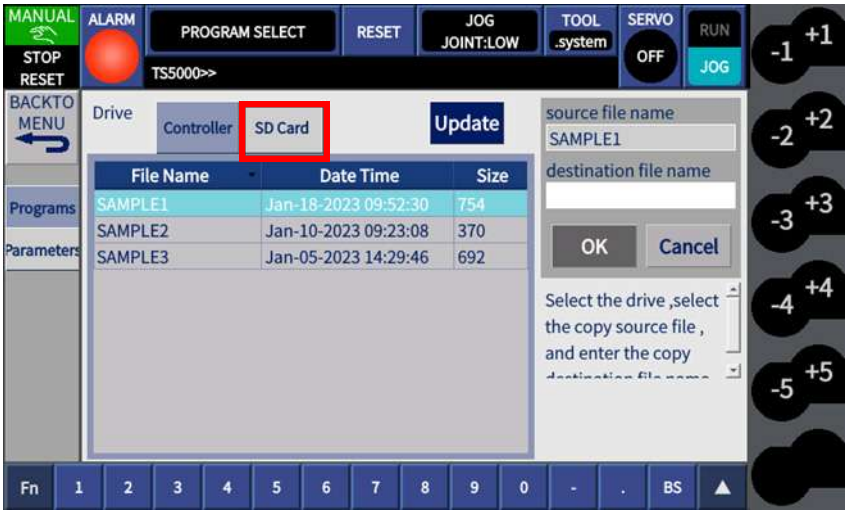
2 Tap the COPY button.

The copy file screen is opened.



3 To copy a file in the SD card, tap the SD Card button in the top section of the SD file list.

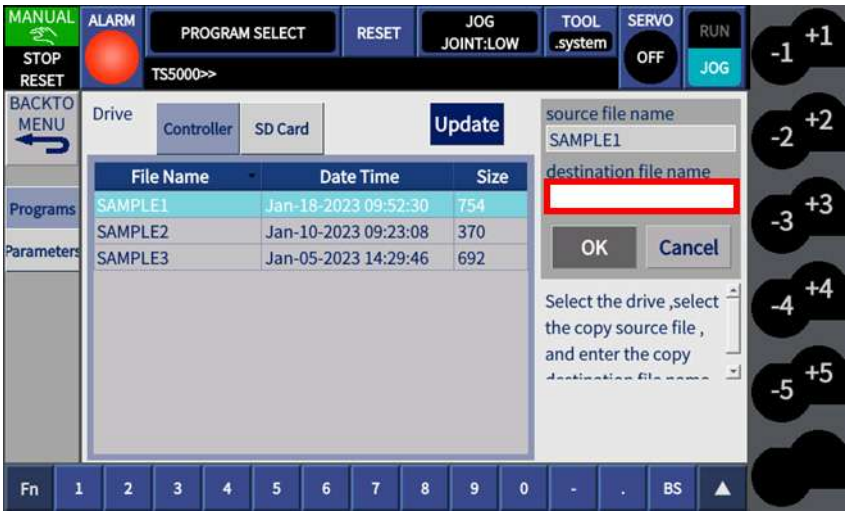
The files in the SD card are displayed in the file list.



4 Enter the name of the copy source file into the text box for "source file name".

> *MEMO*
>
> *Tapping the file list on the left side of the screen enters the file name in the tapped line into the text box for the source file name.*

5 Enter the file name of the copy target to the "destination file name" text box.



6 Tap the OK button.

The file is copied within the same drive.

MEMO

If a file with the name entered in "destination file name" already exists, a message appears asking whether you want to overwrite the file. Tapping OK will overwrite the file.

Copying to Different Drive

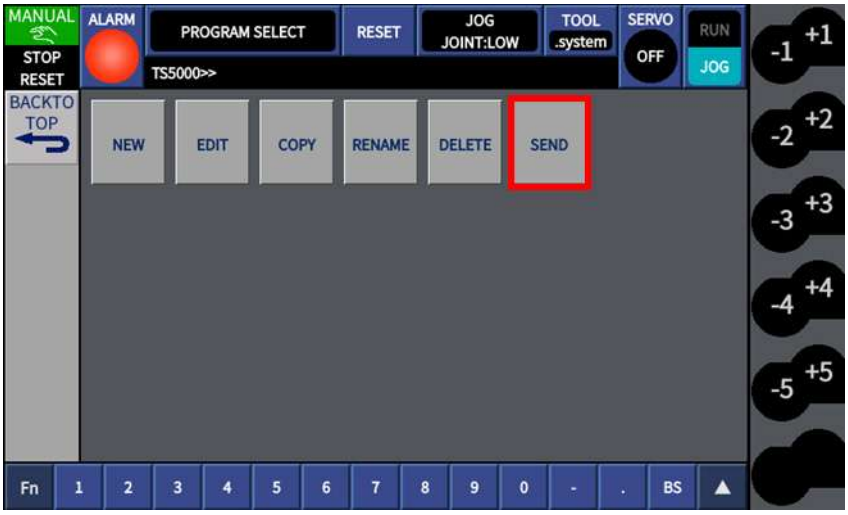
1 Tap the FILE button on the top screen.

The file operation menu screen is opened.



2 Tap the SEND button.

The send file screen is opened.



3 Tap the check box on the left side of the file to be transferred to a different drive.

A check mark is added to the check box.



4 To copy the file from the controller to the SD card, tap the "→" button. To copy the file from the SD card to the controller, tap the "←" button.

The file with the check mark is transferred to the selected drive.



MEMO

If a file with the same name already exists at the selected target drive, the file is not transferred.

If a check mark is added to the "Overwrite same name file" check box, the file at the target drive will be overwritten.

4.5.2 Deleting Files

This describes how to delete files in the controller and SD card.

How to Delete File

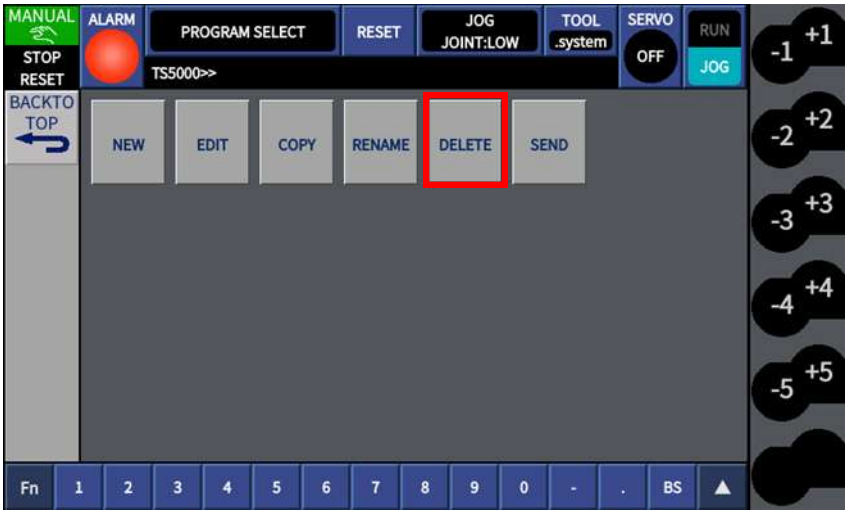
1 Tap the FILE button on the top screen.

The file operation menu screen is opened.



2 Tap the DELETE button.

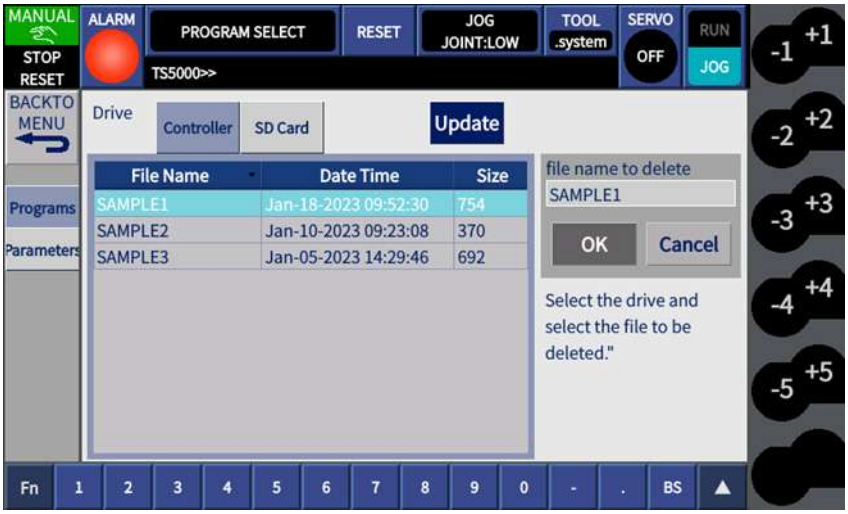
The delete file screen is opened.



3 To delete a file in the SD card, tap the SD Card button in the top section of the SD file list.

The files in the SD card are displayed in the file list.

4 Enter the name of the file to be deleted to the "file name to delete" text box.



MEMO

Tapping the file list on the left side of the screen enters the file name in the tapped line to the text box for the file name to be deleted.

5 Tap the OK button.

A confirmation message appears asking whether you want to delete the file.



6 Tap the OK button.

The file is deleted.

⚠ WARNING

- Do not delete a parameter file. Deleting it may result in an accidental collision with a robot in unexpected motion.
- Deleting the parameter file causes the robot to make motion with the setting that is different from that before deletion.

4.5.3 Renaming Files

How to Rename File

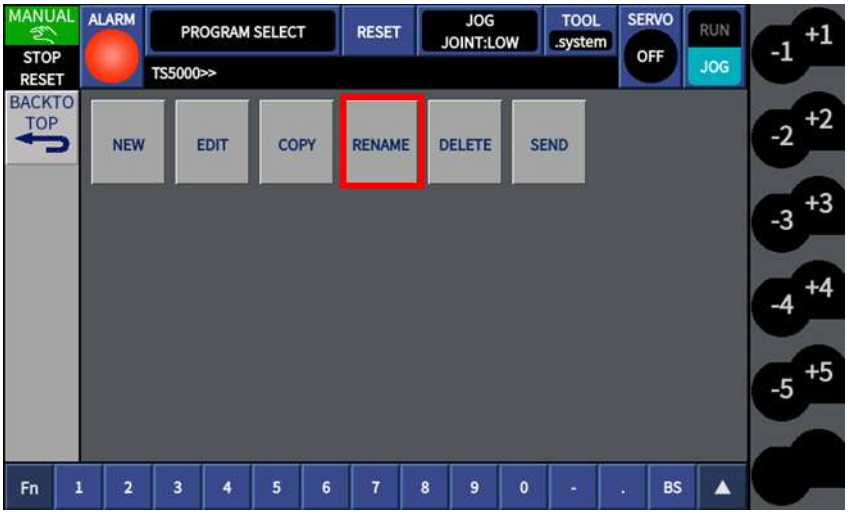
1 Tap the FILE button on the top screen.

The file operation menu screen is opened.



2 Tap the RENAME button.

The rename file screen is opened.



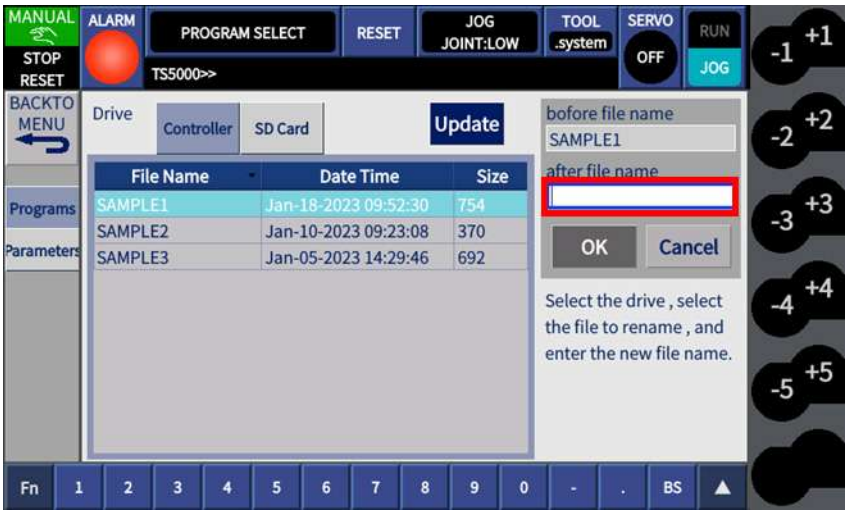
3 To rename a file in the SD card, tap the SD Card button in the top section of the SD file list.

The files in the SD card are displayed in the file list.

4 Enter the name of the file before renaming to the "before file name" text box.

> *MEMO*
>
> *Tapping the file list on the left side of the screen enters the file name in the tapped line into the text box for the before file name.*

5 Enter the name of the file after renaming to the "after file name" text box.



6 Tap the OK button.

The file is renamed.

> *MEMO*
>
> *If a file with the name entered in "after file name" already exists, a message appears asking whether you want to overwrite the file. Tapping OK will overwrite the file.*

⚠ WARNING

●Do not rename a parameter file. Renaming it may result in an accidental collision with a robot in unexpected motion.

• Renaming the parameter file causes the robot to make motion with the setting that is different from that before renaming.

4.5.4 Transferring Files

This section describes how to send and receive files in the controller or SD card.

<Transferring Files>

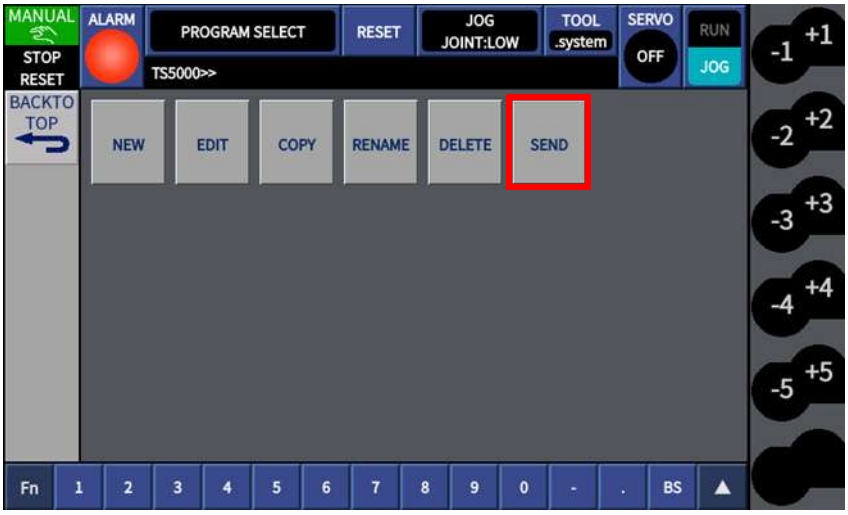
1 Tap the FILE button on the top screen.

The file operation menu screen is opened.



2 Tap the SEND button.

The send file screen is opened.



3 Select the file to be transferred.

Tap the check box next to the file name.

Tapping multiple check boxes can transfer multiple files at a time.



4 Tap the transfer button.

The selected files are transferred.



5. Operations in AUTO Mode

This chapter describes the operations performed in AUTO mode.
AUTO mode is used in performing automatic operation of the robot.
In AUTO mode, TP5000 can only perform "Setting Override", "Stopping and Aborting Program Execution", "Pausing Program Execution", "Monitoring of Execution Programs", and "Display and Overwriting of Global Variable".
When in AUTO mode, operations where the robot moves, such as Servo ON and program execution, and file operations cannot be performed from the TP5000.

MEMO

When in AUTO mode, operations for Servo ON and program execution can be performed using communication commands and interface relay operation.

For details on communication commands, see the separate "STA1164 Communication Manual". For details on interface relays, see the separate "STA1167 Interface Manual" and "STA1168_TCMINI (Internal PLC) Manual".

Use the following procedure to switch from MANUAL mode to AUTO mode.

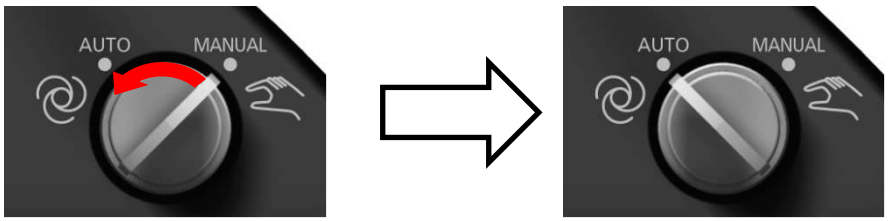
How to Switch to AUTO Mode

1 Insert the key provided with the teach pendant into the master mode key switch.



2 Set the master mode keyswitch to the AUTO position.

Master mode is switched to AUTO.



MEMO

- *When the master mode is switched during Servo ON, the status changes to Servo OFF.*
- *When the master mode is switched while the robot is operating, the robot decelerates to a stop, and the status changes to Servo OFF.*
- *When the master mode keyswitch is switched on other screens than the top screen, an alarm is generated.*

3 Remove the key if you want to prevent other workers from switching the master mode.



5.1 Setting Override

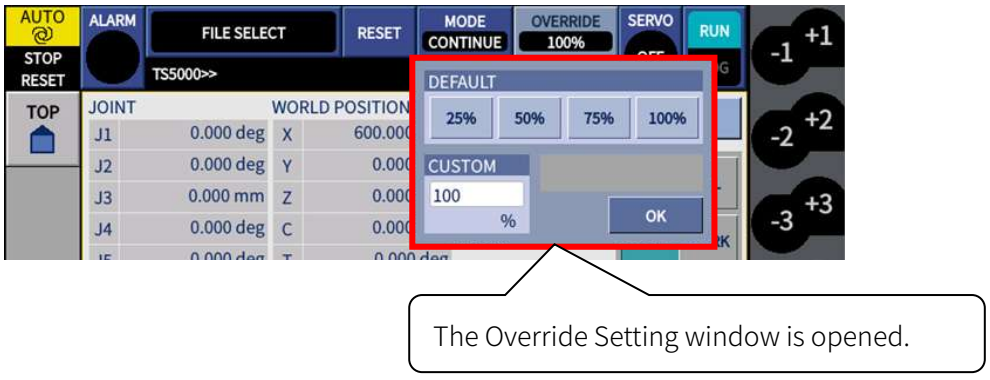
This describes how to set the override.

How to Set Override

1 Tap the override setting button (indicated by red box) in the status bar.

The override setting window is opened.





2 Enter the override value to be set into the text box in the Override Setting window.

Data can be entered into text boxes using one of the two methods below.

Entering Default Override Value

1. Tap one of the override input buttons (indicated by red box).

The value for the tapped button is input to the text box.



MEMO

The value that has already been input to the text box is overwritten.

Entering Custom Value

1. Tap the text box in the override setting window.

The cursor appears in the text box to indicate that the content can be edited.



2. Use the keyboard to enter a custom value.

MEMO

- *In MANUAL mode, the override value can be set from 1 to 20.*
  When a value outside this range is entered into the text box, the override does not change even if the SET button is tapped.
- *The override can be set even during program execution.*

5.2 Stopping and Aborting Program Execution

This describes how to stop program execution.
Program execution can be stopped by a stop or abort operation.

Each procedure is described in detail below.
Stop: Stops program execution once execution of the current command is completed.
Abort: Stops program execution immediately. In this state, the robot decelerates to a stop.

MEMO

Program execution cannot be resumed from TP5000 in AUTO mode.
To resume the program execution, use communication commands or system signals.
For details, refer to the separate "STA1164 Communication Manual" for communication commands and "STA1167 Interface Manual" and "STA1168 TCMINI (Internal PLC) Manual" for system signals (interface relays).

5.2.1 Stopping Program Execution

This describes how to stop program execution.

How to Stop Program Execution

1 Press the displayed operation key that is assigned to "STOP".

The program is stopped once execution of the current command is completed.

MEMO

While a program is stopped, the status indicator changes as shown below.

Screen during program execution



Screen when program execution is stopped



MEMO

In addition to the above operation, the following operations can be used to stop program execution.

• Setting the STOP signal to OFF during program execution

(For details, refer to the STA1167 Interface Manual.)

• Sending the communication command "SOP1" to the HOST port of the controller

(For details, refer to the STA1164 Communication Manual.)

• Executing the STOP command in SCOL2 language

(For details, refer to STA1163 STA1163-0 Robot Language Manual (SCOL2).)

5.2.2 Aborting (BREAK) Program Execution

This describes how to abort program execution.

How to Abort Program Execution

1 Press the displayed operation key that is assigned to "BREAK".

This aborts execution of a program and immediately decelerates the robot in motion to a stop.

MEMO

After aborting, when program execution is performed, program execution is performed again from the execution line where execution was aborted.

MEMO

In addition to the above operation, the following operations can be used to abort program execution.

• Setting the BREAK signal to OFF during program execution

(For details, refer to the STA1167 Interface Manual.)

• Sending the communication command "BRK1" to the HOST port of the controller

(For details, refer to the STA1164 Communication Manual.)

5.3 Pausing (FEEDHOLD) Program Execution

This describes how to pause/resume program execution.

How to Pause Program Execution

1 Press the displayed operation key that is assigned to "FEEDHOLD".

This pauses the program execution.

MEMO

When program execution is paused, the operation status panel in the status bar changes as shown below.

Screen during program execution



Screen while program execution is paused



MEMO

Even while program execution is paused, you can perform the stopping/aborting operation.

If such operation is performed, the program execution stops immediately.

MEMO

In addition to the above operation, the following operation can be used to pause program execution.

• Enabling [P006] and set the FEEDHOLD (register G12C) to OFF during program execution

(For details, refer to STA1166 Parameter Manual and STA1168 TCMINI (Internal PLC) Manual.)

How to Resume Program Execution

1 Press the displayed operation key that is assigned to "FEEDHOLD".

This resumes the program execution.

MEMO

In addition to the above operation, the following operation can be used to resume program execution.

- *Enabling [P006] and set the FEEDHOLD (register G12C) to ON while the program execution is paused.*

(For details, refer to STA1166 Parameter Manual and STA1168 TCMINI (Internal PLC) Manual.)

NOTICE

● Resume program execution in a state where the emergency stop switch can always be accessed.
Failure to do so may cause the robot in unexpected motion to collide with the peripheral device.

5.4 Monitoring of Execution Programs

This describes how to monitor the content of the execution program.

The content of the execution program is displayed on the LDISP screen.

How to Open LDISP Screen

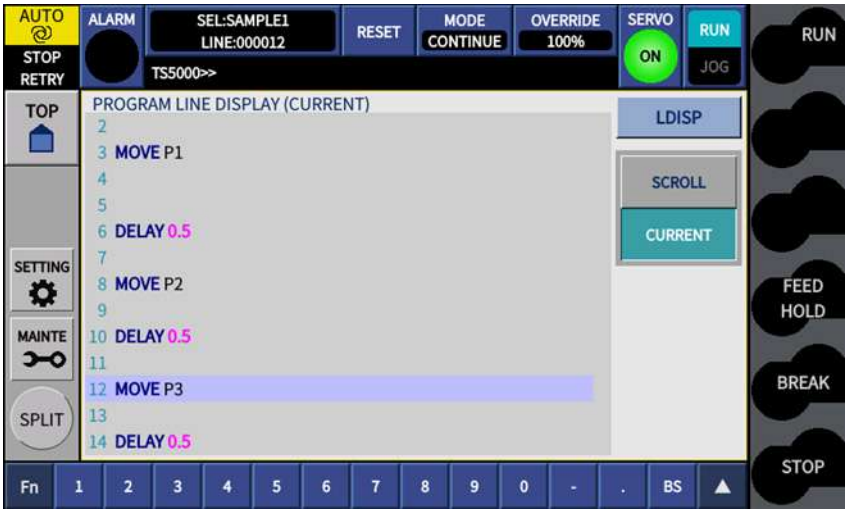
1 Tap the Utility menu of the main window.

The utility menu switching window is displayed.



2 Tap the LDISP menu button.

The LDISP screen is opened in the main window.



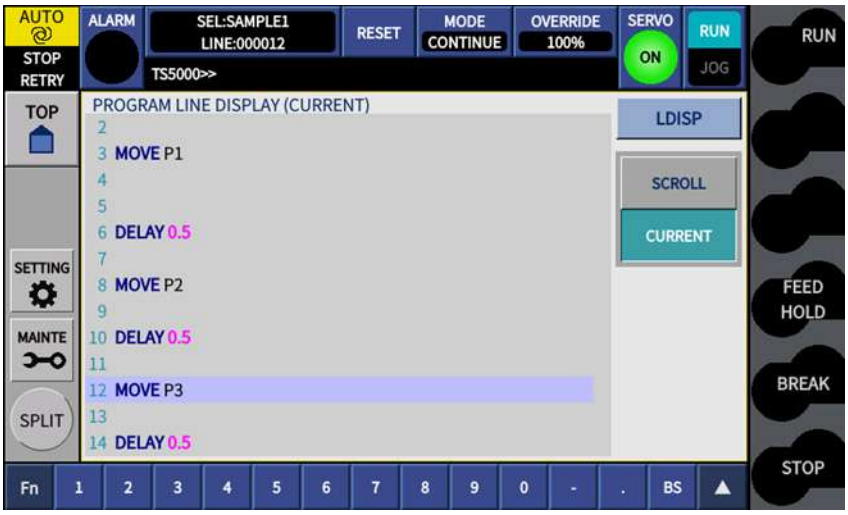
The content of the execution program can be displayed on the LDISP screen using one of the following two methods.

- Display the content in synchronism with the line currently being executed (synchronous display)
- Display any selected line (asynchronous display)

The SCROLL/CURRENT button can be tapped to switch between synchronous display and asynchronous display.

Synchronous Display of Execution Program

In synchronous display, the screen is scrolled based on the line currently being executed.
The line currently being executed is highlighted in blue.



Asynchronous Display of Execution Program

In asynchronous display, the cursor can be moved to scroll the screen.
The line where the cursor is located is highlighted in yellow as shown in the figure below.



5.5 Display and Overwriting of Global Variable

This describes how to display and overwrite the global variables in the execution program.

Global variables are variables defined in the global section (area enclosed by GLOBAL and END) or the data section (area enclosed by DATA and END) in the program file.
Global variables can be displayed and overwritten from the WATCH screen.
For details on the WATCH screen, see "11.8. WATCH Screens" in this manual.

How to Open WATCH Screen

1 Tap the Utility menu of the main window.

The Utility Menu Switching window is displayed.



2 Tap the WATCH menu button.

The WATCH (List) screen is opened in the main window.



Global Variable Details Display Procedure

1 From the list, select the variable whose details you want to view or whose value you want to overwrite.

The background color and text color change for the selected line.

2 Tap the DETAIL button.

Details for the selected global variable are displayed.

* For array-type variables, after editing the text box showing the element number, tap the SHOW button to display the desired variable.



Global Variable Overwrite Procedure

1 Tap the text box.

The cursor appears so that the number can be edited.



2 Use the keyboard to enter the value to be overwritten.

3 After editing, press the Enter key or tap a location on the screen outside the text box.

This saves the edited changes.

* If the value in the text box is outside the range of -9999.999 to 9999.999, the entered value is not saved, and the characters in the text box appear in red.

At this stage, the value for the variable being used in the program is not overwritten yet. To overwrite the value, perform the operation in procedure 5 below.

4 Edit all the elements for the value to be overwritten using procedures 1 to 3.

5 Tap the UPDATE button.

The value of the currently-selected variable is overwritten by the value in the text box.



MEMO

- *To prevent the robot from running out of control due to erroneous input, for position-type and coordinate-type variables, overwriting to element values that exceed ±3 mm (or deg) from their current value is not allowed.*
- *The overwritten value is a temporary change, and the value is reset by performing a program reset. To permanently apply the overwritten value, consider either editing the file or using a memory variable. For details on memory variables, refer to the section "MEMORY" in "STA1163 Robot Language Manual (SCOL2)."*

NOTICE

●Before overwriting a target global variable, confirm the effect that the variable will have on the program. Failure to confirm it may cause the robot to collide with peripheral device.

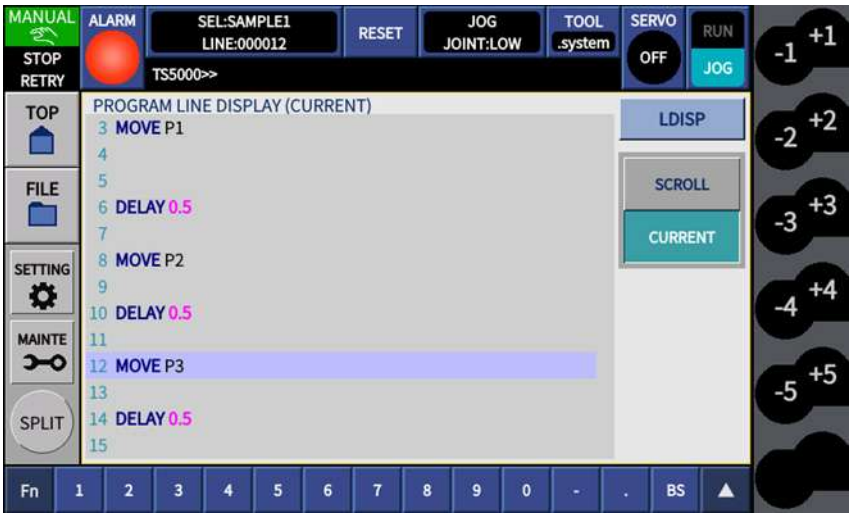
6. Operations in Utility Window

This section describes the information displayed in the Utility window located in the middle of the top screen and file editing screen.

MEMO

The Utility window has two display formats: "Wide view" where a single screen is displayed widely over the entire display area and "Split view" where the display area is split for displaying two screens simultaneously.

• Top screen



• File editing screen



6.1 Changing Display Screens

This describes how to change the display screens in the utility window.

How to Change Utility Window Display Screen

1 Tap the Utility menu button.

The utility menu is displayed.



2 Tap the button of the screen to be displayed in the menu.

The screen changes to the one corresponding to the tapped button.

6.2 Changing Display Format

This describes how to switch the display format between "Wide View" and "Split View".

How to Change Utility Window Display Format

1 Tap the SPLIT button.

In "Wide View", the display format is changed to "Split View". In "Split View", the display format is changed to "Wide View".

• Split View



• Wide View



6.3 Operations Performed on Screens

For a list of utility windows, see “11.1 List of Screens” in Screen Details.

6.3.1 Operations Performed on POS Screen

On the POS screen, you can check the current position and speed of the robot, and the current angle and angular velocity of each joint.

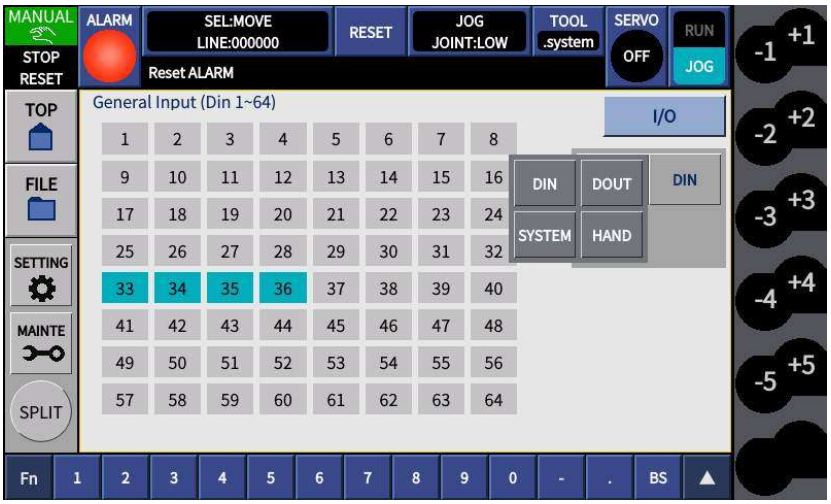
6.3.2 Operations Performed on I/O Screen

On the I/O screen, you can check the statuses of general-purpose I/O, system I/O, and hand I/O, perform ON/OFF control of the general-purpose output signals and hand output signals, and perform signal resetting.

How to Switch Signal to be Displayed

1 Tap the signal switch menu button.

The signal switch menu is displayed.



2 Tap the button for the signal to be displayed.

A screen for the tapped button is opened.



How to Switch Output Signal ON/OFF

1 Tap the location for the number of the signal where the output signal status will be switched on the I/O (DOUT) screen or I/O (HAND) screen.

The cursor moves to the tapped signal number.



2 Tap the ON/OFF selector button.

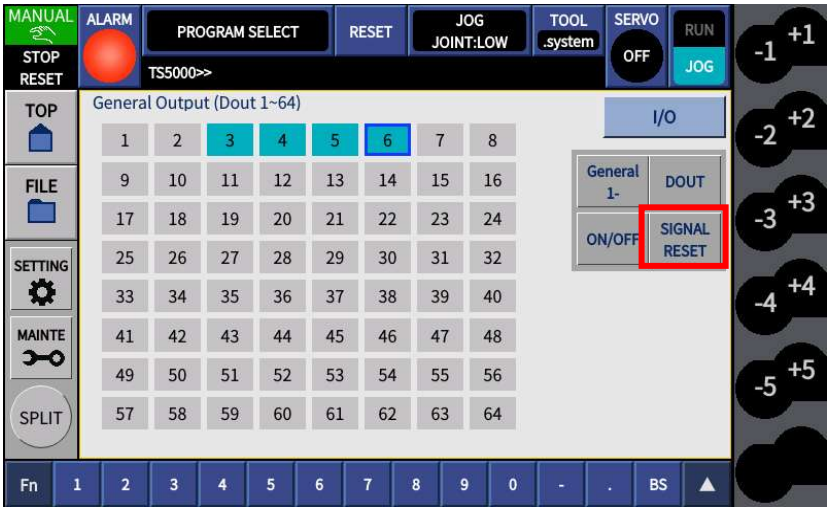
The ON/OFF status of the selected output signal number is switched.

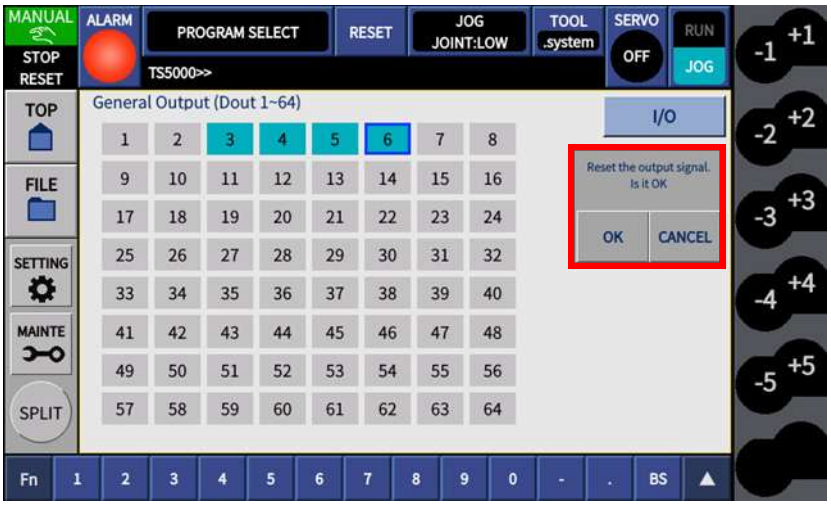


How to Reset Signal

1 Tap the SIGNAL RESET button on the I/O (DOUT) screen.

A signal reset confirmation message appears.





2 Tap the OK button.

All the output signals are reset (set to OFF status).



MEMO

The hand output signals are not reset.

⚠ WARNING

●If the operation is performed to reset output signals collectively, be sure to consider the effect caused by output signal reset. Failure to consider it may cause the connected peripheral device to operate.

6.3.3 Operations Performed on IP STATUS Screen

On the IP STATUS screen, you can confirm the statuses of the communication ports and perform the open/close operation for non-procedural ports.

How to Switch Port Type to be Displayed

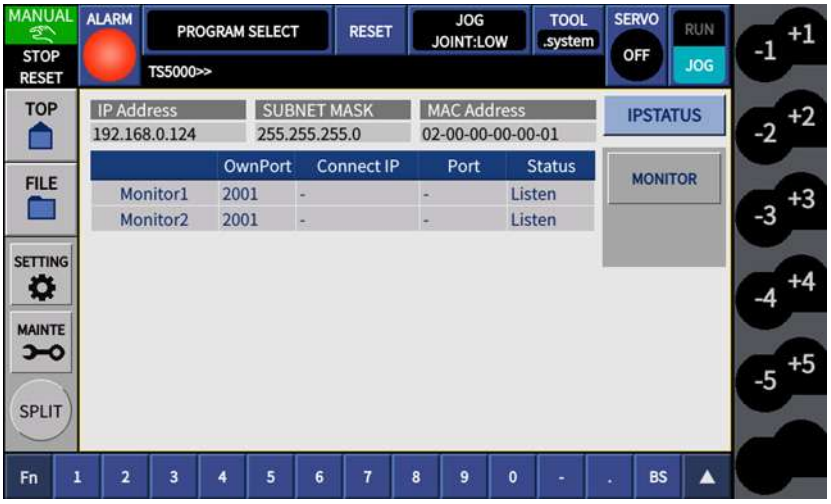
1 Tap the port switch menu button.

The port switch menu is displayed.



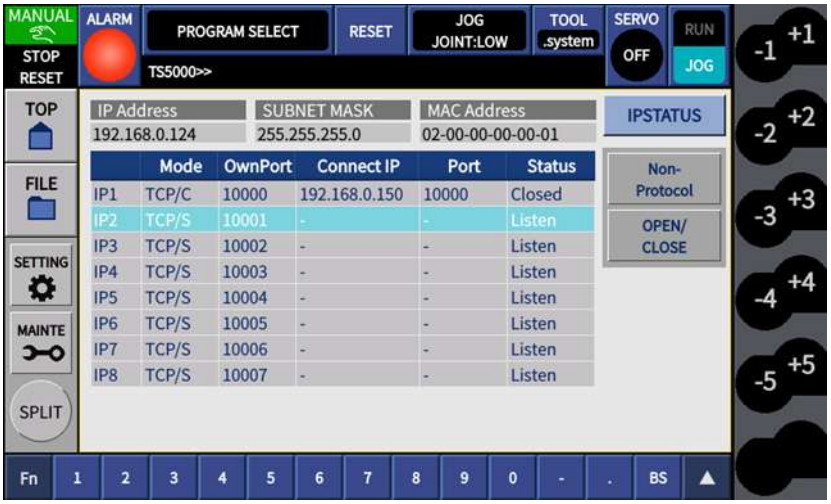
2 Tap the button for the port to be displayed.

A screen for the tapped button is opened.



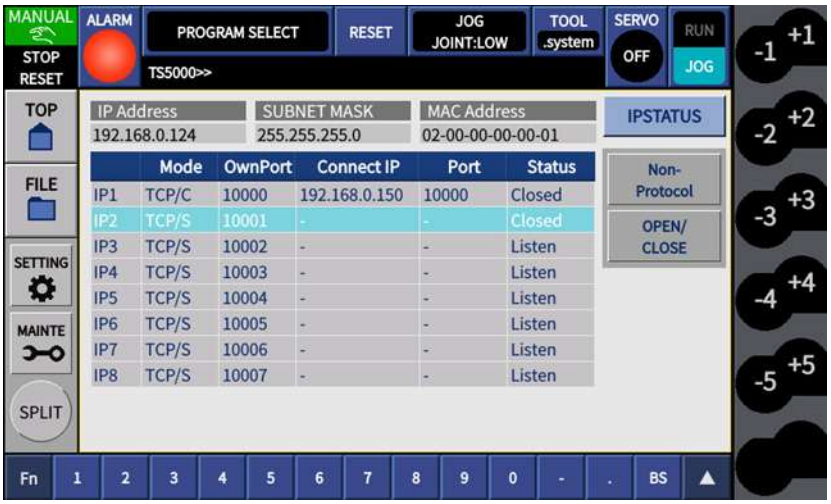
How to Open/Close Communication Port

1 Move the cursor to the line corresponding to the port to be opened or closed.



2 Tap the OPEN/CLOSE selector button.

This switches the OPEN/CLOSE status of the port at the line where the cursor is located.



MEMO

Ports that are set to "non" cannot be opened.

To set the mode, set ETHER.PAR.

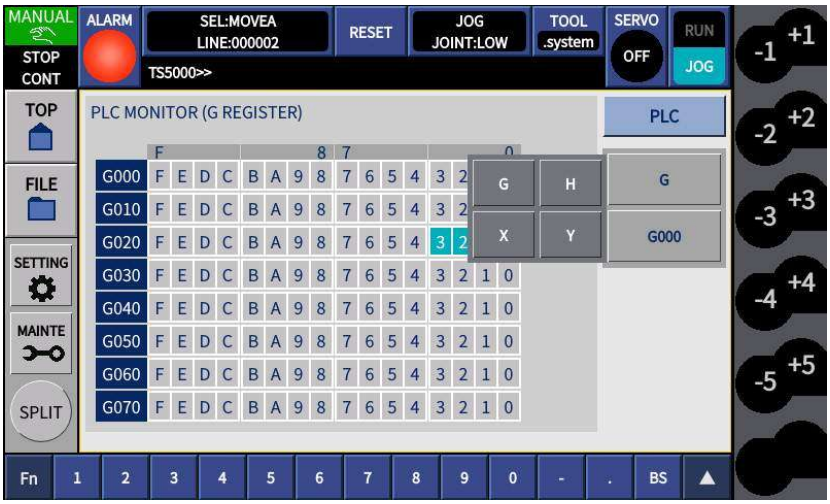
6.3.4 Operations Performed on PLC Screen

On the PLC screen, you can check the status of the TCmini registers.

How to Change Register to be Displayed

1 Tap the register switch menu button.

The register switch menu is displayed.



2 Tap the button for the register to be displayed.

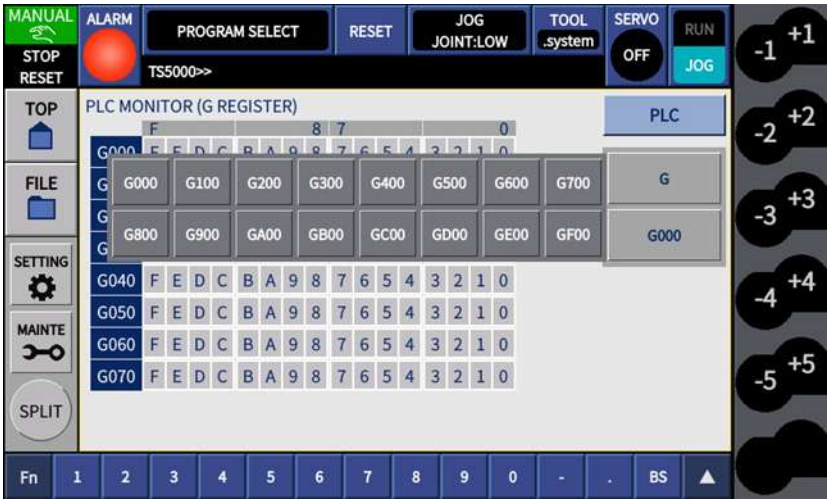
A screen for the tapped button is opened.



How to Change Register Number to be Displayed

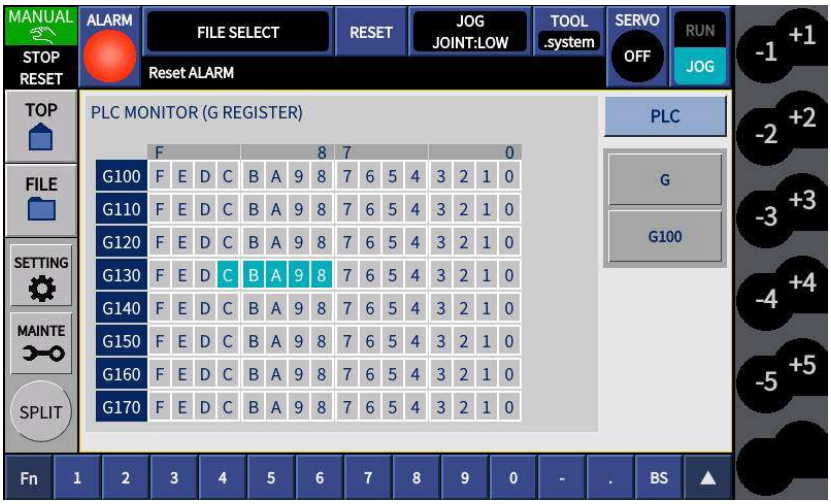
1 Tap the register number switch menu button.

The register number switch menu is displayed.



2 Tap the button for the register number to be displayed.

The register number displayed based on the tapped button is switched.



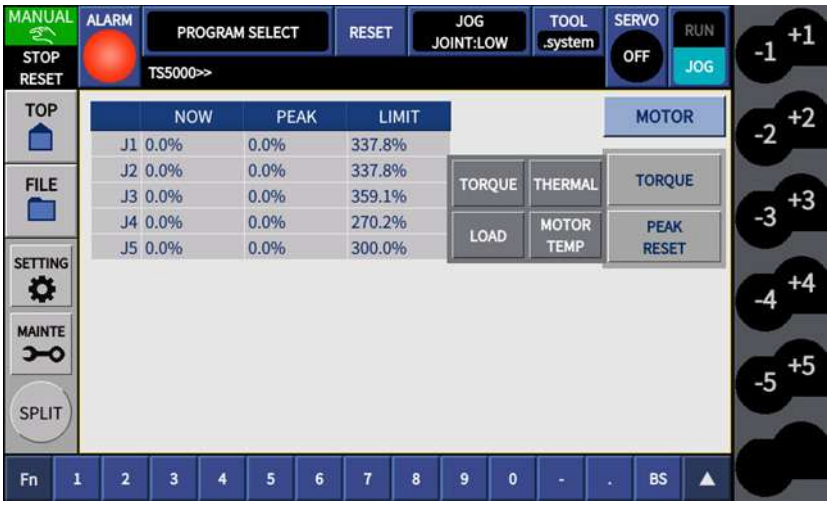
6.3.5 Operations Performed on MOTOR Screen

On the MOTOR screen, you can confirm the statuses of the motors and reset the peak torque.

How to Switch Motor Information to be Displayed

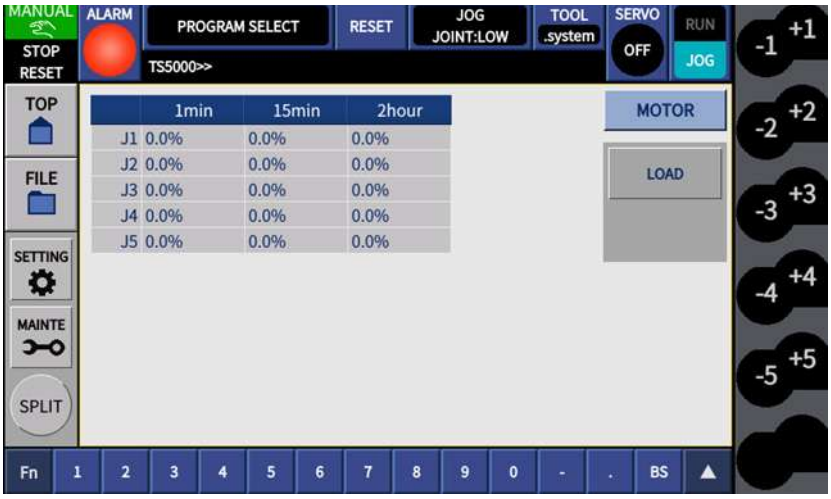
1 Tap the motor information switch menu button.

The motor information switch menu is displayed.



2 Tap the button for the motor information to be displayed.

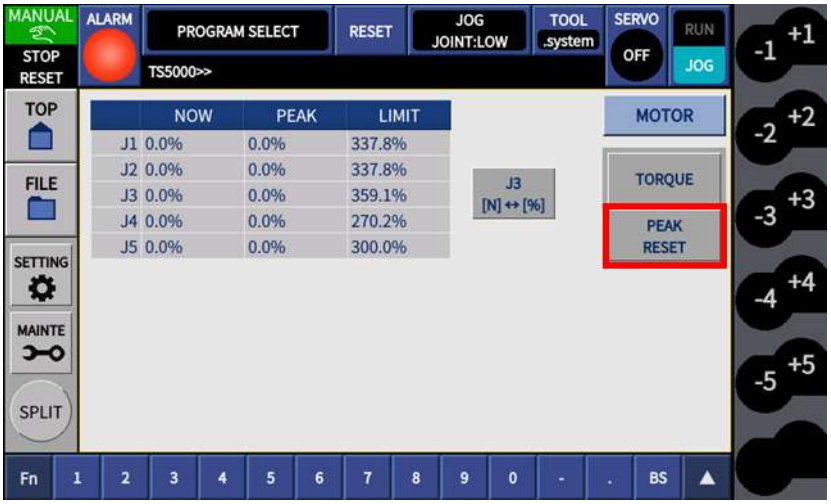
The motor information displayed based on the tapped button is switched.



How to Reset Peak Torque

1 Tap the PEAK RESET button on the MOTOR (Torque) screen.

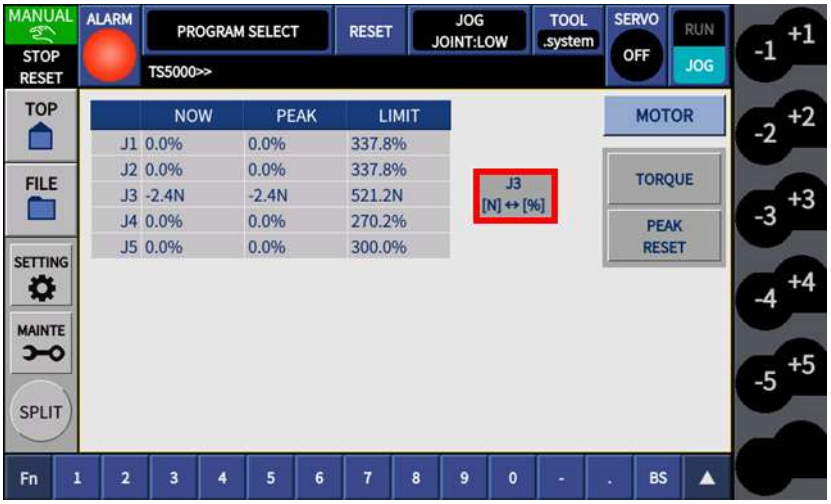
The peak torque is reset.



How to Switch Unit for 3rd Axis to Be Displayed

1 Tap the 3rd axis unit switching button.

This switches the unit for the 3rd axis to be displayed.



6.3.6 Operations Performed on LDISP Screen

On the LDISP screen, you can confirm the contents in the global section and program section of the execution program.

For details on the operating procedures, see "3.2.10 Monitoring of Execution Programs" or "5.4 Monitoring of Execution Programs" in this manual.

6.3.7 Operations Performed on WATCH Screen

On the WATCH screen, you can check the global variables of the execution programs and overwrite values.

For details on the operating procedures, see "3.2.11 Display and Overwriting of Global Variable" or "5.5 Display and Overwriting of Global Variable".

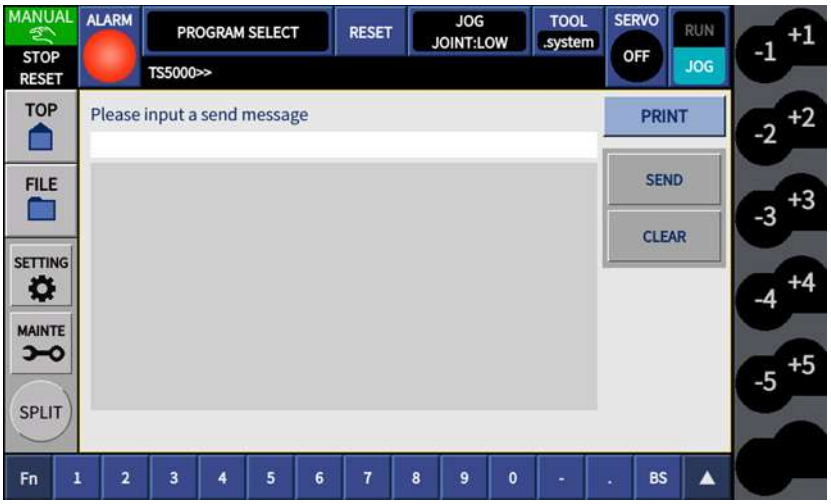
6.3.8 Operations Performed on PRINT/INPUT Screen

On the PRINT/INPUT screen, you can check the character strings output to the teach pendant by the PRINT command in the SCOL2 program and values input by the INPUT command.

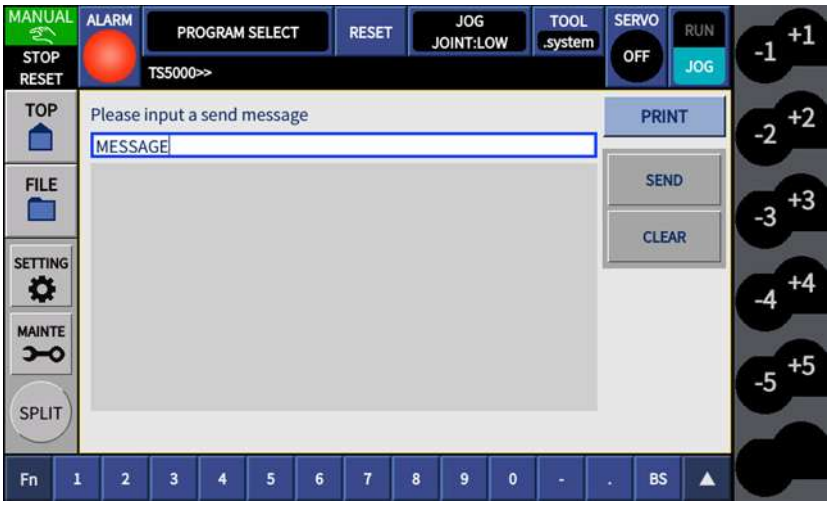
Value Input Procedure Using INPUT Command

1 Tap the text box for input of the message.

The cursor appears so that the content can be edited.



2 Enter the number or character string to be input by the INPUT command.



3 Tap the SEND button.

The character string in the text box is sent to the controller.

> *MEMO*
> *If the controller is not currently executing the INPUT command, the message is not sent even if the SEND button is tapped.*

6.3.9 Operations Performed on DO Screen

Direct execution can be performed on the DO screen.
For details on the operating procedures, see "3.3. Direct Execution" in this manual.

6.3.10 Operations Performed on PRG Screen

On the PRG screen, you can edit the contents in the global section and program section of the program file.
For details on the operating procedures, see "4.2. Editing Global Section and Program Section" in this manual.

6.3.11 Operations Performed on POINT Screen

On the POINT screen, you can edit POINT-type data and TRANS-type data in the data section of the program file.
For details on the operating procedures, see "4.3. Editing Data Section" in this manual.

6.3.12 Operations Performed on TRANS Screen

On the TRANS screen, you can check the TRANS-type data in the data section of the execution program and set the tool coordinates.
For details on the tool coordinate settings and the operating procedures, see "3.3.10 Settings Tool Coordinated" in this manual.

6.3.13 Operations Performed on PAYLOAD Screen

On the PAYLOAD screen, you can edit PAYLOAD-type data in the data section of the program file.
For details on the operating procedures, see "4.3. Editing Data Section" in this manual.

6.3.14 Operations Performed on IP WATCH Screen

On the IPWATCH screen, you can acquire and check the communication log on the non-procedural communication port.

MEMO

This screen is displayed on X8KTA-03A or later versions.

How to Start Communication Log Acquisition

1 Tap the IP switch button.

This displays the IP switch menu.



2 Tap the IP button.

The communication port corresponding to the tapped button is selected.



3 Tap the START button.

This starts communication log acquisition.



How to Stop Communication Log Acquisition

1 Tap the STOP button.

This stops the display of the communication contents of the communication port.

MEMO

When acquiring communication logs for a communication with short communication intervals, the screen display may become slow.

6.3.15 Operations Performed on CONV Screen

On the CONV screen you can perform operations and review data related to the conveyor synchronization function.

For the details of the CONV screen, see the separate "STA1183 Vision + Conveyor Synchronization Manual".

MEMO

This screen is displayed on 8KTA-02A or later versions.

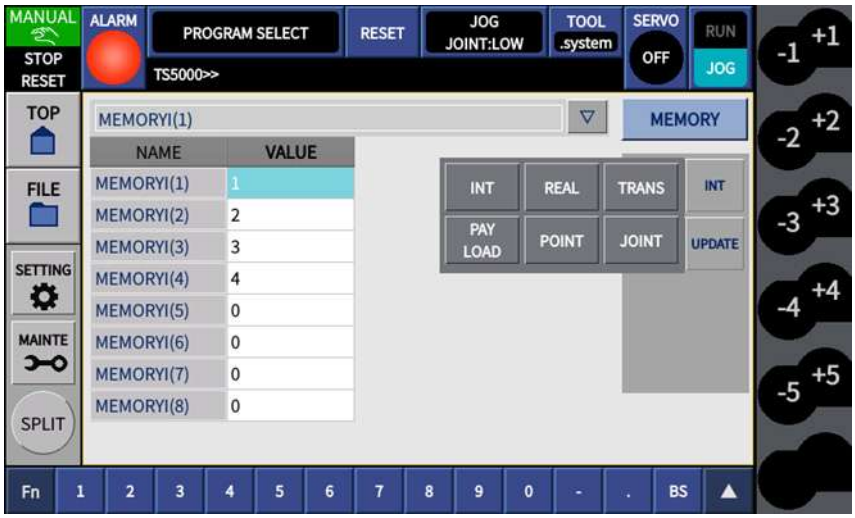
6.3.16 Operations Performed on MEMORY Screen

On the MEMORY screen, you can display and overwrite memory variable information.
Memory variables are stored in the controller's FRAM and their values are not reset when the controller is turned off.
On the MEMORY screen, you can also teach POINT- and JOINT-type memory variables (MEMORYP and MEMORYJ) and move teaching points.

How to Change Memory Variable Type to Be Displayed

1 Tap the variable type selector button.

The variable type selection menu is displayed.



2 Tap the button corresponding to the variable type to be displayed.

The memory variables of the variable type corresponding to the tapped button are displayed.



How to Change Memory Variable Number to Be Displayed

1 Tap the SHOW button.

The variable number selection menu is displayed.



2 Tap the text box.

The cursor appears in the text box to indicate that the content can be edited.

3 Use the keyboard to enter a custom value.



4 Tap the Exec button.

The memory variables of the element numbers that were input in the text box are displayed.



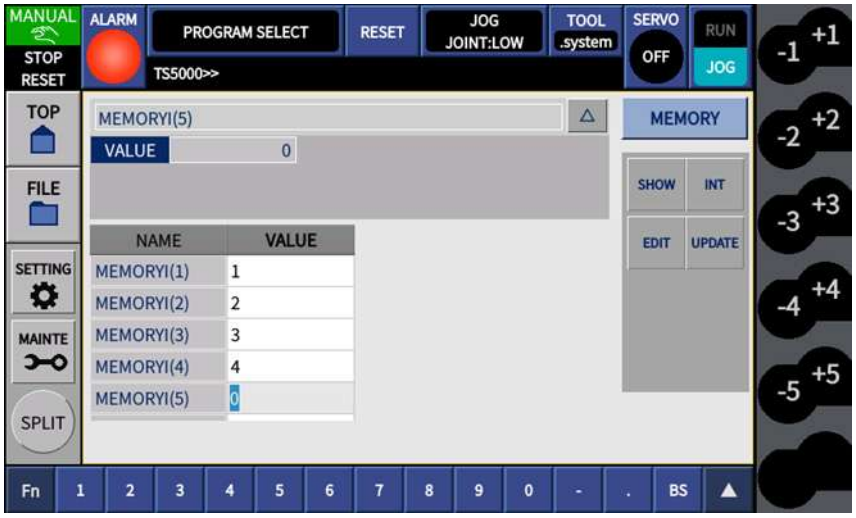
How to Overwrite Memory Variable

1 Tap the cell of the memory variable to be changed.

The cursor moves to the tapped cell.

2 Tap the cell.

The cursor appears in the cell to indicate that the content can be edited.



3 Use the keyboard to enter a custom value.

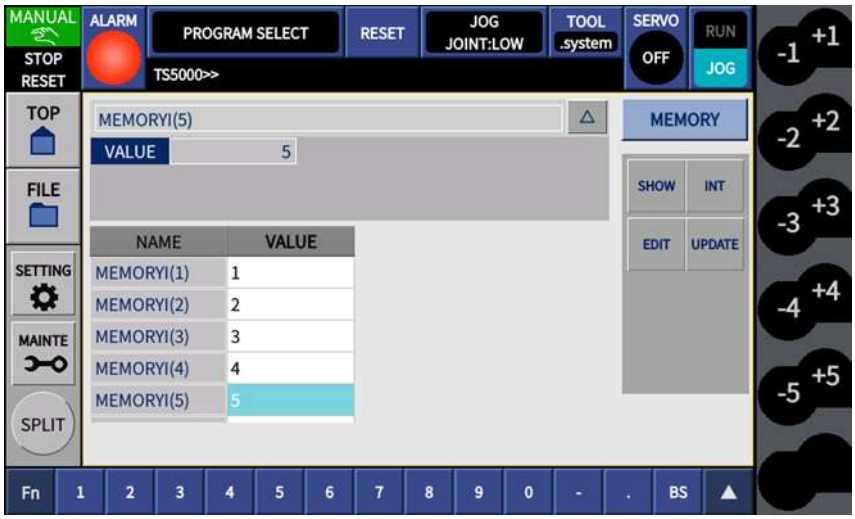


4 After editing is completed, press the Enter key or tap a location on the screen outside the cell being edited.

This saves the edited changes, and the cursor disappears.

5 Tap the UPDATE button.

The value of the memory variable corresponding to the cell where the cursor is located is overwritten by the value in the cell.



NOTICE

● Before overwriting a memory variable, confirm the effect that the target memory variable will have on the program. Failure to confirm it may cause the robot to collide with the peripheral device.

How to Teach POINT-Type Memory Variable

1 On the MEMORYP screen, move the cursor to the variable the current position of which is to be taught.

2 Tap the TEACH button.

The MEMORYP (Teach) screen is displayed.



3 Tap the OK button.

The value of the POINT-type data in the line where the cursor is located is overwritten by the value of the current position of the TCP.

MEMO

The above operation does not overwrite the value of the memory variable.

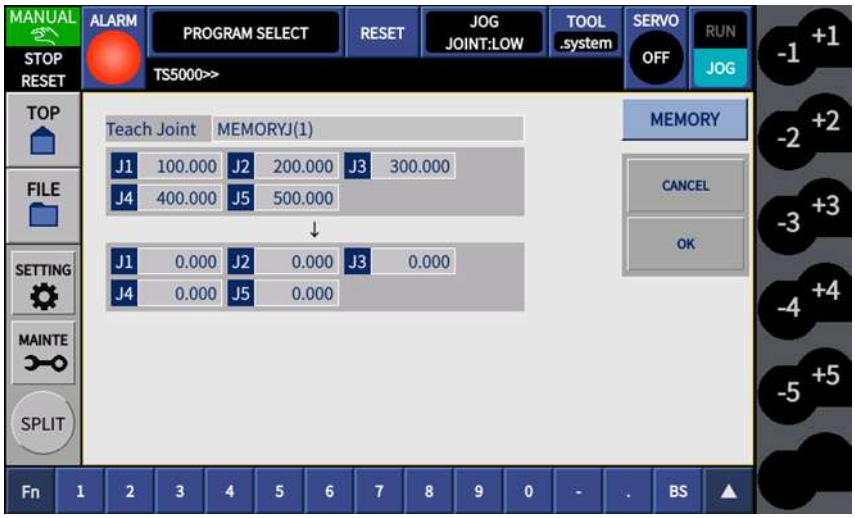
To overwrite memory variables, refer to the procedure How to Overwrite Memory Variable in this section.

How to Teach JOINT-Type Memory Variable

1 On the MEMORYJ screen, move the cursor to the variable the current position of which is to be taught.

2 Tap the TEACH button.

The MEMORYJ (Teach) screen is displayed.



3 Tap the OK button.

The value of the JOINT-type data in the line where the cursor is located is overwritten by the current angle value of each axis.

MEMO

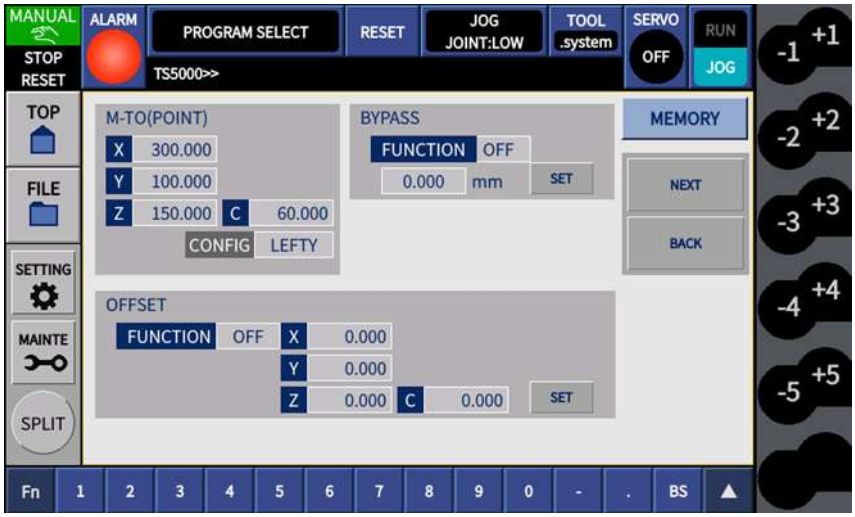
The above operation does not overwrite the value of the memory variable.

To overwrite memory variables, refer to the procedure How to Overwrite Memory Variable in this section.

How to Move Teaching Point to POINT-Type Data

1 Tap the M-TO button on the MEMORYP screen.

The MEMORYP (Teaching Point Movement) screen is opened.

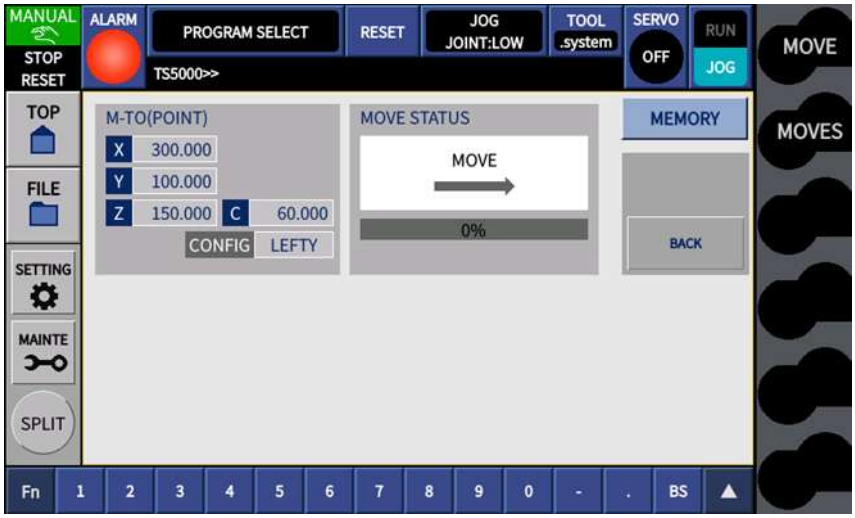


2 Confirm the coordinates of the teaching point, and tap the NEXT button.

The MEMORYP (Teaching Point Movement) screen is opened, and the displayed operation keys are changed.

The robot target position incorporating the offset and bypass settings is displayed on the MEMORYP (Teaching Point Movement) screen.

Check that the displayed target position is correct.



MEMO

The offset and bypass setting operations are the same as those in the POINT (Edit) screen.

For details, see "4.3.5. Moving Teaching Point to Offset Position" and "4.3.6. Moving Teaching Point by Bypass Movement".

3 Tap the SERVO ON button while pressing down the enable switch by one level.

This turns on the servo, and the SERVO ON button indicator is changed.

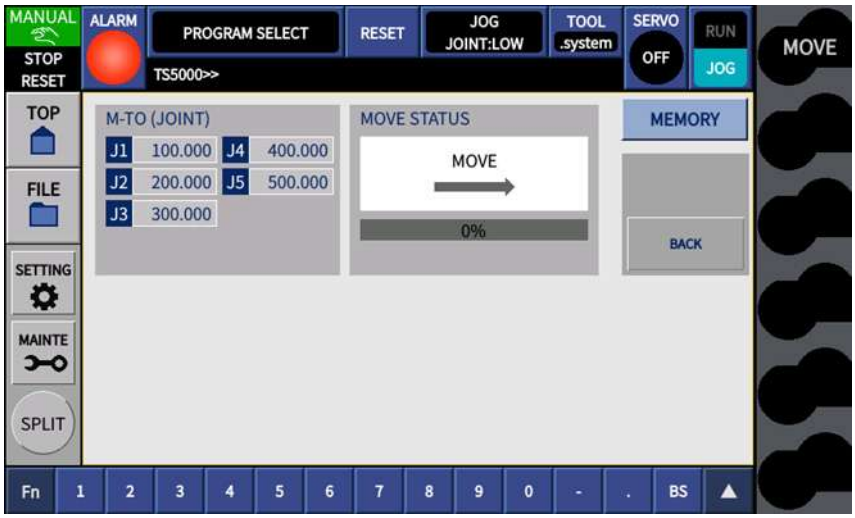
4 Press the MOVE TO key or MOVES TO operation key.

The robot moves toward the target position while the MOVE TO or MOVES TO key is held down.
(The robot performs PTP movement when the MOVE TO key is pressed, and performs linear interpolation movement when the MOVES TO key is pressed.)

How to Move Teaching Point to JOINT-Type Data

1 Tap the M-TO button on the MEMORYJ screen.

The MEMORYJ (Teaching Point Movement) screen is opened, and the displayed operation keys are changed.
The robot target angle is displayed on the MEMORYJ (Teaching Point Movement) screen.
Check that the displayed target angle is correct.



2 Tap the SERVO ON button while pressing down the enable switch by one level.

This turns on the servo, and the SERVO ON button indicator is changed.

3 Press the MOVE TO key operation key.

The robot performs PTP movement toward the target angle while the MOVE TO key is held down.

7. Operations in Alarm Window

This chapter describes the alarm window.

For the descriptions and causes of alarms, see the separate "STA1165 Alarm Manual".

How to Open Alarm Window

1 Tap the ALARM button in the status bar.

The Alarm window opens.

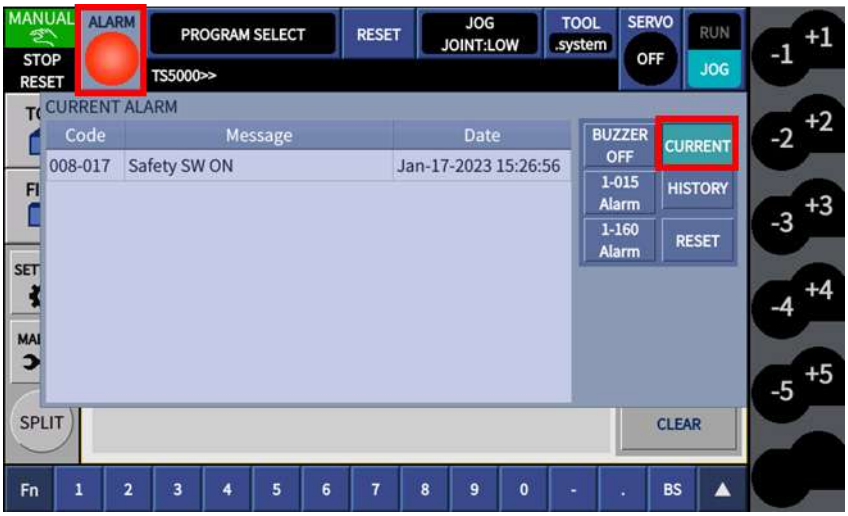
7.1 Current Alarm Screen

7.1.1 This shows the current alarm.

How to Display Current Alarm from Opened Alarm Window

1 Tap the CURRENT button.

The CURRENT ALARM screen is opened.



How to Display Current Alarm when Alarm Window Is Not Opened

1 Tap the ALARM button in the status bar.

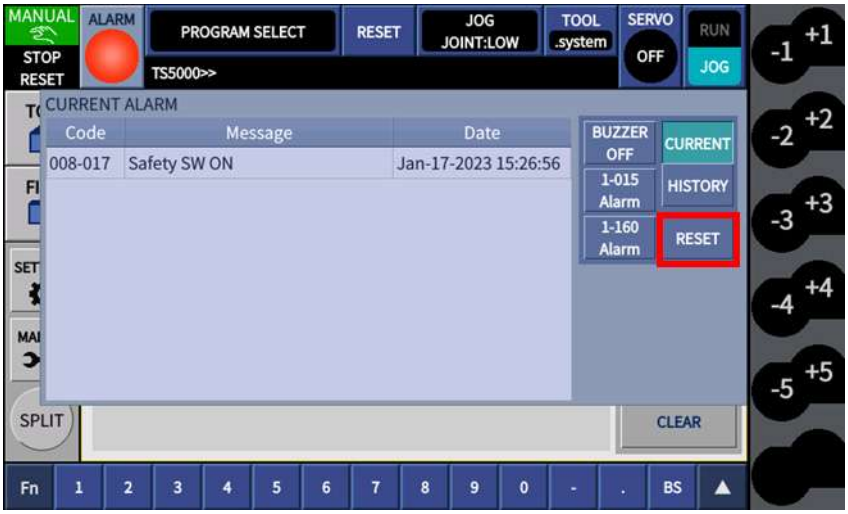
The CURRENT ALARM screen is opened.

7.1.2 Resetting Current Alarm

How to Reset Current Alarm

1 Tap the RESET button on the CURRENT ALARM screen.

The current alarm is reset. If the buzzer is sounding for an alarm, the buzzer sound is stopped.



MEMO

For an alarm whose cause has not been removed before resetting the alarm, the alarm will be generated again even after the alarm is reset.

7.1.3 Stopping the Alarm Buzzer

A buzzer sounds in the TS5000 controller whenever an alarm is generated.
This describes how to stop the alarm buzzer that sounds when an alarm is generated.

How to Stop Buzzer

1 Tap the BUZZER OFF button on the CURRENT ALARM screen.

This stops the alarm buzzer.



MEMO

- *The buzzer is also stopped by resetting the alarm. If an alarm is generated after it is reset, the buzzer will sound again.*
- *Alarms can be set so that no buzzer is sounded. For details, refer to the section "USER.PAR" in "STEA1166 Parameter Manual."*

7.1.4 Viewing Details for "1-015 Parameter Error"

How to Open Parameter Error Details Screen

1 Tap the 1-015 button on the CURRENT ALARM screen.

The parameter error details screen is opened.

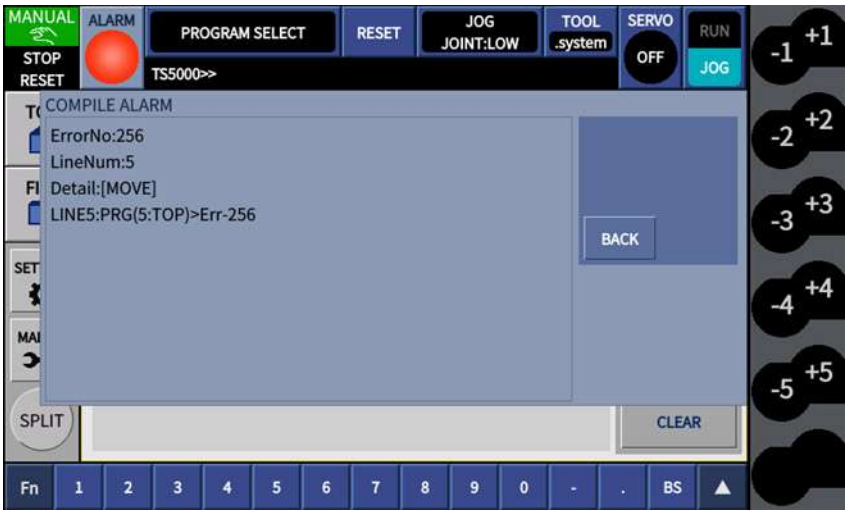


7.1.5 Viewing Details for "1-160 Compile Error"

How to Open Compile Error Details Screen

1 Tap the 1-160 button on the CURRENT ALARM screen.

The compile error details screen is opened.



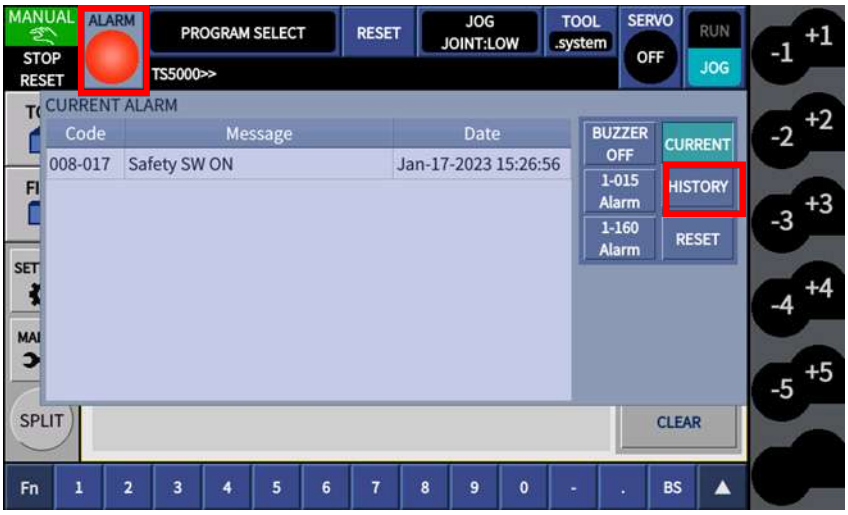
7.2 Alarm History Screen

7.2.1 Opening Alarm History Screen

How to Open Alarm History Screen

1 Tap the HISTORY button on the CURRENT ALARM screen.

The ALARM HISTORY screen is opened.

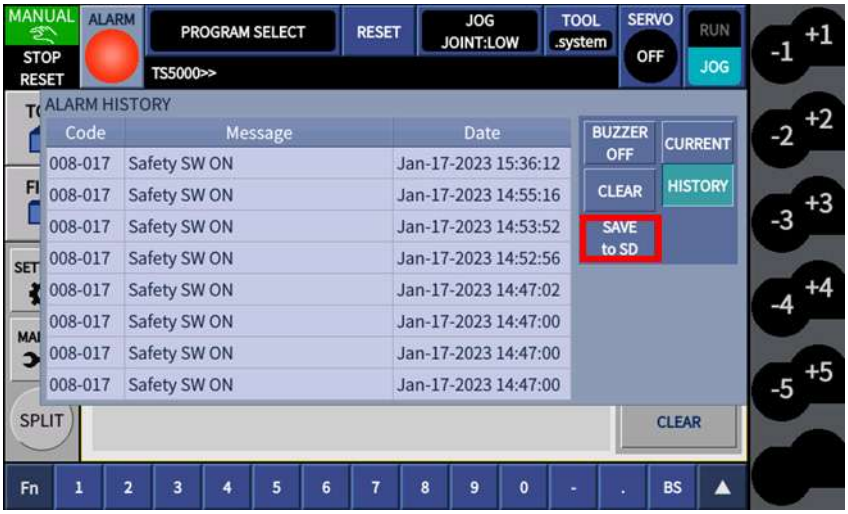


7.2.2 Saving Alarm History to SD Card

How to Save Alarm History to SD Card

1 Tap the SAVE to SD button on the ALARM HISTORY screen.

A save confirmation message appears.



2 Tap the OK button.

The alarm history is saved to an SD card.



MEMO

"ALARM.TXT" is saved to the SD card.

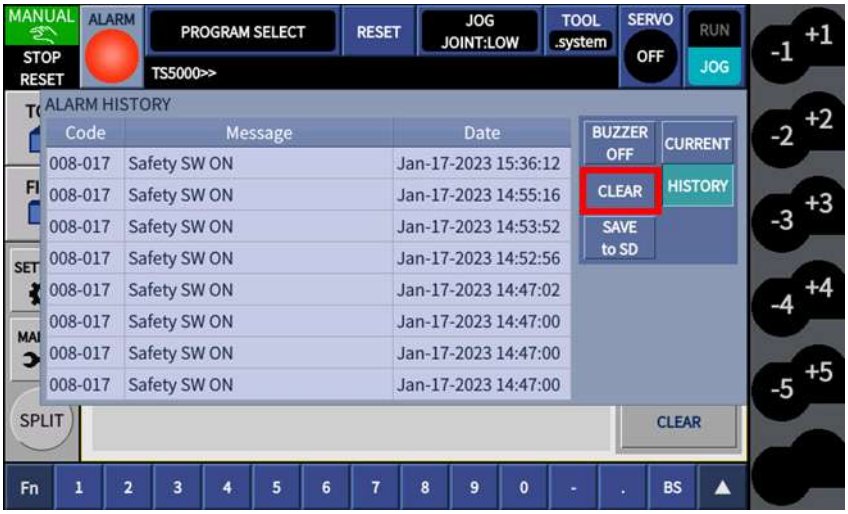
If the file already exists, it is overwritten and saved.

7.2.3 Clearing Alarm History

Alarm History Clear Procedure

1 Tap the CLEAR button on the ALARM HISTORY screen.

A confirmation message appears.



2 Tap the EXE button.

The alarm history is cleared.





8. Robot and Teach Pendant (TP) Setting Operations

This chapter describes how to use the robot and TP setting screens.

8.1 How to Open Robot and TP Setting Screens

How to Open Setting Menu Screen

1 Set the master mode keyswitch to the MANUAL position.

The master mode is switched to MANUAL.

2 Tap the SETTING button on the top screen.

The setting menu screen is opened.



How to Open Setting Screens

3 Tap the button for the screen to be opened on the setting menu screen.

The setting screen for the tapped button is opened.

8.2 Setting Screens

The setting screens are shown in the table below.

For details on each screen, see "10.5. Setting Menu Screen" in this manual.

Table 8.1. Setting Screens

Screen Name	Description
Software limit setting screen	Sets the software limits of each joint axis.
Parameter setting screen	Sets the robot parameters.
TP setting screen	Sets the TP screensaver and backlight.

8.2.1 Operations Performed in Software Limit Setting Screen

8.2.1.1 Setting Software Limit

Software limits are limit values that can be set by the user for each joint axis.

If a joint axis moves outside the software limit range, the alarms "4-42* Joint axis No. * + Software limit" or "4-43* Joint axis No. * - Software limit" will occur.

How to Set Software Limit

1 Tap the cell of the joint axis where the software limit will be changed.

The cursor moves to the tapped cell.

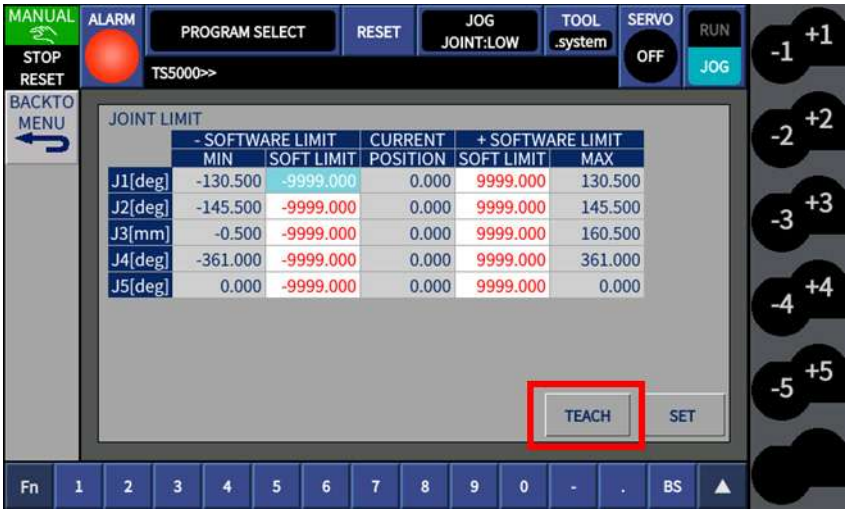
2 Change the value in the selected cell to the desired value for the software limit.

Cell values can be changed in one of the following two ways.

Changing Cell Value to Current Joint Axis Angle

1. Tap the TEACH button.

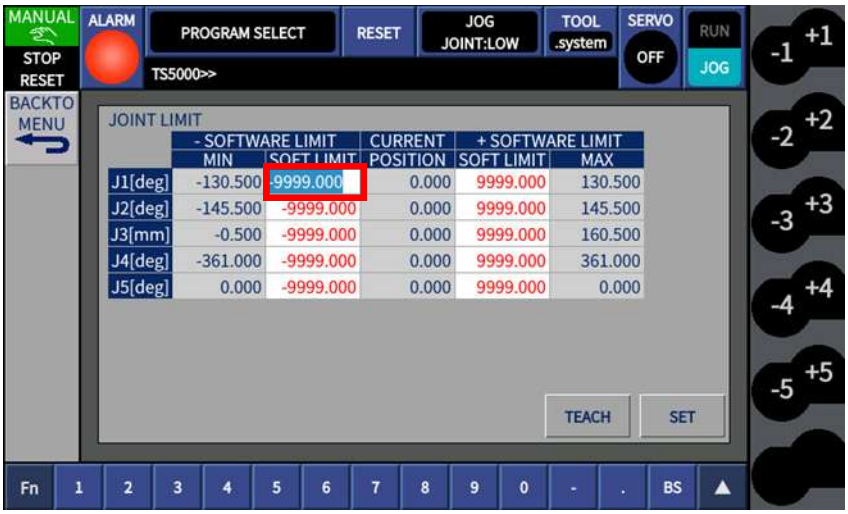
The value in the selected cell changes to the current joint axis angle.



Directly Entering Cell Value

1. **Tap the selected cell.**

   The cursor appears in the cell so that the content can be edited.



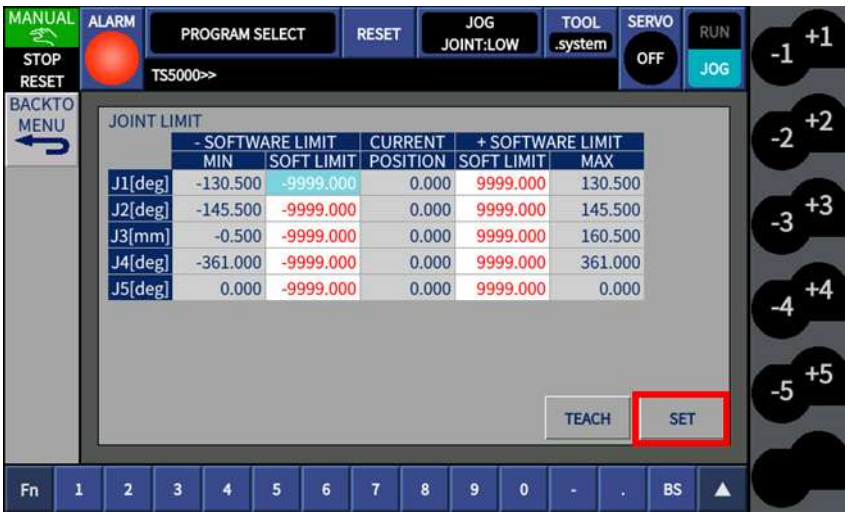
2. **Use the keyboard to enter a custom value.**

3. **After editing is completed, press the Enter key or tap a location on the screen outside the cell being edited.**

   This saves the edited changes, and the cursor disappears.

3 Tap the SET button.

The value in the selected cell is overwritten by the value of joint axis of the corresponding robot.



MEMO

If the input value is outside the range of mechanical limit (maximum value and minimum value), the software limit becomes equal to the value of mechanical limit.

NOTICE

- Be sure to set the software limit inside the mechanical stopper. If the software limit is not set inside the mechanical stopper, the robot in motion collides with the mechanical stopper.
- After setting the software limit, be sure to always check that the software limit alarms (4-42* and 4-43*) are generated correctly when the robot is moved outside the range.

8.2.2 Operations Performed on Parameter Setting Screen

8.2.2.1 Setting the Parameters

This describes how to set the parameters of the controller TS5000.

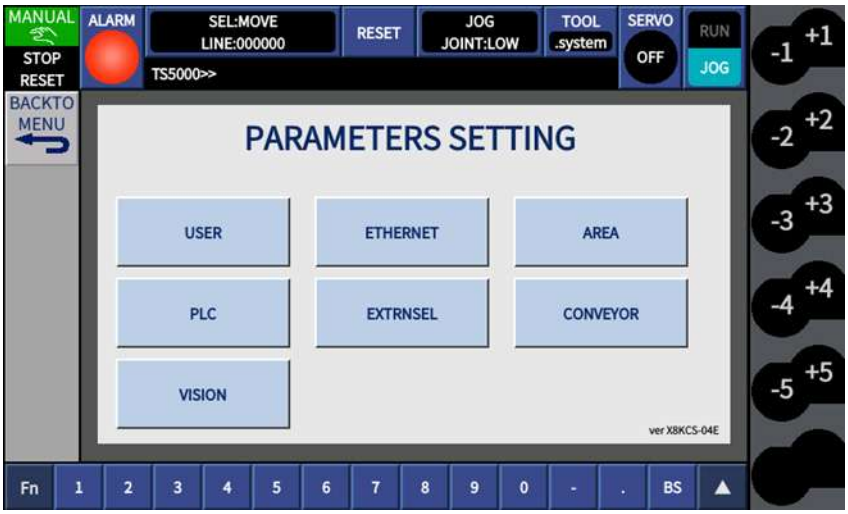
For the parameter types and descriptions, see the separate STA1166 Parameter Manual.

How to Set Parameters

(An example is shown here for setting [U000]LANGUAGE of USER.PAR.)

1 Tap the button for the parameter information to be set on the parameter setting screen.

The parameter selection screen for the parameter file is opened.



2 Tap the button for the parameter to be set.

The parameter details setting screen is opened.



3 Change the setting values of the parameters.

Parameters include one whose setting is selected from multiple options and one for which a value or character string is entered.



4 Tap the OK button.

The parameter is overwritten by the input value.

> *MEMO*
>
> *Some parameters become effective by restarting the controller, and other parameters become effective immediately after setting.*
>
> *For details, see STA1166 Parameter Manual.*

NOTICE

●Do not set values other than those provided in the Parameter Manual. Setting it may cause the robot in unexpected motion to collide with the peripheral device.

8.2.3 Operations Performed in Teach Pendant (TP) Setting Screen

8.2.3.1 Screensaver

The TP5000 includes a screensaver function for preventing screen burn-in and extending the lifespan of the backlight.

The screensaver is activated when no operations are performed for a certain amount of time. Also, the backlight turns off after a certain amount of time has elapsed.

When an operation is performed, the backlight turns on, and the screensaver is deactivated.

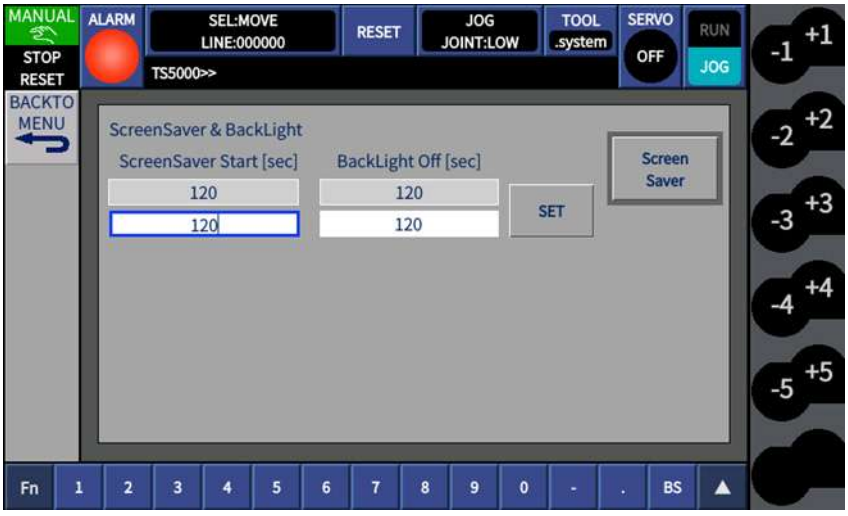
MEMO

An operation refers to tapping the screen, pressing an operation key, switching the master mode key, using the emergency stop switch, or turning on the enable switch.

How to Set Screensaver

1 Tap the ScreenSaver Start text box (screen left side).

The cursor appears so that the content can be edited.



2 Enter the value for the screensaver start time in seconds, and then press the Enter key or tap a location on the screen outside the text box.

This saves the edited changes.

3 Tap the SET button.

The value in the text box is set for the screensaver start time.



MEMO

The screensaver is not activated when a value of 0 is set.

8.2.3.2 Backlight Off Time

How to Set Backlight Off Time

1 Tap the Backlight Off text box (screen right side).

The cursor appears so that the content can be edited.



2 Enter the value for the backlight off time in seconds, and then press the Enter key or tap a location on the screen outside the text box.

This saves the edited changes.

3 Tap the SET button.

The value in the text box is set for the backlight off time.



MEMO

The backlight is not turned off when a value of 0 is set.

8.2.4 Screen Calibration

8.2.4.1 Performing Screen Calibration

This allows you to adjust the touch responding position on the TP5000's LCD panel (screen calibration).

How to Perform Screen Calibration

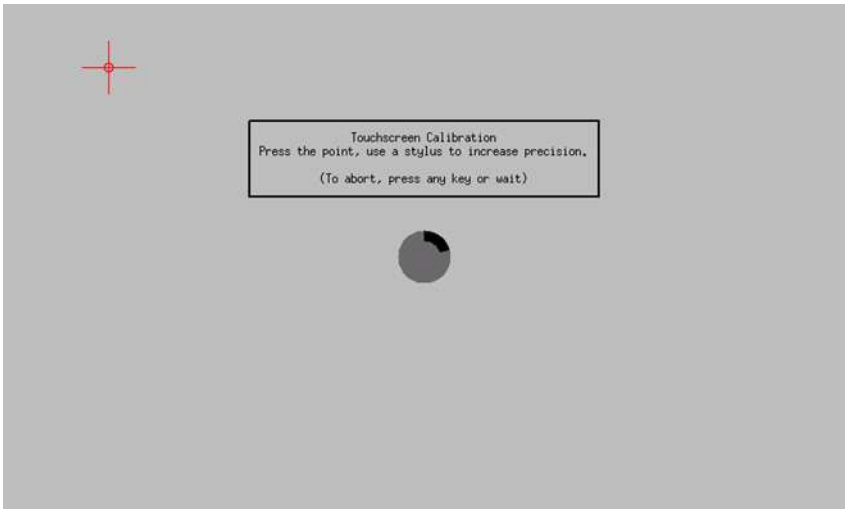
1 Tap the screen calibration button on the setting menu screen.

A confirmation message appears.



2 Tap the OK button.

The screen for screen calibration is displayed.



3 Tap the red mark that appears on the screen four times.

The tap responding position will be adjusted based on the positions where you tap the mark.

MEMO

If you do not tap on the screen for screen calibration for a certain period of time, the screen calibration will be canceled, returning to the original screen.

9. Maintenance Operations

This chapter describes how to use the screens used in robot and controller maintenance.
When actually performing maintenance, be sure to refer to the STA Robot Manual and STA Controller Manual.

9.1 Maintenance Screen Opening Procedure

How to Open Maintenance Menu

1 Tap the MAINTE button on the top screen.

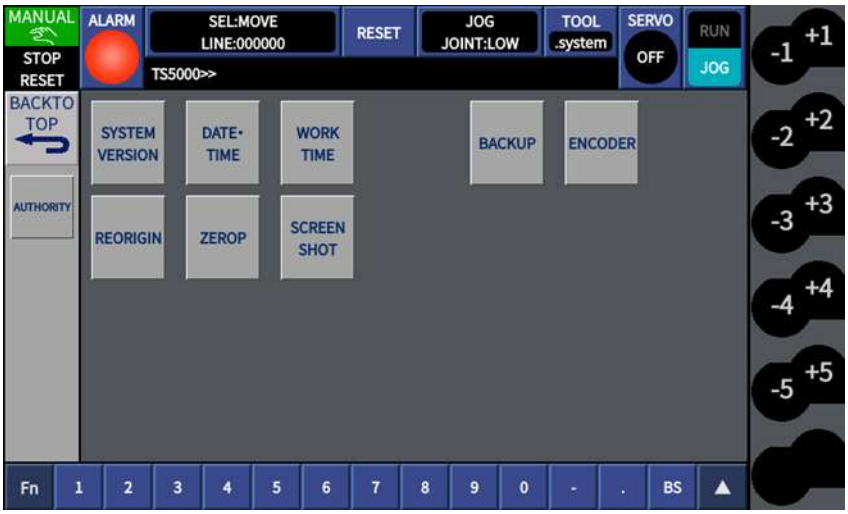
The maintenance menu is displayed.



How to Open Screens

1 Tap the button for the screen to be opened on the maintenance menu screen.

The screen for the tapped button is opened.



9.2 Maintenance Screens

The maintenance screens are shown in the table below.
Some screens cannot be opened in the Servo ON status.
For details on each screen, see "Screen Details: 10.6. Maintenance Menu Screen" in this manual.

Table 8.1. Maintenance Screens

Screen Name	Description	Is Screen Displayable when Servo ON?
Version information screen	Shows the robot and controller models, serial number, version information, TCmini information, etc.	Yes
Date and time setting screen	Shows and enables setting of the date and time in the controller.	Yes
Running time information screen	Shows the running time of the controller.	Yes
Memory variable information screen	Shows the memory variables retained when the controller is powered off and enables overwriting of variables.	Yes
Backup screen	Backs up the files in the controller and restores the files from the backup.	No
Encoder information screen	Shows the encoder information for each joint axis.	Yes
Origin return screen	Returns to the origin for each joint axis.	No
Zero position screen	Sets the zero position for each joint axis.	No
SCREENSHOT Screen	Sets up screenshots.	Yes

9.2.1 Operations Performed on the Version Information Screen

This screen shows the robot and controller model names.
This screen can also open the system information screen and TCmini information screen.

How to Open Version Information Screen

1 Tap the SYSTEM VERSION button on the maintenance menu screen.

The version information screen is opened.



MEMO

When the version information screen is opened, the robot information is displayed.

How to Switch Display Screen

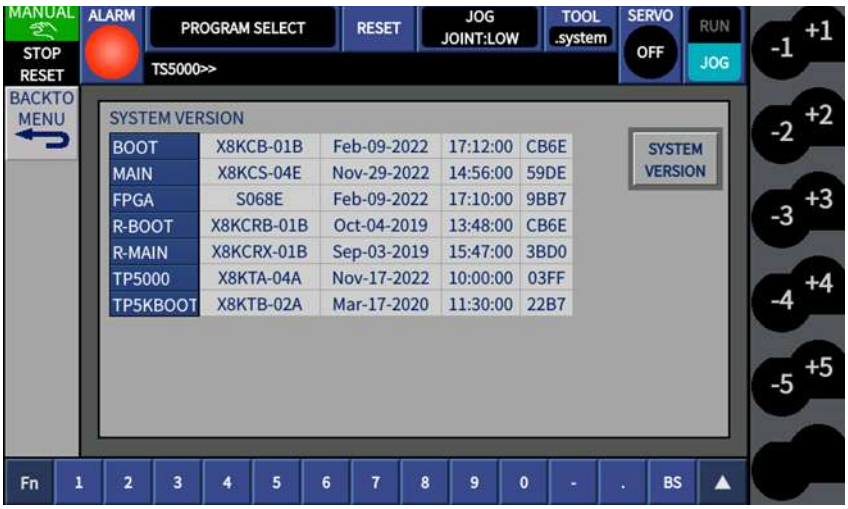
1 Tap the version information switch menu button.

The version information switch menu is displayed.



2 Tap the button for the information to be displayed.

A screen for the tapped button is opened.



9.2.2 Operations Performed on Date and Time Setting Screen

This screen shows the date and time setting in the controller. This screen can also be used to change the date and time in the controller.

How to Change Controller Date and Time

1 Tap the text box to be changed.

The cursor appears in the text box to indicate that the content can be edited.



2 Use the keyboard to enter a custom value.

3 After editing is completed, press the Enter key or tap a location on the screen outside the text box being edited.

This saves the edited changes, and the cursor disappears.

4 Tap the SET button.

The current controller date and time are set in the text boxes.

9.2.3 Running Time Information Screen

This screen shows the controller power-on time and the program run time.

*There are no available operations on this screen.



9.2.4 Encoder Information Screen

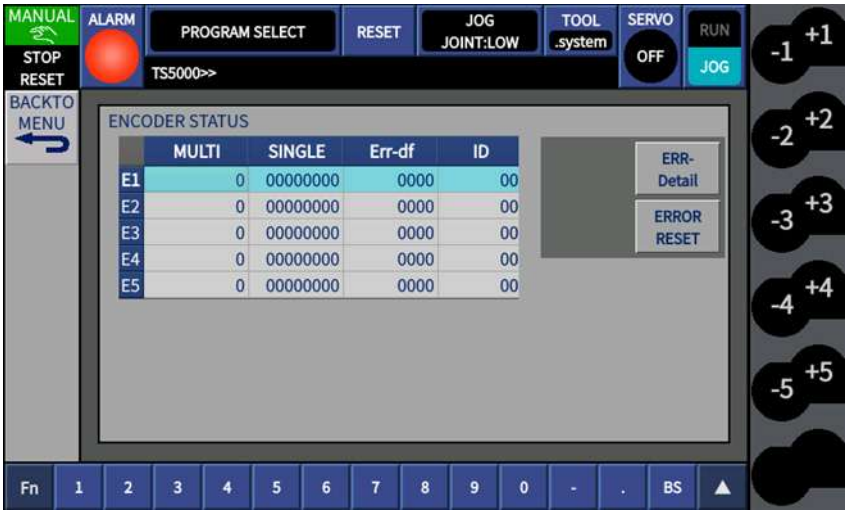
This screen shows the encoder information for each joint axis.

This screen can be used to reset errors that are generated in individual encoders.

How to Reset Encoder Error

1 Tap the line of the joint axis where the error will be reset.

The cursor moves to the tapped line.



2 Tap the ERROR RESET button.

The encoder error for the joint axis is reset for the line where the cursor is located.

3 Tap the OK button.

The encoder error for the joint axis is reset for the line where the cursor is located.



MEMO

If the cause of the error is not removed, the error will be generated again.

9.2.5 Origin Setting Screen

This screen enables setting of the origin and registering of the origin return position for each joint axis.

*This screen is used for maintenance operations. For details on the operations performed on this screen, see the STA Robot Manual.

9.2.6 Zero Position Screen

This screen is used to set the origin for each joint axis.

* This screen is used for maintenance operations. For details on the operations performed on this screen, see the STA Robot Manual.

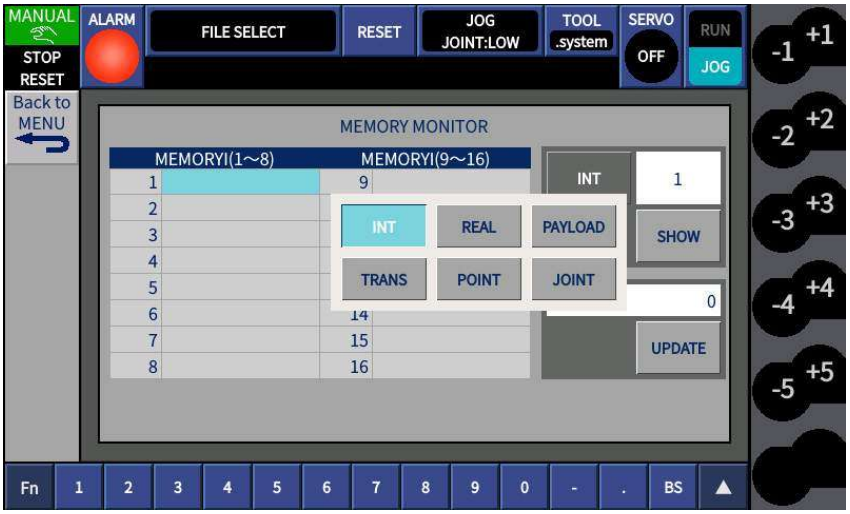
9.2.7 Memory Variable Information Screen

This screen shows memory variable information and enables overwriting of the values.
Memory variables are stored in the controller's FRAM and their values are not reset when the controller is turned off.
This screen is displayed on X8KTA-03B or earlier versions.
On X8KTA-03C or later versions, showing memory variables and overwriting of the values are performed on the MEMORY screen in the utility window.
For details of the MEMORY screen in the utility window, see "6.3.16 Operations Performed on MEMORY Screen".

How to Open Memory Variable Information

1 Tap the variable type selector button.

The variable type selection menu is displayed.

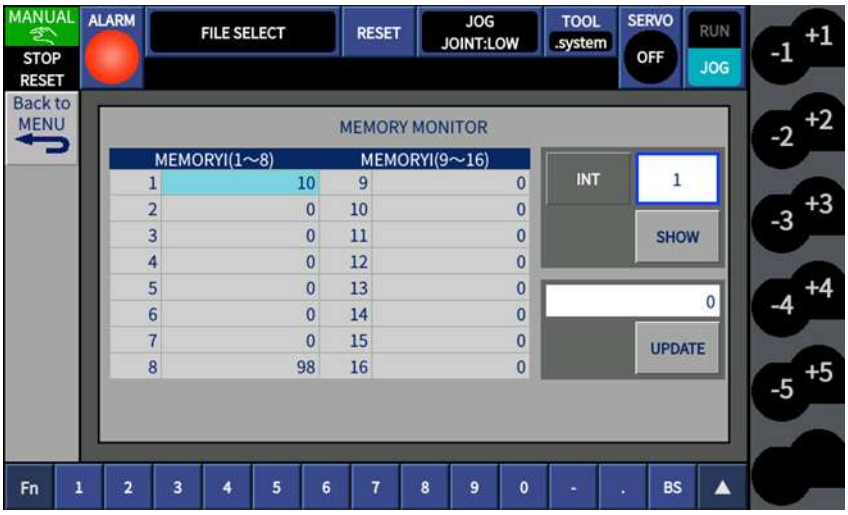


2 Tap the button for the variable type to be displayed.

The variable type selection menu is hidden.

3 Tap the text box for input of the variable number.

The cursor appears in the text box to indicate that the content can be edited.



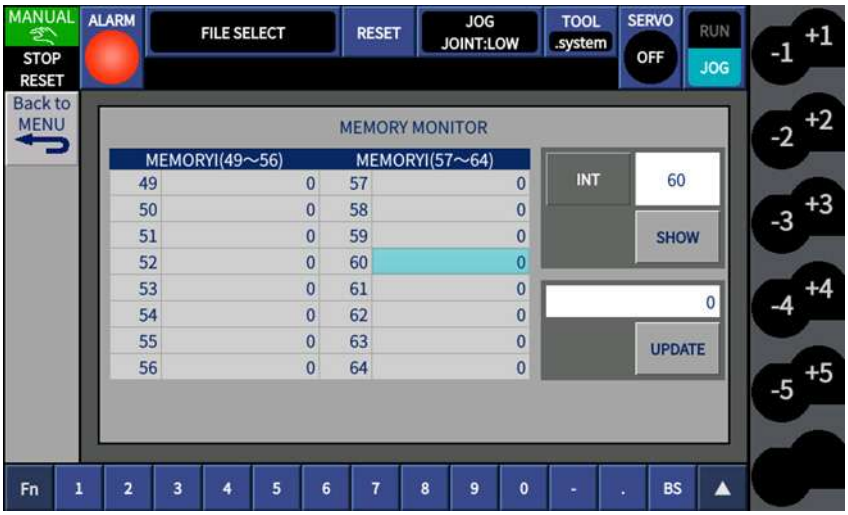
4 Use the keyboard to enter a custom value.

5 After editing is completed, press the Enter key or tap a location on the screen outside the text box being edited.

This saves the edited changes, and the cursor disappears.

6 Tap the SHOW button.

The memory variable is displayed based on the variable type and variable number that have been set.



How to Overwrite Scalar Type Memory Variable

1 Tap the cell of the memory variable to be changed.

The cursor moves to the tapped cell, and the value in the text box changes to the current value for the selected memory variable.

2 Tap the text box.

The cursor appears in the text box to indicate that the content can be edited.



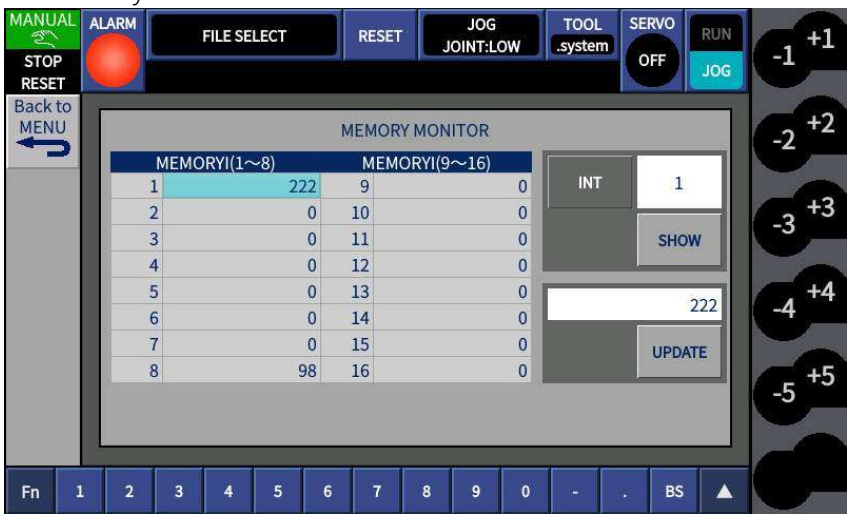
3 Use the keyboard to enter a custom value.

4 After editing is completed, press the Enter key or tap a location on the screen outside the text box being edited.

This saves the edited changes, and the cursor disappears.

5 Tap the UPDATE button.

The value of the memory variable for the cell where the cursor is located is overwritten by the value in the text box.



NOTICE

● Before overwriting a memory variable, confirm the effect that the target memory variable will have on the program. Failure to confirm it may cause the robot to collide with the peripheral device.

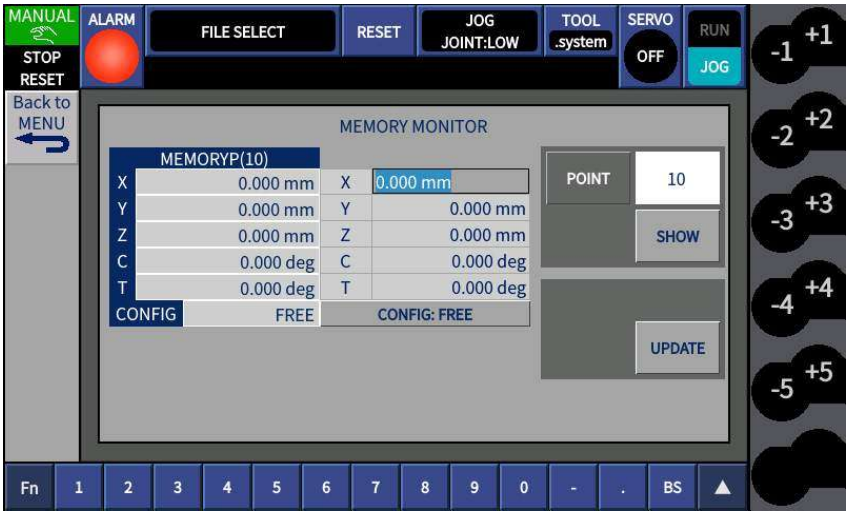
Vector Type Memory Variable Overwrite Procedure

1 Tap the cell of the memory variable element to be changed.

The cursor moves to the tapped cell, and the value in the text box changes to the current value for the selected memory variable.

2 Tap the cell again.

The cursor appears in the cell so that the content can be edited.



3 Use the keyboard to enter a custom value.

4 After editing is completed, press the Enter key or tap a location on the screen outside the cell being edited.

This saves the edited changes, and the cursor disappears.

5 Perform steps 1 to 4 for all elements to be changed.

6 Tap the UPDATE button.

The value of the memory variable is overwritten by the value in the cell.

NOTICE

- Before overwriting a memory variable, confirm the effect that the target memory variable will have on the program. Failure to confirm it may cause the robot to collide with the peripheral device.

9.2.8 Backup Screen

You can restore from files backed up on the SD card.

*This screen is used for maintenance operations. For details on the operations performed on this screen, see the STA Controller Manual.

9.2.9 SCREENSHOT Screen

This screen shows the screen screenshot setting screen.

You can set up to take screenshots.

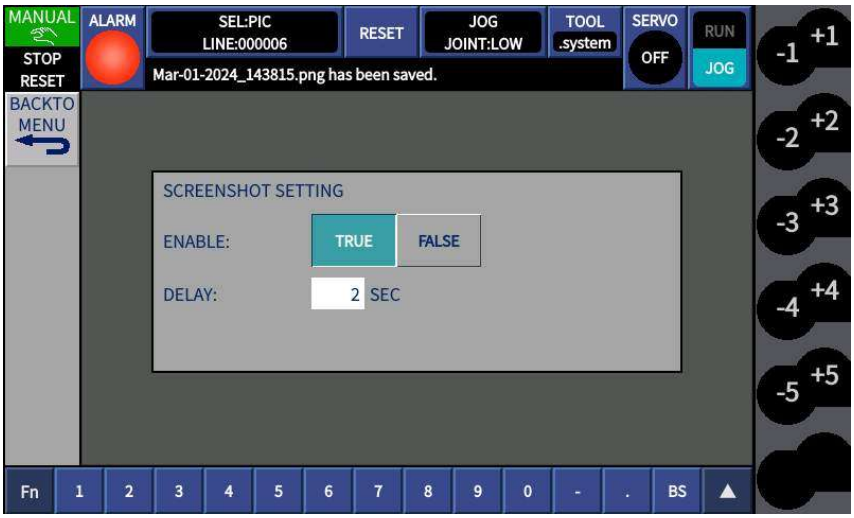
How to Set Up a Screenshot

1 Tap Enable.

If disabled, no image will be saved.

2 Sets the time between tapping and shooting.

Select a cell and enter the time using the keyboard.



MEMO

The setting is canceled when the controller is turned off.

How to Take a Screenshot

3 Insert the SD card into the controller.

4 Display the screen from which you want to take a screenshot.

5 Tap AUTO or MANUAL in the upper left corner of the screen.

Take a screenshot.

6 When the shooting is complete, a message appears on the message panel.

When shooting, the STATUS LED in the upper right corner of the teach pendant lights up red.



7 The captured images are stored in the TP5000-SCEENSHOT folder on the SD card.

MEMO

Screenshots can be taken regardless of master mode (AUTO/MANUAL) or servo ON/OFF.

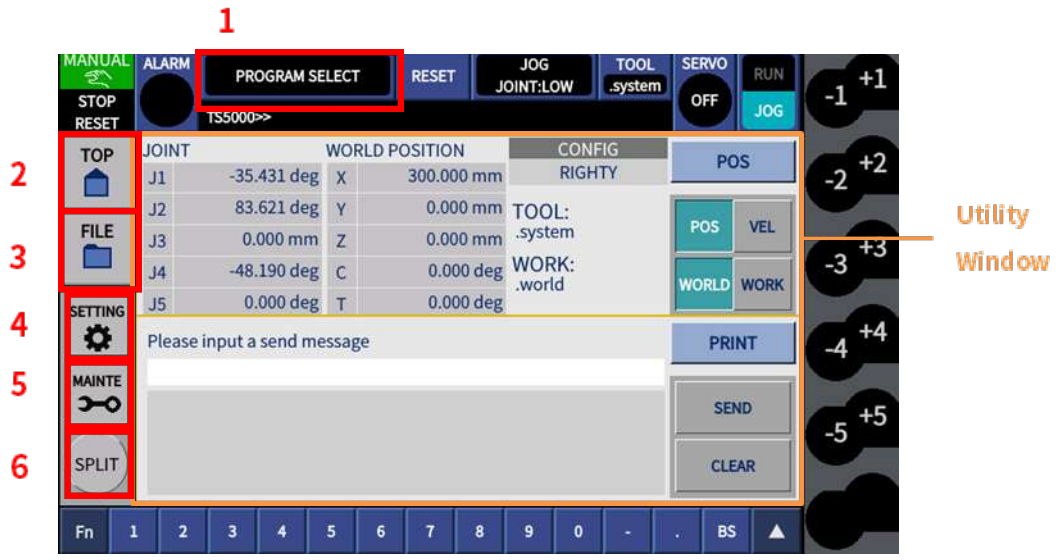
Screen Details

This provides details of each TP5000 screen.

10. Main Window Details

This section describes details of the window other than the utility window.

10.1 Top Screen

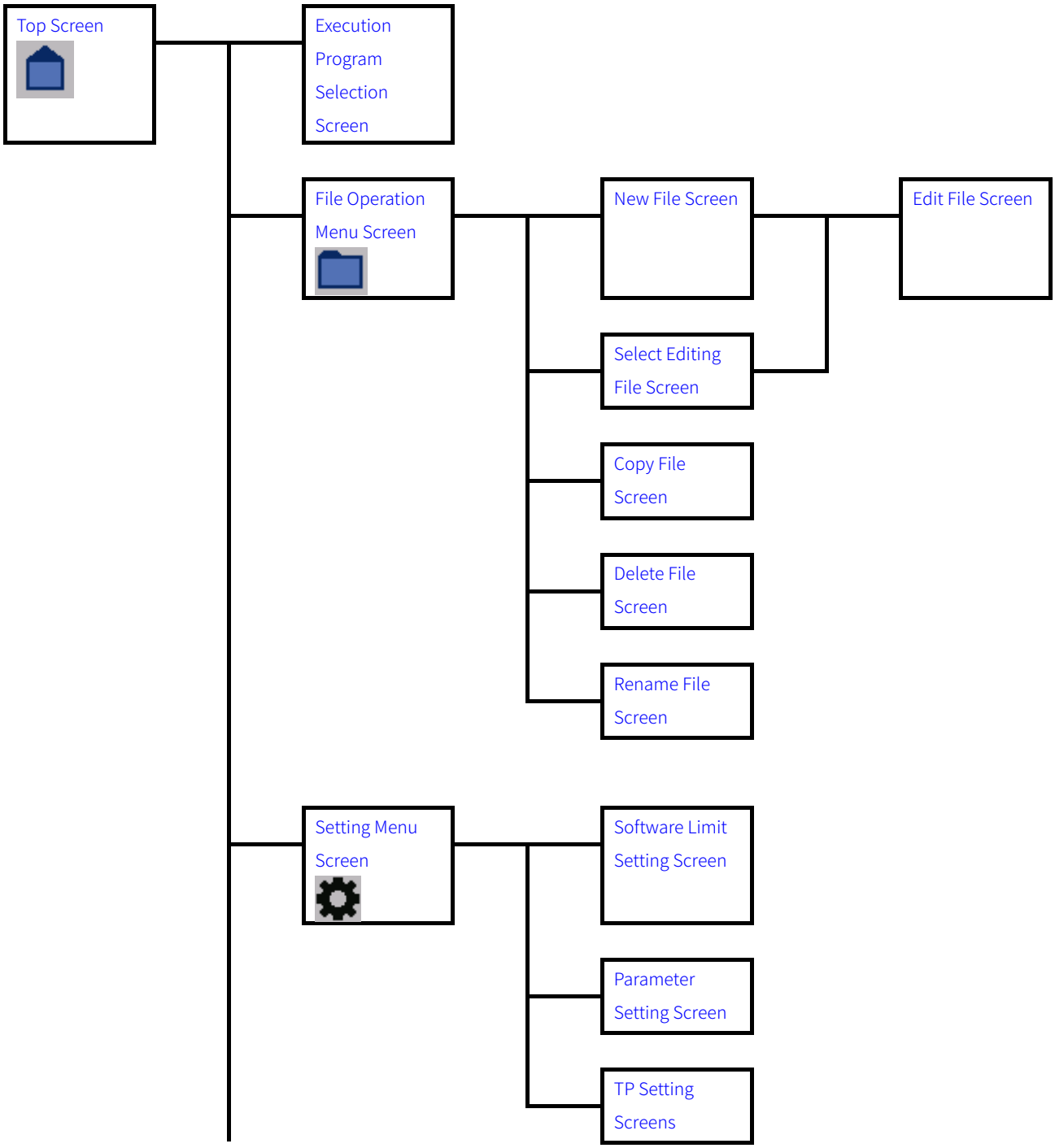


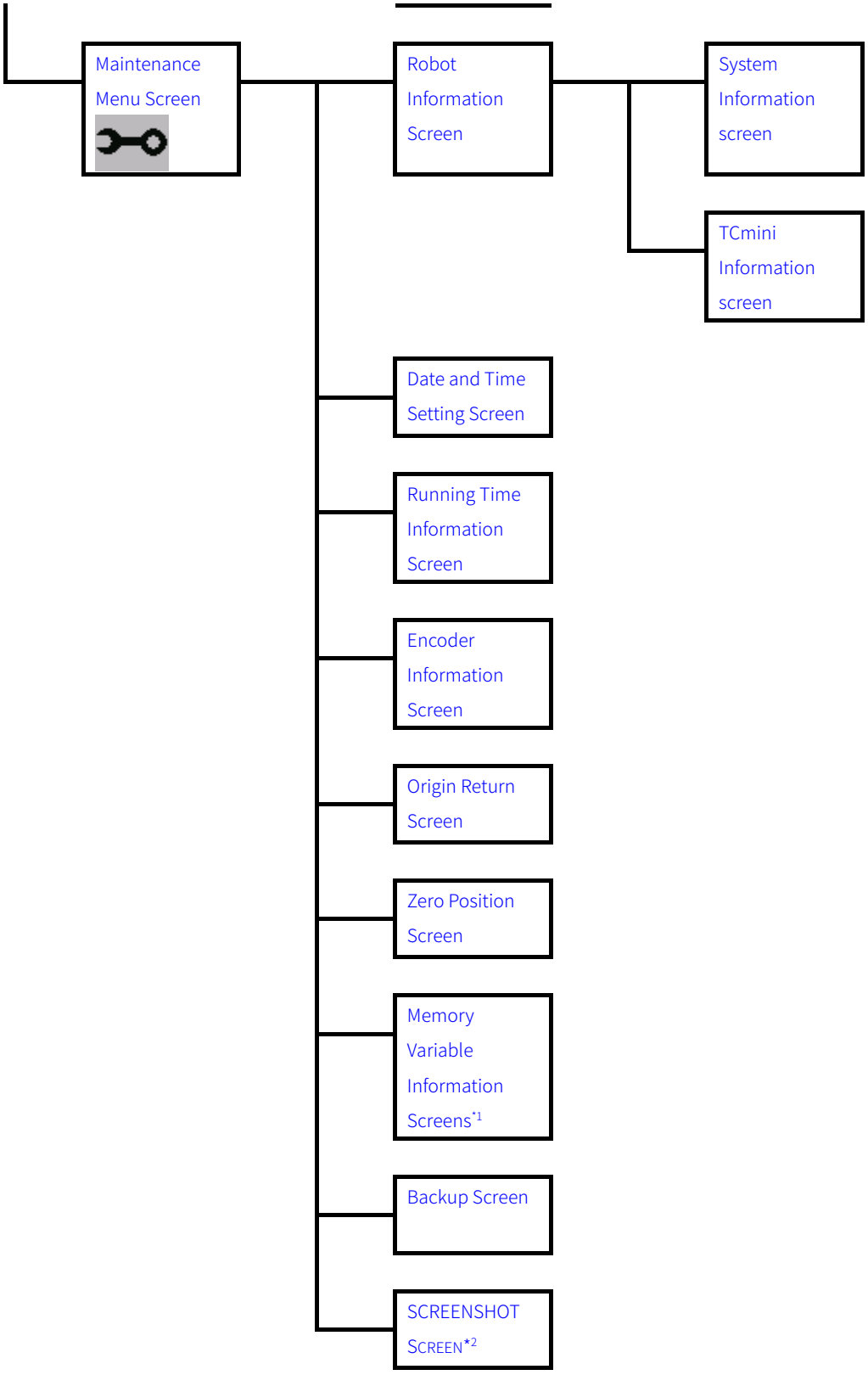
The following table provides descriptions of the screen elements.

No. in Figure	Description	
1	When tapped	The execution program selection screen is opened.
2	When tapped	The displayed Utility window switches to the items at TP5000 startup.
3	When tapped	The file operation menu screen is opened.
4	When tapped	The setting menu screen is opened.
5	When tapped	The maintenance menu screen is opened.
6	When tapped	This switches the display layout for the Utility window.

10.2 Screen Navigation Tree

The screen navigation tree for the TP5000 main window is shown below.

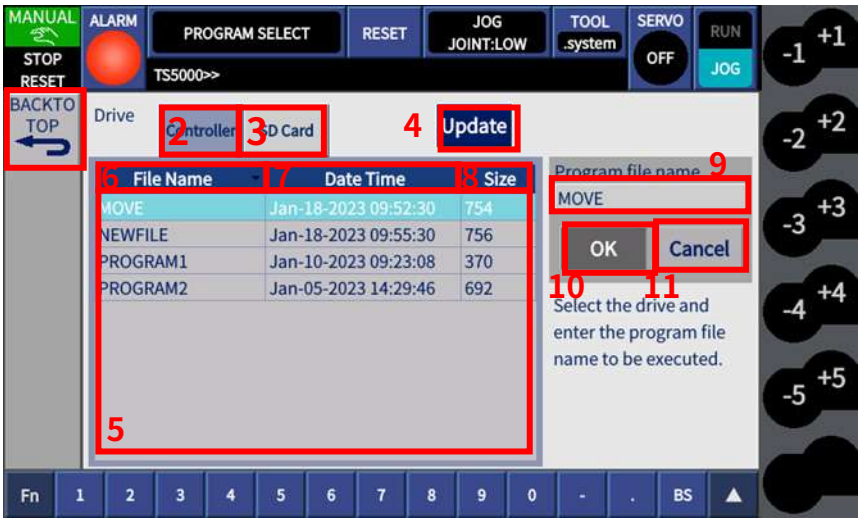




*1 Displayed on X8KTA-03B or earlier versions.

*2 Displayed on X8KTA-04B or later versions.

10.3 Execution Program Selection Screen



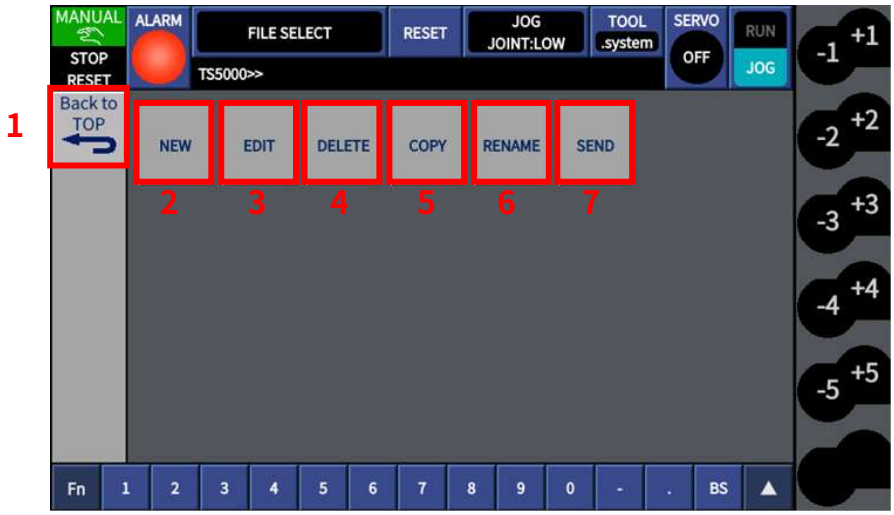
The following table provides descriptions of the screen elements.

No. in Figure	Description	
1	When tapped	The screen returns to the top screen.
2	When tapped	A list of files in the controller is displayed.
3	When tapped	A list of files in the SD card is displayed.
4	When tapped	This updates the displayed file list.
5	Displayed information	A list of the files in the selected medium is displayed.
	When tapped	The cursor moves to the tapped line.
6	When tapped	This sorts the files in alphabetical order by name.
7	When tapped	This sorts the files by the date and time that the file was modified.
8	When tapped	This sorts the files by size.
9	Displayed information	This displays the name of the file at the cursor location in the file list.
10	When tapped	This registers the file at the cursor location in the file list to the execution program and compiles it. Then, the screen returns to the top screen.
11	When tapped	The screen returns to the top screen.

The following table shows the available keys on this screen.

Key Name	Operation When Tapped
↑	Moves the cursor up.
↓	Moves the cursor down.
PageUp	Moves the cursor up by one line.
PageDown	Moves the cursor down by one line.

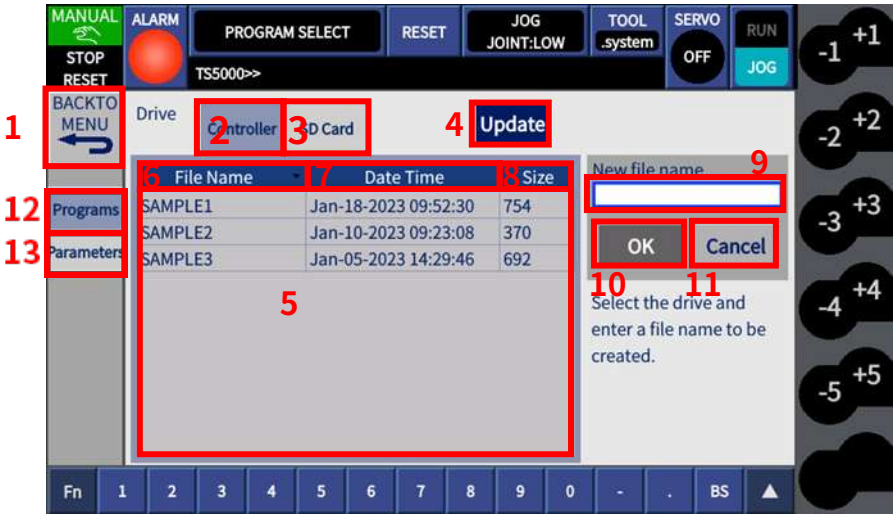
10.4 File Operation Menu Screen



The following table provides descriptions of the screen elements.

No. in Figure	Description	
1	When tapped	The screen returns to the top screen.
2	When tapped	The new file screen is opened.
3	When tapped	The select editing file screen is opened.
4	When tapped	The copy file screen is opened.
5	When tapped	The rename file screen is opened.
6	When tapped	The delete file screen is opened.
7	When tapped	The file transfer screen appears.

10.4.1 New File Screen



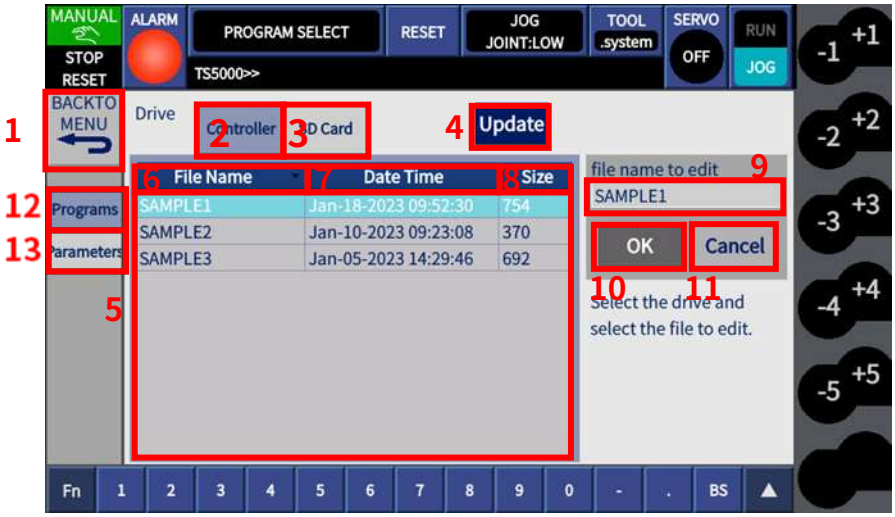
The following table provides descriptions of the screen elements.

No. in Figure	Description	
1	When tapped	The screen returns to the file operation menu screen.
2	When tapped	A list of files in the controller is displayed.
3	When tapped	A list of files in the SD card is displayed.
4	When tapped	This updates the displayed file list.
5	Displayed information	A list of the files in the selected medium is displayed.
	When tapped	The cursor moves to the tapped line.
6	When tapped	This sorts the files in alphabetical order by name.
7	When tapped	This sorts the files by the date and time that the file was modified.
8	When tapped	This sorts the files by size.
9	Input information	This inputs the name of the new file. *When the cursor of the file list is moved, the displayed character string is overwritten by the file name in the line where the cursor is located.
	When tapped	The entry can be edited.
10	When tapped	A file with the name of the text box (in the figure) is temporarily created, and the file edit screen is opened. *A message is displayed if there already is a file name matching the character string in the text box or if the character string does not comply with the file naming rules.
11	When tapped	The screen returns to the top screen.
12	When tapped	Program files are displayed in the file list.
13	When tapped	Parameter files are displayed in the file list.

The following table shows the available keys on this screen.

Key Name	Operation When Tapped
↑	Moves cursor up.
↓	Moves cursor down.
←	The cursor in the text box moves to the left (during edition only).
→	The cursor in the text box moves to the right (during edition only).
Letters and numbers	The tapped letters and numbers at the cursor position are entered (during edition only).
PageUp	Moves cursor up by one line.
PageDown	Moves cursor down by one line.
HOME	The cursor in the text box moves to the left end (during edition only).
END	The cursor in the text box moves to the right end (during edition only).

10.4.2 Select Editing File Screen



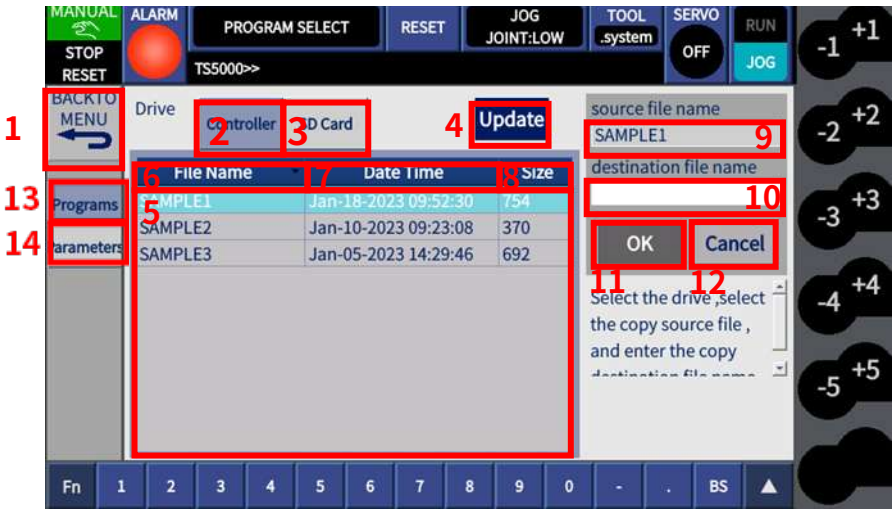
The following table provides descriptions of the screen elements.

No. in Figure	Description	
1	When tapped	The screen returns to the file operation menu screen.
2	When tapped	A list of files in the controller is displayed.
3	When tapped	A list of files in the SD card is displayed.
4	When tapped	This updates the displayed file list.
5	Displayed information	A list of the files in the selected medium is displayed.
	When tapped	The cursor moves to the tapped line.
6	When tapped	This sorts the files in alphabetical order by name.
7	When tapped	This sorts the files by the date and time that the file was modified.
8	When tapped	This sorts the files by size.
9	Displayed information	This displays the name of the file in the line at the cursor location in the file list.
10	When tapped	The file in the line where the cursor is located in the file list is opened, and the file edit screen is opened.
11	When tapped	The screen returns to the top screen.
12	When tapped	Program files are displayed in the file list.
13	When tapped	Parameter files are displayed in the file list.

The following table shows the available keys on this screen.

Key Name	Operation When Tapped
↑	Moves the cursor up.
↓	Moves the cursor down.
PageUp	Moves the cursor up by one line.
PageDown	Moves the cursor down by one line.

10.4.3 Copy File Screen



The following table provides descriptions of the screen elements.

No. in Figure	Description	
1	When tapped	The screen returns to the file operation menu screen.
2	When tapped	A list of files in the controller is displayed.
3	When tapped	A list of files in the SD card is displayed.
4	When tapped	This updates the displayed file list.
5	Displayed information	A list of the files in the selected medium is displayed.
	When tapped	The cursor moves to the tapped line.
6	When tapped	This sorts the files in alphabetical order by name.
7	When tapped	This sorts the files by the date and time that the file was modified.
8	When tapped	This sorts the files by size.
9	Displayed information	This displays the name of the file in the line at the cursor location in the file list. *When the cursor of the file list is moved, the displayed character string is overwritten by the file name in the line where the cursor is located.
10	Input information	This inputs the file name after copying. *When the cursor of the file list is moved, the displayed character string is overwritten by the file name in the line where the cursor is located.
	When tapped	The entry in the text box can be edited.

No. in Figure	Description	
11	When tapped	The content of the file in the line at the cursor location in the file list is copied, and a file with the entered name is created.
12	When tapped	The screen returns to the top screen.
13	When tapped	Program files are displayed in the file list.
14	When tapped	Parameter files are displayed in the file list.

The following table shows the available keys on this screen.

Key Name	Operation When Tapped
↑	Moves the cursor up.
↓	Moves the cursor down.
←	The cursor in the text box moves to the left (during edition only).
→	The cursor in the text box moves to the right. (When editing only)
Letters and numbers	The tapped letters and numbers at the cursor position are entered (during edition only).
PageUp	Moves the cursor up by one line.
PageDown	Moves the cursor down by one line.
HOME	The cursor in the text box moves to the left end (during edition only).
END	The cursor in the text box moves to the right end (during edition only).

10.4.4 Rename File Screen



The following table provides descriptions of the screen elements.

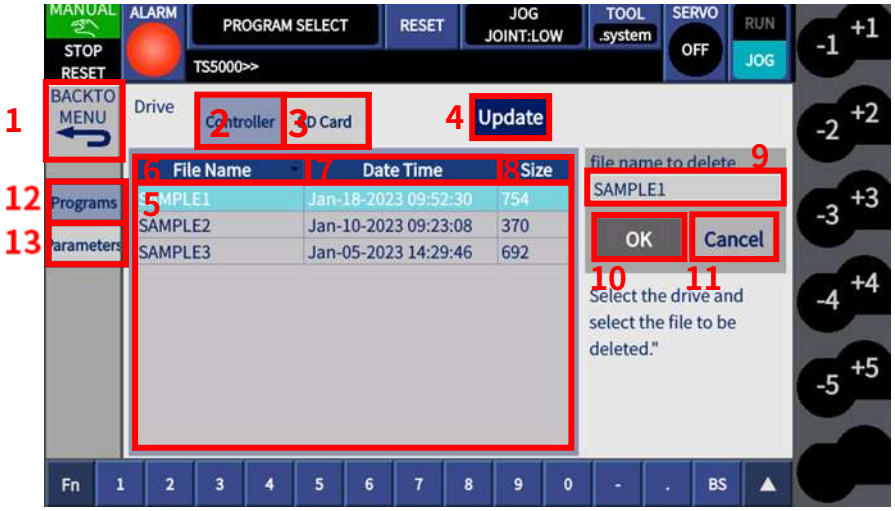
No. in Figure	Description	
1	When tapped	The screen returns to the file operation menu screen.
2	When tapped	A list of files in the controller is displayed.
3	When tapped	A list of files in the SD card is displayed.
4	When tapped	This updates the displayed file list.
5	Displayed information	A list of the files in the selected medium is displayed.
	When tapped	The cursor moves to the tapped line.
6	When tapped	This sorts the files in alphabetical order by name.
7	When tapped	This sorts the files by the date and time that the file was modified.
8	When tapped	This sorts the files by size.
9	Displayed information	This displays the name of the file in the line at the cursor location in the file list. *When the cursor of the file list is moved, the displayed character string is overwritten by the file name in the line where the cursor is located.
10	Input information	Enter the file name after changing. *When the cursor of the file list is moved, the displayed character string is overwritten by the file name in the line where the cursor is located.
	When tapped	The entry in the text box can be edited.

No. in Figure	Description	
11	When tapped	The name of the file in the line at the cursor location in the file list is changed to the file name that was entered.
12	When tapped	The screen returns to the top screen.
13	When tapped	Program files are displayed in the file list.
14	When tapped	Parameter files are displayed in the file list.

The following table shows the available keys on this screen.

Key Name	Operation When Tapped
↑	Moves the cursor up.
↓	Moves the cursor down.
←	The cursor in the text box moves to the left (during edition only).
→	The cursor in the text box moves to the right (When editing only)
Letters and numbers	The tapped letters and numbers at the cursor position are entered (during edition only).
PageUp	Moves the cursor up by one line.
PageDown	Moves the cursor down by one line.
HOME	The cursor in the text box moves to the left end (during edition only).
END	The cursor in the text box moves to the right end (during edition only).

10.4.5 Delete File Screen



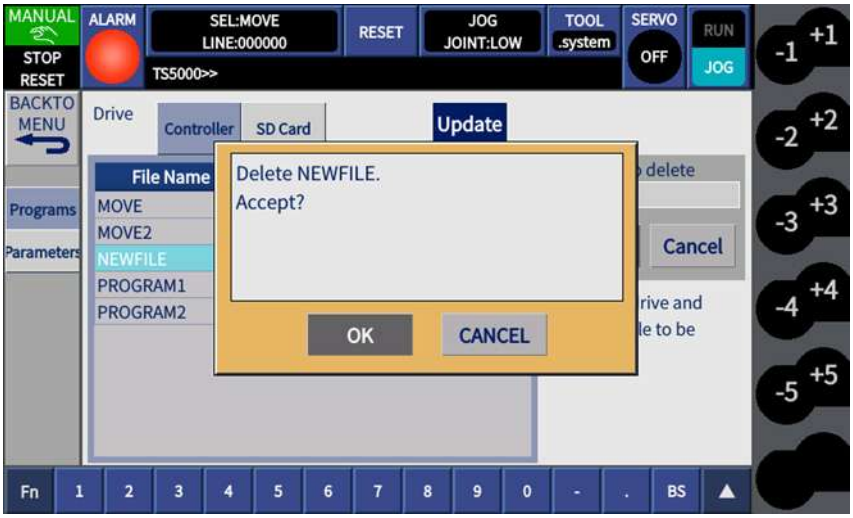
The following table provides descriptions of the screen elements.

No. in Figure	Description	
1	When tapped	The screen returns to the file operation menu screen.
2	When tapped	A list of files in the controller is displayed.
3	When tapped	A list of files in the SD card is displayed.
4	When tapped	This updates the displayed file list.
5	Displayed information	A list of the files in the selected medium is displayed.
	When tapped	The cursor moves to the tapped line.
6	When tapped	This sorts the files in alphabetical order by name.
7	When tapped	This sorts the files by the date and time that the file was modified.
8	When tapped	This sorts the files by size.
9	Displayed information	This displays the name of the file in the line at the cursor location in the file list.
10	When tapped	A file deletion confirmation message appears.
11	When tapped	The screen returns to the top screen.
12	When tapped	Program files are displayed in the file list.
13	When tapped	Parameter files are displayed in the file list.

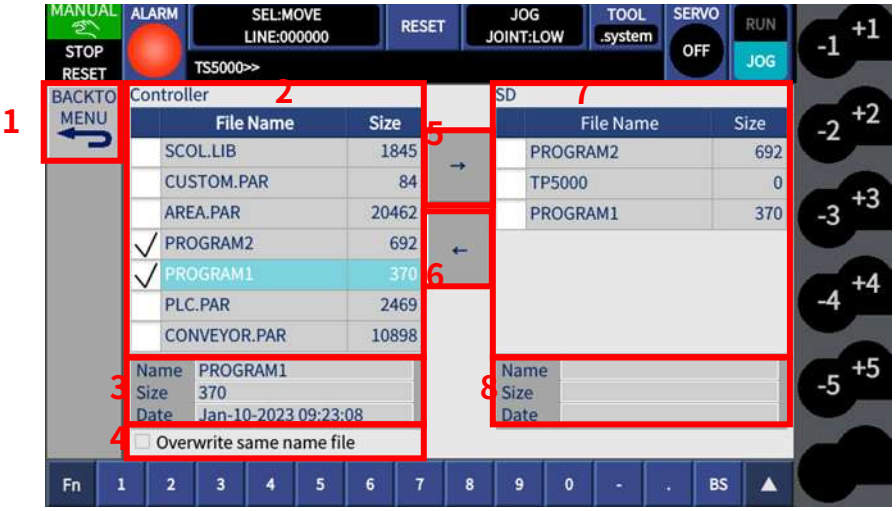
The following table shows the available keys on this screen.

Key Name	Operation When Tapped
↑	Moves the cursor up.
↓	Moves the cursor down.
PageUp	Moves the cursor up by one line.
PageDown	Moves the cursor down by one line.

• File delete confirmation message



10.4.6 Send File Screen



The following table provides descriptions of the screen elements.

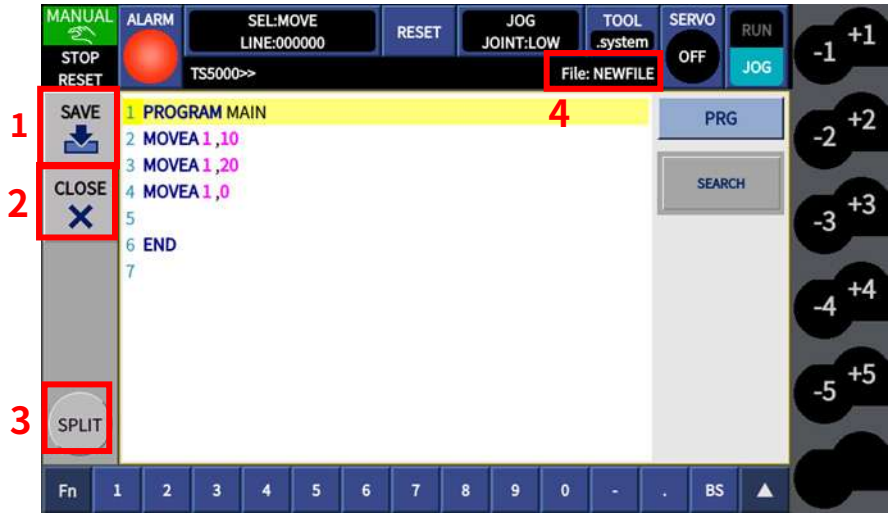
No. in Figure	Description	
1	When tapped	The screen returns to the file operation menu screen.
2	When tapped	Check box: Select the file to be transferred. File name: Display the details of the file.
	Displayed information	A list of files in the controller is displayed.
3	Displayed information	Details of the selected file are displayed.
4	When tapped	When the check box is selected, the file is overwritten during transfer.
5	When tapped	Transfer the selected file from the controller to the SD card.

6	When tapped	Transfer the selected file from the SD card to the controller.
7	When tapped	Check box: Select the file to be transferred. File name: Display the details of the file.
	Displayed information	A list of files in the controller is displayed.
8	Displayed information	The details of the selected file are displayed.

The following table shows the available keys on this screen.

Key Name	Operation When Tapped
↑	Moves the cursor up.
↓	Moves the cursor down.
PageUp	Moves the cursor up by one line.
PageDown	Moves the cursor down by one line.

10.4.7 Edit File Screen



The following table provides descriptions of the screen elements.

No. in Figure	Description	
1	When tapped	This saves the file changes. Then, the edit file screen is opened.
2	When tapped	This terminates the file editing process. If the file is modified, the discard changes (NO SAVE) confirmation screen is opened.
3	When tapped	This switches the display layout for the Utility window.
4	Displayed information	The name of the file being edited is displayed.

10.5 Setting Menu Screen



The following table provides descriptions of the screen elements.

No. in Figure	Description	
1	When tapped	The screen returns to the top screen.
2	When tapped	The software limit setting screen is opened.
3	When tapped	The parameter setting screen is opened.
4	When tapped	The TP setting screen is opened.
5	When tapped	The screen calibration is performed.
6	When tapped	The language is set.

MEMO

Applying the language setting requires the controller to be restarted.

10.5.1 Software Limit Setting Screen

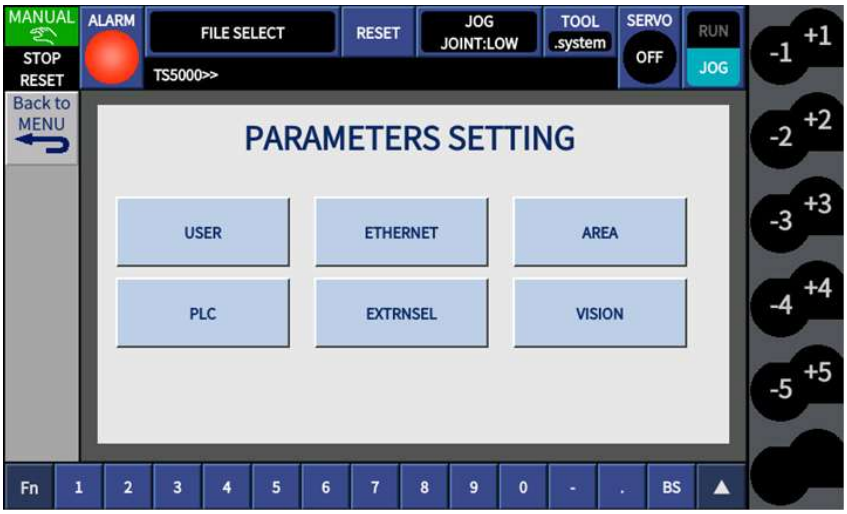


The following table provides descriptions of the screen elements.

No. in Figure	Description	
1	When tapped	The screen returns to the setting menu screen.
2	Displayed information	The minimum value of software limit that can be set is displayed.
3	Displayed information	The minus side software limit set by the user is displayed.
	When tapped	Select the cell. Tapping the selected cell again enables editing.
4	Displayed information	The current angles of each joint axis are displayed.
5	Displayed information	The plus side software limit set by the user is displayed.
	When tapped	Select the cell. Tapping the selected cell again enables editing.
6	Displayed information	The maximum value of software limit that can be set is displayed.
7	When tapped	This enables entry of the current angle for the joint axis in the selected cell.
8	When tapped	This saves the software limit setting.

For details on the software limit settings, see 8.2.1.1 Setting Software Limit.

10.5.2 Parameter Setting Screen



For details on the parameter settings, see 8.2.2.1 Setting the Parameters.

10.5.3 TP Setting Screens

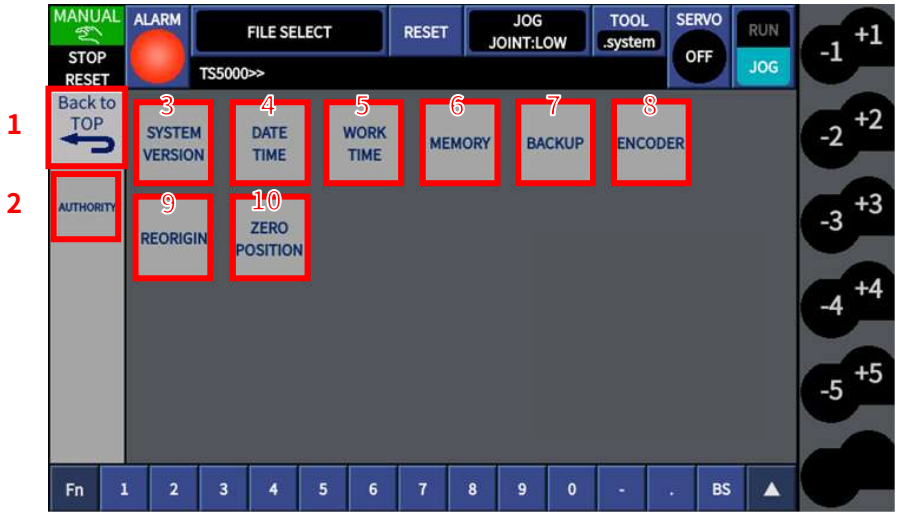
10.5.3.1 Screensaver Setting Screen



The following table provides descriptions of the screen elements.

No. in Figure	Description	
1	When tapped	The screen returns to the setting menu screen.
2	Displayed information	This displays the time until the screensaver is activated.
3	Displayed information	This displays the time until the backlight is turned off.
4		This is the text box for setting the screensaver start time.
	When tapped	The entry in the text box can be edited.
5		This is the text box for setting the backlight off time.
	When tapped	The entry in the text box can be edited.
6	When tapped	This sets the screensaver start time and backlight off time to the values in the text boxes.
7	When tapped	The TP setting menu is displayed. *Only the "Screensaver" button is displayed in the TP setting menu.

10.6 Maintenance Menu Screen



*Screen on X8KTA-03B or earlier



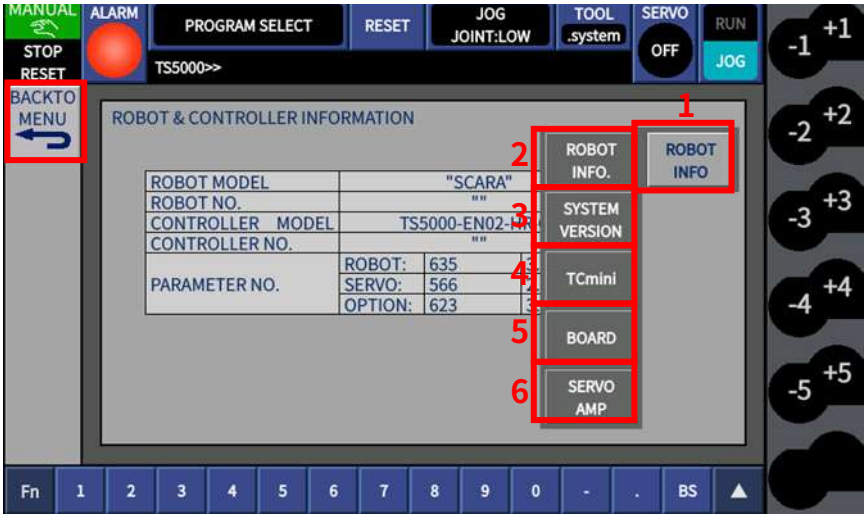
*Screen on X8KTA-04B or later

The following table provides descriptions of the screen elements.

No. in Figure	Description	
1	When tapped	The screen returns to the top screen.
2	When tapped	This displays the user type setting screen. *This button is used for maintenance operations.
3	When tapped	This opens the version information screen.
4	When tapped	This opens the date and time setting screen.
5	When tapped	This opens the running time information screen.
6	When tapped	This opens the memory variable information screen.
7	When tapped	This opens the backup information screen.
8	When tapped	This opens the encoder information screen.
9	When tapped	This opens the origin return screen.
10	When tapped	This opens the zero position screen.
11	When tapped	This screen displays the screenshot setting screen.

10.6.1 Version Information Screen

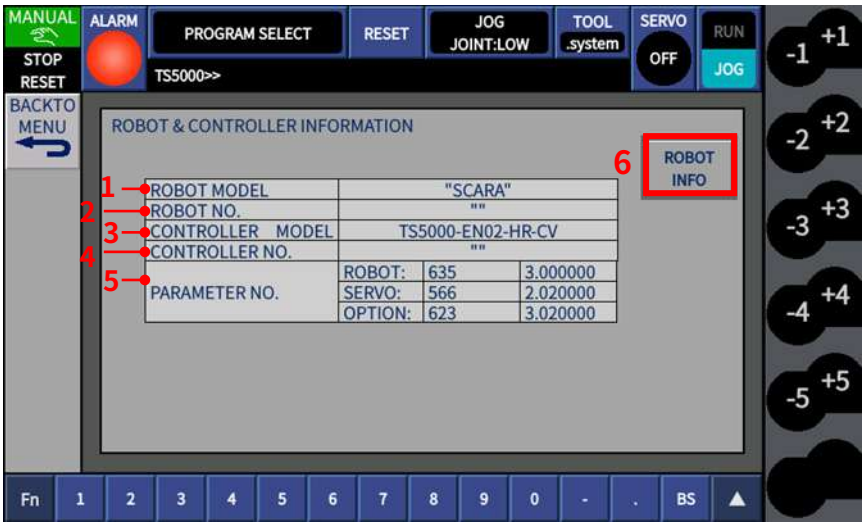
10.6.1.1 Version Information Switch Menu



The following table provides descriptions of the screen elements.

No. in Figure	Description
1	This hides the version information switch menu.
2	This shows the robot information.
3	This shows the system version information.
4	This shows the sequence information.
5	This shows the board information.
6	This shows the servo amplifier information.
7	The screen returns to the maintenance menu screen.

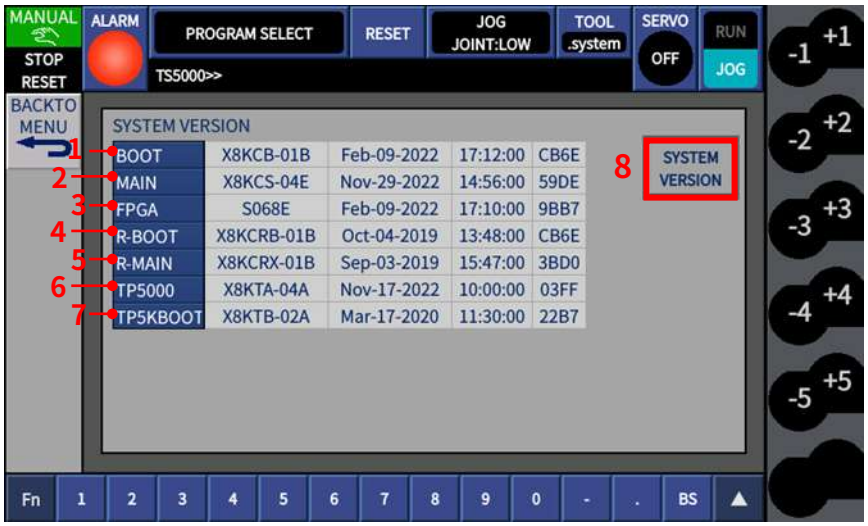
10.6.1.2 Robot Information Screen



The following table provides descriptions of the screen elements.

No. in Figure	Description
1	This shows the robot model name.
2	This shows the serial number from the robot manufacturer.
3	This shows controller model name.
4	This shows controller serial number.
5	This shows the parameter version.
6	When tapped, this displays the version information switch menu.

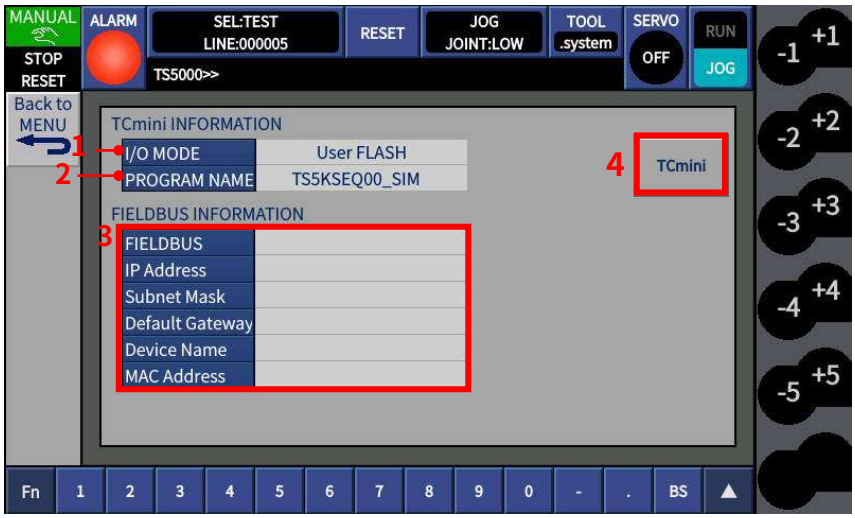
10.6.1.3 System Information Screen



The following table provides descriptions of the screen elements.

No. in Figure	Description
1	This shows the software version of the boot.
2	This shows the software version of the main unit.
3	This shows the software version of the FPGA unit.
4	This shows the software version of the recovery boot unit.
5	This shows the software version of the recovery main unit.
6	This shows the software version of the TP5000 main unit.
7	This shows the software version of the TP5000 boot.
8	When tapped, this displays the version information switch menu.

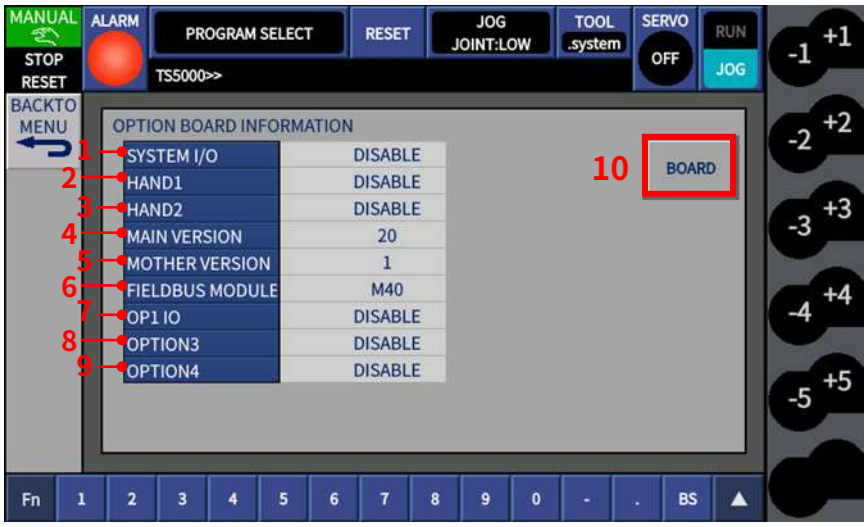
10.6.1.4 TCmini Information Screen



The following table provides descriptions of the screen elements.

No. in Figure	Description
1	This shows the area of the ladder program used in the controller.
2	This shows the name of the ladder program used in the controller.
3	This shows the fieldbus information that is set. *The fieldbus function is currently under development.
4	When tapped, this displays the version information switch menu.

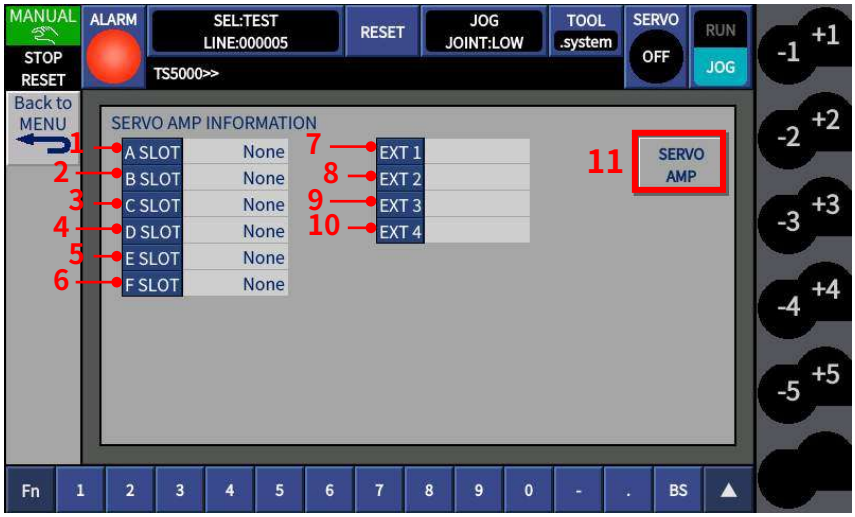
10.6.1.5 Board Information Screen



The following table provides descriptions of the screen elements.

No. in Figure	Description
1	This shows the system I/O board information. ENABLE- P: TYPE P ENABLE-N: TYPE N DISABLE: Not connected ERROR: Error detection
2	This shows the hand I/O board information. ENABLE: Connected DISABLE: Not connected
3	This shows the expansion hand I/O board information. ENABLE: Connected DISABLE: Not connected
4	This shows the main board version.
5	This shows the motherboard version.
6	This shows the fieldbus module information. *The fieldbus function is currently under development.
7	This shows the expansion I/O board information. ENABLE- P: TYPE P ENABLE-N: TYPE N DISABLE: Not connected ERROR: Error detection
8	This shows the Option3 board information. *The Option3 board is currently under development.
9	This shows the Option4 board information. *The Option4 board is currently under development.
10	When tapped, this displays the version information switch menu.

10.6.1.6 Servo Amplifier Information Screen



The following table provides descriptions of the screen elements.

No. in Figure	Description
1	This shows the information on the servo amplifier loaded in slot A.
2	This shows the information on the servo amplifier loaded in slot B.
3	This shows the information on the servo amplifier loaded in slot C.
4	This shows the information on the servo amplifier loaded in slot D.
5	This shows the information on the servo amplifier loaded in slot E.
6	This shows the information on the servo amplifier loaded in slot F.
7	This shows the information on the servo amplifier loaded in slot EXT1.
8	This shows the information on the servo amplifier loaded in slot EXT2.
9	This shows the information on the servo amplifier loaded in slot EXT3.
10	This shows the information on the servo amplifier loaded in slot EXT4.
11	When tapped, this displays the version information switch menu.

10.6.2 Date and Time Setting Screen



The following table provides descriptions of the screen elements.

No. in Figure	Description
1	This shows the current date in the controller.
2	This shows the current date in the controller. *When one of these text boxes is tapped, the entry can be edited.
3	When tapped, the controller date is overwritten by the value in the text box at 2 in the figure.
4	This shows the current time in the controller.
5	This shows the current time in the controller. *When one of these text boxes is tapped, the entry can be edited.
6	When tapped, the controller time is overwritten by the value in the text box at 5 in the figure.

The following table shows the available keys on this screen.

Key Name	Operation When Tapped	
←	When editing	Moves the cursor to the left by one space.
→	When editing	Moves the cursor to the right by one space.
BS key	When editing	Erases the character before the cursor.
ESC	When editing	Erases all the characters in the cell. If no characters are entered in the cell, editing is canceled.
Letters, numbers, and symbols	When editing	This inputs the character corresponding to the key at the cursor location.

10.6.3 Running Time Information Screen



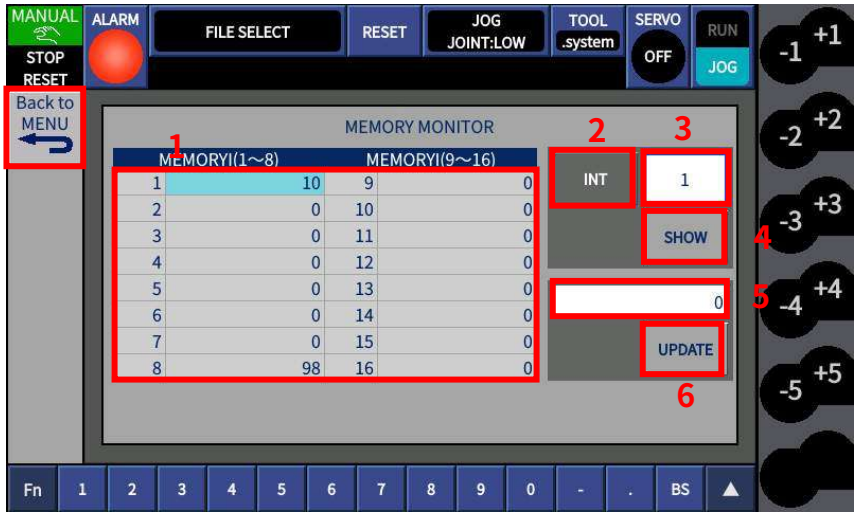
The following table provides descriptions of the screen elements.

No. in Figure	Description
1	This shows the power-on time of the controller.
2	This shows the total servo-on time of the controller.
3	This screen shows the total program run time of the controller.
4	When tapped, the screen returns to the maintenance menu screen.

10.6.4 Memory Variable Information Screens

The memory variable information screens include the scalar-type memory variable information screen and the vector-type memory variable information screen.

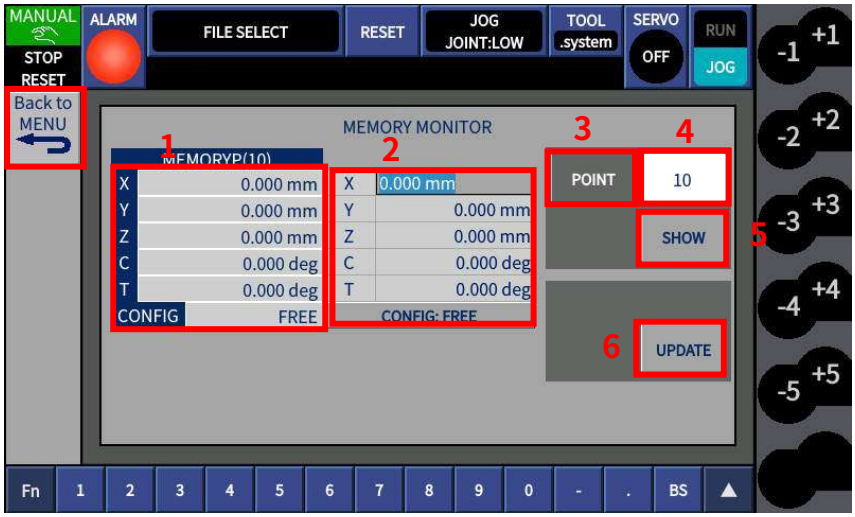
10.6.4.1 Scalar-type Memory Variable Information Screen



The following table provides descriptions of the screen elements.

No. in Figure	Description
1	This shows the current value for the memory variable.
2	When tapped, this displays the variable type change menu.
3	This enters the variable number. When tapped, the entry can be edited.
4	When tapped, this displays the variable type that is set and the memory variable of the variable number.
5	This enters the value of the memory variable to be changed. When tapped, the entry can be edited.
6	When tapped, the value of the memory variable shown by the SHOW button in the top section is set to the value in the text box.
7	When tapped, the screen returns to the maintenance menu screen.

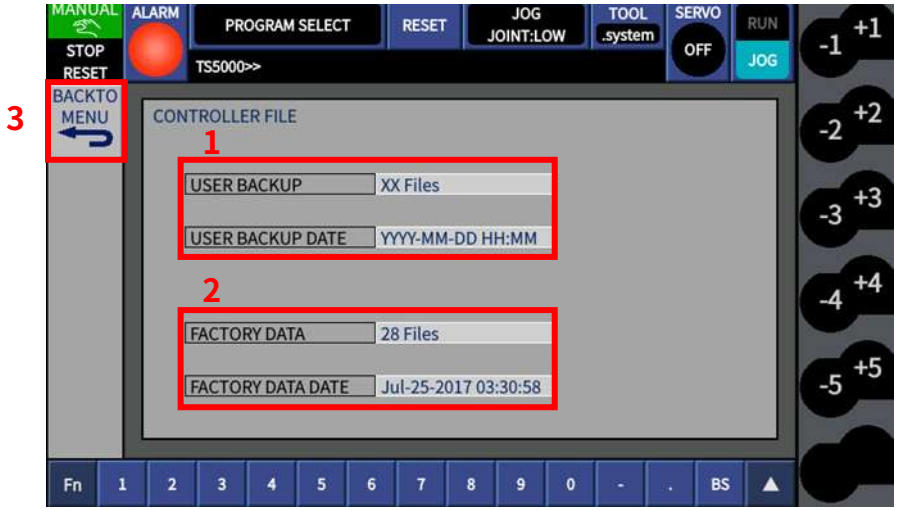
10.6.4.2 Vector-type Memory Variable Information Screen



The following table provides descriptions of the screen elements.

No. in Figure	Description
1	This shows the current value for the memory variable.
2	This enters the value of the memory variable to be changed. When tapped, the cursor moves to the tapped location. When tapped again, the entry can be edited.
3	When tapped, this displays the variable type change menu.
4	This enters the variable number. When tapped, the entry can be edited.
5	When tapped, this displays the variable type that is set and the memory variable of the variable number.
6	When tapped, the value of the memory variable shown by the SHOW button in the top section is set to the value in the text box.
7	When tapped, the screen returns to the maintenance menu screen.

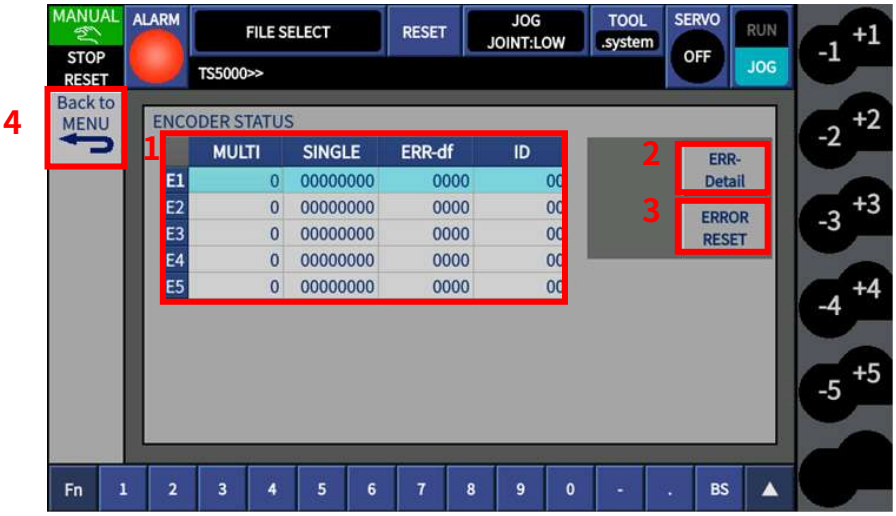
10.6.5 Backup Screen



The following table provides descriptions of the screen elements.

No. in Figure	Description
1	This shows the number of backup files in the user area and the backup date and time.
2	This shows the number of backup files in the manufacturer area and the backup date and time.
3	When tapped, the screen returns to the maintenance menu screen.

10.6.6 Encoder Information Screen



The following table provides descriptions of the screen elements.

No. in Figure	Description
1	This shows the multi-rotational data, single-rotation data, error bits, and encoder ID for each joint axis encoder. *A cursor is added to the selected cell. *When tapped, the cursor moves to the tapped cell.
2	When tapped, this shows the error description for the encoder in the line where the cursor is located.
3	When tapped, this resets the encoder error in the line where the cursor is located.
4	When tapped, the screen returns to the maintenance menu screen.

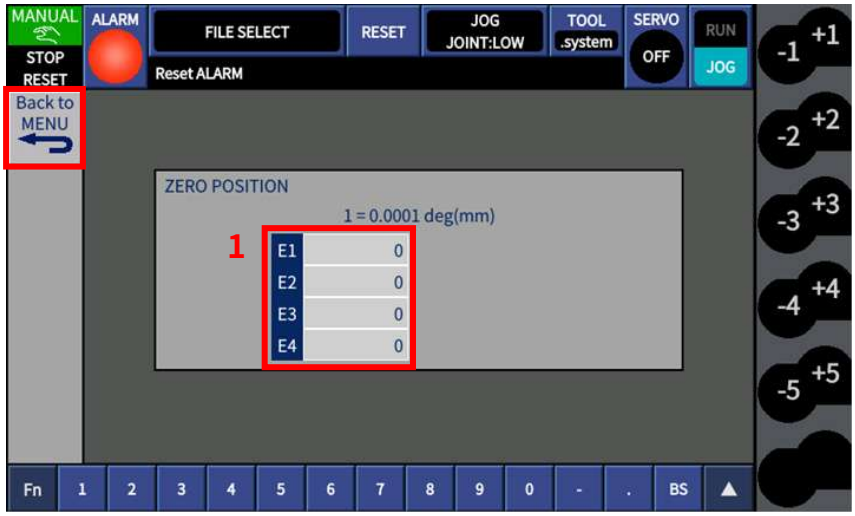
10.6.7 Origin Return Screen



The following table provides descriptions of the screen elements.

No. in Figure	Description
1	This shows the registered origin return position.
2	When tapped, the screen returns to the maintenance menu screen.

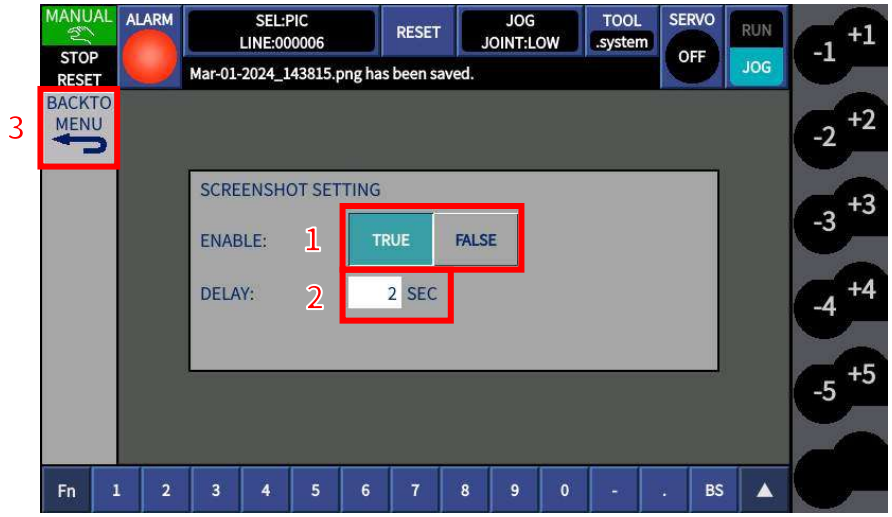
10.6.8 Zero Position Screen



The following table provides descriptions of the screen elements.

No. in Figure	Description
1	This shows the pulse value for the encoder of each joint axis.
2	When tapped, the screen returns to the maintenance menu screen.

10.6.9 SCREENSHOT Screen



The following table provides descriptions of the screen elements.

No. in Figure	Description
1	Enables/disables screenshots.
2	Sets the time between tapping and shooting.
3	Tap to return to the maintenance menu.

11. Utility Window Details

11.1 List of Screens

The Utility window can display the following screens.

The types of screens that can be displayed on the top screen and the file editing screen vary.

Table 11.1. Utility Window Screens

Screen Name	Description	Is Screen Displayable?	
		Top screen	File editing screen
POS	Shows the robot current position and the current angle, speed, and other parameters for the joint axes.	Yes	Yes
I/O	The I/O signal status is displayed. Enables to switch output signal ON/OFF.	Yes	Yes
IP STATUS	Shows the Ethernet communication status. Open/Close	Yes	Yes
PLC	Shows the TCmini status.	Yes	Yes
MOTOR	Shows the motor status.	Yes	Yes
LDISP	Shows the program section of the file registered to the execution program.	Yes	No
WATCH	Shows the values of the global variable for the file registered to the execution program.	Yes	No
PRINT/INPUT	Shows the output to the teach pendant by the PRINT command in the SCOL2 program. Enables input from the teach pendant by the INPUT command in the SCOL2 program.	Yes	No
DO	Enables direct execution.	Yes	No
PRG	Enables editing of the global section and program section.	No	Yes
POINT	Enables editing of the POINT-type data and TRANS-type data of the data section.	Yes *Display of POINT-type data only	Yes
TRANS	Shows the TRANS-type data of the data section.	Yes *Display and Tool Setting only	No
PAYLOAD	Enables editing of PAYLOAD-type data of the data section.	Yes *Display only	Yes
JOINT	Enables editing of the JOINT-type data of the data section.	Yes *Display only	Yes
AUX	Registers the DOUT signal number to switch the signal easily.	Yes	No
IP WATCH	Checks the Ethernet communication contents.	Yes	No
CONV	Displays data related to the conveyor synchronization function.	Yes	Yes
MEMORY	Displays memory variable information.	Yes	Yes

11.2 POS Screen

This shows the robot current position and the current angle, velocity, and other parameters for the joint axes.

11.2.1 POS (Angle and World Position) Screen

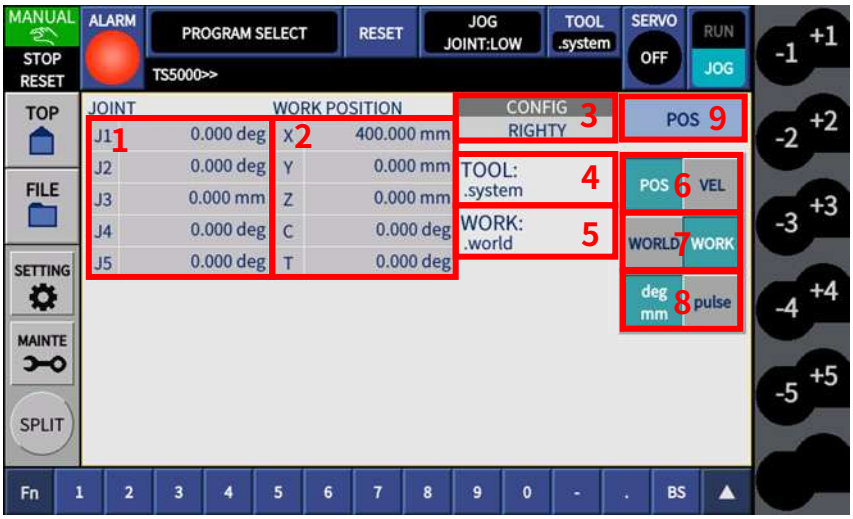
Wide View



The following table provides descriptions of the screen elements.

No. in Figure	Description
1	This shows the current angles of each joint axis.
2	This shows the current positions on the world coordinate system.
3	This shows the current configuration (CONFIG) of the robot.
4	This shows the current tool coordinates of the robot.
5	This shows the current work coordinates of the robot.
6	When tapped, this switches to the POS (angular velocity and world velocity) screen.
7	When tapped, this switches to the POS (angle and work) screen.
8	When tapped, this switches to the POS (pulse) screen.
9	When tapped, this displays the utility menu.

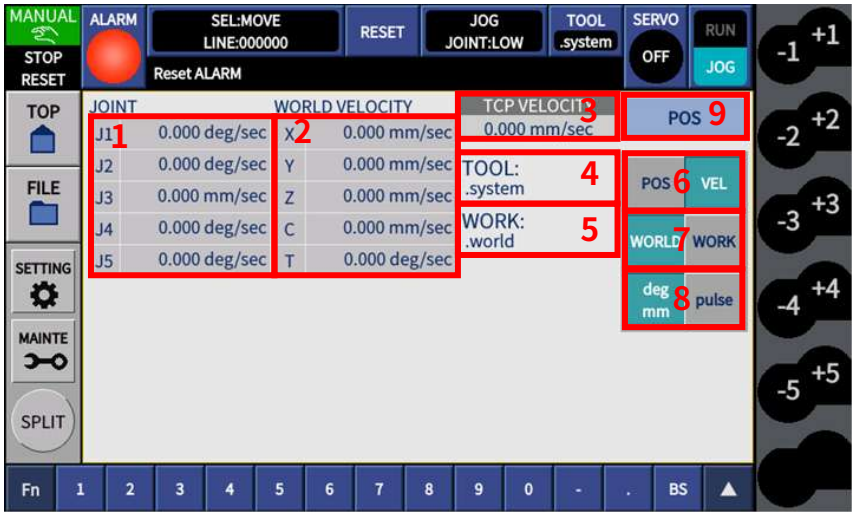
11.2.2 POS (Angle and World Position) Screen



The following table provides descriptions of the screen elements.

No. in Figure	Description
1	This shows the current angles of each joint axis.
2	This shows the current positions on the work coordinate system.
3	This shows the current configuration (CONFIG) of the robot.
4	This shows the current tool coordinates of the robot.
5	This shows the current work coordinates of the robot.
6	When tapped, this switches to the POS (angular velocity and work velocity) screen.
7	When tapped, this switches to the POS (angle and world) screen.
8	When tapped, this switches to the POS (pulse) screen.
9	When tapped, this displays the utility menu.

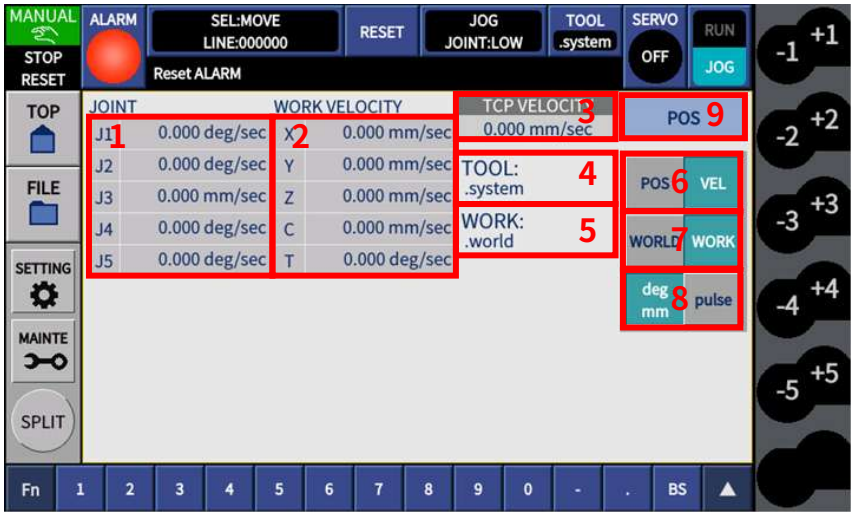
11.2.3 POS (Angular Velocity and World Velocity) Screen



The following table provides descriptions of the screen elements.

No. in Figure	Description
1	This shows the current angular velocity of each joint axis.
2	This shows the current velocity on the world coordinate system.
3	This shows the current velocity of the robot TCP.
4	This shows the current tool coordinates of the robot.
5	This shows the current work coordinates of the robot.
6	When tapped, this switches to the POS (angular velocity and world velocity) screen.
7	When tapped, this switches to the POS (angular velocity and work) screen.
8	When tapped, this switches to the POS (pulse) screen.
9	When tapped, this displays the utility menu.

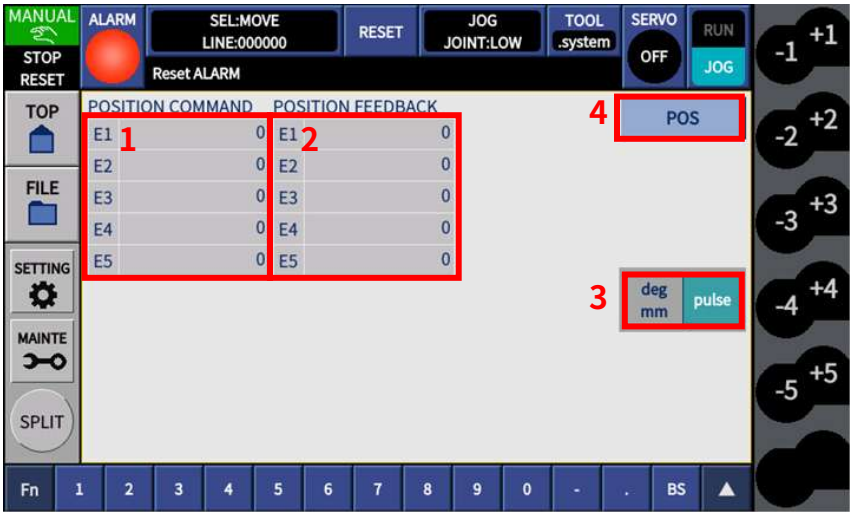
11.2.4 POS (Angular Velocity and Work Velocity) Screen



The following table provides descriptions of the screen elements.

No. in Figure	Description
1	This shows the current angular velocity of each joint axis.
2	This shows the current velocity on the work coordinate system.
3	This shows the current velocity of the robot TCP.
4	This shows the current tool coordinates of the robot.
5	This shows the current work coordinates of the robot.
6	When tapped, this switches to the POS (angle and work velocity) screen.
7	When tapped, this switches to the POS (angular velocity and world) screen.
8	When tapped, this switches to the POS (pulse) screen.
9	When tapped, this displays the utility menu.

11.2.5 POS (Pulse) Screen



The following table provides descriptions of the screen elements.

No. in Figure	Description
1	This shows the pulse command value for each joint axis.
2	This shows the current pulse for each joint axis.
3	When tapped, this switches to the screen before opening the POS (pulse) screen.
4	When tapped, this displays the utility menu.

11.3 I/O Screen

This screen displays the I/O signal status.

The output signals can be switched ON and OFF in this screen.

11.3.1 I/O (DIN) Screen



The following table provides descriptions of the screen elements.

No. in Figure	Description
1	This shows the signal types and signal numbers that are currently being displayed.
2	This shows the signal status. (ON: ■, OFF: □)
3	When tapped, this displays the signal number selection menu.
4	When tapped, this displays the signal selection menu.
5	When tapped, this displays the utility menu.

The following table shows the available keys in this screen.

Key Name	Operation When Tapped
↑	Moves the cursor up.
↓	Moves the cursor down.
←	Moves the cursor to the left.
→	Moves the cursor to the right.
PageUp	This switches the displayed line (only when using split-screen).
PageDown	This switches the displayed line (only when using split-screen).

11.3.2 I/O (DOUT) Screen



The following table provides descriptions of the screen elements.

No. in Figure	Description
1	This shows the signal types and signal numbers that are currently being displayed.
2	This shows the signal status. (ON: ■, OFF: ■) *A cursor is added to the currently-selected signal. (Selection: □) *When tapped, the cursor moves to the tapped location.
3	When tapped, this switches the selected signal ON or OFF.
4	When tapped, this sets all output signals except for hand and system to OFF.
5	When tapped, this displays the signal number selection menu.
6	When tapped, this displays the signal selection menu.
7	When tapped, this displays the utility menu.

The following table shows the available keys in this screen.

Key Name	Operation When Tapped
↑	Moves the cursor up.
↓	Moves the cursor down.
←	Moves the cursor to the left.
→	Moves the cursor to the right.
PageUp	This switches the displayed line (only when using split-screen).
PageDown	This switches the displayed line (only when using split-screen).

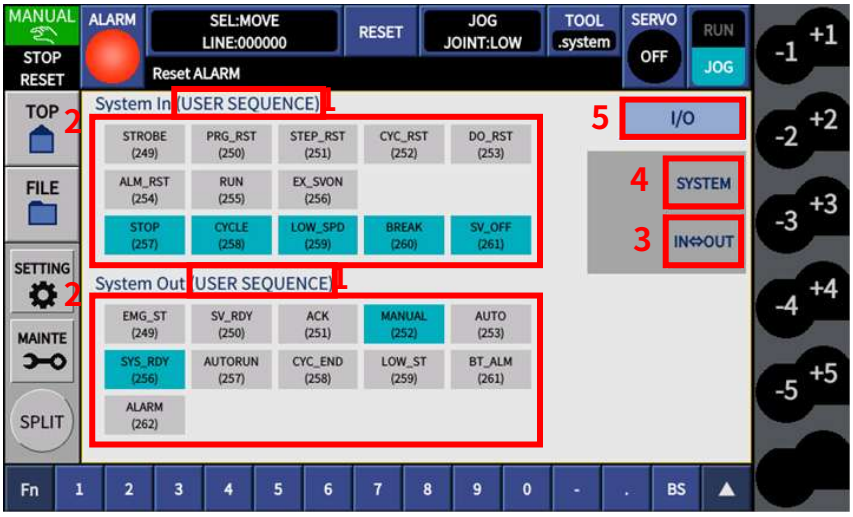
NOTICE

●Before resetting the signal, check the effect of the output signal.

• The suction signal may be turned off, causing the clamped object to drop.

• The robot may collide with the peripheral device.

11.3.3 I/O (SYSTEM) Screen



The following table provides descriptions of the screen elements.

No. in Figure	Description
1	This shows the signals assigned to the system input/output signals in the ladder program. When using a standard ladder program: SYSTEM CONNECTOR When using the customer's ladder program: USER CONNECTOR
2	This shows the status of each system signal. (ON: ■, OFF: ■)
3	When tapped, the display positions of the system input signal and system output signal are interchanged. When used in the split-screen view, the displayed signals are switched.
4	When tapped, this displays the signal selection menu.
5	When tapped, this displays the utility menu.

11.3.4 I/O (HAND) Screen



The following table provides descriptions of the screen elements.

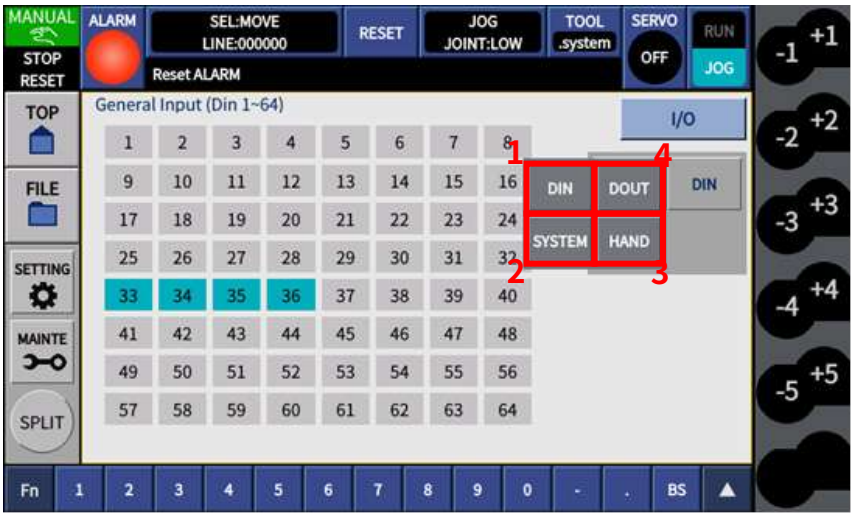
No. in Figure	Description
1	This shows the hand signal status. (ON: ■, OFF: ■) *A cursor is added to the currently-selected hand output signal. (Selection: □) *When tapped, the cursor moves to the tapped location.
2	When tapped, this switches the hand output signal at the cursor location to ON or OFF.
3	When tapped, the display positions of the hand input signal and hand output signal are interchanged. When used in the split-screen view, the displayed signals are switched.
4	When tapped, this displays the signal selection menu.
5	When tapped, this displays the utility menu.

The following table shows the available keys on this screen.

Key Name	Operation When Tapped
↑	Moves the cursor up.
↓	Moves the cursor down.
←	Moves the cursor to the left.
→	Moves the cursor to the right.

11.3.5 Signal Selection Menu

When signal selection menu is displayed



No. in Figure	Description
1	When tapped, this switches to the I/O (DOUT) screen.
2	When tapped, this switches to the I/O (SYSTEM) screen.
3	When tapped, this switches to the I/O (HAND) screen.
4	When tapped, this closes the signal selection menu.

11.3.6 Signal Number Selection Menu

When signal number selection menu is displayed

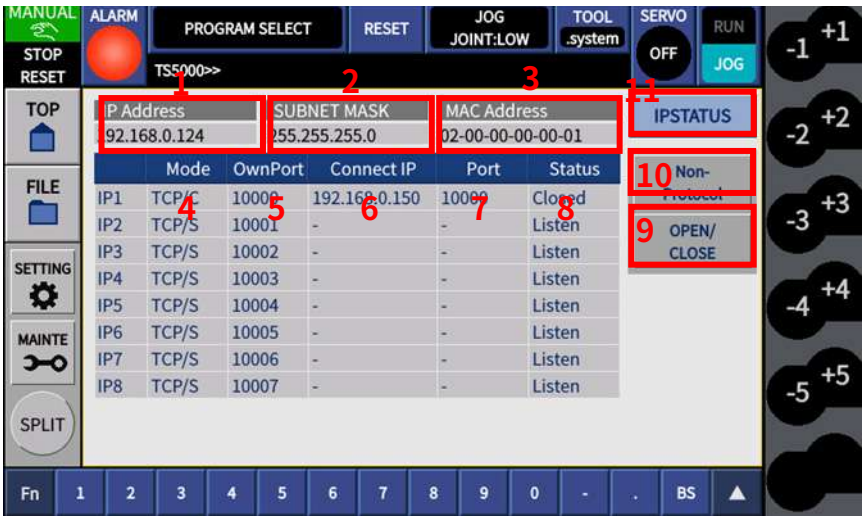


No. in Figure	Description
1	When tapped, the displayed signal number is switched to the signal number corresponding to the button.
2	When tapped, this closes the signal number selection menu.

11.4 IP STATUS Screen

11.4.1 IP STATUS (Non-Procedural Port) Screen

This screen shows the Ethernet communication status.



The following table provides descriptions of the screen elements.

No. in Figure	Description
1	This shows the IP address of the controller.
2	This shows the subnet mask of the controller.
3	This shows the default gateway of the controller.
4	This shows the communication mode of each IP. TCP/S: TCP communication server TCP/C: TCP communication client UDP: UDP communication non: Not used
5	This shows the port number of each IP.
6	This shows the IP address where each IP is connected.
7	This shows the connection port number of each IP.
8	This shows the status of each IP. For a detailed description of the meaning of each status, refer to a book or other reference that explains TCP/IP (available commercially).
9	When tapped, this switches the OPEN/CLOSE status for the port at the line where the cursor is located.
10	When tapped, this displays the port switch menu.
11	When tapped, this displays the utility menu.

The following table shows the available keys in this screen.

Key Name	Operation When Tapped
↑	Moves the cursor up.
↓	Moves the cursor down.

NOTICE

- Before closing the port, confirm the effect on the peripheral devices. Failure to do so may cause the robot in unexpected motion to collide with the peripheral device.

11.4.2 Port Switch Menu



No. in Figure	Description
1	When tapped, this switches the displayed port to a non-procedural port.
2	When tapped, this switches the displayed port to a monitor port.
3	When tapped, this switches the displayed port to a regular monitor port.
4	When tapped, this switches the displayed port to a dump port.
5	When tapped, this switches the displayed port to a control communication port.
6	When tapped, this hides the port switch menu.

MEMO

For description of each port, refer to "STA1164 Communication manual."

11.5 PLC Screen

This screen shows the TCmini status.

11.5.1 PLC (G Register) Screen



The following table provides descriptions of the screen elements.

No. in Figure	Description
1	This shows the types for the displayed registers.
2	This shows the numbers for the displayed registers.
3	This shows the statuses of the registers. (0: ■ 1: □)
4	When tapped, this displays the register selection menu.
5	When tapped, this displays the G register number selection menu.
6	When tapped, this displays the utility menu.

11.5.2 Register Selection Menu



No. in Figure	Description
1	When tapped, this switches to the PLC (G register) screen.
2	When tapped, this switches to the PLC (H register) screen.
3	When tapped, this switches to the PLC (X register) screen.
4	When tapped, this switches to the PLC (Y register) screen.
5	When tapped, this closes the register selection menu.

11.5.3 Register Number Selection Menu



No. in Figure	Description
1	When tapped, the displayed register number is switched to the register number corresponding to the button.
2	When tapped, this closes the register number selection menu.

11.6 MOTOR Screen

This screen shows the motor status.

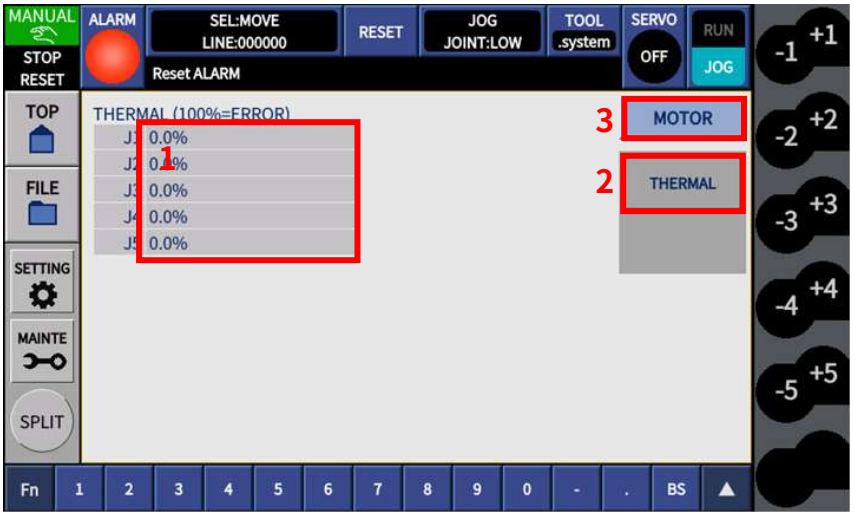
11.6.1 MOTOR (Torque) Screen



The following table provides descriptions of the screen elements.

No. in Figure	Description
1	This shows the current torque value for each joint axis.
2	This shows the torque peak value for each joint axis.
3	This shows the torque limit value for each joint axis.
4	When tapped, this switches the unit for the 3rd axis to be displayed.
5	When tapped, this resets the torque peak value for all joint axes.
6	When tapped, this displays the MOTOR screen selection menu.
7	When tapped, this displays the utility menu.

11.6.2 MOTOR (Thermal) Screen



The following table provides descriptions of the screen elements.

No. in Figure	Description
1	This shows the thermal value for each joint axis.
2	When tapped, this displays the MOTOR screen selection menu.
3	When tapped, this displays the utility menu.

11.6.3 MOTOR (Load) Screen



The following table provides descriptions of the screen elements.

No. in Figure	Description
1	This shows the 1-minute load factor for each joint axis.
2	This shows the 15-minute load factor for each joint axis.
3	This shows the 2-hour load factor for each joint axis.
4	When tapped, this displays the MOTOR screen selection menu.
5	When tapped, this displays the utility menu.

11.6.4 MOTOR (Temperature) Screen



The following table provides descriptions of the screen elements.

No. in Figure	Description
1	This shows the current motor temperature of each joint axis.
2	This shows the motor temperature limit value for each joint axis.
3	When tapped, this displays the MOTOR screen selection menu.
4	When tapped, this displays the utility menu.

11.7 LDISP Screens

These screens show the program sections of the files registered to the execution program.

11.7.1 LDISP (Synchronous Display) Screen



The following table provides descriptions of the screen elements.

No. in Figure	Description
1	This shows the content of the execution program. The line currently being executed is highlighted in blue.
2	When tapped, this switches to the LDISP (Asynchronous Display) screen.
3	When tapped, this displays the utility menu.

11.7.2 LDISP (Asynchronous Display) Screen



The following table provides descriptions of the screen elements.

No. in Figure	Description
1	This shows the content of the execution program. The line currently being selected is highlighted in yellow. When tapped, the cursor moves to the tapped line. The line currently being executed is highlighted in blue.
2	When tapped, this switches to the LDISP (Synchronous Display) screen.
3	When tapped, this displays the utility menu.
4	When tapped, this opens the search window.

The following table shows the available keys in this screen.

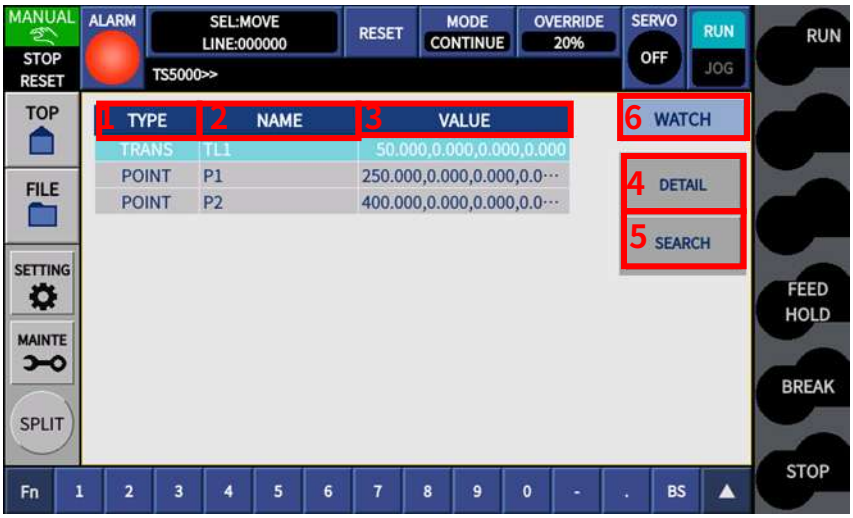
Key Name	Operation When Tapped
↑	Moves the cursor up.
↓	Moves the cursor down.
PageUp	Moves the cursor up by one page.
PageDown	Moves the cursor down by one page.

11.8 WATCH Screens

These screens show the global variables of the files registered to the execution program.

11.8.1 WATCH (List) Screen

This shows a list of the global variables of the execution program.



The following table provides descriptions of the screen elements.

No. in Figure	Description
1	When tapped, this sorts the variables by variable type.
2	When tapped, this sorts the variables in alphabetical order by name.
3	This shows the type, name, and value of the global variable. *A cursor is added to the currently-selected variable. *When tapped, the cursor moves to the tapped line.
4	When tapped, this switches to the selected WATCH (Details) screen.
5	When tapped, this opens the search window.
6	When tapped, this displays the utility menu.

The following table shows the available keys on this screen.

Key Name	Operation When Tapped
↑	Moves the cursor up.
↓	Moves the cursor down.
PageUp	Moves the cursor up by one page.
PageDown	Moves the cursor down by one page.

11.8.2 WATCH (Details) Screen



The following table provides descriptions of the screen elements.

No. in Figure	Description
1	This shows the name of the currently-displayed global variable.
2	When the currently-displayed global variable is array type, this shows the element number.
3	When the currently-displayed global variable is array type, tapping this shows the variables of the element numbers that were input.
4	When tapped, this enables editing of the element value.
5	When tapped, this switches to the WATCH (List) screen.
6	When tapped, the variable value is overwritten by the value in the text box on the left side.
7	This shows the current value of the variable.
8	When tapped, this displays the utility menu.

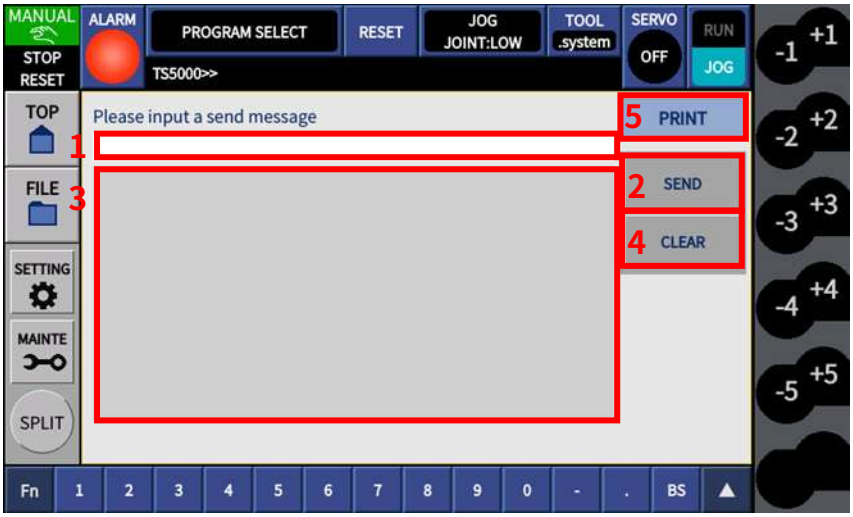
The following table shows the available keys on this screen.

Key Name	Operation When Tapped
←	The cursor in the text box moves to the left (during edition only).
→	The cursor in the text box moves to the right (during edition only).
Letter and numbers	The tapped letters and numbers at the cursor position are entered (during edition only).

11.9 PRINT/INPUT Screen

This screen shows the output to the teach pendant by the PRINT command in the SCOL2 program.
It allows the user to input from the teach pendant by the INPUT command in the SCOL2 program.

11.9.1 PRINT/INPUT Screen



The following table provides descriptions of the screen elements.

No. in Figure	Description
1	When tapped, this enables editing of the value in the text box.
2	When tapped, the character string in the text box is sent to the controller. *This can be tapped only when the controller is in the INPUT wait status.
3	This shows the character string that was sent to the teach pendant (TP) by the PRINT command from the controller. *Up to 64 previous lines of character strings can be displayed.
4	When tapped, this clears the displayed character string that was sent from the controller.
5	When tapped, this displays the utility menu.

The following table shows the available keys in this screen.

Key Name	Operation When Tapped
←	The cursor in the text box moves to the left (during edition only).
→	The cursor in the text box moves to the right (during edition only).
Letters and numbers	The tapped letters and numbers at the cursor position are entered (during edition only).

NOTICE

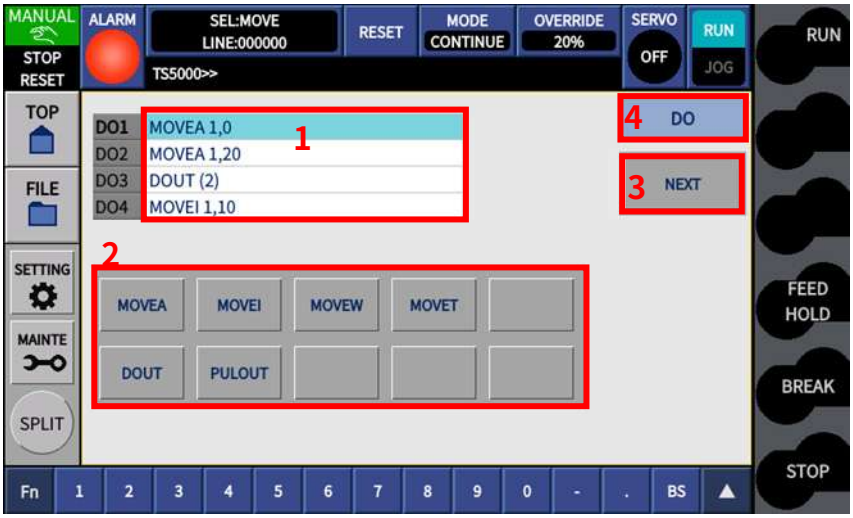
●Before sending INPUT data, check that the numbers and character string are correct. Failure to check it may cause the robot in unexpected motion to collide with the peripheral device.

11.10 DO Screen

This screen enables direct execution.

Before using direct execution, be sure that you understand the information in "3.3. Direct Execution" in this manual.

11.10.1 (DO) Setting Screens



The following table provides descriptions of the screen elements.

No. in Figure	Description
1	When tapped, this enables editing of the value in the text box.
2	When tapped, the displayed movement command is input to the text box currently being edited.
3	When tapped, this opens the DO (Execution) screen.
4	When tapped, this displays the utility menu.

The following table shows the available keys in this screen.

Key Name	Operation When Tapped
←	The cursor in the text box moves to the left (during edition only).
→	The cursor in the text box moves to the right (during edition only).
Letters and numbers	The tapped letters and numbers at the cursor position are entered (during edition only).

MEMO

The character strings in the text boxes are saved to the controller.

11.10.2 DO (Execution) Screen

The displayed operation keys change as shown below while the DO (execution) screen is opened.



The following table provides descriptions of the screen elements.

No. in Figure	Description
1	When tapped, this switches to the DO (Setting) screen.
2	When tapped, this displays the utility menu.

The following table shows the available keys on this screen.

Key Name	Operation When Tapped
Operation keys	When pressed, direct execution is performed for the character string in the text box assigned to the respective key.

MEMO

If you leave the DO (Execution) Screen and tap DO in the utility menu again, the DO (Setting) screen is opened instead of DO (Execution).

11.11 PRG Screen

This screen enables editing of the global section and program section.

For details on editing of the global section and program section, see "4.2. Editing Global Section and Program Section" in this manual.

11.11.1 PRG Screen



The following table provides descriptions of the screen elements.

No. in Figure	Description
1	When tapped, this displays the search menu.
2	When tapped, this displays the utility menu.

The following table shows the available keys on this screen.

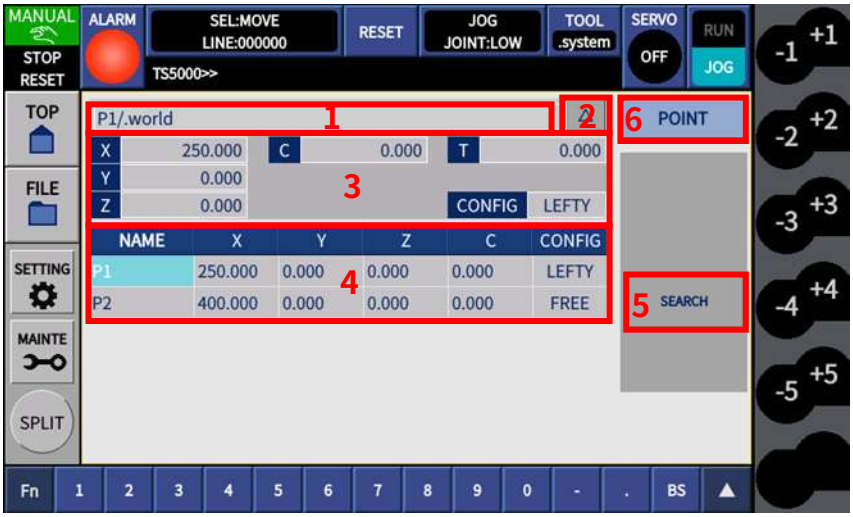
Key Name	Operation When Tapped
↑	Moves the cursor up by one line.
↓	Moves the cursor down by one line.
PageUp	Moves the cursor to previous page.
PageDown	Moves the cursor to next page.
←	Moves the cursor to previous character. If the cursor is at the top of the line, it moves to the end of the line above it.
→	Moves the cursor to next character. If the cursor is at the end of the line, it moves to the top of the line below it.
Enter key	This enters a line feed code to the cursor position.
BS key	This erases the character before the cursor.
INS key	This switches the character input mode.
Letters, numbers, and symbols	This enters the character corresponding to the key at the cursor location.

11.12 POINT Screen

This screen enables editing of the POINT-type data and TRANS-type data of the data section.
For details on editing of the data section, see "4.3. Editing Data Section" in this manual.

11.12.1 POINT (List) Screen

When POINT is tapped in the utility menu on the top screen, this screen is opened.



The following table provides descriptions of the screen elements.

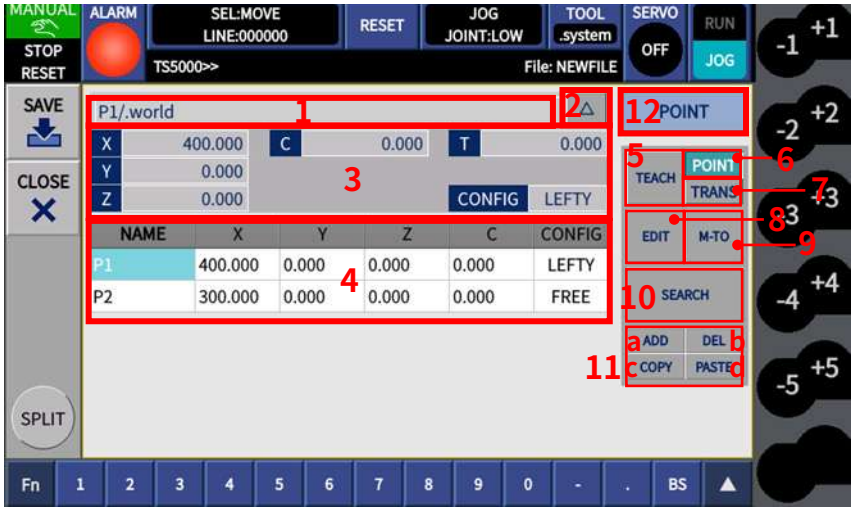
No. in Figure	Description
1	This shows the selected variable name and work name.
2	When tapped, this turns display ON or OFF for the details of the selected variable.
3	This shows the details of the selected variable.
4	This shows the variable name and value for the POINT-type data in the execution program. *A cursor is added to the selected line. *When tapped, the cursor moves to the tapped line.
5	When tapped, this displays the search menu.
6	When tapped, this displays the utility menu.

The following table shows the available keys on this screen.

Key Name	Operation When Tapped
↑	Moves the cursor up by one line.
↓	Moves the cursor down by one line.
←	This switches the displayed columns to NAME, X, Y, Z, C, and T.
→	This switches the displayed columns to NAME, Y, Z, C, T, and CONFIG.
PageUp	Moves the cursor to previous page.
PageDown	Moves the cursor to next page.

11.12.2 POINT (Edit) Screen

When POINT is tapped in the utility menu in the file edit screen, this screen is opened.



The following table provides descriptions of the screen elements.

No. in Figure	Description
1	This shows the selected teaching point and work name.
2	When tapped, this turns display ON or OFF for the details of the selected variable.
3	This shows the details of the selected variable.
4	This shows the variable name and value for the POINT-type data in the file being edited. *A cursor is added to the selected cell. *When tapped, the cursor moves to the tapped cell. *When the selected cell is tapped, the value in the cell can be edited.
5	When tapped, this switches to the TEACH (Edit) screen.
6	When tapped, this opens the POINT (Teach) screen.
7	When tapped, this switches to the TRANS (Edit) screen.
8	When tapped, the value in the selected cell can be edited.
9	When tapped, this opens the POINT (M-TO Setting) screen.
10	When tapped, this displays the search menu.
11	Performs line operation corresponding to each button. a. Adds one line under the selected line. b. Deletes the selected line. c. Copies data of the selected line. d. Overwrites data copied to the selected line.
12	When tapped, this displays the utility menu.

The following table shows the available keys in this screen.

Key Name	Operation When Tapped	
↑	Moves the cursor up by one cell.	
↓	Moves the cursor down by one cell.	
←	For operations other than editing	Moves the cursor one cell to the left. If the cell is in the Y column, the displayed columns are switched to NAME, X, Y, Z, C, and T, and the cursor moves to the X column.
	When editing	Moves the cursor to the left by one space.
→	For operations other than editing	Moves the cursor one cell to the right. If the cell is in the T column, the displayed columns are switched to NAME, Y, Z, C, T, and CONFIG, and the cursor moves to the CONFIG column.
	When editing	Moves the cursor to the right by one space.
PageUp	Moves the cursor to previous page.	
PageDown	Moves the cursor to next page.	
BS key	When editing	This erases the character before the cursor.
ESC	When editing	This erases all the characters in the cell. If no characters are entered in the cell, editing is canceled.
Letters, numbers, and symbols	When editing	This enters the character corresponding to the key at the cursor location.

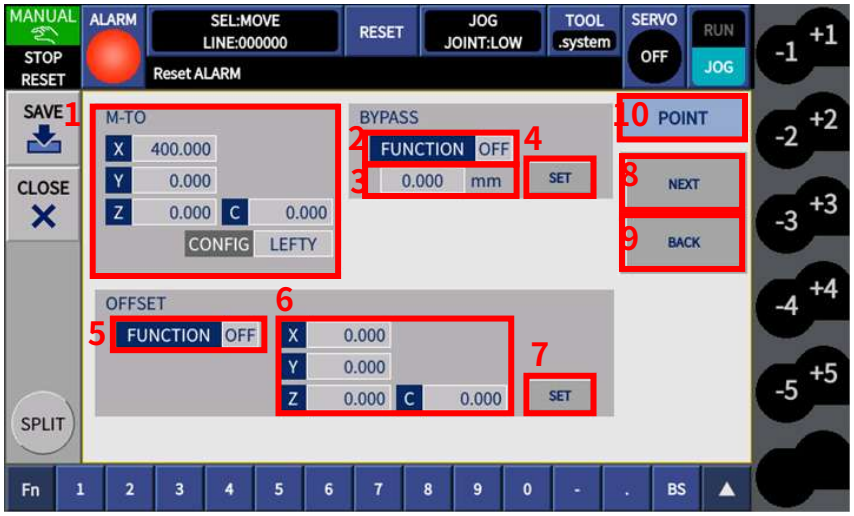
11.12.3 POINT (Teach) Screen



The following table provides descriptions of the screen elements.

No. in Figure	Description
1	This shows the selected variable name.
2	This shows the value of the selected variable.
3	This shows the value (current value of robot TCP) that is overwritten by the selected variable.
4	This shows the variable name and value of the TRANS-type data that is set in the tool coordinates.
5	When tapped, this returns to the POINT (Edit) screen without changing the value.
6	When tapped, this overwrites the value of the selected variable by the current value of the robot TCP and returns to the POINT (Edit) screen.
7	When tapped, this displays the utility menu.

11.12.4 POINT (Teaching Point Movement Setting) Screen



The following table provides descriptions of the screen elements.

No. in Figure	Description	
1	This shows the name and value of the selected variable.	
2	This shows the enable/disable setting for the bypass function.	
3	This shows the Z-axis offset amount for the bypass function.	
4	When tapped, this opens the POINT (Bypass Setting) screen.	
5	This shows the enable/disable setting for the offset function.	
6	This shows the offset amount.	
7	When tapped, this opens the POINT (Offset Setting) screen.	
8	When the bypass function is disabled	When tapped, this switches to the POINT (Teaching Point Movement Execution) screen.
	When the bypass function is enabled	When tapped, this switches to the POINT (Bypass Execution) screen.
9	When tapped, this switches to the POINT (Edit) screen.	
10	When tapped, this displays the utility menu.	

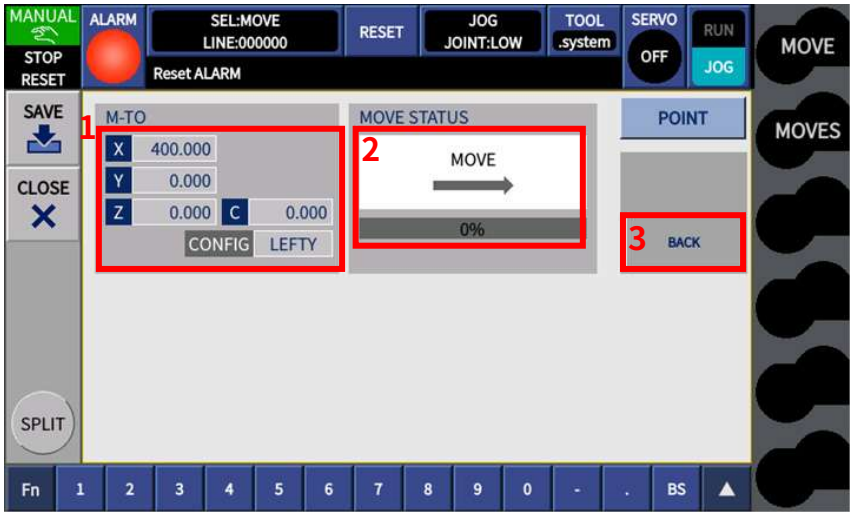
11.12.5 POINT (Offset Setting) Screen



The following table provides descriptions of the screen elements.

No. in Figure	Description
1	When tapped, this disables the offset function.
2	When tapped, this enables the offset function.
3	When tapped, this enables editing of the offset amount.
4	When tapped, this cancels the enable/disable setting and editing of the offset amount and returns to the POINT (Edit) screen.
5	When tapped, this saves the enable/disable setting and the edited offset amount and returns to the POINT (Edit) screen.
6	When tapped, this displays the utility menu.

11.12.6 POINT (Teaching Point Movement Execution) Screen



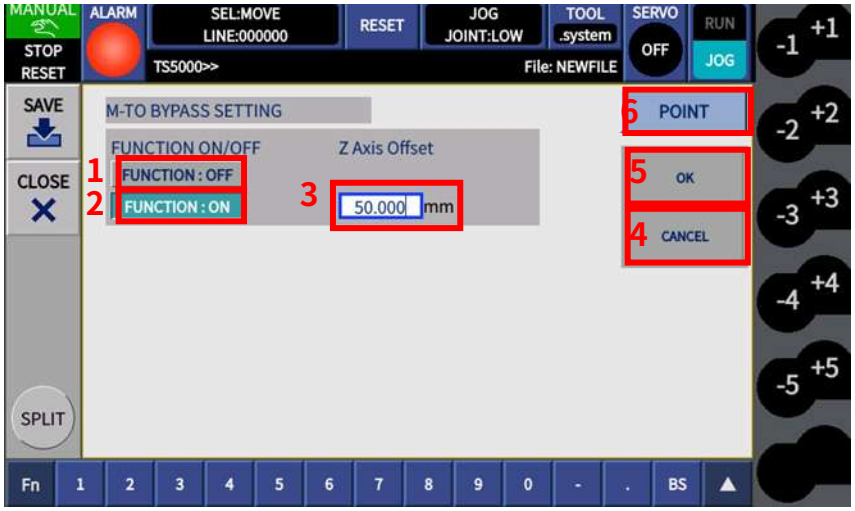
The following table provides descriptions of the screen elements.

No. in Figure	Description
1	This shows the target position of the teaching point movement.
2	This shows the amount moved from starting of robot movement until the target position.
3	When tapped, the screen returns to the POINT (Teaching Point Movement Setting) screen.

The following table shows the available keys on this screen.

Key Name	Operation When Tapped
MOVE key	When this key is held down, the robot TCP continues to move to the target position by PTP movement.
MOVES key	When this key is held down, the robot TCP continues to move to the target position by CP movement.

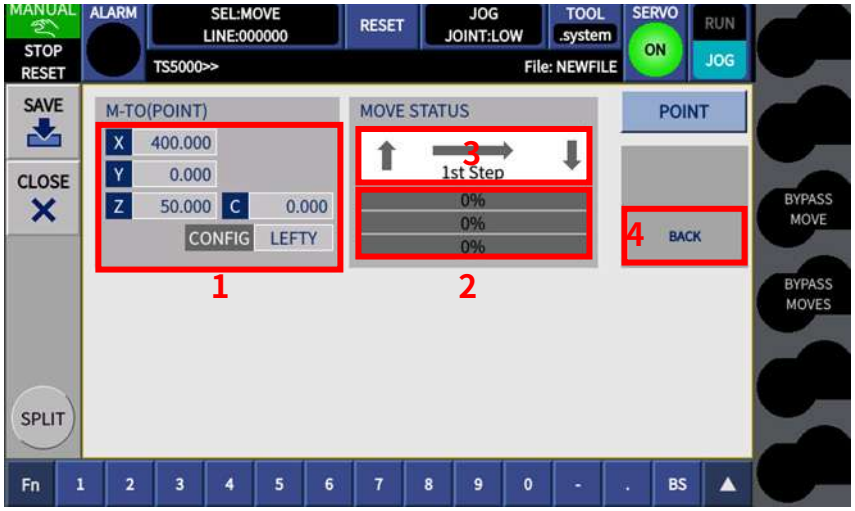
11.12.7 POINT (Bypass Setting) Screen



The following table provides descriptions of the screen elements.

No. in Figure	Description
1	When tapped, this disables the bypass function.
2	When tapped, this enables the bypass function.
3	When tapped, this enables editing of the Z-axis offset amount.
4	When tapped, this cancels the enable/disable setting and editing of the Z-axis offset amount and returns to the POINT (Edit) screen.
5	When tapped, this saves the enable/disable setting and the edited Z-axis offset amount and returns to the POINT (Edit) screen.
6	When tapped, this displays the utility menu.

11.12.8 POINT (Bypass Execution) Screen



The following table provides descriptions of the screen elements.

No. in Figure	Description
1	This shows the target position for each movement in the bypass movement.
2	This shows the amount moved from starting of robot movement until the target position of each movement.
3	This shows the current movement step.
4	When tapped, the screen returns to the POINT (Teaching Point Movement Setting) screen.

The following table shows the available keys on this screen.

Key Name	Operation When Tapped
MOVE key	When this key is held down, the robot TCP continues to move to the target position by PTP movement.
MOVES key	When this key is held down, the robot TCP continues to move to the target position by CP movement.

11.13 TRANS Screen

11.13.1 TRANS (Edit) Screen



The following table provides descriptions of the screen elements.

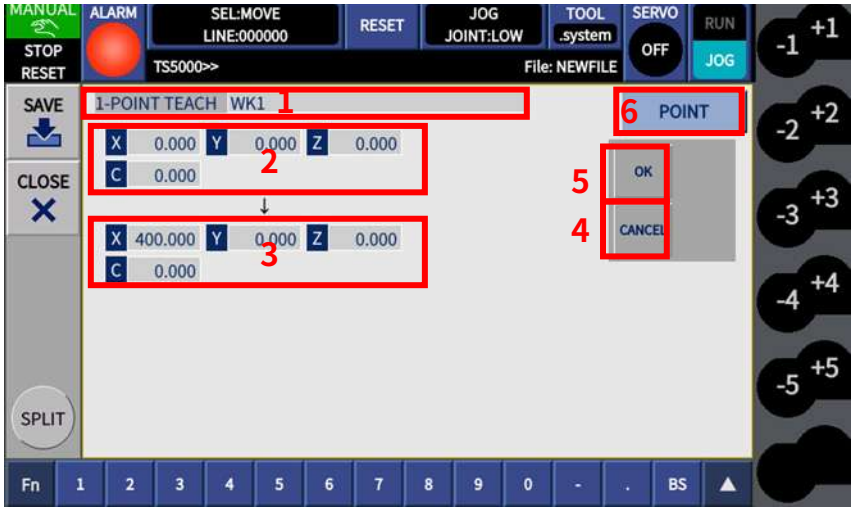
No. in Figure	Description
1	This shows the selected variable name.
2	When tapped, this turns display ON or OFF for the details of the selected variable.
3	This shows the details of the selected variable.
4	This shows the variable name and value for the TRANS-type data in the file being edited. *A cursor is added to the selected cell. *When tapped, the cursor moves to the tapped cell. *When the selected cell is tapped, the value in the cell can be edited. *A "t" is added for TRANS-type data that is registered in the tool coordinates, and a "w" is added for TRANS-type data that is registered in the work coordinates.
5	When tapped, this registers the TRANS-type data in the line where the cursor is located to the work coordinates and switches to the POINT (Edit) screen.
6	When tapped, this displays the POINT (Teach) teach menu.
7	When tapped, this displays the TRANS teach menu.
8	When tapped, this registers the TRANS-type data in the line where the cursor is located to the tool coordinates. *If the TRANS-type data in the line where the cursor is located is already registered to the tool coordinates, the registered tool coordinates are cleared.
9	When tapped, this displays the search menu.
10	Performs line operation corresponding to each button. a. Adds one line under the selected line. b. Delete the selected line. c. Copies data of the selected line.

	d. Overwrites data copied to the selected line.
11	When tapped, this displays the utility menu.

The following table shows the available keys on this screen.

Key Name	Operation When Tapped	
↑	Moves cursor up by one cell.	
↓	Moves cursor down by one cell.	
←	For operations other than editing	Moves the cursor one cell to the left. If the cell is in the Y column, the displayed columns are switched to NAME, X, Y, Z, C, and T, and the cursor moves to the X column.
	When editing	Moves the cursor to the left by one space.
→	For operations other than editing	Moves the cursor one cell to the right. If the cell is in the T column, the displayed columns are switched to NAME, Y, Z, C, T, and CONFIG, and the cursor moves to the CONFIG column.
	When editing	Moves the cursor to the right by one space.
PageUp	Moves the cursor to previous page.	
PageDown	Moves the cursor to next page.	
BS key	When editing	This erases the character before the cursor.
ESC	When editing	This erases all the characters in the cell. If no characters are entered in the cell, editing is canceled.
Letters, numbers, and symbols	When editing	This enters the character corresponding to the key at the cursor location.

11.13.2 TRANS (1-POINT TEACH) Screen



The following table provides descriptions of the screen elements.

No. in Figure	Description
1	This shows the selected variable name.
2	This shows the value of the selected variable.
3	This shows the value (current value of robot TCP) that is overwritten by the selected variable.
4	When tapped, this returns to the POINT (Edit) screen without changing the value.
5	When tapped, this overwrites the value of the selected variable by the current value of the robot TCP and returns to the POINT (Edit) screen.
6	When tapped, this displays the utility menu.

11.13.3 TRANS (2-POINT TEACH) Screen



The following table provides descriptions of the screen elements.

No. in Figure	Description
1	This shows the selected variable name.
2	This shows the coordinate values of the origin and a point on the coordinate axis used for 2-point teaching. *A cursor is added to the selected cell. *When tapped, the cursor moves to the tapped cell. *When the selected cell is tapped, the value in the cell can be edited.
3	When tapped, this overwrites the value of the line where the cursor is located by the current value of the robot TCP.
4	This opens the TRANS (Read Teaching Point) screen.
5	When tapped, this calculates the coordinate value of the TRANS-type data from the origin and a point on the coordinate axis, overwrites the data for the line where the cursor is located in the TRANS (Edit) screen with the calculated coordinate value, and returns to the TRANS (Edit) screen.
6	When tapped, the screen returns to the TRANS (Edit) screen without changing the value.
7	When tapped, this displays the utility menu.

The following table shows the available keys in this screen.

Key Name	Operation When Tapped	
↑	Moves cursor up by one cell.	
↓	Moves cursor down by one cell.	
←	Moves cursor up by one cell.	
→	Moves cursor down by one cell.	
PageUp	Moves cursor to previous page.	
PageDown	Moves cursor to next page.	
BS key	When editing	This erases the character before the cursor.
ESC	When editing	Erases all the characters in the cell. If no characters are entered in the cell, editing is canceled.
Letters, numbers, and symbols	When editing	This enters the character corresponding to the key at the cursor location.

11.13.4 TRANS (Tool Measurement) Screen



The following table provides descriptions of the screen elements.

No. in Figure	Description
1	This shows the selected variable name.
2	This shows the coordinate values for the four points used in tool measurement. *A cursor is added to the selected cell. *When tapped, the cursor moves to the tapped cell. *When the selected cell is tapped, the value in the cell can be edited.

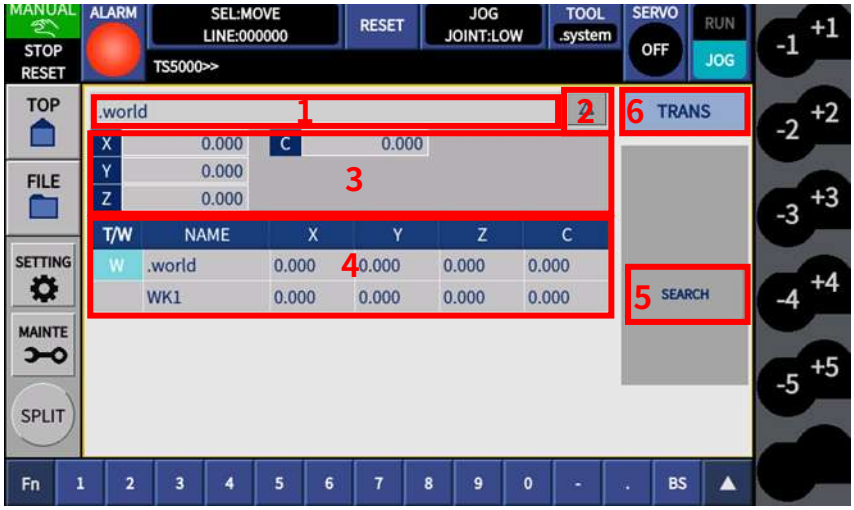
No. in Figure	Description
3	When tapped, this overwrites the value of the line where the cursor is located by the current value of the robot TCP.
4	When tapped, this returns to the TRANS (Edit) screen without changing the value.
5	When tapped, this calculates the coordinate value of the TRANS-type data from the four points, overwrites the data for the line where the cursor is located in the TRANS (Edit) screen with the calculated coordinate value, and returns to the TRANS (Edit) screen.
6	This opens the TRANS (Read Teaching Point) screen.
7	When tapped, this displays the utility menu.

The following table shows the available keys on this screen.

Key Name	Operation When Tapped	
↑	Moves the cursor up by one cell.	
↓	Moves the cursor down by one cell.	
←	Moves the cursor up by one cell.	
→	Moves the cursor down by one cell.	
PageUp	Moves the cursor to previous page.	
PageDown	Moves the cursor to next page.	
BS key	When editing	This erases the character before the cursor.
ESC	When editing	This erases all the characters in the cell. If no characters are entered in the cell, editing is canceled.
Letters, numbers, and symbols	When editing	This enters the character corresponding to the key at the cursor location.

11.13.5 TRANS (List) Screen

When TRANS is tapped in the utility menu in the top screen, this screen is opened.



The following table provides descriptions of the screen elements.

No. in Figure	Description
1	This shows the selected variable name.
2	When tapped, this turns display ON or OFF for the details of the selected variable.
3	This shows the details of the selected variable.
4	This shows the variable name and value for the TRANS-type data in the execution program. *A cursor is added to the selected line. *When tapped, the cursor moves to the tapped line. * "T" is added for TRANS-type data that is registered in the tool coordinates, and "W" is added for TRANS-type data that is registered in the work coordinates.
5	When tapped, this displays the search menu.
6	When tapped, this displays the utility menu.

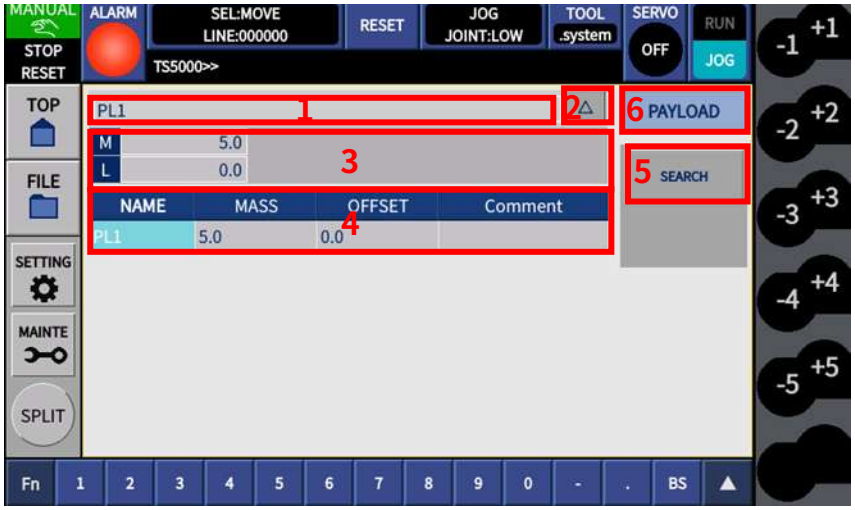
The following table shows the available keys in this screen.

Key Name	Operation When Tapped
↑	Moves the cursor up by one line.
↓	Moves the cursor down by one line.
PageUp	Moves the cursor to previous page.
PageDown	Moves the cursor to next page.

11.14 PAYLOAD Screens

11.14.1 PAYLOAD (List) Screen

When PAYLOAD is tapped in the utility menu in the top screen, this screen is opened.



The following table provides descriptions of the screen elements.

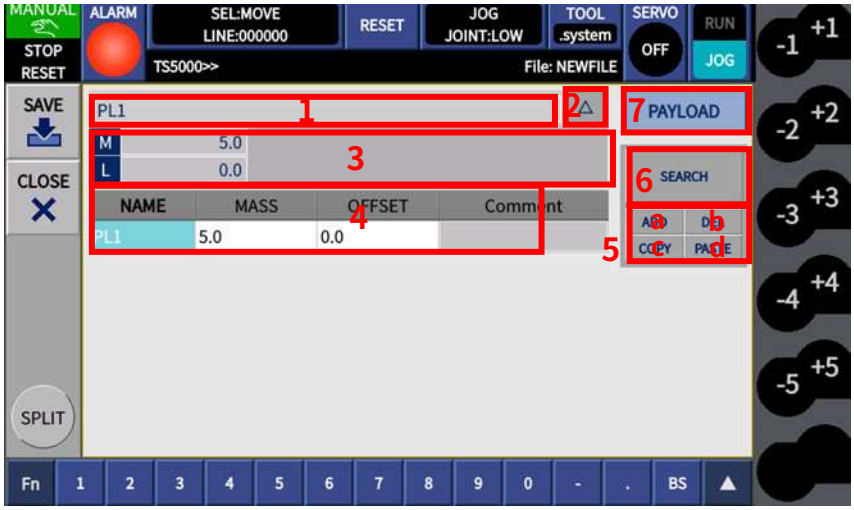
No. in Figure	Description
1	This shows the selected variable name.
2	When tapped, this turns display ON or OFF for the details of the selected variable.
3	This shows the details of the selected variable.
4	This shows the variable name and value for the PAYLOAD-type data in the execution program. *A cursor is added to the selected line. *When tapped, the cursor moves to the tapped line.
5	When tapped, this displays the search menu.
6	When tapped, this displays the utility menu.

The following table shows the available keys on this screen.

Key Name	Operation When Tapped
↑	Moves the cursor up by one line.
↓	Moves the cursor down by one line.
PageUp	Moves the cursor to previous page.
PageDown	Moves the cursor to next page.

11.14.2 PAYLOAD (Edit) Screen

When PAYLOAD is tapped in the utility menu in the file edit screen, this screen is opened.



The following table provides descriptions of the screen elements.

No. in Figure	Description
1	This shows the selected variable name.
2	When tapped, this turns display ON or OFF for the details of the selected variable.
3	This shows the details of the selected variable.
4	This shows the variable name and value for the PAYLOAD-type data in the file being edited. *A cursor is added to the selected cell. *When tapped, the cursor moves to the tapped cell. *When the selected cell is tapped, the value in the cell can be edited.
5	Performs line operation corresponding to each button. a. Adds one line under the selected line. b. Deletes the selected line. c. Copies data of the selected line. d. Overwrites data copied to the selected line.
6	When tapped, this displays the search menu.
7	When tapped, this displays the utility menu.

The following table shows the available keys on this screen.

Key Name	Operation When Tapped	
↑	Moves the cursor up by one cell.	
↓	Moves the cursor down by one cell.	
←	For operations other than editing	Moves the cursor one cell to the left.
	When editing	Moves the cursor to the left by one space.
→	For operations other than editing	Moves the cursor one cell to the right.
	When editing	Moves the cursor to the right by one space.
PageUp	Moves cursor to previous page.	
PageDown	Moves cursor to next page.	
BS key	When editing	This erases the character before the cursor.
ESC	When editing	This erases all the characters in the cell. If no characters are entered in the cell, editing is canceled.
Letters, numbers, and symbols	When editing	This enters the character corresponding to the key at the cursor location.

11.15 JOINT Screen

11.15.1 JOINT (List) Screen

When JOINT is tapped in the utility menu on the top screen, this screen is opened.



The following table provides descriptions of the screen elements.

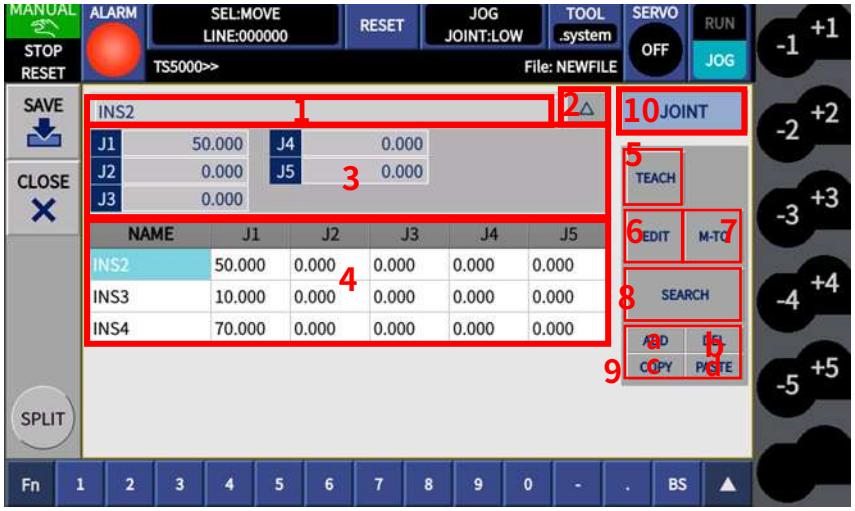
No. in Figure	Description
1	This shows the selected variable name.
2	When tapped, this turns display ON or OFF for the details of the selected variable.
3	This shows the details of the selected variable.
4	This shows the variable name and value for the JOINT-type data in the execution program. *A cursor is added to the selected cell. *When tapped, the cursor moves to the tapped cell.
5	When tapped, this displays the search menu.
6	When tapped, this displays the utility menu.

The following table shows the available keys on this screen.

Key Name	Operation When Tapped
↑	Moves the cursor up by one line.
↓	Moves the cursor down by one line.
←	Switches the displayed column to one left column.
→	Switches the displayed column to one right column.
PageUp	Moves the cursor to previous page.
PageDown	Moves the cursor to next page.

11.15.2 JOINT (Edit) Screen

When JOINT is tapped in the utility menu on the file edit screen, this screen is opened.



The following table provides descriptions of the screen elements.

No. in Figure	Description
1	This shows the selected variable name.
2	When tapped, this turns display ON or OFF for the details of the selected variable.
3	This shows the details of the selected variable.
4	This shows the variable name and value for the JOINT-type data in the file being edited. *A cursor is added to the selected cell. *When tapped, the cursor moves to the tapped cell. *When the selected cell is tapped, the value in the cell can be edited.
5	When tapped, this switches to the JOINT (Teach) screen.
6	When tapped, the value in the selected cell can be edited.
7	When tapped, this opens the JOINT (M-TO Setting) screen.
8	When tapped, this displays the search menu.
9	Performs line operation corresponding to each button. a. Adds one line under the selected line. b. Deletes the selected line. c. Copies data of the selected line. d. Overwrites data copied to the selected line.
10	When tapped, this displays the utility menu.

The following table shows the available keys on this screen.

Key Name	Operation When Tapped	
↑	Moves the cursor up by one cell.	
↓	Moves the cursor down by one cell.	
←	Moves the cursor one cell to the left.	
→	Moves the cursor one cell to the right.	
PageUp	Moves cursor to previous page.	
PageDown	Moves cursor to next page.	
BS key	When editing	This erases the character before the cursor.
ESC	When editing	This erases all the characters in the cell. If no characters are entered in the cell, editing is canceled.
Letters, numbers, and symbols	When editing	This enters the character corresponding to the key at the cursor location.

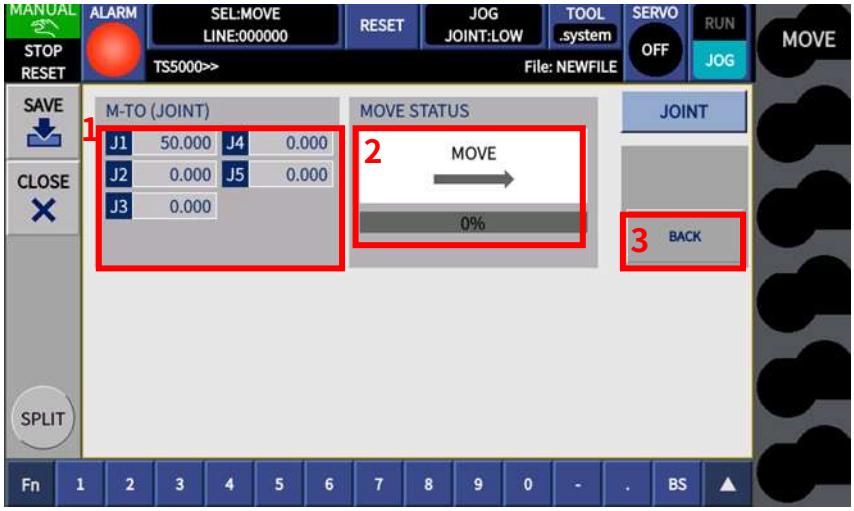
11.15.3 JOINT (Teach) Screen



The following table provides descriptions of the screen elements.

No. in Figure	Description
1	This shows the selected variable name.
2	This shows the value of the selected variable.
3	This shows the value (current value of robot joint axis) that is overwritten by the selected variable.
4	When tapped, this returns to the JOINT (Edit) screen without changing the value.
5	When tapped, this overwrites the value of the selected variable by the current value of robot joint axis and returns to the JOINT (Edit) screen.
6	When tapped, this displays the utility menu.

11.15.4 JOINT (Teaching Point Movement Execution) Screen



The following table provides descriptions of the screen elements.

No. in Figure	Description
1	This shows the target position of the teaching point movement.
2	This shows the amount moved from starting of robot movement until the target position.
3	When tapped, the screen returns to the JOINT (Teaching Point Movement Setting) screen.

The following table shows the available keys on this screen.

Key Name	Operation When Tapped
MOVE key	When this key is held down, the robot TCP continues to move to the target position by PTP movement.

11.16 AUX Screen

This screen shows and operates AUX registered to user parameter [U018] AUX SIGNAL.

11.16.1 AUX (List) Screen

An AUX list is displayed.



The following table provides descriptions of the screen elements.

No. in Figure	Description
1	This shows the AUX numbers and registered names. *A cursor is added to the selected AUX. *When tapped, the cursor moves to the tapped line.
2	This shows the registered DOUT signal number. When the registered number is 0, this shows "-."
3	This switches the signal to ON or OFF.
4	When tapped, this switches to the AUX (Detail) screen of the selected AUX.
5	When tapped, this updates the window.
6	When tapped, this displays the utility menu.

The following table shows the available keys on this screen.

Key Name	Operation When Tapped
↑	Moves the cursor up.
↓	Moves the cursor down.
PageUp	Moves the cursor up by one page.
PageDown	Moves the cursor down by one page.

NOTICE

●Before switching the signal to ON or OFF, check the signal number. Failure to check it may cause the robot in unexpected motion to collide with the peripheral device and workpiece.

11.16.2 AUX (Details) Screen



The following table provides descriptions of the screen elements.

No. in Figure	Description
1	The name to be set for the selected AUX is entered.
2	Set the first signal number.
3	Set the second signal number. To use as single solenoid, enter 0. To use as double solenoid, enter a numeric value.
4	This displays the currently set AUX.
5	This switches the currently set AUX to ON or OFF.
6	When tapped, this returns to the AUX (List) screen.
7	When tapped, this registers the AUX name, first signal number, and second signal number.
8	When tapped, this displays the utility menu.

The following table shows the available keys on this screen.

Key Name	Operation When Tapped
←	The cursor in the text box moves to the left (during editing only).
→	The cursor in the text box moves to the right (during editing only).
Letter and numbers	This enters the tapped letters and numbers at the cursor location (during editing only).

11.17 IP WATCH Screen

This section displays the contents of communication used over each IP.

11.17.1 IP WATCH Screen

The IP WATCH screen is opened.



The following table provides descriptions of the screen elements.

No. in Figure	Description
1	This displays a list of communication data. I/O: Display the transmission/reception direction. Communication data: Display the communication contents. *When tapped, the cursor moves to the tapped line. *The data in the lower part is latest.
2	This displays the controller, IP address of IP, and communication direction in the selected line.
3	This displays the details of communication contents in the selected line.
4	When tapped, this displays the IP switch menu.
5	When tapped, this switches between start and stop of log acquisition. *When log acquisition is not started, log is not acquired.
6	When tapped, this displays the utility menu.

The following table shows the available keys on this screen.

Key Name	Operation When Tapped
↑	Moves the cursor up.
↓	Moves the cursor down.

11.18 CONV Screen

This screen displays data related to the conveyor synchronization function.

This screen is displayed on 8KTA-02A or later versions.

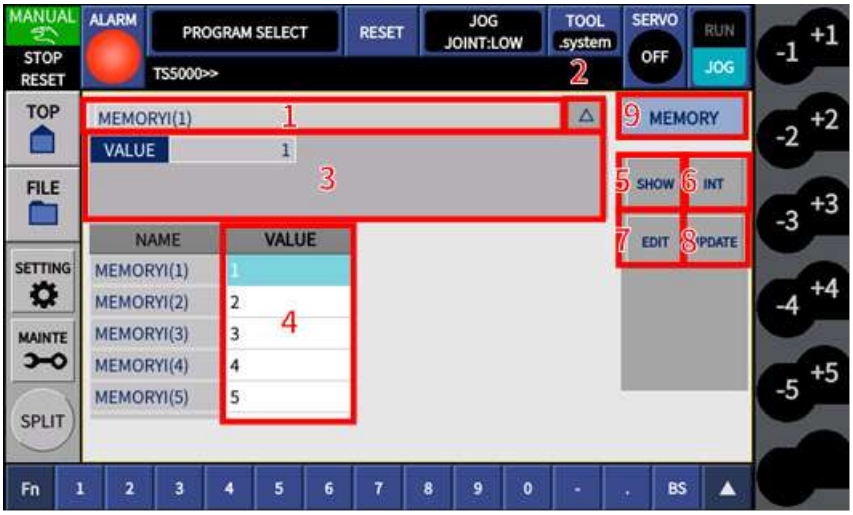
For the details of the CONV screen, see the separate "STA1183 Vision + Conveyor Synchronization Manual".

11.19 MEMORY Screen

This screen displays memory variable information.

This screen is displayed on X8KTA-03C or later versions.

11.19.1 MEMORYI Screen



The following table provides descriptions of the screen elements.

No. in Figure	Description
1	Displays the name of the selected memory variable.
2	When tapped, this turns display ON or OFF for the details of the selected variable.
3	Displays the details of the selected memory variable.
4	Displays the INT-type memory variables. *A cursor is added to the selected cell. *When tapped, the cursor moves to the tapped cell. *When the selected cell is tapped, the value in the cell can be edited.
5	When tapped, the variable number selection menu is displayed.
6	When tapped, the variable type selection menu is displayed.
7	When tapped, the value in the selected cell can be edited.

No. in Figure	Description
8	When tapped, the value of the selected memory variable is overwritten by the value in the cell.
9	When tapped, this displays the utility menu.

The following table shows the available keys on this screen.

Key Name	Operation When Tapped	
↑	Moves the cursor up.	
↓	Moves the cursor down.	
←	When editing	Moves the cursor to the left by one space.
→	When editing	Moves the cursor to the right by one space.
PageUp	Moves cursor to previous page.	
PageDown	Moves cursor to next page.	
BS key	When editing	This erases the character before the cursor.
ESC	When editing	This erases all the characters in the cell. If no characters are entered in the cell, editing is canceled.
Letters, numbers, and symbols	When editing	This enters the character corresponding to the key at the cursor location.

11.19.2 MEMORYR Screen



The following table provides descriptions of the screen elements.

No. in Figure	Description
1	Displays the name of the selected memory variable.
2	When tapped, this turns display ON or OFF for the details of the selected variable.
3	Displays the details of the selected memory variable.
4	Displays the REAL-type memory variables. *A cursor is added to the selected cell. *When tapped, the cursor moves to the tapped cell. *When the selected cell is tapped, the value in the cell can be edited.
5	When tapped, the variable number selection menu is displayed.
6	When tapped, the variable type selection menu is displayed.
7	When tapped, the value in the selected cell can be edited.
8	When tapped, the value of the selected memory variable is overwritten by the value in the cell.
9	When tapped, this displays the utility menu.

The following table shows the available keys on this screen.

Key Name	Operation When Tapped	
↑	Moves the cursor up.	
↓	Moves the cursor down.	
←	When editing	Moves the cursor to the left by one space.
→	When editing	Moves the cursor to the right by one space.
PageUp	Moves cursor to previous page.	
PageDown	Moves cursor to next page.	
BS key	When editing	This erases the character before the cursor.
ESC	When editing	This erases all the characters in the cell. If no characters are entered in the cell, editing is canceled.
Letters, numbers, and symbols	When editing	This enters the character corresponding to the key at the cursor location.

11.19.3 MEMORT Screen



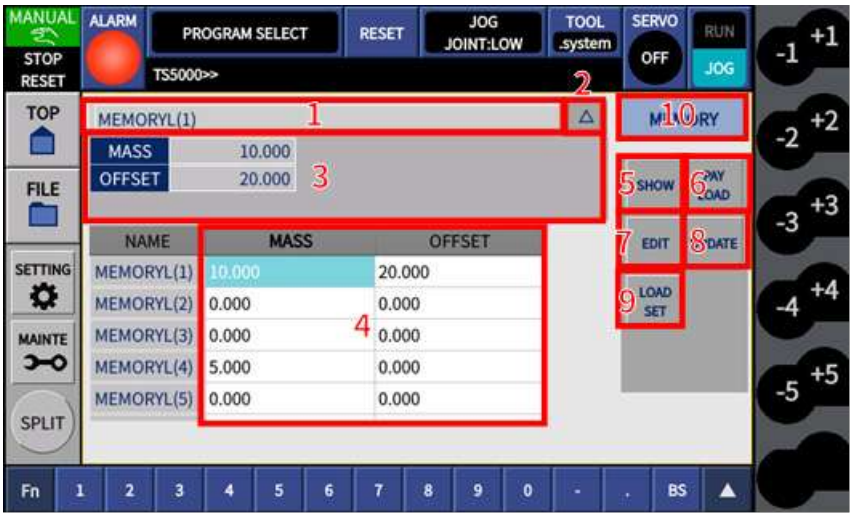
The following table provides descriptions of the screen elements.

No. in Figure	Description
1	Displays the name of the selected memory variable.
2	When tapped, this turns display ON or OFF for the details of the selected variable.
3	Displays the details of the selected memory variable.
4	Displays the TRANS-type memory variables. *A cursor is added to the selected cell. *When tapped, the cursor moves to the tapped cell. *When the selected cell is tapped, the value in the cell can be edited.
5	When tapped, the variable number selection menu is displayed.
6	When tapped, the variable type selection menu is displayed.
7	When tapped, the value in the selected cell can be edited.
8	When tapped, the value of the selected memory variable is overwritten by the value in the cell.
9	When tapped, the TRANS-type memory variables in the line where the cursor is located are set to the work coordinates.
10	When tapped, the TRANS-type memory variables in the line where the cursor is located are set to the tool coordinates.
11	When tapped, this displays the utility menu.

The following table shows the available keys on this screen.

Key Name	Operation When Tapped	
↑	Moves the cursor up by one cell.	
↓	Moves the cursor down by one cell.	
←	For operations other than editing	Moves the cursor one cell to the left.
	When editing	Moves the cursor to the left by one space.
→	For operations other than editing	Moves the cursor one cell to the right.
	When editing	Moves the cursor to the right by one space.
PageUp	Moves cursor to previous page.	
PageDown	Moves cursor to next page.	
BS key	When editing	This erases the character before the cursor.
ESC	When editing	This erases all the characters in the cell. If no characters are entered in the cell, editing is canceled.
Letters, numbers, and symbols	When editing	This enters the character corresponding to the key at the cursor location.

11.19.4 MEMORYL Screen



The following table provides descriptions of the screen elements.

No. in Figure	Description
1	Displays the name of the selected memory variable.
2	When tapped, this turns display ON or OFF for the details of the selected variable.
3	Displays the details of the selected memory variable.
4	Displays the PAYLOAD-type memory variables. *A cursor is added to the selected cell. *When tapped, the cursor moves to the tapped cell. *When the selected cell is tapped, the value in the cell can be edited.
5	When tapped, the variable number selection menu is displayed.
6	When tapped, the variable type selection menu is displayed.
7	When tapped, the value in the selected cell can be edited.
8	When tapped, the value of the selected memory variable is overwritten by the value in the cell.
9	When tapped, the PAYLOAD-type memory variables in the line where the cursor is located are set to the manual guidance load.
10	When tapped, this displays the utility menu.

The following table shows the available keys on this screen.

Key Name	Operation When Tapped	
↑	Moves the cursor up by one cell.	
↓	Moves the cursor down by one cell.	
←	For operations other than editing	Moves the cursor one cell to the left.
	When editing	Moves the cursor to the left by one space.
→	For operations other than editing	Moves the cursor one cell to the right.
	When editing	Moves the cursor to the right by one space.
PageUp	Moves cursor to previous page.	
PageDown	Moves cursor to next page.	
BS key	When editing	This erases the character before the cursor.
ESC	When editing	This erases all the characters in the cell. If no characters are entered in the cell, editing is canceled.
Letters, numbers, and symbols	When editing	This enters the character corresponding to the key at the cursor location.

11.19.5 MEMORYP Screen



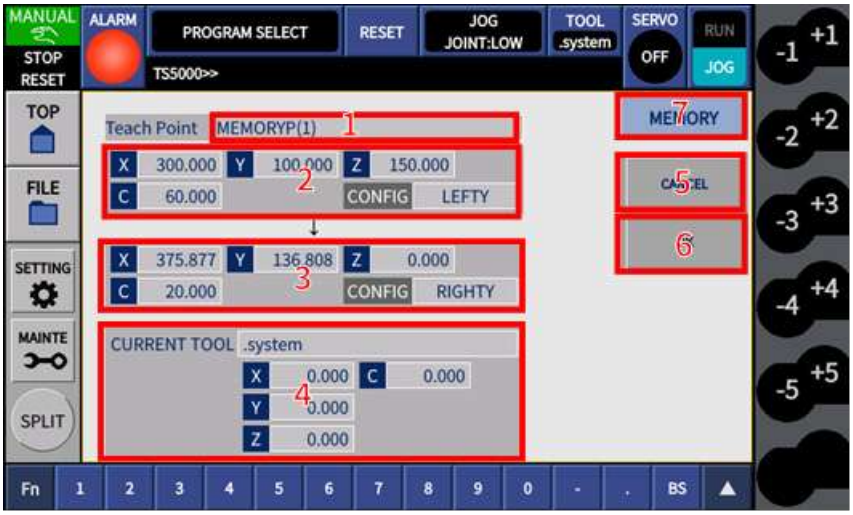
The following table provides descriptions of the screen elements.

No. in Figure	Description
1	Displays the name of the selected memory variable.
2	When tapped, this turns display ON or OFF for the details of the selected variable.
3	Displays the details of the selected memory variable.
4	Displays the POINT-type memory variables. *A cursor is added to the selected cell. *When tapped, the cursor moves to the tapped cell. *When the selected cell is tapped, the value in the cell can be edited.
5	When tapped, the variable number selection menu is displayed.
6	When tapped, the variable type selection menu is displayed.
7	When tapped, the value in the selected cell can be edited.
8	When tapped, the value of the selected memory variable is overwritten by the value in the cell.
9	When tapped, the MEMORYP (Teach) screen is displayed.
10	When tapped, the MEMORYP (Teaching Point Movement) screen is displayed.
11	When tapped, this displays the utility menu.

The following table shows the available keys on this screen.

Key Name	Operation When Tapped	
↑	Moves the cursor up by one cell.	
↓	Moves the cursor down by one cell.	
←	For operations other than editing	Moves the cursor one cell to the left.
	When editing	Moves the cursor to the left by one space.
→	For operations other than editing	Moves the cursor one cell to the right.
	When editing	Moves the cursor to the right by one space.
PageUp	Moves cursor to previous page.	
PageDown	Moves cursor to next page.	
BS key	When editing	This erases the character before the cursor.
ESC	When editing	This erases all the characters in the cell. If no characters are entered in the cell, editing is canceled.
Letters, numbers, and symbols	When editing	This enters the character corresponding to the key at the cursor location.

11.19.6 MEMORYP (Teach) Screen



The following table provides descriptions of the screen elements.

No. in Figure	Description
1	Displays the name of the selected memory variable.
2	Displays the details of the selected memory variable.
3	Displays the current value of the robot TCP.
4	This shows the variable name and value of the TRANS-type data that is set in the tool coordinates.
5	When tapped, this returns to the MEMORYP screen without changing the value.
6	When tapped, this overwrites the value of the selected cell by the current value of the robot TCP and returns to the MEMORYP screen.
7	When tapped, this displays the utility menu.

11.19.7 MEMORYP (Teaching Point Movement) Screen



The following table provides descriptions of the screen elements.

No. in Figure	Description	
1	Displays the name of the selected memory variable.	
2	This shows the enable/disable setting for the bypass function.	
3	This shows the Z-axis offset amount for the bypass function.	
4	When tapped, this opens the MEMORYP (Bypass Setting) screen.	
5	This shows the enable/disable setting for the offset function.	
6	This shows the offset amount.	
7	When tapped, this opens the MEMORYP (Offset Setting) screen.	
8	When the bypass function is disabled	When tapped, this switches to the MEMORYP (Teaching Point Movement Execution) screen.
	When the bypass function is enabled	When tapped, this switches to the MEMORYP (Bypass Execution) screen.
9	When tapped, this switches to the MEMORYP screen.	
10	When tapped, this displays the utility menu.	

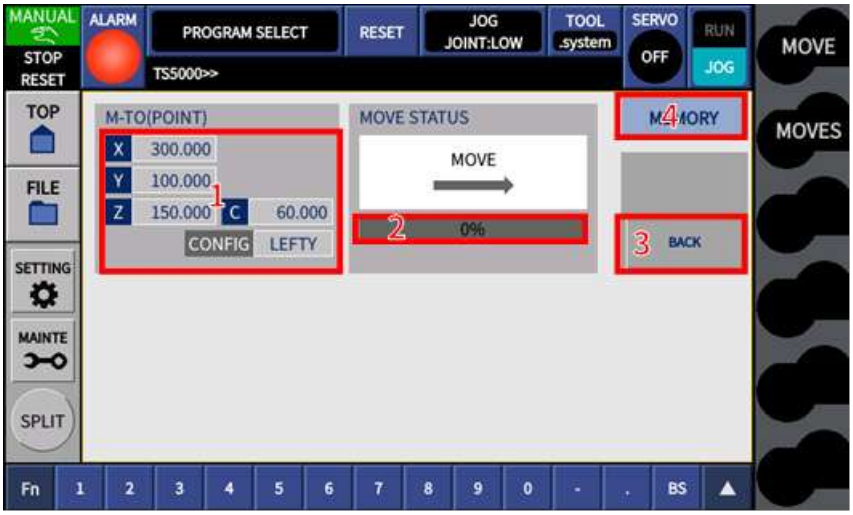
11.19.8 MEMORYP (Offset Setting) Screen



The following table provides descriptions of the screen elements.

No. in Figure	Description
1	When tapped, this disables the offset function.
2	When tapped, this enables the offset function.
3	When tapped, this enables editing of the offset amount.
4	When tapped, this saves the enable/disable setting and the edited offset amount and returns to the MEMORYP (Teaching Point Movement) screen.
5	When tapped, this cancels the enable/disable setting and editing of the offset amount and returns to the MEMORYP (Teaching Point Movement) screen.
6	When tapped, this displays the utility menu.

11.19.9 MEMORYP (Teaching Point Movement Execution) Screen



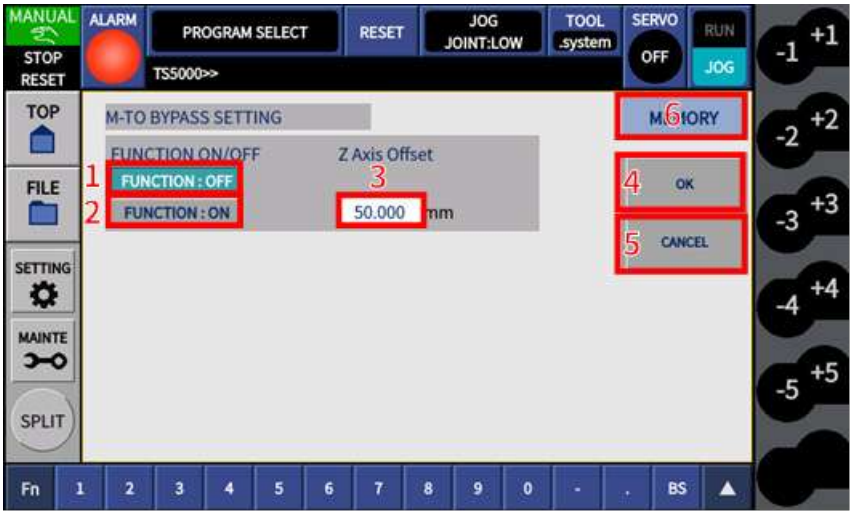
The following table provides descriptions of the screen elements.

No. in Figure	Description
1	This shows the target position of the teaching point movement.
2	This shows the amount moved from starting of robot movement until the target position.
3	When tapped, the screen returns to the MEMORYP (Teaching Point Movement Setting) screen.
4	When tapped, this displays the utility menu.

The following table shows the available keys on this screen.

Key Name	Operation When Tapped
MOVE key	While this key is held down, the robot TCP continues to perform PTP movement toward the target position.
MOVES key	While this key is held down, the robot TCP continues to perform CP movement toward the target position.

11.19.10 MEMORYP (Bypass Setting) Screen



The following table provides descriptions of the screen elements.

No. in Figure	Description
1	When tapped, this disables the bypass function.
2	When tapped, this enables the bypass function.
3	When tapped, this enables editing of the Z-axis offset amount.
4	When tapped, this saves the enable/disable setting and the edited Z-axis offset amount and returns to the MEMORYP (Teaching Point Movement) screen.
5	When tapped, this cancels the enable/disable setting and editing of the Z-axis offset amount and returns to the MEMORYP (Teaching Point Movement) screen.
6	When tapped, this displays the utility menu.

11.19.11 MEMORYP (Bypass Execution) Screen



The following table provides descriptions of the screen elements.

No. in Figure	Description
1	This shows the target position for each movement in the bypass movement.
2	This shows the amount moved from starting of robot movement until the target position of each movement.
3	This shows the current movement step.
4	When tapped, the screen returns to the MEMORYP (Teaching Point Movement) screen.

The following table shows the available keys on this screen.

Key Name	Operation When Tapped
BYPASS MOVE key	While this key is held down, the robot TCP continues to perform PTP movement to bypass toward the target position.
BYPASS MOVES key	While this key is held down, the robot TCP continues to perform CP movement to bypass toward the target position.

11.19.12 MEMORYJ Screen



NAME	J1	J2	J3	J4	J5
MEMORYJ(1)	100.000	200.000	300.000	400.000	500.000
MEMORYJ(2)	50.000	20.000	0.500	0.500	0.000
MEMORYJ(3)	-66.565	76.656	43.100	-10.093	0.000
MEMORYJ(4)	0.000	0.000	0.000	0.000	0.000
MEMORYJ(5)	0.000	0.000	0.000	0.000	0.000

The following table provides descriptions of the screen elements.

No. in Figure	Description
1	Displays the name of the selected memory variable.
2	When tapped, this turns display ON or OFF for the details of the selected variable.
3	Displays the details of the selected memory variable.
4	Displays the variable name and value for the JOINT-type data in the file being edited. *A cursor is added to the selected cell. *When tapped, the cursor moves to the tapped cell. *When the selected cell is tapped, the value in the cell can be edited.
5	When tapped, the variable number selection menu is displayed.
6	When tapped, the variable type selection menu is displayed.
7	When tapped, the value in the selected cell can be edited.
8	When tapped, the value of the selected memory variable is overwritten by the value in the cell.
9	When tapped, the display switches to the MEMORYJ (Teach) screen.
10	When tapped, the MEMORYJ (Teaching Point Movement) screen is displayed.
11	When tapped, this displays the utility menu.

The following table shows the available keys on this screen.

Key Name	Operation When Tapped	
↑	Moves the cursor up by one cell.	
↓	Moves the cursor down by one cell.	
←	Moves the cursor one cell to the left.	
→	Moves the cursor one cell to the right.	
PageUp	Moves cursor to previous page.	
PageDown	Moves cursor to next page.	
BS key	When editing	This erases the character before the cursor.
ESC	When editing	This erases all the characters in the cell. If no characters are entered in the cell, editing is canceled.
Letters, numbers, and symbols	When editing	This enters the character corresponding to the key at the cursor location.

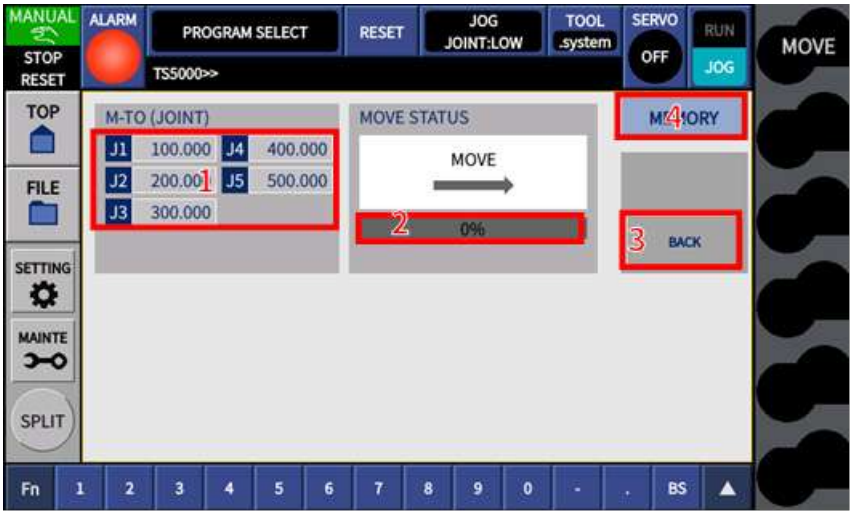
11.19.13 MEMORYJ (Teach) Screen



The following table provides descriptions of the screen elements.

No. in Figure	Description
1	This shows the selected variable name.
2	This shows the value of the selected variable.
3	Displays the current value of the robot joint axis.
4	When tapped, this returns to the MEMORYJ screen without changing the value.
5	When tapped, this overwrites the value of the selected variable by the current value of the robot joint axis and returns to the MEMORYJ screen.
6	When tapped, this displays the utility menu.

11.19.14 MEMORYJ (Teaching Point Movement) Screen



The following table provides descriptions of the screen elements.

No. in Figure	Description
1	This shows the target position of the teaching point movement.
2	This shows the amount moved from starting of robot movement until the target position.
3	When tapped, this returns to the MEMORYJ screen.
4	When tapped, this displays the utility menu.

The following table shows the available keys on this screen.

Key Name	Operation When Tapped
MOVE key	While this key is held down, the robot TCP continues to perform PTP movement toward the target position.

11.19.15 Variable Type Selection Menu

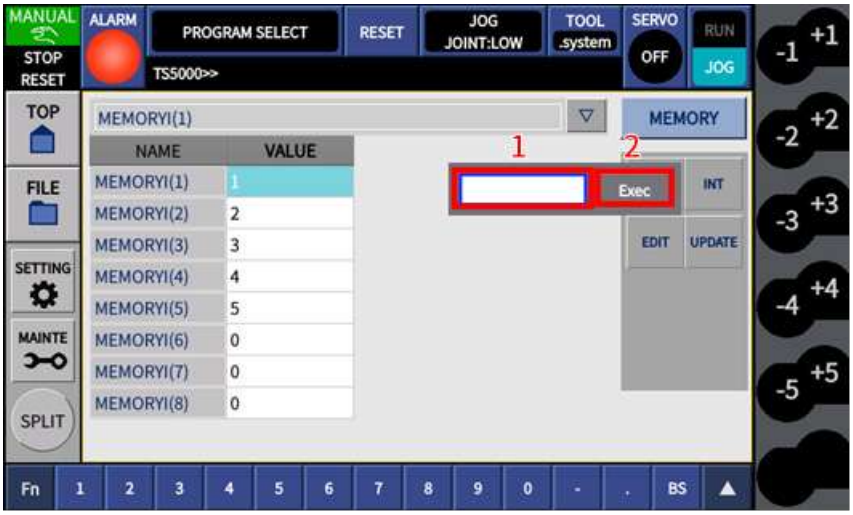
When variable type selection menu is displayed



No. in Figure	Description
1	When tapped, this switches to the MEMORYI screen.
2	When tapped, this switches to the MEMORYR screen.
3	When tapped, this switches to the MEMORYT screen.
4	When tapped, this switches to the MEMORYL screen.
5	When tapped, this switches to the MEMORYP screen.
6	When tapped, this switches to the MEMORYJ screen.
Other than above	When tapped, this closes the variable type selection menu.

11.19.16 Variable Number Selection Menu

When variable number selection menu is displayed



The following table provides descriptions of the screen elements.

No. in Figure	Description
1	Enter the variable number. When tapped, the content can be edited.
2	When tapped, the memory variable of the set variable number is displayed.
Other than above	When tapped, this closes the variable number selection menu.

The following table shows the available keys on this screen.

Key Name	Operation When Tapped
←	The cursor in the text box moves to the left (during edition only).
→	The cursor in the text box moves to the right (during edition only).
Letters and numbers	The tapped letters and numbers at the cursor position are entered (during edition only).

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